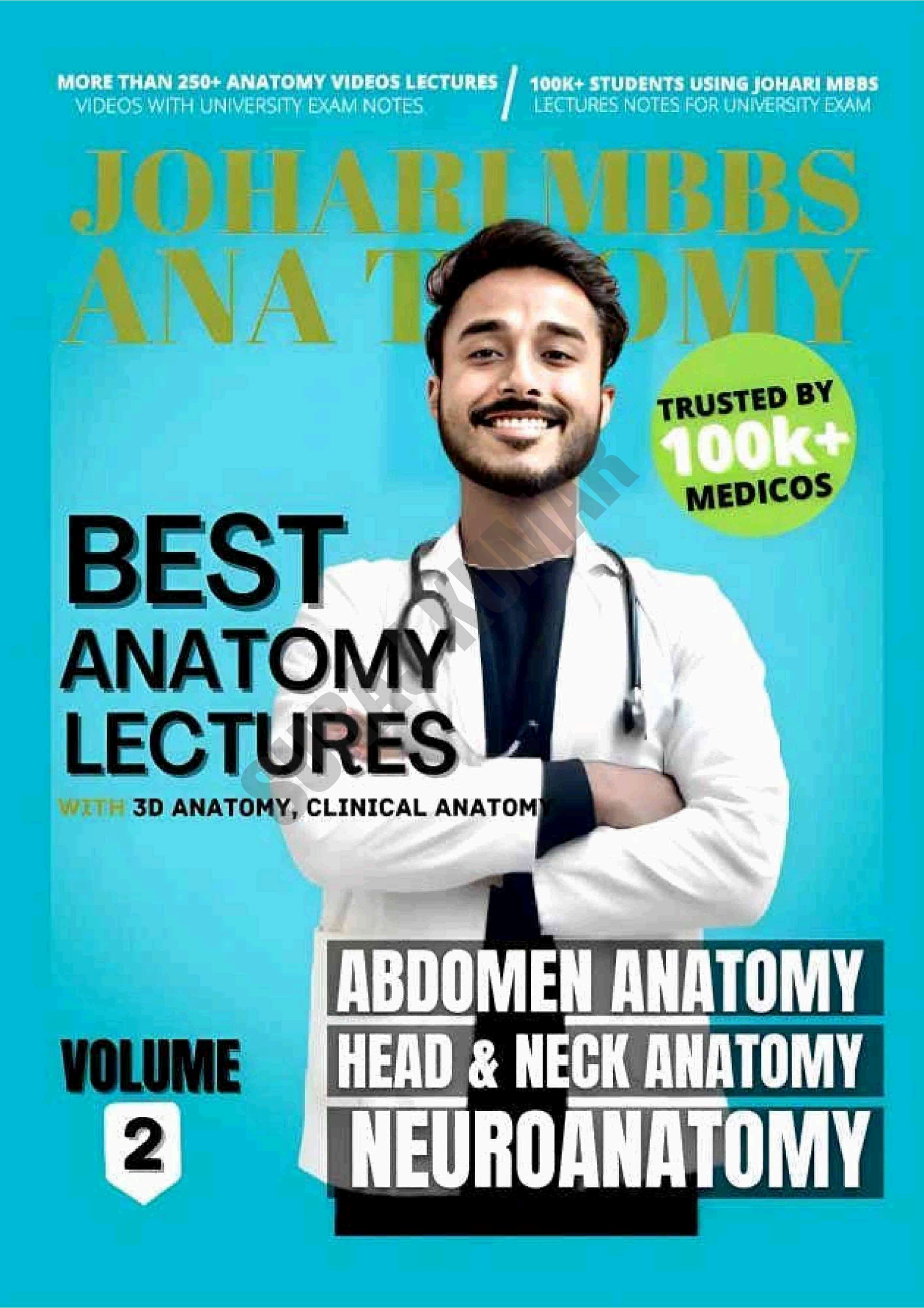
MORE THAN 250+ ANATOMY VIDEOS LECTURES
VIDEOS WITH UNIVERSITY EXAM NOTES

100K+ STUDENTS USING JOHARI MBBS
LECTURES NOTES FOR UNIVERSITY EXAM

JOHARI MBBS ANATOMY



TRUSTED BY 100k+ MEDICOS

BEST ANATOMY LECTURES

WITH 3D ANATOMY, CLINICAL ANATOMY

ABDOMEN ANATOMY
HEAD & NECK ANATOMY
NEUROANATOMY

VOLUME 2

ABDOMEN IMPS

JOHARI MBBS | YOUTUBE

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RECTUS SHEATH

SCAN QR WITH ORBOT APP TO DIRECTLY OPEN VIDEO LINK

1. IT IS AN APONEUROTIC SHEATH COVERING THE RECTUS ABDOMINIS.
2. IT HAS TWO PARTS - ANTERIOR & POSTERIOR

*FEATURES: -

ANT WALL - IT IS COMPLETE COVERING THE MS FROM END TO END.
- ITS COMPOSITION IS VARIABLE AS DESCRIBED BELOW.
-IT IS FIRMLY ADHERENT TO THE TENDINOUS INTERSECTIONS OF THE RECTUS MS.

POST WALL - IT IS INCOMPLETE, BEING DEFICIENT ABOVE THE COSTAL MARGIN & BELOW THE ARCUATE LINE.
- ITS COMPOSITION IS VARIBLE AS DESCRIBE BELOW.

*FORMATION: -

1.ABOVE THE COSTAL MARGIN -

ANT WALL- EXT OBLIQUE APONEUROSIS.
POST. WALL - IT IS DEFICIENT, THE RECTUS MS RESTS DIRECTLY ON 5TH 6TH & 7TH COSTAL CARTILAGE.

2 BETWEEN THE COSTAL MARGIN & THE ARCUATE LINE -

ANT WALL - EXT. OBLIQUE APONEUROSIS & ANT APONEUROSIS OF THE INT OBLIQUE.
POST WALL- POST LAMINA OF THE APONEUROSIS OF THE INTERNAL OBLIQUE & APONEUROSIS OF THE TRANSVERSUS MS.

3.BELOW THE ARCUATE LINE -

- ANT WALL- APONEUROSES OF ALL THE THREE FLAT MS OF THE ABDOMEN.APONEUROSES OF THE TRANSVERSUS & INTERNAL OBLIQUE ARE FUSED BUT THE EXTERNAL OBLIQUE APONEUROSIS REMAIN SEPARATE.
-POST WALL- IT IS DEFICIENT. --->RECTUS MS - FASCIA TRANSVERSALIS

*CONTENT -

1.MUSCLE - RECTUS ABDOMINIS & PYRAMIDALIS
2.ARTERY - SUP. EPIGASTRIC ARTERY. ---->ENTER SHEATH BY PASSING BETWEEN COSTAL & XIPHOID
-INF.EPIGASTRIC ARTERY----->ENTER SHEATH BY INFRONT OF ARCUTE LINE
3.VEIN - SUP. EPIGASTRIC VEIN & INF. EPIGASTRIC VEIN
4.NERVES - TERMINAL PART OF THE LOWER SIX THORACIC NERVES.

*FUNCTION - CHECK BOWING OF RECTUS MS. - DURING CONTRACTION.
- MAINTAIN STRENGTH OF ANT. ABDOMINAL WALL.

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INGUINAL CANAL

*OBLIQUE INTERMUSCULAR PASSAGE IN THE LOWER PART OF THE ANTERIOR ABDOMINAL WALL

*SITUATED --->JUST ABOVE MEDIAL HALF OF INGUINAL LIGAMENT

*DIRECTED DOWNWARD, FORWARD, MEDIAL.

*EXTEND - FROM DEEP INGUINAL RING TO SUPERFICIAL INGUINAL RING.

DEEP INGUINAL RING - OVAL OPENING IN FASICA TRANSVERSALIS
-SITUATED 1.2CM ABOVE THE MIDINGUINAL POINT.
-IMMIDIATELY---->LATERAL TO STEM OF INFERIOR EPIGASTRIC ARTERY

SUPERIOR INGUINAL RING - TRIANGULAR OPENING - IN EXTERNAL OBLIQUE APONEUROSIS
- IT IS LIKE AN OBTUSE ANGLE TRIANGLE
BASE -->FORM BY PUBIC CREST
TO MARGIN OF TRIANGULAR ---->LATERAL & MEDIAL REFERRED AS CRURA
BEYOND THE APEX OF TRIANGLE -->2 CRURA-> UNITED BY INERCRURAL FIBERS

*BOUNDARIS -

ANTERIOR WALL - WHOLE EXTEND - SKIN & SUPERFICIAL FASICA
- EXTERNAL OBLIQUE APONEUROSIS
- LATERAL 1/3RD - INTERNAL OBLIQUE MUSCLE

POSTERIOR WALL - WHOLE EXTEND - FASCIA TRANSVERSALIS
- EXTRAPERITONEAL TISSUE
- PARIETAL PERITONEUM
-MEDIAL 2/3RD - CONJOINT TENDON
- MEDIAL END BY THE REFLECTED PART OF INGUINAL LIGAMENT

ROOF - ARCHED FIBER - INTERNAL OBLIQUE
-TRANSVERSUS ABDOMINIS MUSCLE

FLOOR - GROOVED UPPER SURFACE OF INGUINAL LIGAMENT
- MEDIAL END BY LACUNAR LIGAMENT

SEX DIFFERENCE - INGUINAL CANAL IS LARGE IN MALE & SMALLER IN FEMALE

*STRUCTURES PASSING THROUGH THE CANAL -

1.SPERMATIC CORD IN MALES, ROUND LIGAMENT OF THE UTERUS IN FEMALE.
2.ILIOINGUINAL NERVE ENTERS THE CANAL THROUGH THE INTERVAL BETWEEN THE EXTERNAL & INTERNAL OBLIQUE MUSCLE & PASSES OUT THROUGH THE SUPERFICIAL INGUINAL RING.

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SPERMATIC CORD

*CONSTITUENT OF SPERMATIC CORD -

1. DUCTUS DEFERENS
2. TESTICULAR AND CREMASTERIC ARTERIES & DUCTUS DEFERENS ARTERY
3. PAMPINIFORM PLEXUS OF VEINS
4. LYMPH VESSEL
5. GRENITAL BRANCH-->OF GENITOFORMAL NERVE
   -PLEXUS OF SYMPATHETIC NERVES
6. REMAINS OF PROCESSUS VAGINALIS

*COVERING OF SPERMATIC CORD - FROM WITHIN OUTWARD FOLLOWS -

1. INTERNAL SPERMATIC FASCIA - DERIVED FROM-FASCIA TRANSVERSALIS
2. CREMASTRIC FASCIA - DERIVED FROM - INTERNAL OBLIQUE TRANSVERSUS ABDOMINIS MUSCLES
   -MADE UP OF - MUSCLE LOOPS -->CONSTITUTING CREMASTER MUSCLE
3. EXTERNAL SPERMATIC FASCIA - DERIVED FROM - EXTERNAL OBLIQUE APONEUROSIS

MECHANISM OF INGUINAL CANAL

*PRESENCE OF INGUINAL CANAL IS CAUSE OF WEAKNESS IN LOWER PART OF ANTERIOR ABDOMINAL WALL.
*WEAKNESS IS COMPENSATED BY FOLLOWING FACTORS.

1. OBLIQUITY OF INGUINAL CANAL -
   -2 INGUINAL RING DO NOT LIE OPPOSITE TO EACH OTHER
   -THEREFORE WHEN INTRA-ABDOMINAL PRESSURE RISES, THE ANT.& POST. WALL ARE APPROXIMATED.
   -THIS IS KNOWN AS FLAP VALVE MECHANISM
2. SUPERFICIAL RING GUARDED - FROM BEHIND - CONJOINT TENDON
   -REFLECTED PART OF INGUINAL LIGAMENT
3. DEEP INGUINAL RING GUARDED - FROM FRONT - INTERNAL OBLIQUE
4. SHUTTER MECHANISM OF INTERNAL OBLIQUE MUSCLE -
   - MUSCLE HAS TRIPLE RELATION TO INGUINAL CANAL.
   - IT FORM ANTERIOR, POSTERIOR WALL & ROOF OF CANAL
   - WHEN IT CONTRACTS THE ROOF IS APPROXIMATED TO FLOOR LIKE SHUTTER.
5. CONTRACTION OF CREMASTER/BALL VALVE MECHANISM - HELP SPERMATIC CORD TO PLUG SUPERFICIAL INGUINAL RING.
6. SPLIT VALVE MECHANISM - CONTRACTION OF EXTERNAL OBLIQUE RESULT IN APPROXIMATION OF 2 CRURA OF SUPRFICIAL INGUINAL RING.

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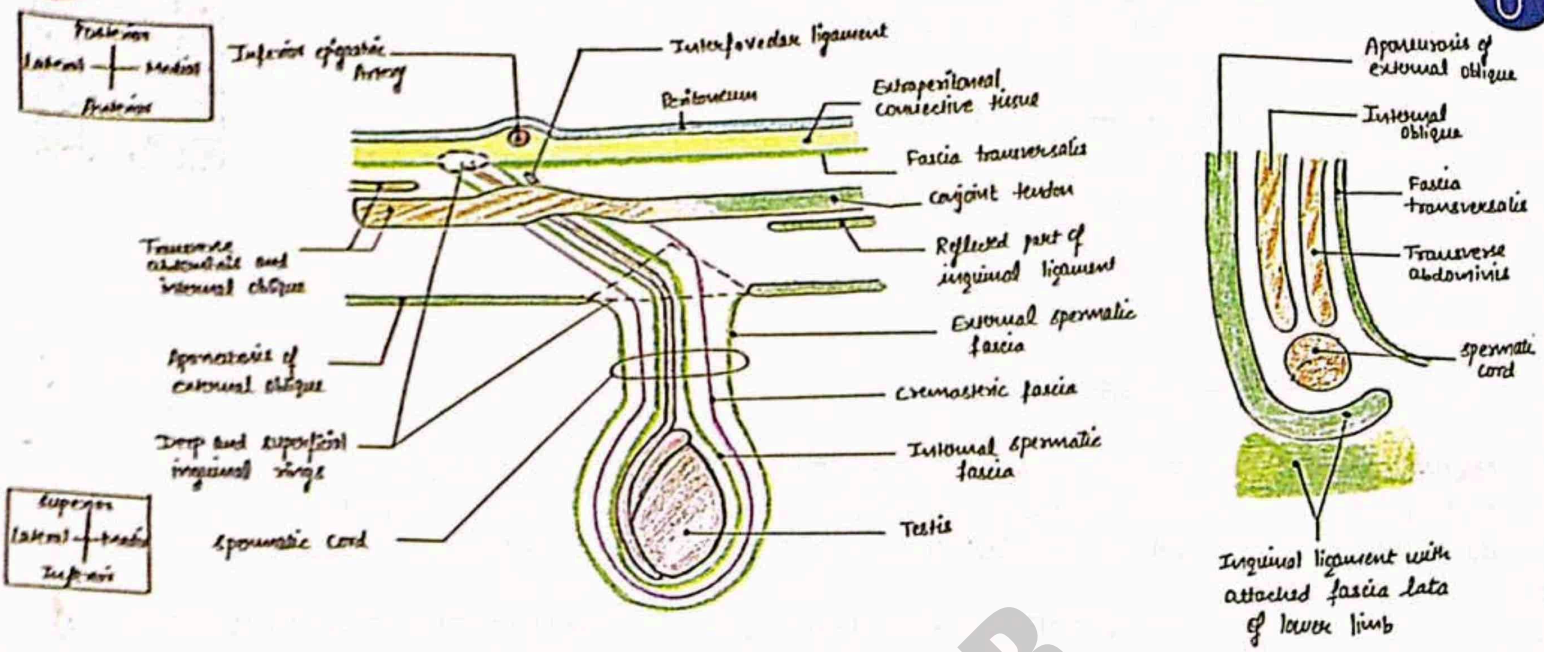
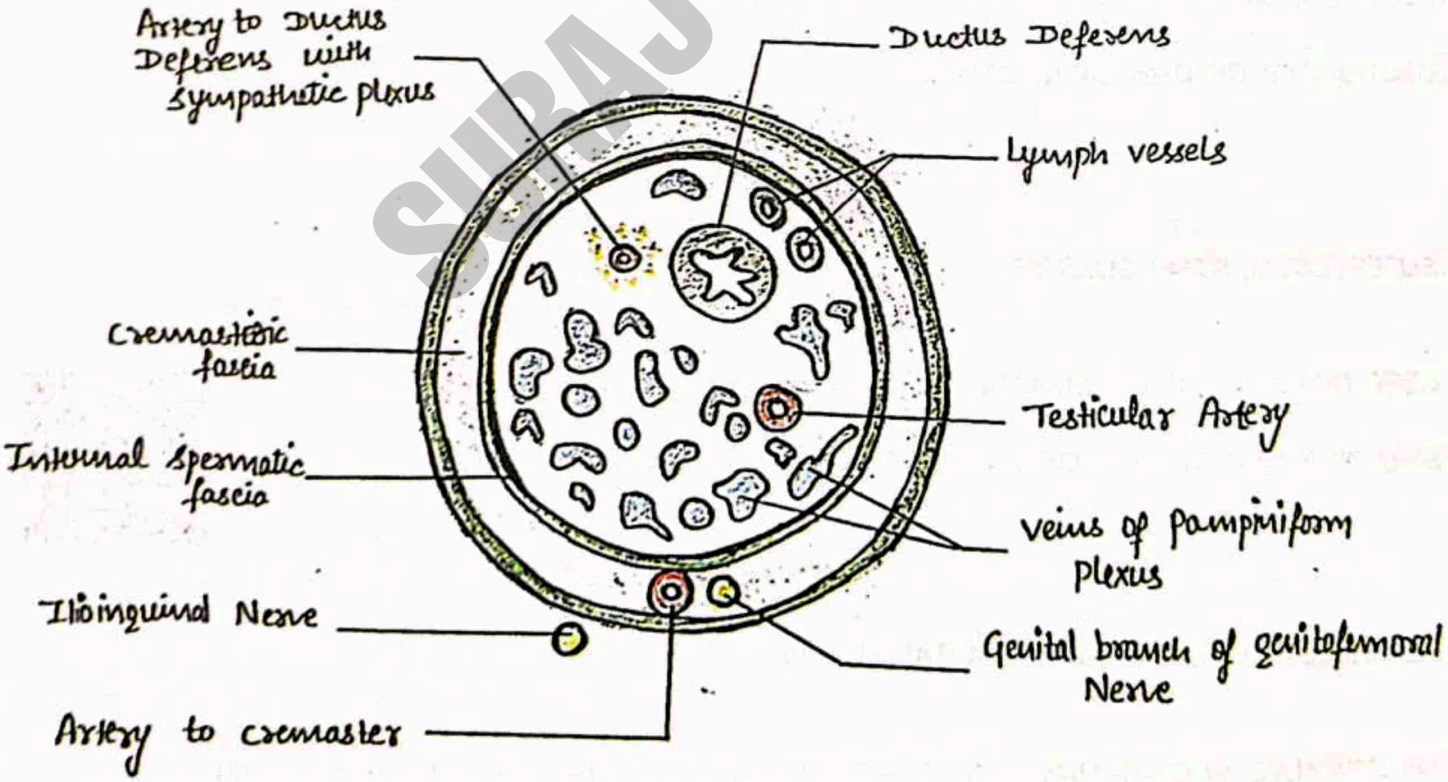


Boundaries of inguinal canal ⓐ Anterior and posterior walls in a horizontal section and
ⓑ the roof and floor in a sagittal section.
Fascia lata attached to inguinal ligament.

INGUINAL CANAL



SPERMATIC CORD

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CLINICAL ANATOMY

HERNIA

*HERNIA IS A PROTRUSION OF ANY ABDOMINA CONTENT THROUGH ANY OF ITS WALL- THIS IS CALLED EXTERNAL HERNIA.
*INTESTINE PROTRUBES INTO THE NO ENTRY ZONE WITHIN ABDOMINALL CAVITY ITSELF-THIS CONDITION IS CALLED INTERNAL HERNIA.
*HERNIA CONSIST OF--→SAC, CONTENT, COVERING.

TYPE OF HERNIA - INTERNAL HERNIA & EXTERNAL HERNIA.

INGUINAL HERNIA - PROTUSION OF LOOP OF INTESTINE THROUGH INTESTINAL WALL IS CALLED INGUINAL HERNIA.

INDIRECT INGUINAL HERNIA - WHEN THE PROTUSION OCCUR THROUGH THE DEEP INGUINAL RING, INGUINAL CANAL, SUPERFICIAL INGUINAL RING INTO SCROTUM, IT IS CALLED INDIRECT HERNIA.

COVERING - EXTRAPERITONEAL TISSUE
-INTERNAL SPERMATIC FASCIA
-CREMASTRIC FASCIA
-EXTERNAL SPERMATIC FASCIA
-SKIN

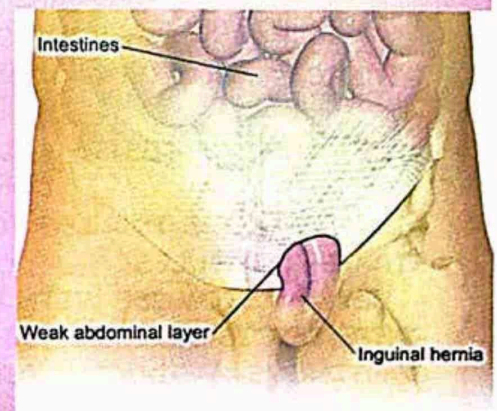
DIRECT HERNIA - WHEN THE PROTRUSION OCCUR THROUGH WEAK POSTERIOR WALLL OF INGUINAL CANAL OR TRIANGLE OF HESSELBACH THIS IS CALLED DIRECT HERNIA.

*HESSELBACH - BOUNDED BY INFERIOR EPIGASTRIC ARTERY, LATERAL BORDER OF RECTUS ABDOMINIS & INGUINAL LIGAMENT.
*THIS AREA IS DIVIDED INTO MEDIAL AND LATERAL PART BY PASSAGE OF OBLITERATED UMBILICAL ARTERY.

COVERING -

COVERINGS OF LATERAL DIRECT INGUINAL HERNIA -
1.EXTRAPERITONEAL TISSUE
2.FASCIA TRANSVERSALIS
3.CREMASTERIC FASCIA
4.EXTERNAL SPERMATIC FASCIA
5.SKIN

COVERINGS OF MEDIAL DIRECT INGUINAL HERNIA -
1.EXTRAPERITONEAL TISSUE
2.FASCIA TRANSVERSALIS
3.CONJOINT TENDON
4.EXTERNAL SPERMATIC FASCIA
5.SKIN



Inguinal Hernia

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TESTIS

*MALE GONAD - SUSPENDED IN THE SCROTUM BY SPERMATIC CORD.
*LIES OBLIQUETY, SO THAT UPPER POLE - FORWARD - LATERALLY.
*LEFT TETIS IS SLIGHTLY LOWER THAN RIGHT

*SHAPE & SIZE -
1.OVAL IN SHAPE - 3.75CM LONG, 2.CM BROAD, 1.8CM THICK FROM SIDE TO SIDE
2.WEIGHT ABOUT - 10 TO 15g

*EXTERNAL FEATURE -
1.TWO POLE - UPPER POLE & LOWER POLE ARE CONVEX & SMOOTH
2.TWO BORDER - ANTERIOR BORDER - CONVEX & SMOOTH
-FULLY COVERED - BY TUNICA UAGINALIS
- POSTERIOR BORDER - STRAIGHT
- PARTIALLY COVERED BY TUNICA VAGINALIS
- EPIDIDYMIS LIE - ALONG LATERAL PART OF POSTERIOR BORDER
3.TWO SURFACE - MEDIAL SURFACE & LATERAL SURFACE - CONVEX & SMOOTH

""MEDIAL SURFACE OF EPIDIDYMIS IS SEPERATED FROM TESTIS BY AN EXTENSION OF CAVITY OF TUNIICA VAGINALIS - THIS CALLED - SINUS OF EPIDIDYMIS""

*COVERINGS OF TESTIS -
*TESTIS IS COVERED BY LAYER OF SCROTUM
*IN ADDITION, ALSO COVERED BY 3 COATS - TUNICA VAGINALI - PARIETAL
- VISCERAL
- TUNICA ALBUGINEA
- TUNICA VASCULOSA

1.TUNICA VAGINALIS - REPRESENT LOWER PERSISTENT PORTION OF PROCESSUS VAGINALIS.
- COVER WHOLE TESTIS - EXCEPT FOR POSTERIOR BORDER
2.TUNICA ALBUGINEA - DENSE WHILE FIBROUS COAT COVERING THE TESTIS ALL AROUND.
- COVERED BY VISCERAL LAYER OF T.V EXCEPT - POSTERIORLY WHERE TESTICULAR VESSEL & NERVE ENTER
-POSTERIOR BORDER OF TUNICA ALBUGINEA IS THICKENED
--->FORM INCOMPLETE VERTICAL SEPTUM - "MEDIASTINUM TESTIS"
-NUMEROUS SEPTA EXTEND FROM MEDIASTINUM
3.TUNICA VASCULOSA - INNER MOST - VASCULAR COAT

*STRUCTURE OF TESTIS - GLANDULAR PART - CONSIST OF 200 TO 300 LOBULES --> CONTAIN 2-3 SEMINIFEROUS TUBULES.
-SEMINIFEROUS TUBULES JOIN TOGETHER AT APICES OF LOBULES TO FORM 20-30 STRAIGHT TUBULES.
-THEY ANASTOMOSE WITH EACH OTHER - TO FORM RETE TESTIS.
-RETE TESTIS GIVE RISE 12-30 EFFERENT DUCTULES---->ENTER EPIDIDYMIS CONTINOUS--->DUCTUS DIFERENS.

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*ARTERIAL SUPPLY -
-TESTICULAR ARTERY----→BRANCH OF ABDOMINAL AORTA---→GIVEN OFF AT LEVEL OF L2.
----→DESCEND ON POSTERIOR ABDOMINAL WALL--→ TO REACH WHERE DEEP INGUINAL RING--→ENTER SPERMATIC CORD
-AT POSTERIOR BORDER OF TESTIS - IT DIVIDES INTO BRANCHES.
-SOME SMALL BRANCHES ---→ENTER POSTERIOR BORDER
-WHILE LARGE BRANCHES ---→MEDIAL & LATERAL ---→PIERCE TUNICA ALBUGINEA

*VEIN SUPPLY - VEIN - EMERGING FROM THE TESTIS FORM - PAMPINI FORM PLEXUS.
-ANTERIOR PART OF PLEXUS ARRANGED AROUND TESTICULAR ARTERY.
-MIDDLE PART OF PLEXUS ARRANGED TESTICULAR ARTERY-DUCTUS DEFERNS
-POSTERIOR PART IS - ISOLATED
-PLEXUS CONDENSES-→INTO 4 VEIN---→AT SUPERFICIAL INGUINAL RING
-→INTO 2 VEIN-→DEEP INGUINAL RING.
-ONE VEIN---→DRAIN INTO INFERIOR VENA CAVA ON RIGHT SIDE
-OTHER VEIN---→DRAIN INTO LEFT RENAL VEIN ON LEFT SIDE

*NERVE SUPPLY - SYM. - T10 OF SPINAL CORD
-NERVE ARE BOTH AFFERENT FOR TESTICULAR SENSATION & EFFERENT TO BLOOD VESSEL (VASOMOTOR).

CLINICAL ANATOMY

1.UNDESCENDED TESTIS OR CRYPTORCHIDISM : THE ORGAN MAY LIE IN THE LUMBER, ILIAC, INGUINAL, OR UPPER SCROTAL REGION. THE IMPORTANT FEATURES OF AN UNDESCENDED TESTIS ARE AS FOLLOWS.
A.THE TESTIS MAY COMPLETE ITS DESCENT AFTER BIRTH.
B.SPERMATOGENESIS MAY FAIL TO OCCUR IN IT.
C.A MALINGNANT TUMOUR IS MORE PRONE TO DEVELOP IN IT.
D.THE CONDITION CAN BE SURGICALLY CORRECTED.

EPTOPIC TESTIS :
1.TESTIS MAY OCCUPY AN ABNORMAL POSITION DUE TO DEVIATION FROM THE NORMAL ROUTE OF DESCENT.
2.IT MAY BE UNDER THE SKIN OF THE LOWER PART OF THE ABDOMIN, UNDER THE SKIN OF THE FRONT OF THE THIGH, IN THE FEMORAL CANAL, UNDER THE SKIN OF THE PENIS & IN THE PERINEUM BEHIND THE SCROTUM.

VARICOCOELE - IT PRODUCED BY DIALATATION OF PEMPINIFORM PLEXUS, IT IS USUALLYLEFT SIDE.

HERMAPHRODITISM - TRUE HERMAPHRODITISM - BOTH TESTIS & OVARY ARE PRESENT.

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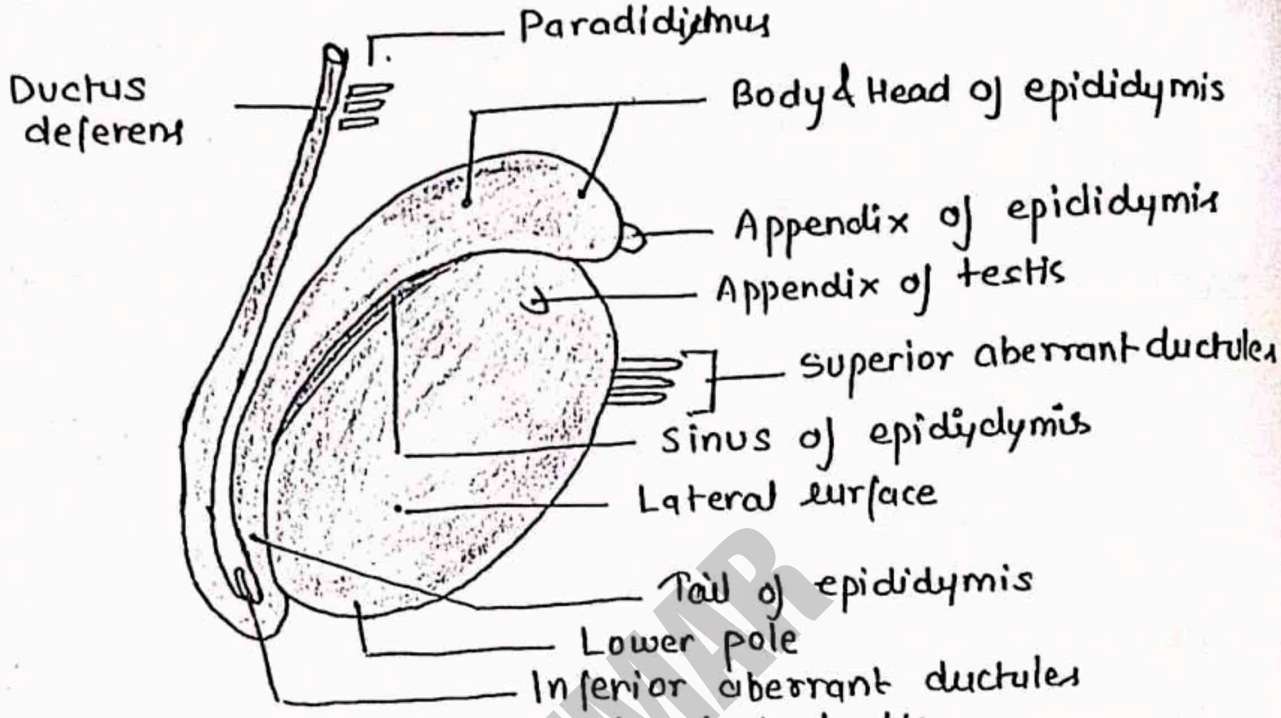
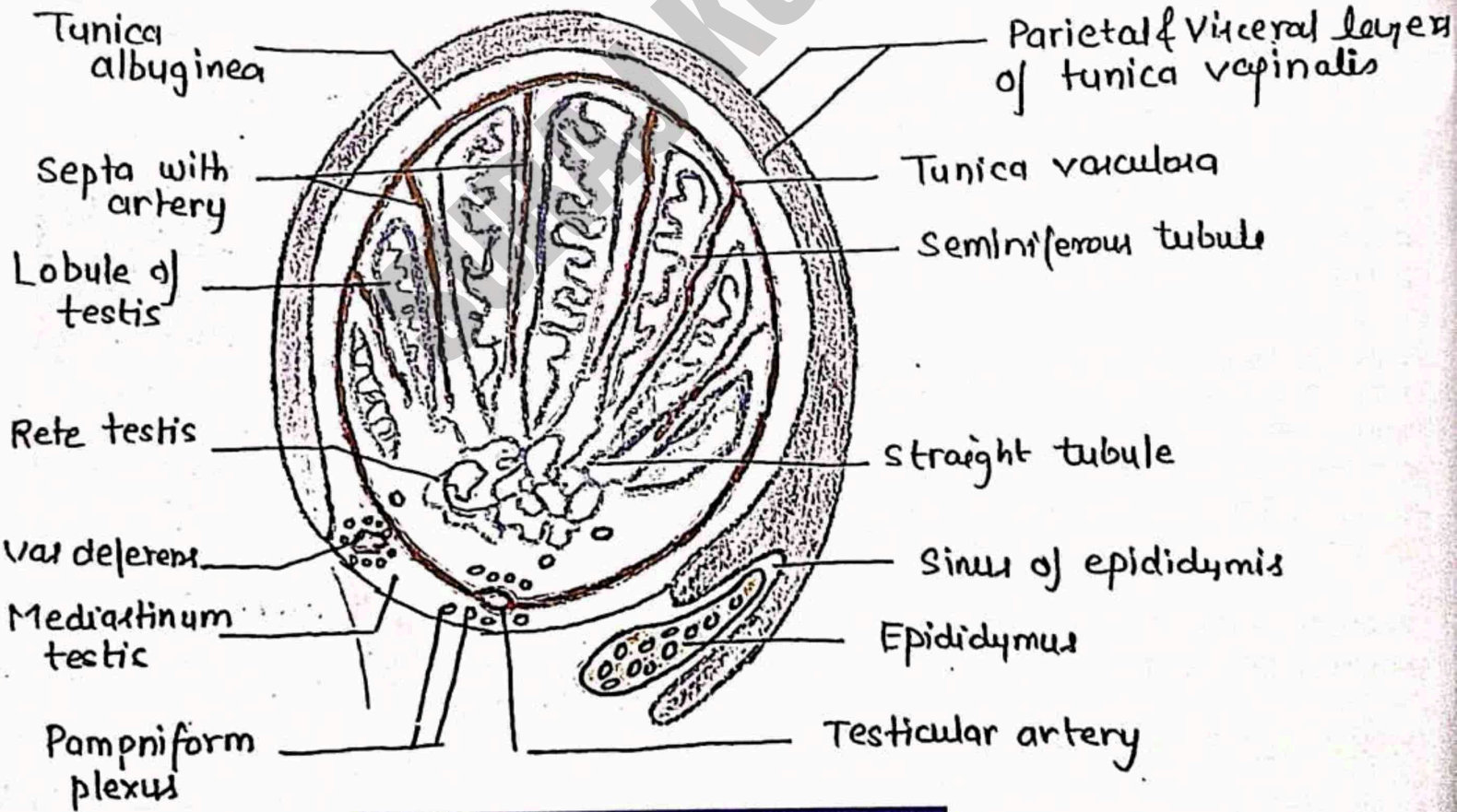


∵ Lateral view of the right testis



TESTIS

STOMACH

*DEFINITION - A STOMACH IS--->MUSCULAR BAG--->FORMING WIDEST PART OF DIGESTIVE TUBE.
-ACT AS--> RESERVIOR OF FOOD->HELP IN DIGESTION OF CARBOHYDRATES, PROTIEN & FATS.

LOCATION - UPPER LEFT PART OF ABDOMEN.
-OCCUPYING - EPIGASTRIC, UMBILICAL & LEFT HYPOCHONDRIAC REGION.
SHAPE - IN NORMAL ACTIVE PERSON SHAPE IS "J-SHAPED"
SIZE & CAPACITY - ABOUT 25CM LONG
-CAPACITY - 30ML AT BIRTH
-1000ML AT PUBERTY
-2LITRES OR MORE IN ADULTS.

*EXTERNAL FEATURES -
-TWO ORIFICES - CARDIAC ORIFICE - IS JOINED BY - LOWER END OF OESOPHAGUS.
-BEHIND THE LEFT 7TH COASTAL CARTILAGE.
-2.5CM FROM STERNUM.
-AT LEVEL OF T11
-PYLORIC ORIFICE - OPEN INTO DEODENUM
-AT LEVEL OF L1
-ITS POSITION IS INDICATED ON THE SURFACE OF STOMACH: BY PREPYLORIC VEIN
-BY CIRCULAR GROOVE
- TWO CURVATURES - LESSER CURVATURE - CONCAVE
- FORM RIGHT BORDER OF STOMACH
- PROVIDES ATTACHMENT OF LESSER OMENTUM
- GREATER CURVATURE - CONVEX
- FORM LEFT BORDER OF STOMACH
- PROVIDES ATTACHMENT TO GREATER OMENTUM, GASTROSPLENIC LIGAMENT, GASTROPHRENIC LIGAMENT.

- TWO SURFACE - ANTERIOR - FACE - FORWARDS & UPWARDS
POSTERIOR - FACE - BACKWARD & DOWNWARD
-TWO PARTS - CARDIAC--->FURTHER SUB DIVIDED INTO--->FUNDUS & BODY
-PYLORIC - PYLORIC ANTRUM & PYLORIC CANAL.

*RELATION OF STOMACH -

-PERITONEAL RELATIONS - LINE BY PERITONEUM ON BOTH ITS SURFACES.
1.AT LESSER CURVATURE - LAYER OF PERITONEUM LINING ANTERIOR & POSTERIOR SURFACE MEET.
- AND BECOME CONTINOUS WITH LESSER OMENTUM
2.AT GREATER - TWO LAYER MEET --->TO FORM GASTROSPLENIC LIGAMENT.
-VISCERAL RELATION - ANTERIOR SURFACE - LIVER

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-DIAPHRAGM
-TRANSVERSE COLON
-LEFT PLEURA
-PERICARDIUM
-6-9 RIBS

- POSTERIOR SURFACE - DIAPHRAGM
- LEFT KIDNEY
- LEFT SUPRARENAL GLAND
- PANCREAS
- TRANSVERSE MESOCOLON
- SPLENIC FLEXURE OF COLON
- SPLENIC ARTERY

*BLOOD SUPPLY - STOMACH IS SUPPLIED ALONG-

1.LESSER CURVATURE BY - LEFT GASTRIC ARTERY-→BRANCH COELIAC TRUNK
- RIGHT GASTRIC ARTERY-→PROPER HEPATIC ARTERY

2.GREATER CURVATURE BY - RIGHT GASTROEPIPLOIC A. →GASTRODUODENAL ARTERY
- LEFT GASTROEPIPLOIC A.-→SPLENIC

3.FUNDUS - 5-7 SHORT GASTRIC ARTRIES -→BRANCH OF SPLENIC.

*VEIN - RIGHT & LEFT GASTRIC VEIN -→DRAIN INTO PORTAL VEIN
-RIGHT GASTRO EPIPLOIC END-→SUPERIOR MESENTERIC VEIN
-LEFT GASTROEPIPLOIC END VEIN -→TERMINATE IN SPLENIC VEIN &SHORT GASTRIC VEINS

*LYMPHATIC DRAINAGE - DIVIDED INTO 4X LYMPHATIC TERRITORIES -

1) AREA(A) - UPPER PART OF LEFT 1/3RD -→DRAIN INTO PANCREATICOSPLENIC NODE- TO REACH COELIAC NODES

2) AREA(B) - RIGHT 2/3RD ----→DRAIN INTO LEFT GASTRIC NODE - TO REACH COELIAC NODE

3) AREA(C) - LOWER PART OF LEFT 1/3RD -→RIGHT GASTROEPIPLOIC NODE-→LYING ALONG ARTERY OF SAME NODE
---→DRAIN INTO - SUBPYLORIC NODE, HEPATIC NODE, COELIAC NODE

4) AREA(D) - PYLORIC PART -→DRAIN INTO DIFFERENT DIRECTION-→INTO PYLORIC, HEPATIC, GASTRIC NODES-→TO REACH COELIAC NODES

*LYMPH FROM ALL AREA OF STOMACH ULTIMATELY REACH - COELIAC NODE.

*NERVE SUPPLY - SYMPATHETIC NERVE -→DERIVED FROM T6-10SPINAL SEGMENT VIA GREATER SPLACHNIC NERVE, COELIAC & HEPATIC PLEXUS
-PARASYMPATHETIC NERVE - →DERIVED FROM VAGI THROUGH OESOPHAGEAL PLEXUS, GASTRIC NERVE.
-ANTERIOR GASTRIC NERVE -→CONTAIN MAINLY LEFT VEGAL FIBER
-POSTERIOR GASTRIC NERVE -→RIGHT VEGAL FIBERS

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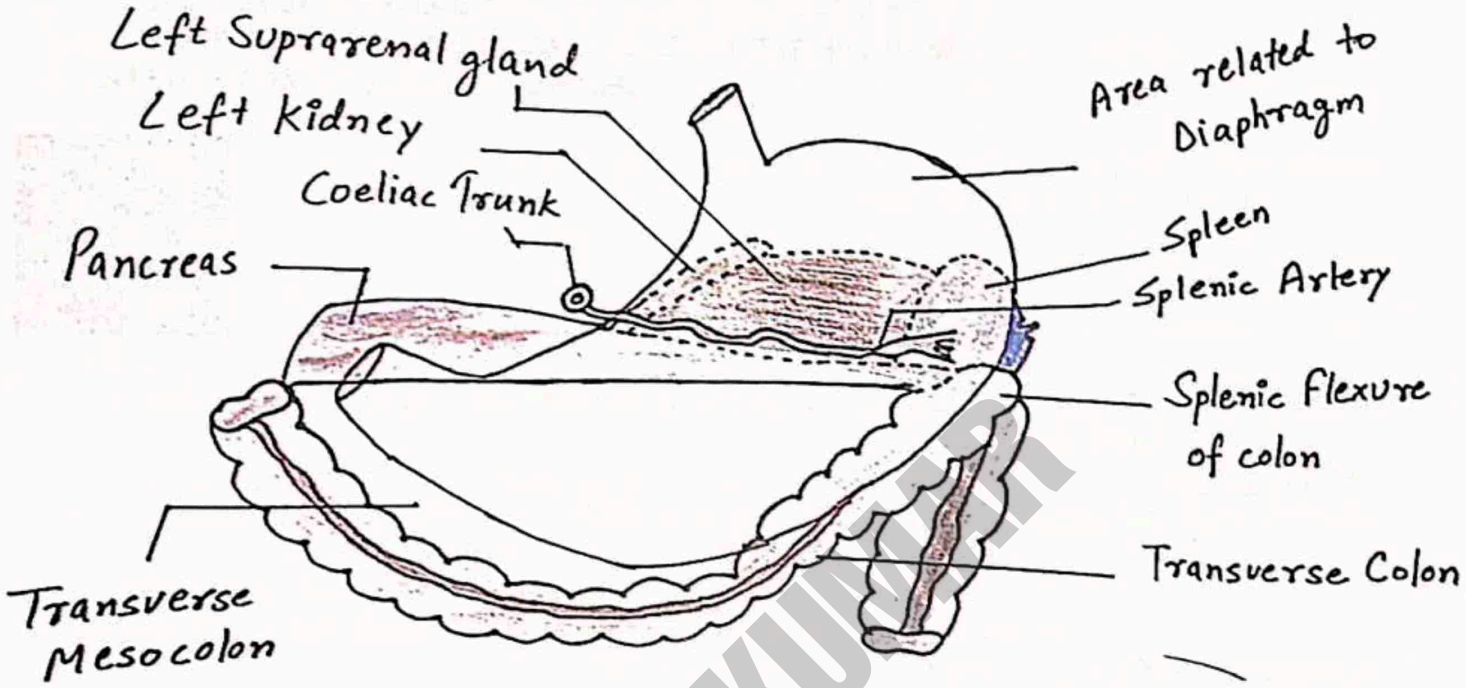
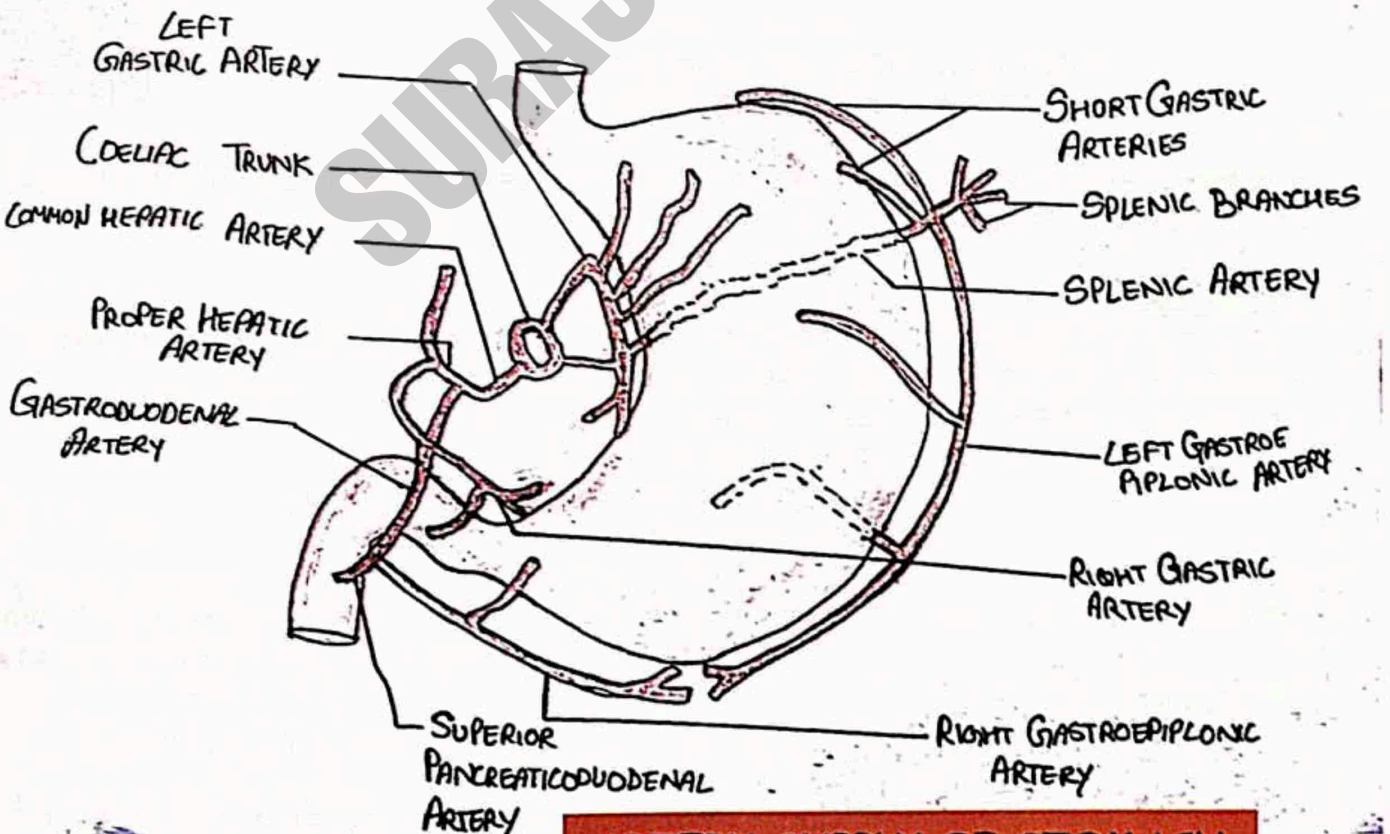


POSTERIOR RELATION OF STOMATCH



ARTERY SUPPLY OF STOMACH

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DUODENUM

*DEFINITION & LOCATION - SHORT, WIDEST & MOST FIX PART OF SMALL INTESTINE
-EXTEND FROM PYLORUS TO DUODENOJEJUNAL FLEXURE
-LIE ABOVE THE LEVEL OF UMBILICUS - OPPOSITE FIRST, SECOND * THIRD LUMBAR VERTEBRA.

*PARTS - 25CM LONG & DIVIDED INTO 4x PART
1.FIRST OR SUPERIOR PART, 5CM LONG
2.SECOND OR DESCENDING PART, 7.5CM LONG
3.THIRD OR HORIZONTAL PART, 10CM LONG
4.FOURTH OR ASCENDING PART, 2.5CM LONG

1) FIRST PART - BEGIN AT PYLORUS TO MEET SECOND PART AT SUPERIOR DUODENAL FLEXURE.

*RELATION -
-PERITONEAL RELATION - PROXIMAL 2.5CM MOVABLE--->ATTACHMENT LESSER OMENTUM ABOVE
--->GREATER OMENTUM BELOW
-DISTAL 2.5CM FIXED-RETROPERITONEAL-COVER ONLY ANTERIORLY
-VISCERAL - ANTERIOR - QUADRATE LOBE OF LIVER
- GALL BLADDER
- POSTERIORLY - GASTRODUODENAL ARTERY
- BILE DUCT
- PARIETAL VEIN
- SUPERIORLY - EPIPLOIC FORAMEN
- INFERIORLY - HEAD & NECK OF PANCREAS

2) SECOND PART - BEGIN AT SUPERIOR DUODENAL FLEXURE TO REACH LOWER BORDER OF 3RD LUMBAR VERTEBRA.

*RELATION -
-PERITONEAL RELATION - RETROPERITONEAL - ANTERIOR SURFACE COVERED WITH PERITONEUM-EXCEPT-MIDDLE-->WHERE COLON RELATED
-VISCERAL - ANTERIORLY - RIGHT LOBE OF LIVER
- TRANSVERSE COLON
- ROOT OF THE TRANSVERSE MESOCOLON
- SMALL INTESTINE

3) THIRD PART - BEGIN AT INFERIOR DUONDENAL FLEXURE - AND BY JOINING 4TH PART IN FRONT OF ABDOMINAL AORTA.

RELATION -
-PERITONEAL RELATION - RETROPERITONEAL-ANTERIOR SURFACE IS COVERS BY PERITONEUM
-EXCEPT - MEDIAN PLANE - IT IS CROSSED BY SUPERIOR

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MESCENTERIC VESSEL

-VISCERAL RELATION - ANTERIORLY - SUPERIOR MESCENTERIC VESSEL
-ROOT OF MESCENTERY
- POSTERIORLY - RIGHT URETER
- RIGHT PSOAS MAJOR
- TESTICULAR OR OVARIAN VESSEL
- INFERIOR VENA CAVA
- ABDOMINAL AORTA
- SUPERFICIALY - HEAD OF PANCREAS
- INFERIORLY - COLIS OF JEJUNUM

4)FOURTH PART - RUN UPWARD UPTO UPPER BORDER OF 2ND LUMBAR VERTEBRA
- TO BECOME CONTINOUS WITH JEJUNUM AT DUODENOJEJUNAL FLEXURE

*RELATION -

-PERITONEAL RELATION - MOSTLY RETROPERITONEAL - COVERED WITH PERITONEUM ONLY ANTERIORLY.

-VISCERAL RELATION - ANTERIORLY - TRANSVERSE COLON
- TRANSVERSE MESOCOLON
- LESSER SAC
- STOMACH
- POSTERIORLY - LEFT SYMPATHETIC CHAIN
- LEFT RENAL ARTERY
- LEFT GONADAL ARTERY
- INFERIOR MESCENTRIC VEIN
- TO RIGHT - ATTACHMENT OF UPPER PART OF ROOT OF MESCENTERY
-TO LEFT - LEFT KIDNEY
- LEFT URETER

*ARTERIAL SUPPLY-DEVELOP PARTLY FROM--->FOREGUT - OPENING OF BILE DUCT INTO -->2ND PART -REPRESENT JUNCTION BETWEEN FOREGUT & MID GUT
- UP THE LEVEL OF OPENING -->SUPPLIED BY SUPERIOR PANCREATICODUODENAL ARTERY
-BELOW--> INFERIOR PANCREATICODUODENAL ARTERY

*VENOUS DRAINAGE - VEINS OF DUODENUM DRAIN INTO - SPLENIC SUPERIOR MESCENTRIC PORTAL VEN

*NERVE SUPPLY - SYMPATHETIC NERVE - T9-10 SPINAL
-PARASYMPATHETIC - FROM VAGUS PASS THROUGH COELIAC PLEXUS.

CLINICAL ANATOMY

1.PEPTIC ULCER CAN OCCUR IN THE SIDE OF PEPSIN & HCL ACID, NAMELY THE STOMACH AND FIRST PART OF DUODENUM.
2.GASTRIC ULCER OCCUR MAINLY ALONG THE LESSER CURVATURE.
3.CONGENITAL STENOSIS & OBSTRUCTION OF THE 2ND PART OF THE DUODENUM MAY OCCUR AT THE SITE OF THE OPENING OF THE BLUE DUCT.

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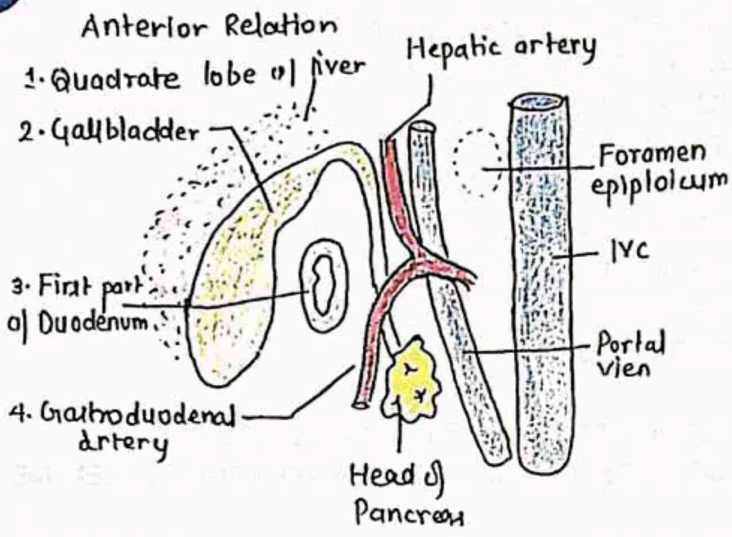
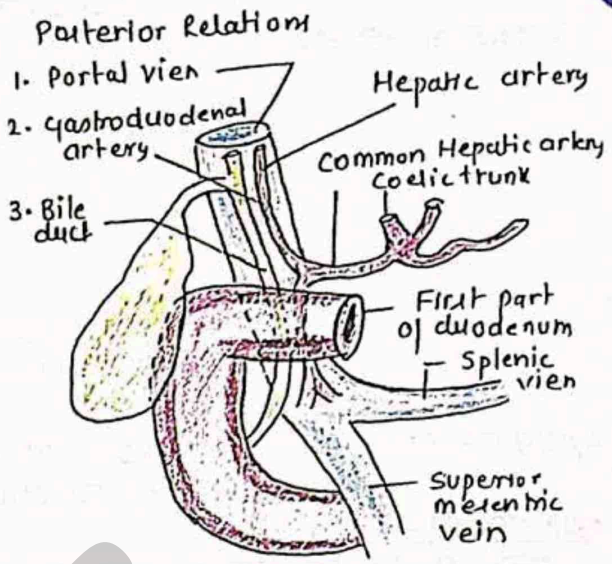
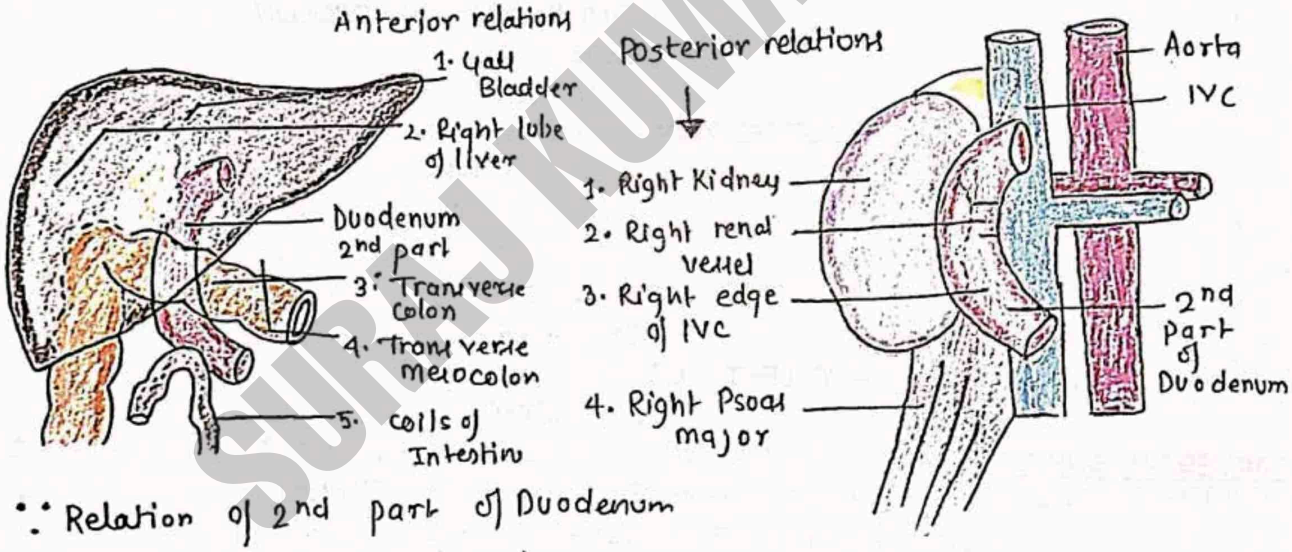
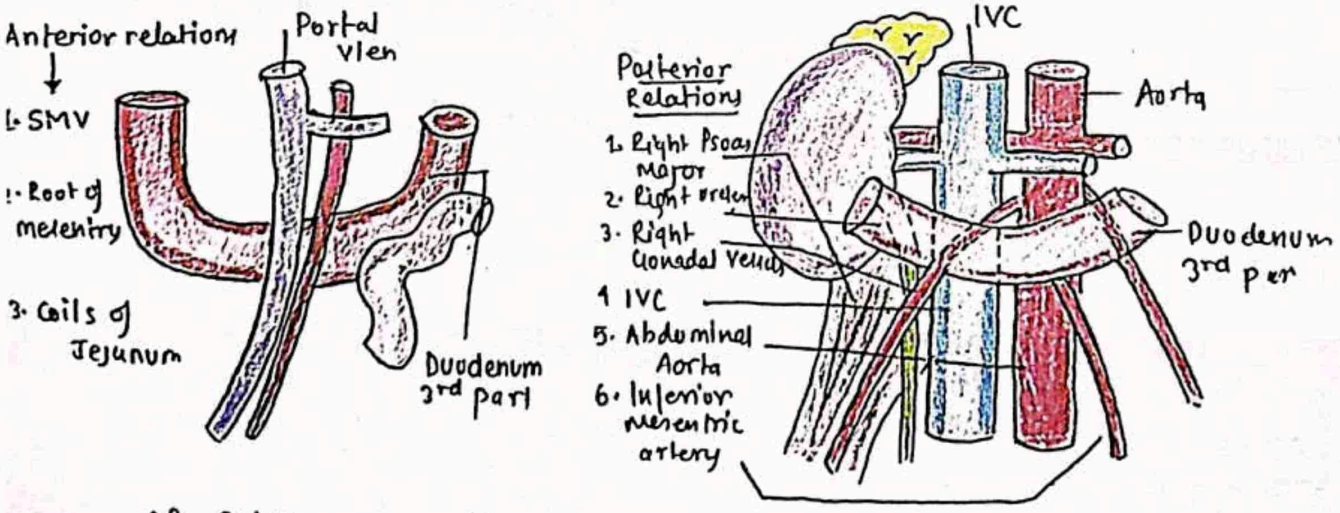




∴ Relation of First part of Duodenum



∴ Relation of 2nd part of Duodenum



∴ Relation of 3rd part of Duodenum

RELATION OF PARTS OF DUODENUM

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CAECUM

*FEATURES -

- LARGE BLIND SAC FORMING THE COMMENCEMENT OF LARGE INTESTINE
- SITUATED IN - RIGHT ILIAC FOSSA
- ABOVE THE LATERAL HALF OF INGUINAL LIGAMENT
- COMMUNICATION- SUPERIALLY - ASCENDING COLON
  MEDIALLY -CAECOCOLIC JUNCTION WITH ILEUM
  POSTEROMEDIALLY- APPEN DISC

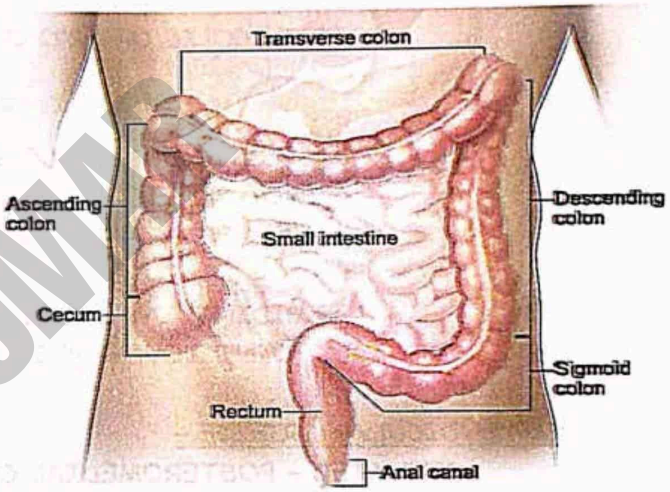
DIMENSION- 6CM LONG AND 7.5 CM BROAD (GREATER WIDTH THAN LENGTH)

*RELATION-

ANTERIOR - COIL OF INTESTINE
-ANTERIOR ABDOMINAL WALL
POSTERIOR - RIGHT PSOAS AND ILIACUS

NERVE-GENITOFEMORAL NERVE
- FEMORAL NERVE
- LATERAL CUTANEOUS NERVE OF THIGH

VESSEL- TESTICULAR OR OVARIAN



*ARTERIAL SUPPLY -
BRANCHES OF ILEOCOLIC ARTERY

*VEIN SUPPLY - SUPERIOR MESENTERIC VEIN

*NERVE - SYM: T11 TO L1
- PARA SYM:- VAGUS NERVE

*ILIOCAECAL VALVE- LOWER END OF ILEUM OPEN POSTEROMEDIAL - CAECOCOLIC JUNCTION
- ILEOCAECAL OPENING GUARDED, ILEOCAECAL VALUE
-TWO FRENULA - FORMED BY FUSION OF LIPS AT THE END OF APERTURE
-THESE ARE -→LEFT OR ANTERIOR FRENULA
--→ RIGHT OR POSTERIOR FRENULA
-VALVE IS ACTIVELY CLOSED BY SYMPATHETIC NERVE → CAUSE TONIC CONTRACTION OF ILEACAECAL SPHINCTER

*FUNCTION -
- PREVENT REFLEX. FROM CAECUM TO ILEUM
- REGULATE PASSAGE OF ILEAL CONTENTS INTO CAECUM

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VERMIFORM APPENDIX

*WORM LIKE DIVERTICULUM - ARISING FROM POSTEROMEIAL WALL OF CAECUM.
*ABOUT 2CM BELOW ILEOCAECAL ORIFICE.

*DIMENSIONS - LENGTH VARIES FROM 2 TO 20 CM WITH AN AVERAGE OF 9 CM
-LONGER IN CHILDREN THAN IN ADULT.
*POSITION - LIE IN RIGHT ILIAC FOSSA →BASE →IS FIXED
TIP → DIRECTION
-POSITION ARE OFTEN COMPARED TO HOUR HAND OF CLOCK.
-->PARACOLIC OR 11'O CLOCK POSITION - APPENDIX MAY PASS UPWARD
-->RETROCOLIC OR 12'O CLOCK POSITION - APPENDIX MAY LIE BEHIND CAECUM
- COMMONEST POSITION - 65%
--->SPLENIC OR 2'O CLOCK POSITION - APPENDIX POINT TOWARDS SPLEEN
--->PROMONTERIC OR 3'O CLOCK POSITION - HORIZONTALLY
---->PELVIC OR 4'O CLOCK POSITION - MAY DESCEND INTO PELVIS
-COMMON POSITION-30%
---->MIDINGUINAL OR 6'O CLOCK POSITION - BELOW THE CAECUM & MAY POINT TOWARDS INGUINAL LIGAMENT.

*APPENDICULAR ORIFICE -
-SITUATED - POSTEROMEDIAL OF CAECUM 2CM BELOW THE ILEOCAECAL ORIFICE
-OCCASIONALLY GUARDED BY - SEMILUNAR FOLD OF THE MUCOUS MEMBRANE, KNOWN AS 'VALVE OF GERLACH'
-ORIFICE IS MOVED ON SURFACE -→BY POINT SITUATED 2CM BELOW
→TRANSTUBERCULAR & RIGHT LATERAL PLANES
-MC BURNEY'S POINT - SITE OF MAX.TENDERNESS IN APPENDICITIS
- POINT LIE AT JUNCTION OF LATERAL 1/3RD AND MEDIAL 2/3RD OF LINE JOINING THE UMBILICUS TO RIGHT ANTERIOR,SUPERIOR ILAC SPINE.

*PERITONEAL RELATION - APPENDIX IS SUSPENDED BY SMALL, TRIANGULAR FOLD OF PERITONEUM CALLED MESOAPPENDIX.

*BLOOD SUPPLY - APPENDICULAR ARTERY -->IS BRANCH OF LOWER DIVISION-→OF ILIOCOLIC ARTERY
-RUN BEHIND - TERMINAL PART OF ILEUM & ENTERS THE MESOAPPENDIX
-BLOOD DRAINED BY - APPENDICULAR VEIN
- ILEOCOLIC VEIN
- SUPERIOR MESCENTRIC VEIN--->TO PORTAL VEIN

*NERVE - SYMPATHETIC NERVE- T9 TO T10 SEGMENT THROUGH THE COELIAC PLEXUS
PARASYMPATHETIC NERVE--->DERIVED FROM VAGUS

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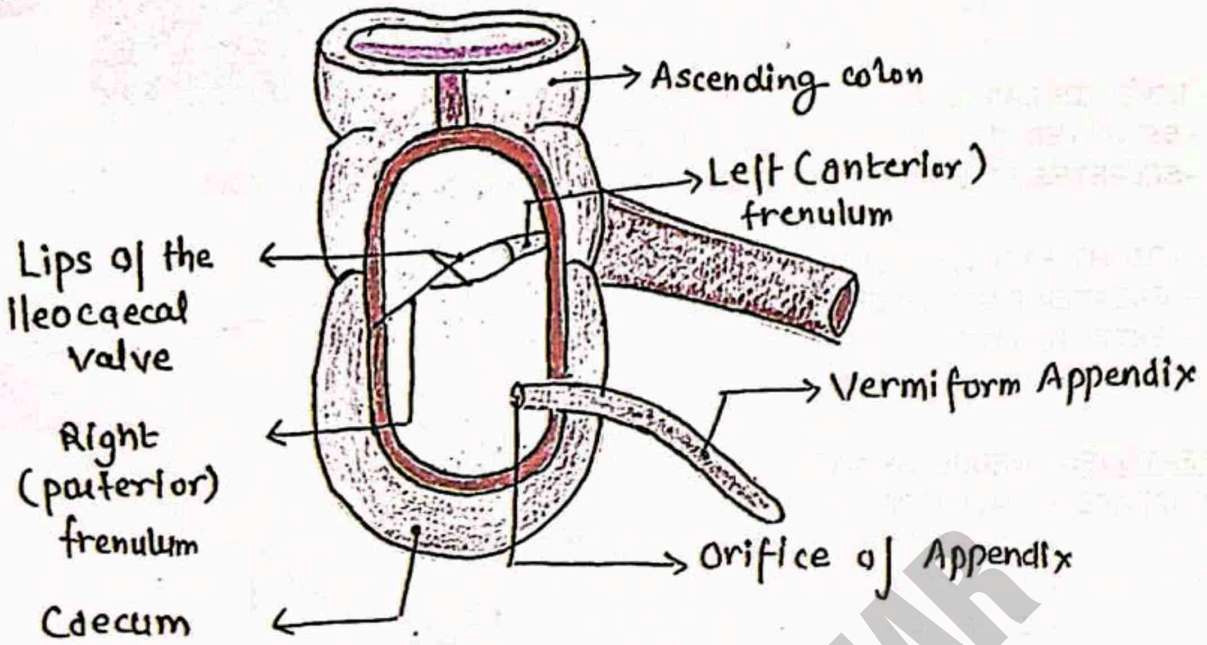
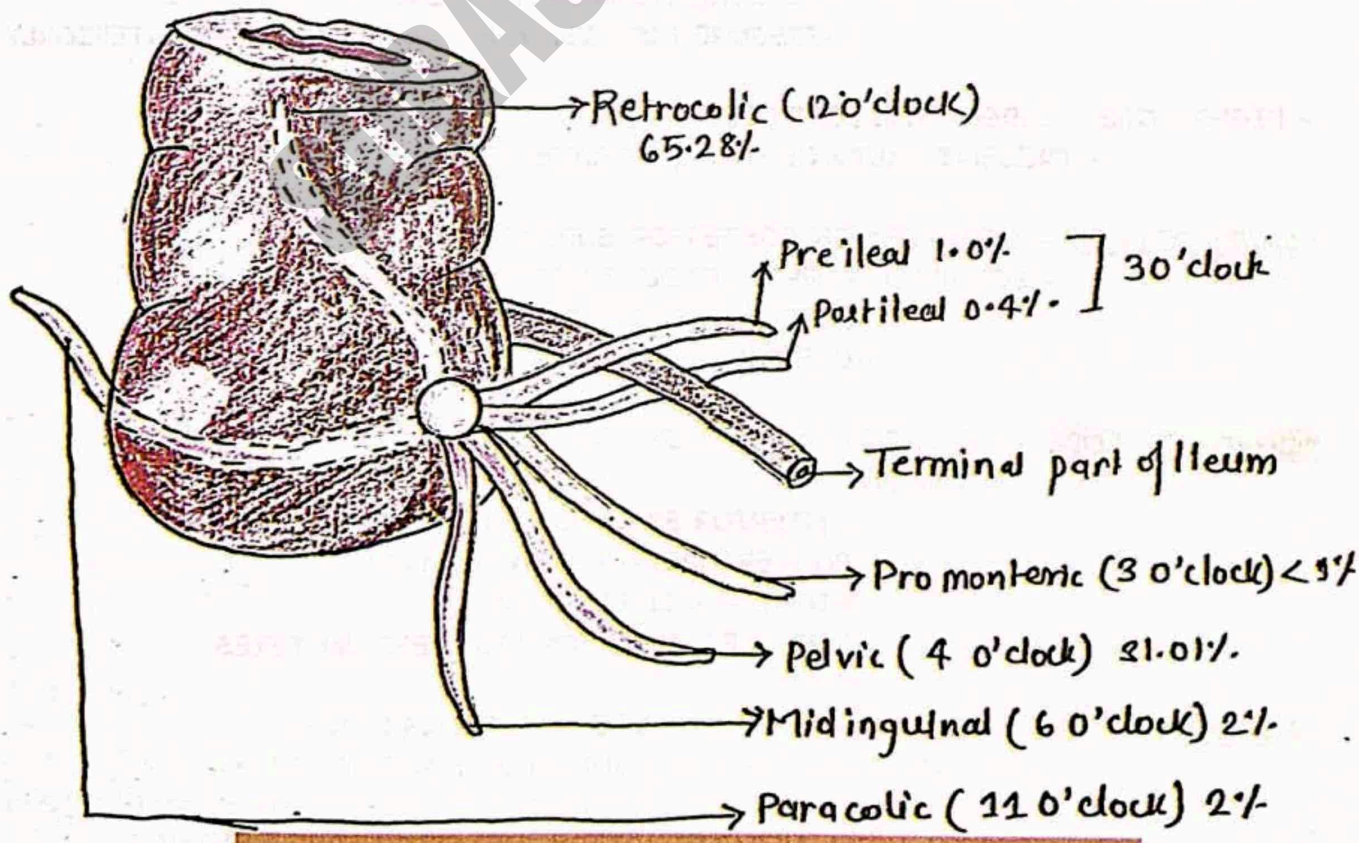


INTERIOR OF CAECUM



POSITION OF VERMIFORM APPENDIX

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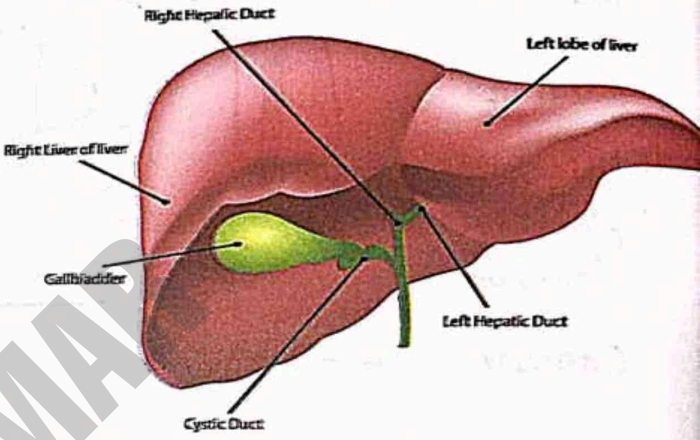
LIVER ANATOMY

*FEATURES:- LIVER IS LARGE AND SOLID GLAND
-SITUATED IN RIGHT UPPER QUADRANT OF ABDOMINAL CAVITY
-SECRETES BILE AND PERFORMS VARIOUS METABOLIC FUNCTION

*LOCATION:- -RIGHT HYPOCHONDRIUM
- GREATER PART IN EPIGASTRIUM
- EXTEND INTO LEFTHYPOCHONDRIUM



*EXTERNAL FEATURE:- WEDGE-SHAPED.
-FIVE SURFACE - ANTERIOR
- POSTERIOR
- SUPERIOR
- INFERIOR
- RIGHT

- ONE PROMINENT BORDER - INFERIOR BORDER
- SEPARATES ANTERIOR FROM INFERIOR SURFACE

- TWO LOBES - DIVIDED INTO- RIGHT & LEFT LOBES BY FALCIFORM LIGAMENT---→ ANTERIORLY & SUPERIORLY
-FISSURE LIGAMENTUM TERES - INFERIORLY
-FISSURE FOR LIGAMENTUM VENOSUM - POSTERIORLY

- RIGHT LOBE - LARGER THE LEFT LOBE
- PRESENT CAUDATE AND QUADRATE LOBES

*CAUDATE LOBE - SITUATED ON POSTERIOR SURFACE
- BOUNDED→RIGHT > GROOVE FOR THE INFERIOR VENA CAVA
-LEFT→ FISSURE FOR LIGAMENTUM VENOSUM
-INFERIORLY----→ PORTA HEPATIS.

*QUADRATE LOBE - SITUATED ON ---→ INFERIOR SURFACE
- BOUNDED ---→
ANTERIOR BY INFERIOR BORDER
POSTERIORLY→ PORTA HEPATIS
RIGHT - GALL BLADDER
LEFT - FISSURE FOR LIGAMENTUM TERES

*PORTA HEPATIS- DEEP, TRANSVERSE FISSURE ABOUT 5CM LONG
- SITUATED ON, INFERIOR SURFACE OF RIGHT LOBE OF LIVER
- LIE B/W CAUDATE LOBE ABOVE AND QUADRATE LO BE BELOW
- PORTAL VEIN, HEPATIC ARTERY, HEPATIC NERVE - ENTER

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*LIGAMENT :- LIGAMENT OF LIVER ARE 2X TYPE - FALSE & TRUE

*FALSE LIGAMENT - ARE ACTUALLY PERITONEAL FOLDS

1.FALCIFORM LIGAMENT - SICKLE-SHAPED FOLD OF PERITONEUM
-CONNECT THE LIVER TO UNDER SURFACE OF DIAPHRAGM AND ANTERIOR ABDOMINAL WALL.

2.CORONARY LIGAMENT -TRIANGLE FOLD OF PERITONEUM CONNECTING-BARE AREA TO DIAPHRAGM
- CONSIST OF TWO LAYER - UPPER AND LOWER

3.RIGHT TRIANGULAR LIGAMENT- SMALL TRIANGULAR FOLD OF PERITONEUM
-WHICH CONNECTING RIGHT LATERAL SURFACE OF LIVER TO DIAPHRAGM

4.LEFT TRIANGULAR LIGAMENT- SMALL TRIANGULAR FOLD OF PERITONEUM
-WHICH CONNECT UPPER SURFACE OF LEFT LOBE TO DIAPHRAGM

5.LESSER OMENTUM- FOLD OF PERITONEUM CONNECTING LESSER CURVATURE OF STOMACH ----->VISCERAL SURFACE OF LIVER
-HEPATIC ATTACHMENT IS -J SHAPE
- VERTICAL LIMB ATTACH, MARGIN OF FISSURE FOR LIGAMENT VENOSUM
- HORIZONTAL LIMB----> MARGIN OF PORTA HEPATIS.

*TRUE LIGAMENTS - THEY ARE ACTUALLY REMNANTS OF FETAL STRUCTURE

-LIGAMENTUM TERES HEPATIS- REMNANT OF LEFT UMBILICAL VEIN

-LIGAMENTUM VENOSUM - REMNANT OF DUCTUS VENOSUS
- WHICH IN FETAL LIFE CONNECT LEFT BRANCH OF PORTAL VEIN WITH INTERIOR VENA CAVA

*RELATION :-

*PERITONEAL RELATION - MOST OF LIVER IS COVERD BY PERITONEUM
- AREA NOT COVERED BY PERITONEUM ARE AS FOLLOWS -

1.TRIANGULAR BARE AREA - ON THE POSTERIOR SURFACE OF RIGHT LOBE
- LIMITED BY SUPERIOR & INFERIOR LAYER OF CORONARY LIGAMENT

2.GROOVE FOR INFERIOR VENA CAVA - POSTERIOR SURFACE OF RIGHT LOBE

3.FOSSA FOR GALLBLADDER - LIE ON INFERIOR SURFACE OF RIGHT LOBE

4.AREA OF ATTACHMENT OF LESSER OMENTUM

*VISCERAL RELATION -

1) ANTERIOR SURFACE - RELATED TO XIPHOID PROCESS

2) DIAPHRAGM ON EACH SIDE -->SEPARATE PLEURA- ABOVE LEVEL OF LINE DRAWN FROM.
-XIPHISTERNAL JOINT TO 10TH RIBS IN MID AXILLARY LINE
--->LUNGS - ABOVE LEVEL OF LINE FROM SAME JOINT TO 8TH RIB.

3) POSTERIOR SURFACE-->IS TRIANGULAR -BARE AREA->RIGHT SUPRARENAL

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-GROOVE FOR INFERIOR VENA CAVA--->UPPER PART OF VESSEL
-CAUDATE LOBE - CRURA OF DIAPHRAGM

-FISSURE FOR LIGAMENTUM VENOSUM - CONTAIN 2 LAYER OF LESSER OMENTUM

-POSTERIOR SURFACE OF LEFT LOBE-->OESOPHAGEAL IMPRESSION

4) INFERIOR SURFACE - INFERIOR SURFACE OF LEFT LOBE-->GASTRIC IMPRESSION
-FISSURE FOR LIGAMENTUM TERE - PASS INFERIOR TO PORTA HEPATITIS FROM BORDER
-QUADRATE LOBE->RELATED PYLORUS FIRST PART OF DUODENUM
-FOSSA FOR GALL BLADDER->LIE TO RIGHT OF QUADRATE LOBE
-->RIGHT OF THIS FOSSA-->BEAR COLIC IMPRESSION

5) RIGHT SURFACE - QUADRILATERAL & CONVEX
-RELATED TO DIAPHRAGM OPPOSITE 7TH TO 11TH RIBS IN MIDAXILLARY LINE
-DIAPHRAGM SEPARATE-->PLEURA UP TO 10TH RIB
-->LUNGS UPTO 8TH RIB

6) SUPERIOR SURFACE - QUADRILATERAL & CONCAVITY IN MIDDLE->IS CARDIAC IMPRESSION
-EACH SIDE - CONVEX-->TO FIT THE DOME OF DIAPHRAGM

***BLOOD SUPPLY -**

*LIVER IS HIGHLY VASCULAR ORGAN
*RECEIVE BLOOD FROM 2x SOURCE->ARTERIAL BLOOD->IS SUPPLY BY ->HEPATIC ARTERY(20%)
----->VENOUS->IS SUPPLY BY -PORTAL VEIN(80%)

***VENOUS DRAINAGE** - BY 3 LARGE HEPATIC VEIN - LEFT HEPATIC VEIN
-MIDDLE HEPATIC VEIN
-RIGHT HEPATIC VEIN
- 3 VEIN MAY ENTER IUC INDEPENDENTLY
- BUT LEFT & MIDDLE VEIN USUALLY JOIN, SO THAT ONLY 2 MAJOR VEIN JOINT THE IUC

***NERVE SUPPLY -**

*SYMPATHETIC FIBRES-->DERIVED FROM COELIAC PLEXUS
*RUN ALONG LESSER OMENTUM & ENTER PORTA HEPATITIS

*PARASYMPATHETIC FIBRE ->DERIVED FROM HEPATIC BRANCH OF ANTERIOR VAGAL TRUNK
*REACH PORTA HEPATIC THROUGH LESSER OMENTUM

***HEPATIC SEGMENTS - STRUCTURAL UNIT OF LIVER -**

- RIGHT LOBE - DIVIDED INTO ->ANTERIOR & POSTERIOR PART
- LEFT LOBE-DIVIDED INTO-->MEDIAL & LATERAL PARTS

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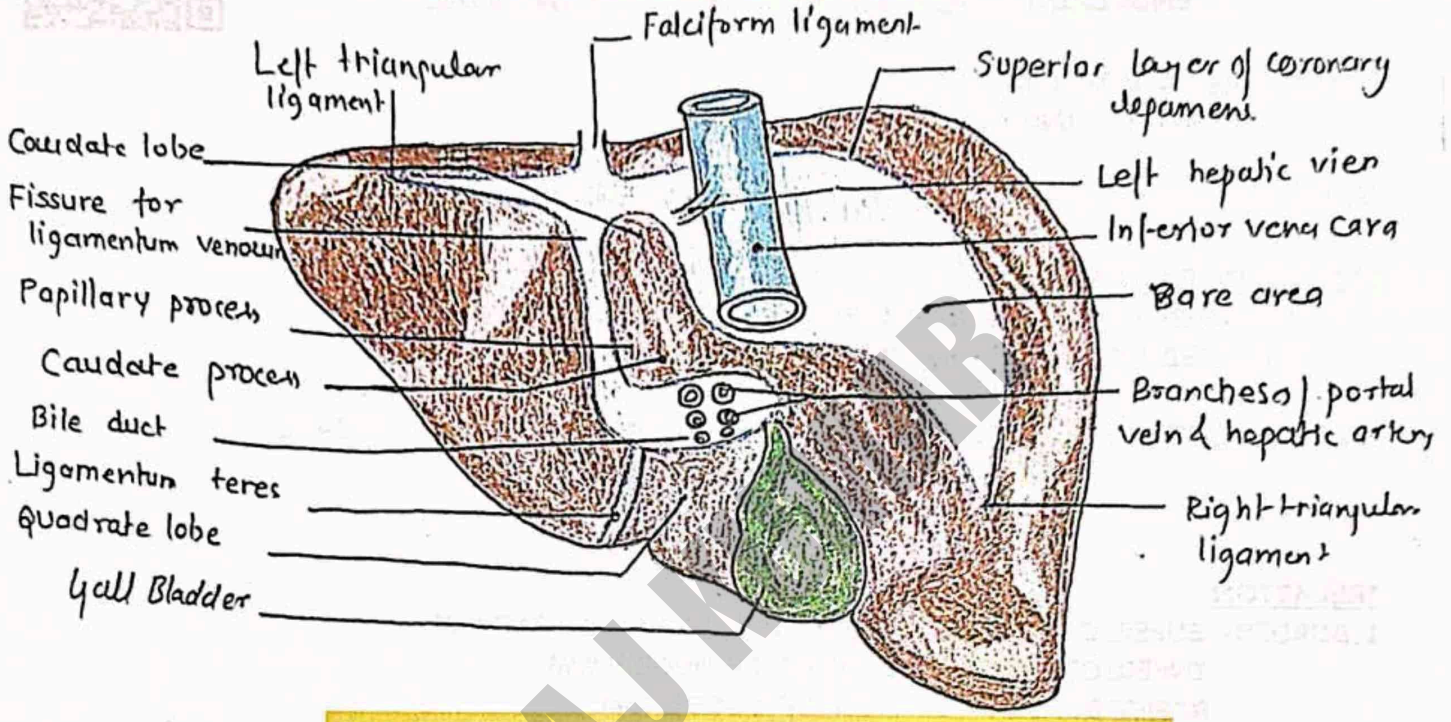
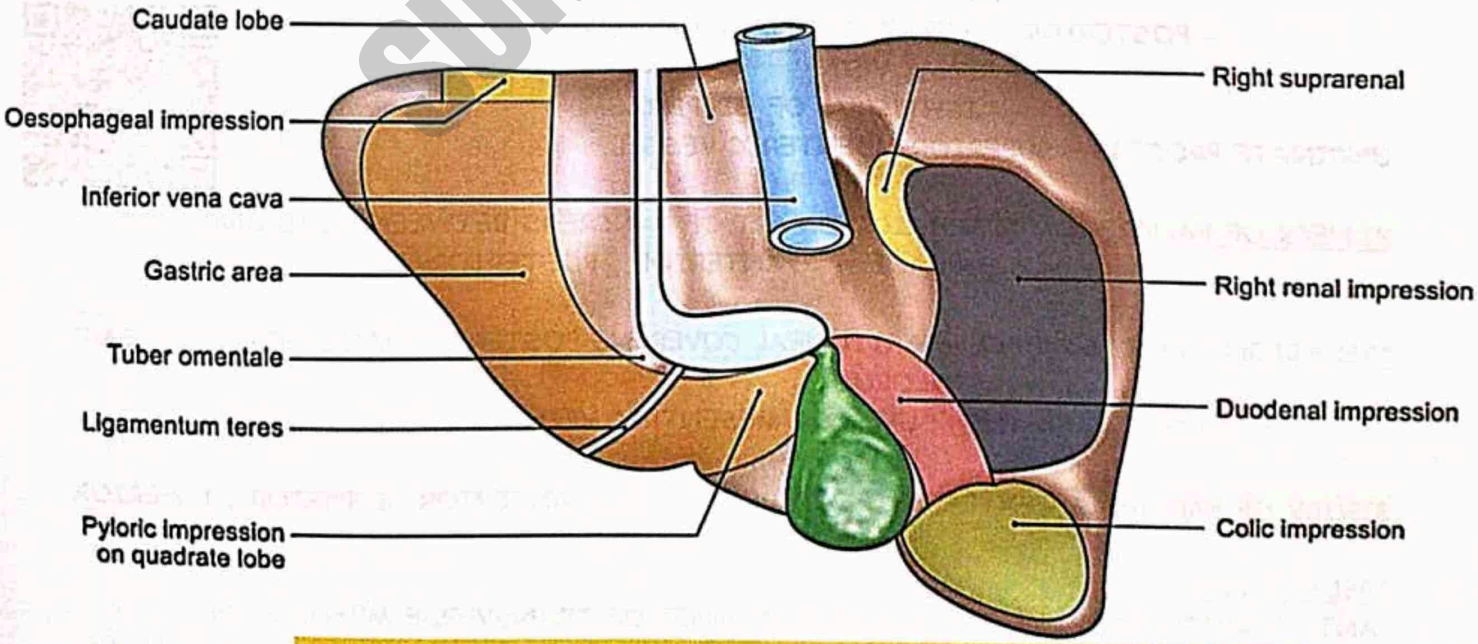
- EACH OF THESE PART FURTHUR DIVIDED INTO UPPER & LOWER PART
-VEIN DRAINING THE HEPATIC SEGMENTS ARE INTERSEGMENTAL
--->THEY DRAIN MORE THAN ONE SEGMENT

-FISSURE FOR LIGAMENTUM TERE & LIGAMENTUM VENOSUM-->MAKE JUNCTION BETWEEN MEDIAL & LATERAL SEGMENT OF LEFT LOBE.



LIVER SEEN FROM BEHIND & BELOW



RELATION OF INFERIOR SURFACE OF LIVER

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PANCREAS ANATOMY

*FEATURES- GLAND THAT IS PARTLY EXOCRINE AND PARTLY ENDOCRINE
- EXOCRINE - SECRETS - DIGESTIVE PANCREATIC JUICE
- ENDOCRINE - SECRETS - HORMONE E.G - INSULINE

*LOCATION - LIE IN POSTERIOR ADBOMINAL WALL
-AT 1ST AND 2ND LUMBAR VERTEBRA

*PANCREAS- DIVIDED INTO HEAD , NECK , BODY AND TAIL

1) HEAD OF PANCREAS :-
- ENLARGED FLATTENED RIGHT END OF PANCREAS
- SITUATED WITHIN C - SHAPE CURVE OF DUODENUM

*EXTERNAL FEATURE - 3 BORDER - SUPERIOR, INFERIOR, RIGHT LATERAL
- 2 SURFACE - ANTERIOR & POSTERIOR
- ONE PROCESS - UNCINATE PROCESS

*RELATION
1.BORDER- SUPERIOR - OVERLAPPED BY 1ST PART OF DUODENUM
INFERIOR BORDER - 3RD PART OF DUODENUM
RIGHT BORDER - 3RD PART OF DUODENUM

2.SURFACE - ANTERIOR - FRIST PART OF DUODENUM
- TRANSVERSE COLON & JEJUNUM
- POSTERIOR - INFERIOR VENA CAVA
- RENAL VEIN
- RIGHT CRUS OF DIAPHRAGM
UNCINATE PROCESS - SUPERIOR MSENTERIC VESSEL

2) NECK OF PANCREAS - CONSTRICTED PART OF PANCREAS BETWEEN HEAD AND BODY
- 2 SURFACE --->ANTERIOR , POSTERIOR

*RELATION - ANT. SURFACE - PERITONEAL COVERING POSTERIOR WALL OF LESSER SAC
-PYLORUS
-POST. SURFACE - SUPERIOR MESENTRIC VEIN

3)BODY OF PANCREAS - EXTERNAL FEATURE-3 BORDER-ANTERIOR, SUPERIOR , INFERIOR

*RELATION -
ANT. BORDER - PROVIDE ATTACHMENT TO ROOF OF TRANSVERSE MESOCOLON
SUP. BORDER - COELIAC TRUNK, HEPATIC ARTERY, SPLENIC ARTERY

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INF. BORDER - SUPERIOR MESENTRIC VESSEL

3.SRUFACE - ANT. SURFACE - COVERED BY PERITONEUM
- LESSER SAC & STOMACH

POST. SURFACE - DEVOID OF PERITONEUM
-AORTA
-LEFT SUPRARENAL GLAND & LEFT KIDNEY
-SPLENIC VEIN

INFERIOR SURFACE - COVERED BY PERITONEUM
-DUODENOJEJUNAL FLEXURE

4)TAIL OF PANCREAS

- LEFT END OF PANCREAS
- LIE IN LIENORENAL LIGAMENT

***DUCTS OF PANCREAS -2 DUCTS - MAIN & ACCESSARY PANCREATIC DUCT**

1) MAIN PANCREATIC DUCT - 3MM IN DIAMETER
-LIE NEAR POSTERIOR SURFACE OF PANCREASE
- BEGIN AT TAIL, RUN TOWARD RIGHT & BEND AT NECK DOWNWAR TO RIGHT IN THE HEAD
-WITH IN HEAD OF PANCREAS - PANCREATIC DUCT RELATE TO BILE DUCT
-2X DUCT ENTER THE WALL OF SECOND PART OF DUODENUM
-JOIN TO FORM HEPATO PANCREATIC AMPULLA OF VATER
-WHICH OPEN--→ AS MAJOR DUODENAL PAPILLA
-8 TO 10 CM DISTAL TO PYLORUS

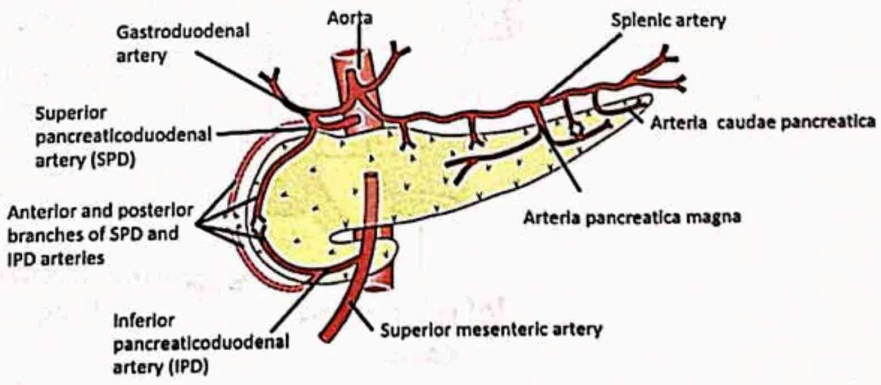
2) ACCESSARY PANCREATIC DUCT - BEGIN IN LOWER PART OF HEAD, CROSS FRONT OF MAIN DUCT
- OPEN INTO DUODENUM AT MINAR DUODENAL PAPILLA
-6 TO 18 CM DISTAL TO PYLORUS

***ARTERIAL SUPPLY:-**

-MAINLY BY- PANCREATIC BRANCHES OF SPLENIC ARTERY
- SUPERIOR PANCREATICODUODENAL ARTERY
- INFERIOR PANCREATICO DUODENAL ARTERY

***VENOUS DRAINAGE :-**

- SPLENIC VEIN
- SUPERIOR MESENTERIC VEIN
- PORTAL VEIN



*NERVE SUPPLY:-

-PARASYMPATHETIC NERVE
- VAGUS NETVE
-SYMPATHETIC NETUE -
SPLANCHNIC NERVE

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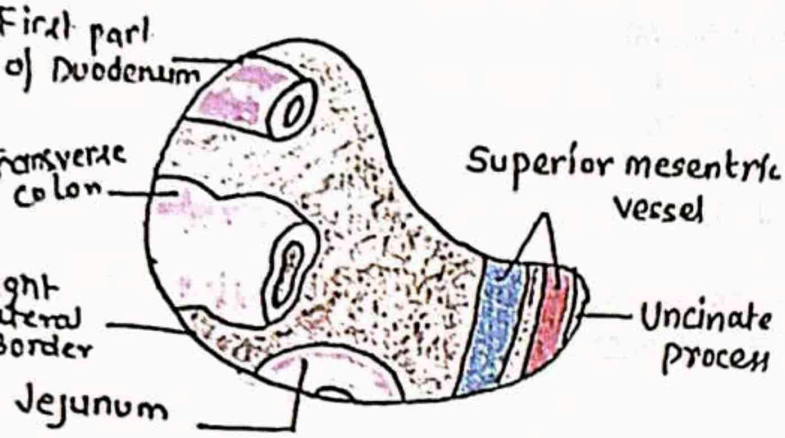
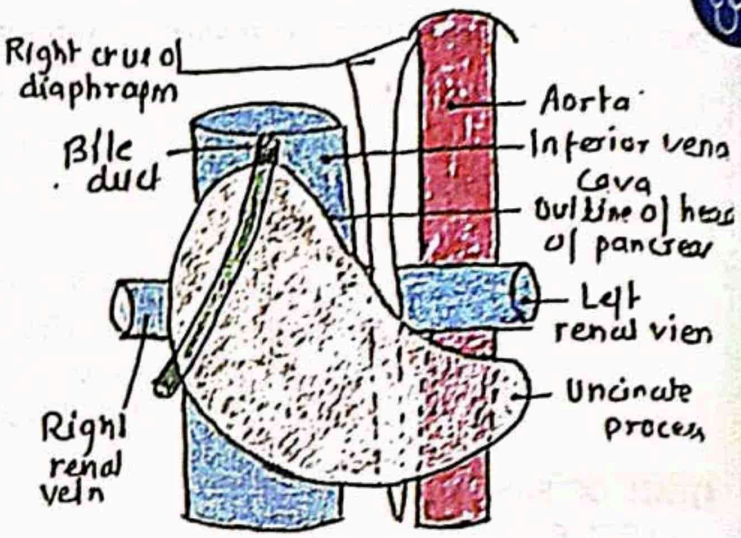
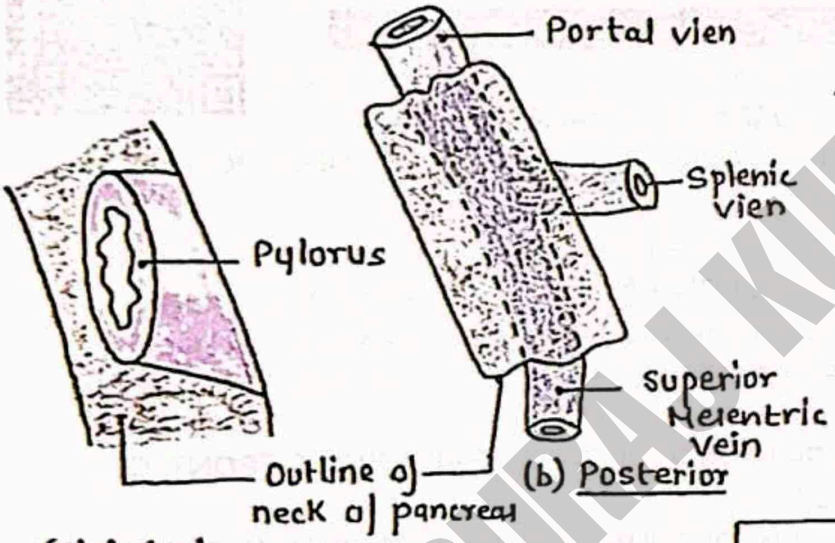
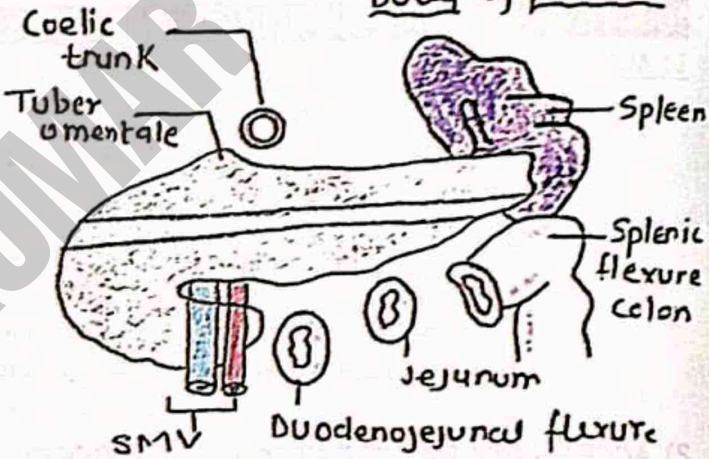


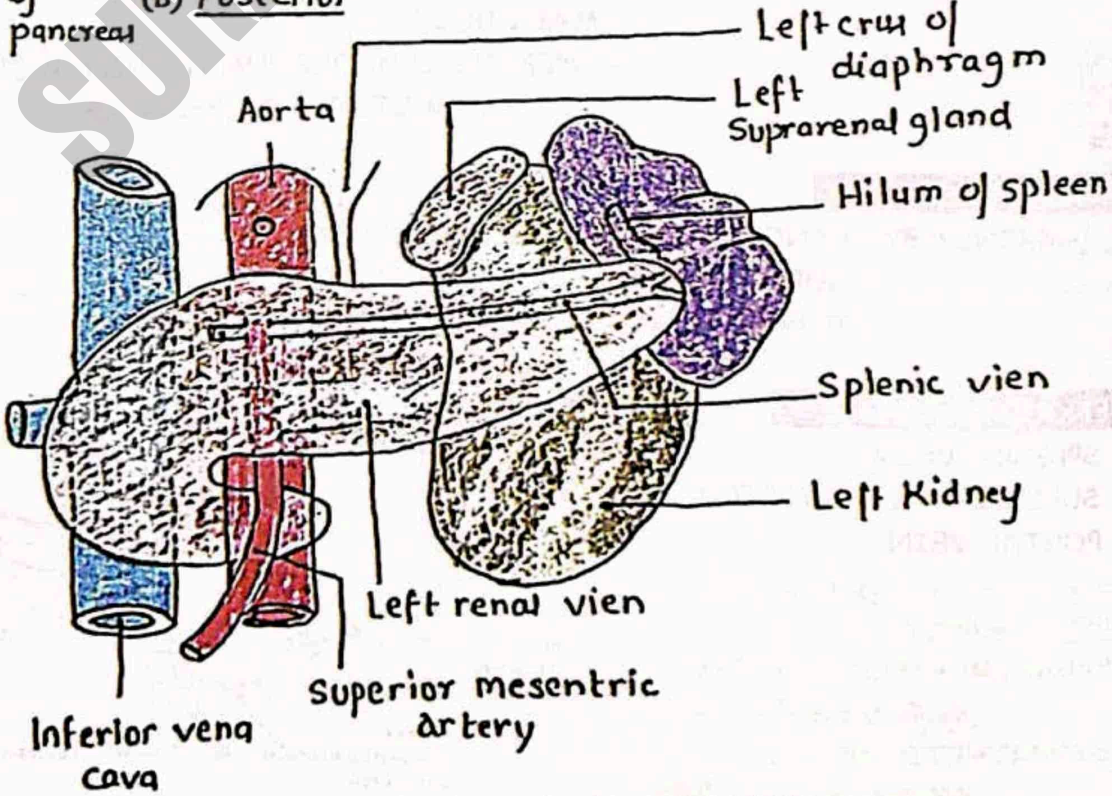


∵ Relation of the neck of pancreas



∵ Relation of anterior surface of body of pancreas





∵ Posterior relation of the body of the pancreas

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KIDNEY

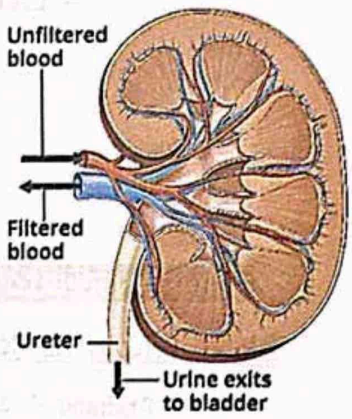
*DEFINITION - PAIR OF EXCRETORY ORGANS
-SITUATED ON POSTERIOR ABDOMINAL WALL
-EACH SIDE OF VERTEBRAL COLUMN BEHIND PERITONEUM

*LOCATION -
- OCCUPY EPIGASTRIC HYPOCHONDRIAC LUMBAR & UMBLICAL REGION
- EXTEND - UPPER BORDER OF T2 TO CENTER BODY OF L3
- RIGHT KIDNEY IS SLIGHTLY LOWER THAN LEFT

*SHAPE - EACH KIDNEY IS ABOUT - 11CM LONG 6CM BROAD 3CM THICK
-WEIGHT - 150g - IN MALE
-135g IN FEMALE

*EXTERNAL FEATURE - BEAN SHAPE
1 2 POLES-UPPER POLE- BROAD, CLOSE CONTACT WITH SUPRARENAL GLAND
2 2 SURFACE - ANTERIOR SURFACE - IRREGULAR
-POSTERIOR SURFACE - FLAT
3. 2 BORDER - LATERAL BORDER - CONVEX
-MEDIAL BORDER - CONCAVE, MIDDLE PART SHOW DEPRESSION - HILUM
4 HILUM - FOLLOWING STRUCTURE SEEN IN HILUM
-FROM ANTERIOR SIDE TO POSTERIOR -RENAL VEIN, RENAL ARTERY, RENAL PELVIS

RELATION OF KIDNEY - RETROPERITONEAL ORGAN
(PERITONEUM COVERED ANTERIOR ONLY)



RELATION COMMON TO 2 KIDNEY -
*UPPER POLE - SUPRARENAL GLAND
*MEDIAL BORDER - SUPRARENAL GLAND, URETER
*POSTERIOR SURFACE - DIAPHRAGM
-MEDIAL & LATERAL ARCUATE LIGAMENTS
-PSOAS MAJOR
-QUADRATUS LUMBORUM
-TRANSVERSUS ABDOMINIS
-SUBCOSTAL VESSELS
-SUBCOSTAL, ILIOHYPOGASTRIC & ILIOINGUINAL NERVES.

OTHER RELATION OF RIGHT KIDNEY -
*ANTERIOR RELATIONS - RIGHT SUPRARENAL GLAND
-LIVER
-SECOND PART OF DUODENUM
-HEPATIC FLEXURE OF COLON
-SMALL INTESTINE

OTHER RELATION TO LEFT KIDNEY -

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ANTERIOR RELATIONS - LEFT SUPRARENAL GLAND
-SPLEEN
-STOMACH
-PANCREAS
-SPLENIC VESSELS
-SPLENIC FLEXURE & DESCENDING COLON
-JEJUNUM

*CAPSULES OR COVERINGS OF KIDNEY -

1.FIBROUS CAPSULE - THIN MEMBRANE - CLOSELY INVEST THE KIDNEY
-NORMALLY IT CAN EASILY STRIPPED OFF
-BUT IN CERTAIN DISEASE IT BECOME ADHERENT & CANNOT STRIPPED.

2.PERINEPHRIC FAT - LAYER OF ADIPOSE TISSUE

3.RENAL FASCIA - FIBROAREOLAR SHEATH-→CONSIST OF 2 LAYER→ANTERIOR - FASCIA OF GEROTA.
→POSTERIOR - FASCIA OF ZUCRERKANDL'S

*EXTENSION - SUPERIORLY - 2 LAYER ENCLOSE SUPRARENAL GLAND
-THEN FUSE EACH OTHER & BECOME CONTINOUS AS DIAPHRAGMATIC FASCIA
- INFERIORLY - 2 LAYER REMAIN SEPARATE & ENCLOSE URETER
-ANTERIOR - LOST IN EXTRAPERITONEAL TISSUE OF ILIAC FOSSA
-POSTERIOR - BLEND WITH FASCIA ILIAC.
- LATERALLY - 2 LAYER UNITE FIRMLY & CONTINOUS WITH FASCIA TRANSVERSALIS
- MEDIALLY - ANTERIORLY LAYER PASSES IN FRONT OF KIDNEY AND MERGE WITH CONNECTIVE TISSUE SURROUND AORTA AND INFERIOR VENA CAVA
-POSTERIOR LAYER PASSES BEHIND KIDNEY AND ATTACH TO FASCIA COVERING QUADRATUS LUMBORUM AND PSOAS MAJOR

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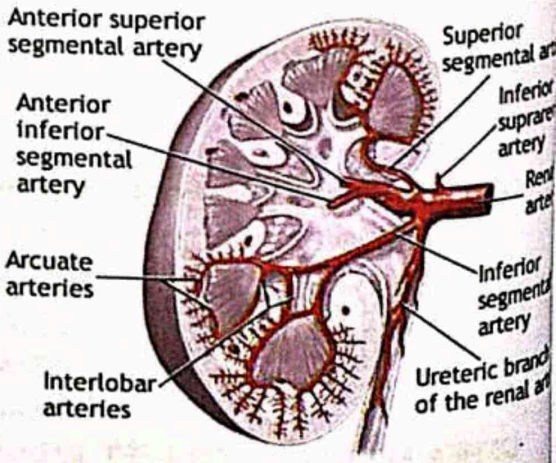
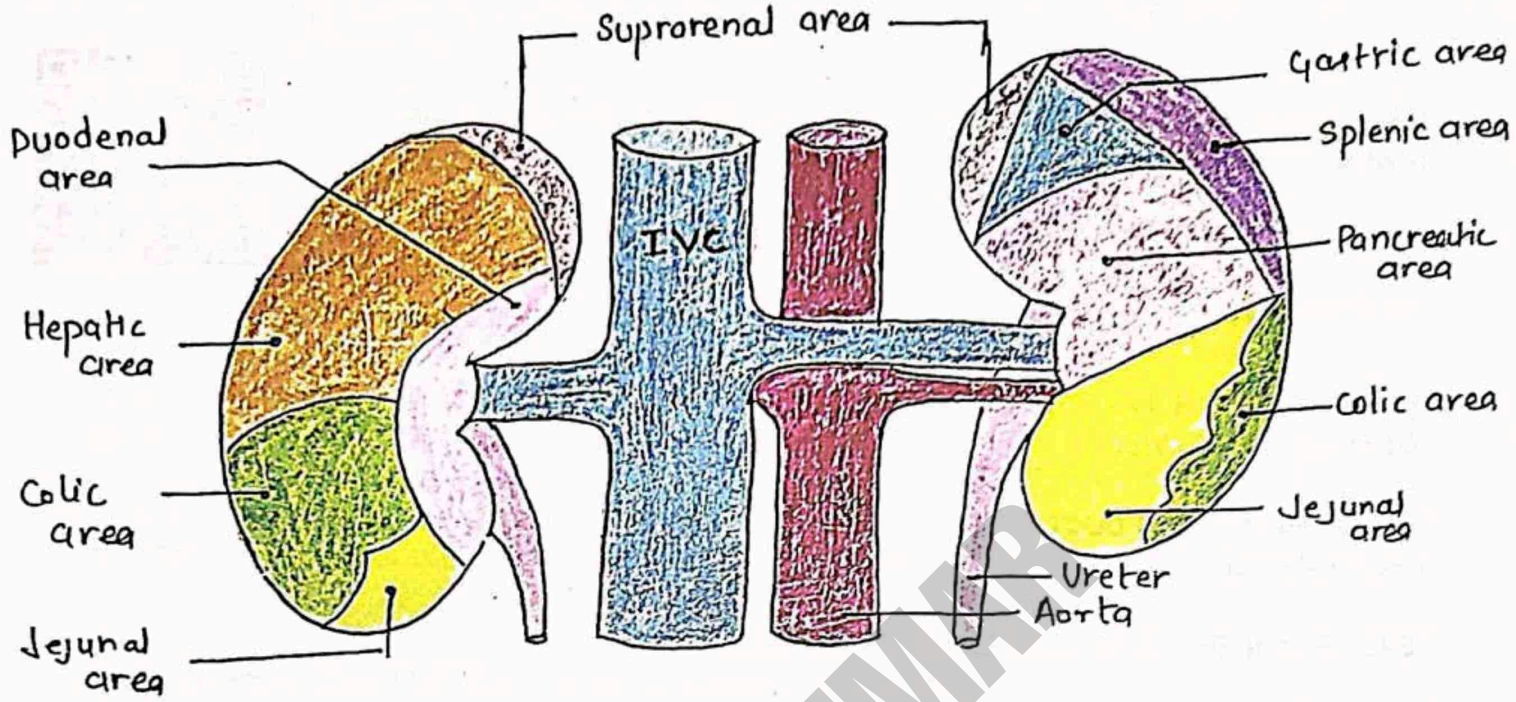
4) PARANEPHRIC FAT - LAYER OF FAT LYING OUTSIDE THE RENAL FASCIA

Blood Supply of Kidney :-

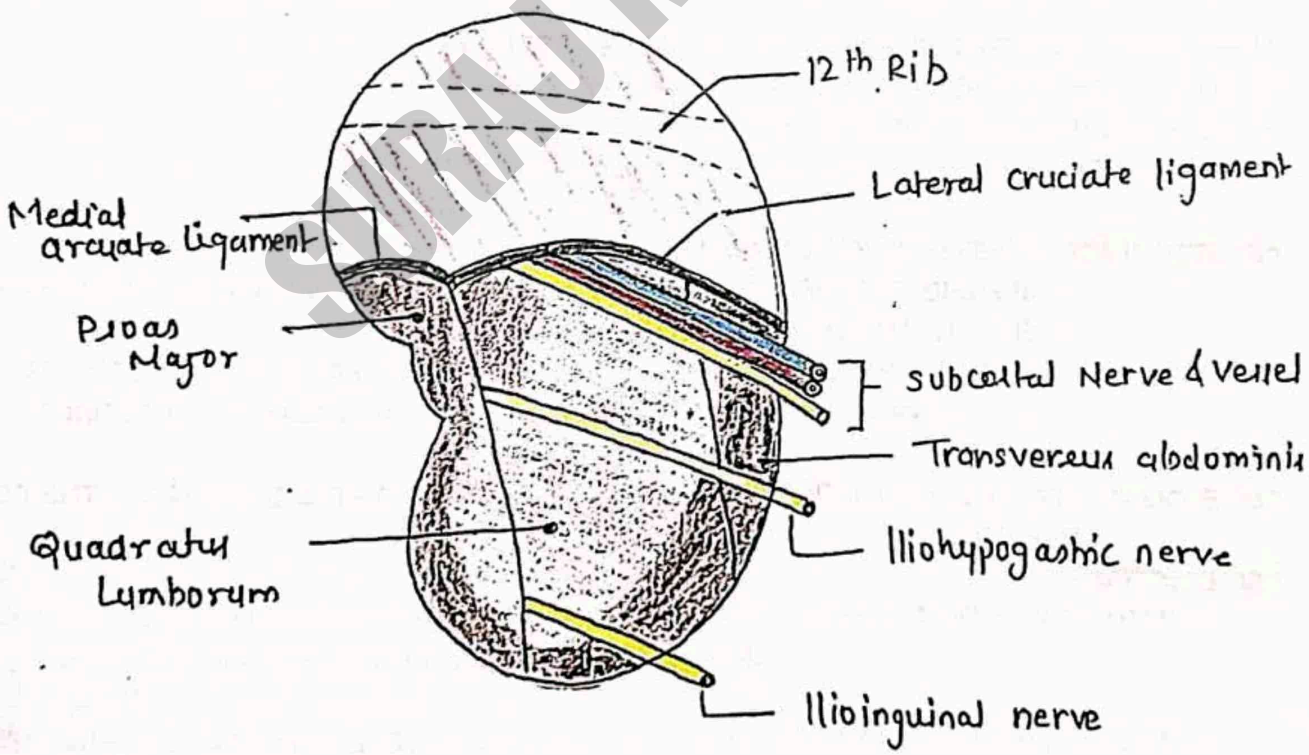
Abdominal aorta
↓
Renal artery
↓
5- segmental artery
↓
Each segmental artery
↓
Lobar artery
↓
interlobar artery
↓
arcuate artery
↓
Interlobula artery
↓
Afferent arteriole → Glomerulus → Efferent arteriole

inferior vena cava
↑
Renal vein
↑
5x segmental vein
↑
segmental vein
↑
Lobar vein
↑
interlobar vein
↑
Arcuate vein
↑
interlobular vein
↑
Peritubular Plexus
↑





∴ Anterior relations of Kidneys



∴ Posterior relation of Kidney

KIDNEY

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EXTRAHEPATIC BILIARY APPARATUS

*EBA COLLECTS BILE FROM LIVER-->STORE IT IN GALL BLADDER-> TRANSMIT IT TO->SECOND PART OF DUODENUM.

*APPARATUS CONSIST OF - RIGHT & LEFT HEPATIC DUCT
-COMMON HEPATIC DUCT
-GALL BLADDER, CYSTIC DUCT & BILE DUCT

*RIGHT & LEFT HEPATIC DUCTS - EMERGE AT PORTA HEPATIS FROM THE RIGHT & LEFT LOBES OF THE LIVER.

*COMMON HEPATIC DUCT - FORMED BY UNION OF RIGHT & LEFT HEPATIC DUCT NEAR THE RIGHT END OF PORTA HEPATIS

*GALL BLADDER - PEAR SHAPED - RESERVOIR OF BILE
-SITUATED IN FOSSA ->ON THE INFERIOR SURFACE OF RIGHT LOBE OF LIVER
-GALL BLADDER IS DIVIDED INTO - FUNDUS
-BODY
-NECK

1) FUNDUS - PROJECT BEYOND THE INFERIOR BORDER OF LIVER
2) BODY - LIE IN THE FOSSA FOR THE GALL BLADDER ON THE LIVER
3) NECK - NARROW UPPER END OF GALL BLADDER
-SITUATED NEAR THE RIGHT END OF PORTA HEPATIS

*CYSTIC DUCT - CYSTIC DUCT IS ABOUT 3-4CM LONG
- BEHIND THE NECK OF GALL BLADDER - RUN DOWNWARD, BACKWARD
- END BY JOINING THE COMMON HEPATIC
- MUCOUS MEMBRANE OF CYSTIC DUCT FORM FROM 5-12 CRECENTIC FOLD ---->ARRANGED SPIRALLY->CALLED SPIRAL VALVE OF HEISTER

*BILE DUCT -FORMED BY UNION OF CYSTIC & COMMON HEPATIC DUCT NEAR THE PORTA HEPATIS.

*SPHINCTER -

- SPHINTER CHOLEDOCHUS - TERMINAL PART OF BILE DUCT - SURROUND SMOOTH MUSCLE -THAT FORM SPHINCTER CHOLEDOCUS.

- SPHINCTER PANCREATICUS - ANOTHER LESS DEVELOPED SPHINCTER, AROUND TERMINAL PART OF PANCREATIC DUCT IS THE SPHINCTER PANCREATICUS.

- SPHINCTER AMPULLAE - THIS IS THIRD SPHINCTER SURROUNDS THE HEPATOPANCREATIC AMPULLA.

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PORTAL VEIN

*LARGE VEIN WHICH COLLECT BLOOD FROM - ABDOMINAL PART OF ALIMENTARY TRACT
-GALL BLADDER
- PANCREAS
- SPLEEN

*IN LIVER, THE PORTAL VEIN BREAK UP INTO SINUSOIDS - WHICH DRAINED BY - HEPATIV VEIN TO INFERIOR VENA CAVA.

*IT IS CALLED PORTAL VEIN BECAUSE - ITS MAIN TRIBUTARY, SUPERIOR MESCENTRIC VEIN, BEGIN IN ONE SET OF CAPILLARIES (IN THE GUT) & PORTAL ONE END IN ANOTHER SET OF CAPILLARIES IN LIVER.

*FORMATION - PORTAL VEIN ABOUT 8CM LONG.
- FORMED BY UNION OF - SUPERIOR MESCENTRIC VEIN BEHIND NECK OF PANCREAS & SPLENIC VEIN.

*COURSE - RUN UPWARD - FIRST BEHIND THE NECK OF PANCREAS
-NEXT BEHIND THE FIRST PART OF DUODENUM
-LASTLY - FREE MARGIN OF LESSER OMENTUM
-DIVIDED INTO -INFRADUODENAL
-RETRODUODENAL
-SUPRADUODENAL PART

-TERMINATION - VEIN END AT RIGHT END OF PORTA HEPATIS BY DIVIDING INTO-
→RIGHT & LEFT BRANCH

*RELATIONS -
- INFRADUODENAL PART - ANTERIORLY - NECK OF PANCREAS
- POSTERIORLY - INFERIOR VENA CAVA
- RETRODUODENAL PART - ANTERIORLY - FIRST PART OF DUODENUM
-BILE DUCT
-GASTRODUODENAL ARTERY
- POSTERIORLY -INTERIOR VENA CAVA
- SUPRADUODENAL PART - ANTERIORLY - HEPATIC ARTERY
-BILE DUCT
- POSTERIORLY - INFERIOR VENA CAVA

*BRANCHES →RIGHT BRANCH - SHORTER & WIDER
→LEFT BRANCH - LONGER & NARROWER

*TRIBUTARIES - PORTAL VEIN RECEIVE THE FOLLOWING VEIN :-
1.LEFT GASTRIC
2.RIGHT GASTRIC
3.SUPERIOR PANCREATICODUODENAL
4.CYSTIC VEIN
5.PARAUMBILICAL VEIN.

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Portosystemic Communications...

Position :-	Portal vein :-	Systemic vein
◦- Lower end of oesophagus	◦- Left gastric	◦- oesophageal vein
◦- Lower end of rectum	◦- Superior rectal	◦- Middle and inferior rectal vein
◦- Umbilicus	◦- Paraumbilical	Above - Superior epigastric lateral thoracic
		Below - Superior epigastric inferior epigastric
◦- Posterior abdominal wall	◦- splenic	Sides - Posterior intercastal
◦- falciform ligament	◦- Paraumbilical	◦- Left renal vein
◦- Ligamentum venosum	◦- Left branch of portal	◦- Diaphragmatic
		◦- Inferior vena cava

CLINICAL ANATOMY

1.PORTAL PRESSURE : NORMAL PRESSURE IN THE PORTAL VEIN IS ABOUT 5-15MM HG IT IS USUALLY MEASURED BY SPLENIC PUNCTURE & RECORDING THE INTRASPLENIC PRESSURE.

2.PORTAL HYPERTENSION(PRESSURE ABOVE 40MM HG): IT CAN BE USED BY THE FOLLOWING :-

a.CIRRHOSIS OF LIVER, IN WHICH THE VASCULAR BED OF LIVER IS MARKEDLY OBLITERATED.

b.BANTI'S DISEASE

c.THROMBOSIS OF PORTAL VEIN.

3.THE EFECT OF PORTAL HYPERTENSION ARE AS FOLLOWS :-

1.CONGESTIVESPLENOMEGALY.

2.COLLATERAL CIRCULATION THROUGH THE PORTOSYSTEMIC COMMUNICATIONS.IT FORMS:-CAPUT MEDUSAE,OESOPHAGEAL VARICES,HAEMORRHOIDS IN ANAL CANAL.

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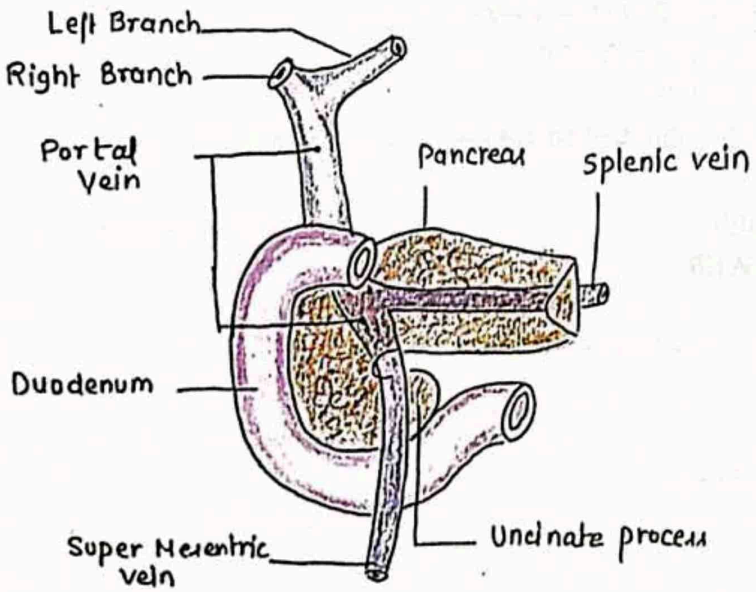
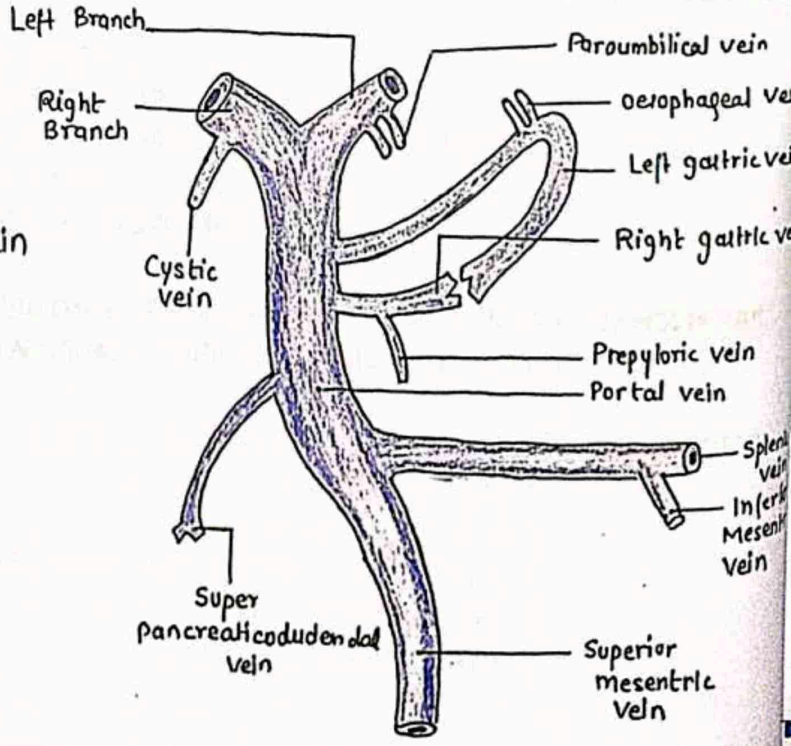


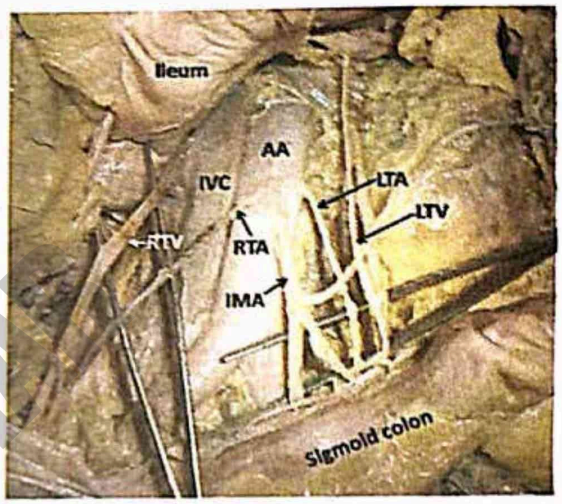


ABDOMINAL AORTA

*COURSE :-

1. BEGIN AT THE AORTIC OPENING OF DIAPHRAGM
2. IT RUN DOWN WARD AND END IN FRONT OF LOWER PART OF BODY OF VERTEBRA L4.

*RELATIONS :-



1 ANTERIORLY - COELIAC AND AORTIC PLEXUSES
- BODY OF PANCREAS
- THIRD PART OF DUODENUM

2.POSTERIORLY - BODY OF UPPER 4X LUMBAR VERTEBRA
- ANTERIOR LONGITUDINAL LIGAMENT

3.TO RIGHT SIDE - INFERIOR VENA CAVA,
- RIGHT CRUS OF DIAPHRAGM
- CISTERNA CHYLI

4.TO LEFT SIDE - LEFT CRUS OF DIAPHRAGM
- PANCREAS
- 4TH PART OF DUODENUM

*BRANCHES

1) VENTRAL BRANCHES- 1. COELIAC TRUNK→ LEFT GASTRIC ARTERY
→ COMMON HEPATIC ARTERY
→ SPLENIC BRANCH

2.SUPERIOR MESENTERIC ARTERY → INFERIOR PANCREATICODUODENAL
→MIDDLE COLIC
→RIGHT COLIC
→ILEO COLIC
-12-15 JEJUNAL
→ ILEAL BRANCHES

3.INFERIOR MESENTERIC ARTERY --LEFT COLIC
→ SIGMOID ARTERY
→ SUPERIOR RECTAL

2) LATERAL BRANCHES - INFERIOR PHRENIC ARTERIES
- MIDDLE SUPRARENAL ARTERIES
- RENAL ARTERIES
- TESTICULAR OR OVARIAN ARTERIES

3) DORSAL BRANCHES - LUMBAR ARTERY
- MEDIAN SACRAL ARTERY

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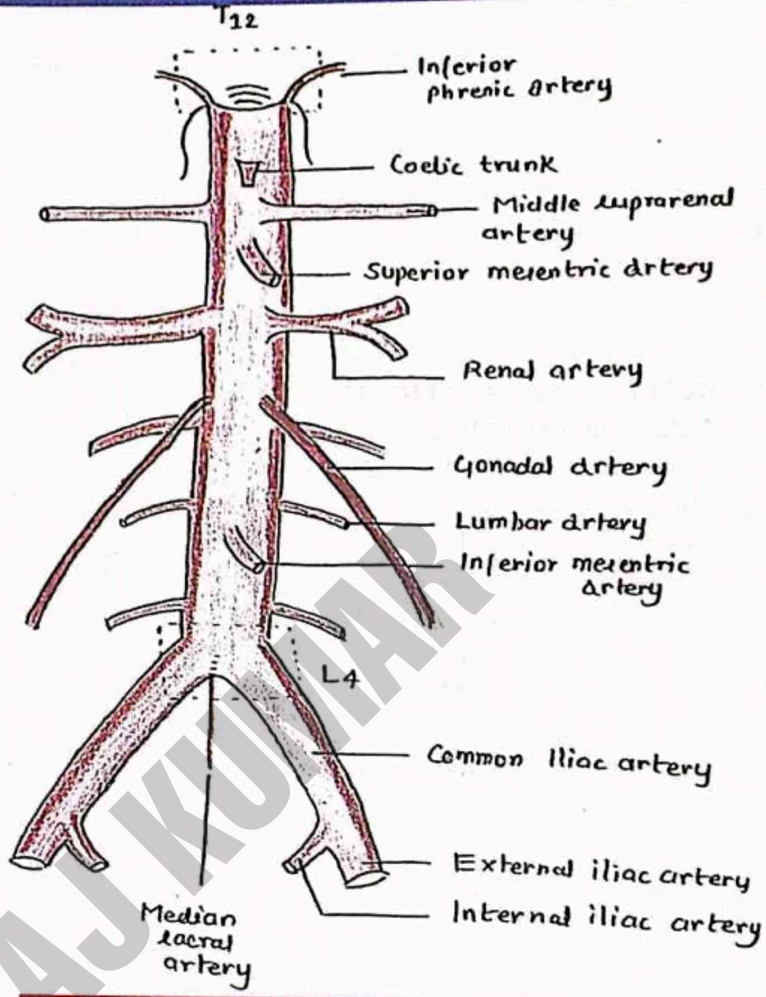
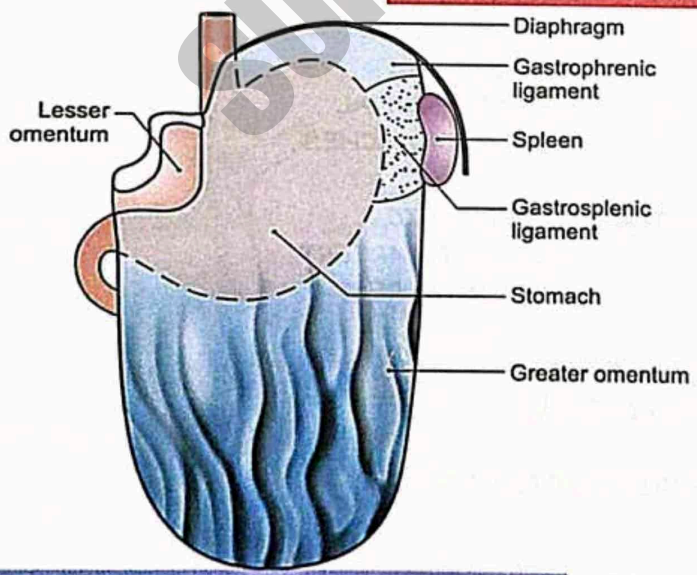
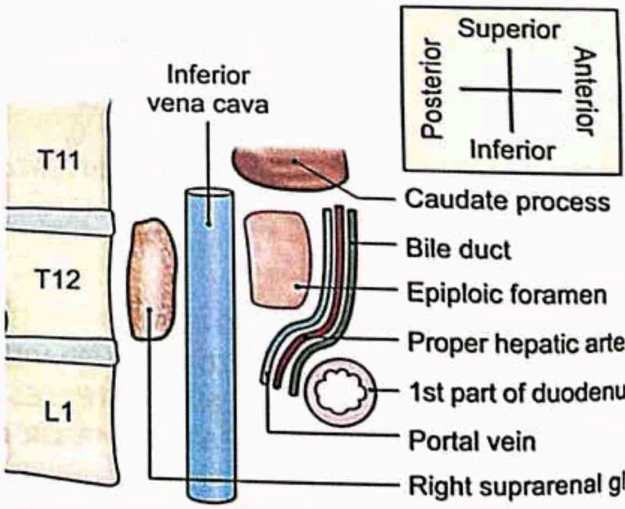


ABDOMINAL AORTA



GREATER OMENTUM



BOUNDARIES EPIPLOIC FORAMEN

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GREATER OMENTUM

*FOLD OF PERITONEUM - WHICH HANGS DOWN FROM-GREATER CURVATURE OF STOMACH.
*MADE UP OF FOUR LAYERS OF PERITONEUM- ALL WHICH ARE FUSED TOGETHER
*AND FORM THIN FENESTRATED MEMBRANE CONTAINING VARIABLE-QUANTITIES OF
FAT
-SMALL ARTERIES & VEIN

*ATTACHMENT - ANTERIOR 2x LAYER DESCEND FROM GREATER CURVATURE OF STOMACH TO A, VARIABLE EXTEND.
- AND FOLD THEMSELVE TO FORM POSTEOIR TWO LAYER
- FOLDING OF OMENTUM IS SUCH THAT FIRST LAYER BECOME- FOURTH LAYER & 2ND LAYER-→ 3RD LAYER
- IN UPPER PART, FOURTH LAYER IS PARTIALLY FUSED TO ANTERIOR SURFACE OF TRANSVERSE COLON & TRANSVERSE MESOCOLON.

*CONTENT - RIGHT & LEFT GASTROPIPLOIC VESSELS
-FAT
*FUNCTIONS - STORE HOUSE OF FAT
- PROTECT PERITONEAL CAVITY AGAINST INFECTION BECAUSE OF PRESENCE OF MACRO PHAGES.
- COLLECTION OF MACROPHAGES FROM SMALL, DENSE PATCH KNOWN - AS MILKY SPOT.
- IT ALSO LIMIT SPREAD OF INFECTION BY MOVING TO SITE OF INFECTION & SEALING IT OFF.
- GREATER OMENTUM IS ALSO KNOWN AS POLICEMEN OF ABDOMIN.

LESSER OMENTUM

1) FOLD OF PERITONEUM WHICH EXTEND-FROM- LESSER CURVATURE OF STOMACH TO FIRST 2CM OF DUODENUM TO THE LIVER
2) PORTION OF L.O B/W STOMACH AND LIVER- HEPATOGASTRIC LIGAMENT
3) PORTION OF L.O B/W DUODENUM AND LIVER- HEPATODUODENAL LIGAMENT

*ATTACHMENTS:-
1) INFERIORLY- LESSER CURVATURE OF STOMACH
- UPPER BARDER OF FIRST 2CM OF DUODENUM

2) SUPERIORLY - LIVER ATTACHMENT AS AN INVERTED 'L'
-VERTICAL LIMB OF L'- IS FISSURE FAR LIGAMENTUM VENOSUM
-HARIZONTAL LIMB - MARGIN OF PORTA HEPATIS.

*CONTENTS - PROPER HEPATIC ARTERY

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- PORTAL VEIN
- BILE DUCT
- LYMPH NODES
- HEPATIC PLEXUS OF NERVE

- ALONG THE- LESSER CURVATURE OF STOMACH - CONTAIN - RIGHT & LEFT GASTRIC VESSELS.
-GASTRIC GROUP OF LYMPH NODE
-BRANCHES FROM GASTRIC NERVE

EPIPLOIC FORAMEN/OMENTAL AND WINSLOW FORAMEN

-VERTICAL SLIT-LIKE OPENING THROUGH WHICH - LESSER SAC COMMUNICATES WITH GREATER SAC.
-FORAMEN IS SITUATED BEHIND THE RIGHT FREE MARGIN OF LESSER OMENTUM AT LEVEL OF T12 VERTEBRA.

BOUNDARIES :-
1) ANTERIORLY: LESSER OMENTUM → CONTAINING - PORTAL VEIN
- PROPER HEPATIC ARTERY
- BILE DUCT

2) POSTERIORLY: - INFERIOR VENA CAVA
- RIGHT SUPRARENAL GLAND
- T12 VERTEBRA
3) SUPERIORLY - CAUDATE PROCESS OF LIVER
4) INFERIORLY - FIRST PART OF DUODENUM

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CLINICAL ANATOMY

CLINICAL ANATOMY OF ISCHIONAL FOSSA

1.PROLAPSE OF RECTUM : THE ISCHIONAL FAT ACTS AS A CUSHION LIKE SUPPORT TO THE RECTUM & ANAL CANAL. LOSS OF THIS FAT IN DEBILITATING DISEASES LIKE DIARRHOEA IN CHILDREN MAY RESULT IN PROLAPSE OF RECTUM.

2.BOTH THE PERIANAL & ISCHIOANAL ARE COMMON SIDE OF ABSCESSES.

3.ISCHIONAL HERNIA : THE OCCASIONAL GAP BETWEEN THE TENDINOUS ORIGIN OF LEVATOR ANI & THE OBTURATOR FASCIA IS KNOWN AS HIATUS OF SCHWALBE. RARELY, PELVIC ORGANS MAY HERNIATE THROUGH THIS GAP INTO THE ISCHIONAL FOSSA, RESULTING IN AN ISCHIONAL HERNIA.

ISCHIONAL FOSSA

-WEDGE-SHAPED FOSSA SITUATED - EACH SIDE ANAL CANAL BELOW→ PELVIC DIAPHRAGM.

- BASE - DOWNWARD, TOWARD SURFACE
- APEX - UPWARD

*BOUNDARIES:-

1) BASE- FORMED BY SKIN OF PERINEUM
2) APEX -FORMED BY - LINE WHERE- OBTURATOR FASCIA MEET INFERIOR FASCIA OF PELVIC DIAPHRAGM
3) ANTERIORLY- POSTERIOR BORDER OF PERINEAL MEMBRANE
4) POSTERIARLY - GLUTEUS MAXIMUS
   -SACROTUBEROUS LIGAMENT
5) LATERAL WALL - OBTURATOR INTERNUS
   -ISCHIAL TUBEROSITY
6) MEDIAL - LEVATOR ANI
   - EXTERNAL ANAL SPHINCTER

*SPACE AND CANNALS OF FOSSA:

1) PERIANAL SPACE - PERIANAL FOSSA FORM OF SEPTUM PASSES LATERALLY LONGITUDINAL COAT OF ANAL CANAL

↳ EXTEND-MEDIALLY FROM WHITE LINE OF HILTON
LATERALLY→ PUDENDAL CANAL
↳ SEPARATES SHALLOW SUBCUTANEOUS PERIANAL SPACE FROM DEEP ISCHIOANAL SPACE
↳ FAT IN TIGHTLY ARRANGED IN SMALL LOCULI FORMED BY → COMPLETE SEPTA

2) ISCHIOANAL SPACE - LARGE AND DEEP
   - FAT- IN LOSSELY ARRANGED IN LARGE LOCULI FORMED- BY→INCOMPLETE DELICATE SEPTA.
   -LUNATE FASCIA-ARCHES OVER THE ISCHIAL FAT
3) PUDENDAL CANNAL- FASCIAL CANNAL IN LATERAL WALL OF IF
   - ENCLOSING - PUDENDAL NERVE
   - INTERNAL PUDENDAL VESSEL

SUPERIOR SURFACE -> MALE - COMPLETELY COVERED BY PERITONEUM
->FEMALE - PERITONEUM COVERED GREATER POINT EXCEPT SMALL AREA NEAR POSTERIOR BORDER.

*CONTENTS OF ISCHIONAL FOSSA -

1.ISCHIOANAL PAD OF FAT
2.INFERIOR RECTAL NERVE & VESSEL

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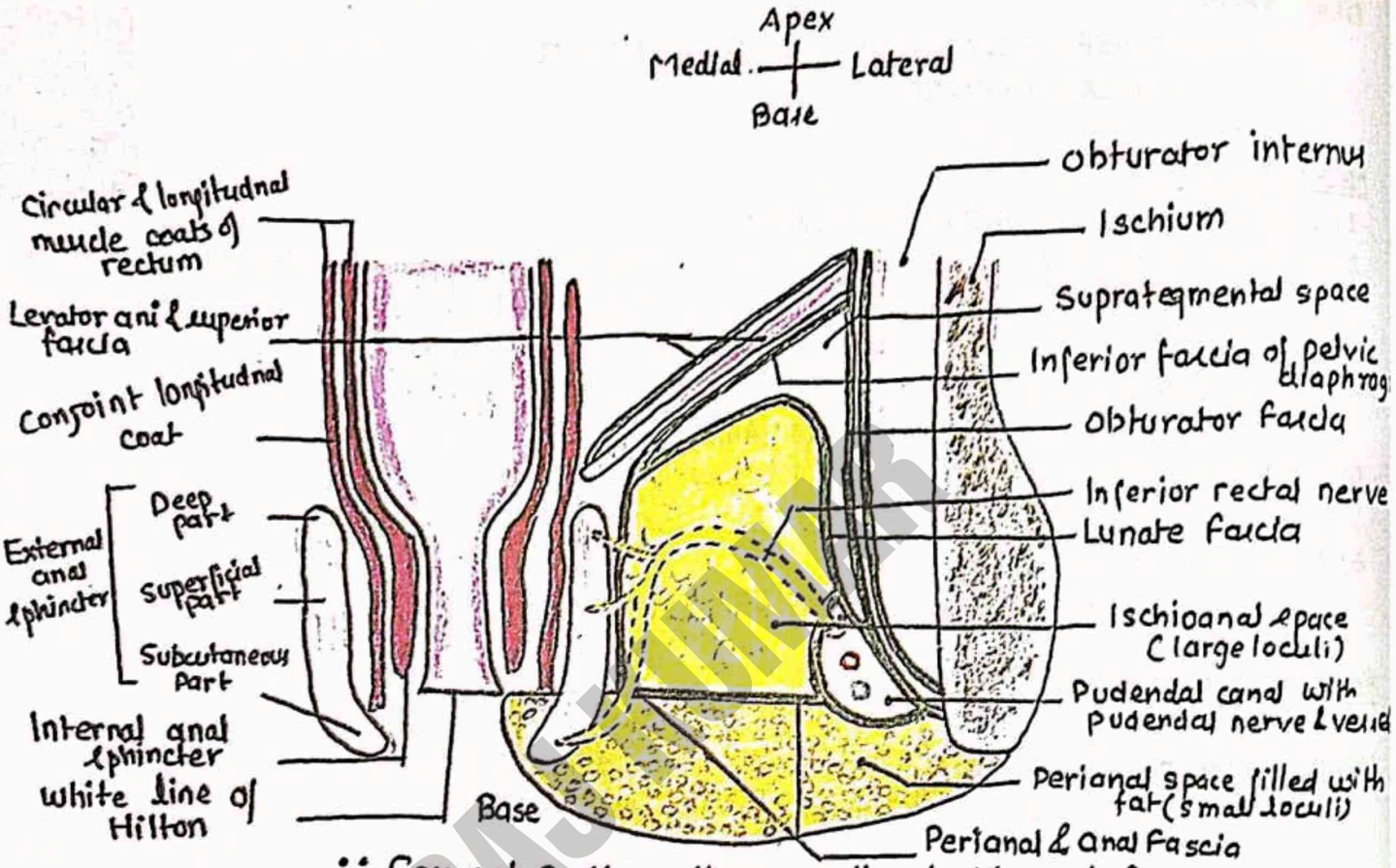
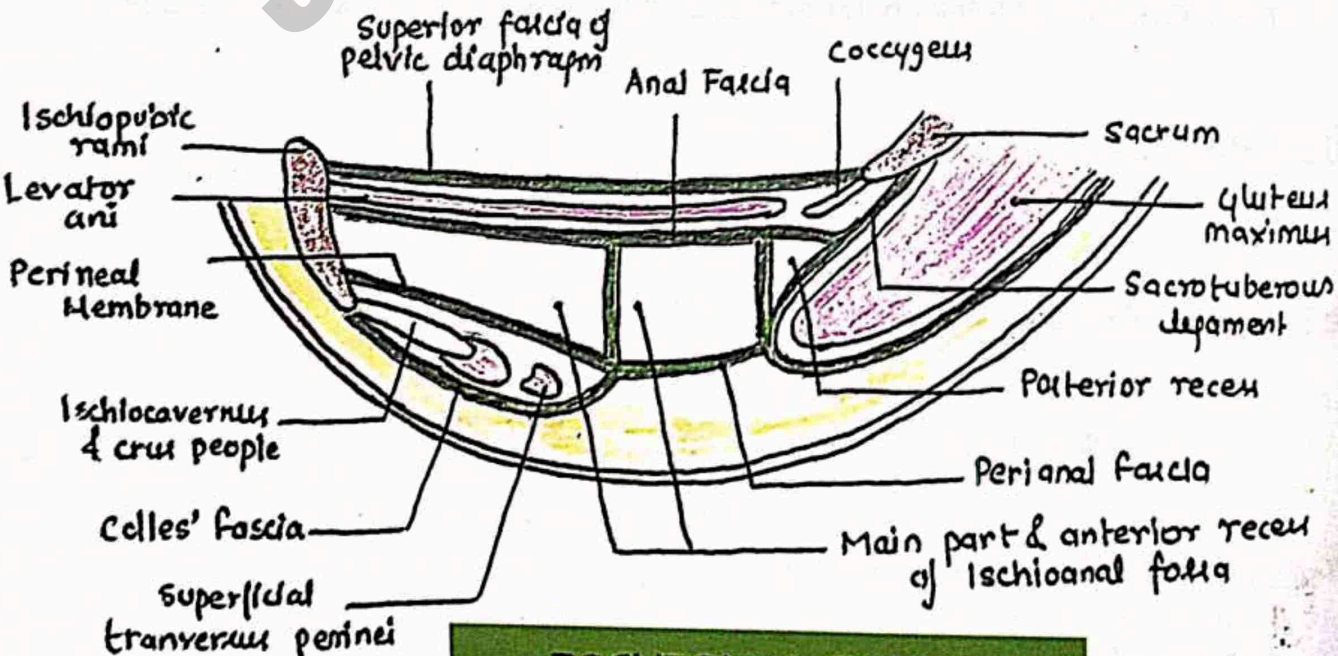
3.PUDENDAL CANAL
4.POSTERIOR SCROTAL OR POSTERIOR LABIAL NERVE & VESSEL
5.PERINEAL BRANCH OF 4TH SACRAL(S4)
6.PERFORATING CUTANEOUS BRANCHES OF NERVE (S2, S3)



∵ Coronal section through the Ischioanal fossa



ISCHIONAL FOSSA

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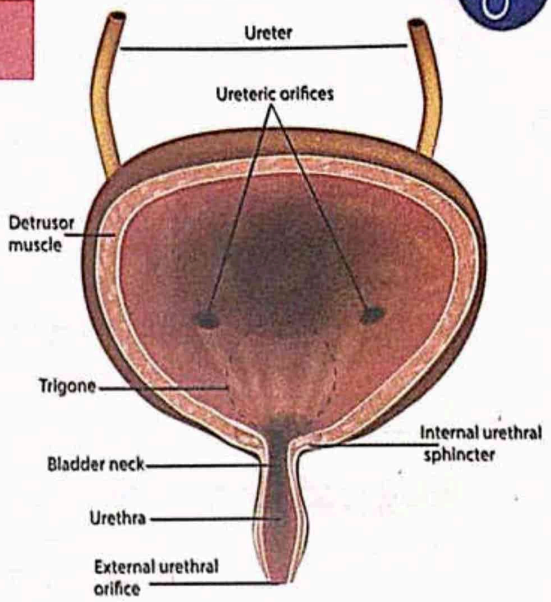
URINARY BLADDER



*FEATURE:-

- MUSCULAR RESERVOIR OF URINE
- LIE IN ANTERIOR PART OF PELVIC CAVITY
- SHAPE- EMPTY → TETRAHEDRAL
  FULL → OVOID

*EXTERNAL FEATURE :-

1) EMPTY BLADDER -APEX
   - BASE OR FUNDUS
   - NECK
   - 3 SURFACE -SUPERIOR
     -RIGHT AND LEFT INFEROLATERAL
   - 4 BORDER - 2 LATERAL , 1 ANTERIOR , 1 POSTERIOR

2) FULL BLADDER- APEX
   - NECK
   - TWO SUFACE - ANTERIOR OR POSTERIOR

*RELATION-

1) APEX - MEDIAN UMBILICAL LIGAMENT

2) BASE - FEMALE- UTERINE CERVIX
   -VAGINA

   - MALE- UPPER-SEPARATED FROM RECTUM BY RECTOVESICAL POUCH
     LOWER - SEMINAL VESICLE , VAS-DEFERENS

3) NECK - LOWER MOST FIXED PART OF BLADDER
   MALE- SMOOTH MUSCLE SURROUND BLADDER NECK AND PREPROSTATIC URETHRA→
   PREPROSTATIC → DEVOID OF PARASYMPATHETIC CHOLINERGIC NERVE

   FEMALE- RELATED TO PELVIC FASCIA → SURROUNDING THE UPPER PART OF URETHRA

4) SUPERIOR SURFACE -
MALE - COMPLETELY COVERED BY PERITONEUM

FEMALE - PERITONEUM COVER GREATER PART EXCEPT SMALL AREA NEAR POSTERIOR BORDER.

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5) INFEROLATERAL SURFACE - DEVOID OF PERITONEUM
- MALE- PUBIS
- PUBOPROSTATIC LIGAMENT
- RETROPUBIC FAT
- LEVATOR ANI
- OBTURATAN INTERNUS

-FEMALE- SAME EXCEPT - PUBOPROSTATIC LIGAMENT

*LIGAMENT:-

TRUE LIGAMENT - CONDENSATION OF PELVIC FASCIA AROUND NECK AND BASE OF BLADDER
- LATERAL TRUE LIGAMENT
- MEDIAL PUBOPROSTATIC LIGAMENT OR PUBOVESICAL LIGAMENT
- MEDIAN UMBILICAL LIGAMENT
- POSTERIOR LIGAMENT WITH VENOUS PLEXUS.

FALSE LIGAMENT- THEY ARE PERITONEAL FOLD
- MEDIAN UMBILICAL FOLD
- MEDIAL UMBILICAL FOLD
- LATERAL FALSE LIGAMENT
- POSTERION FALSE LIGAMENT

*ARTERIAL SUPPLY - MAIN- SUPERIOR, INFERIOR VESICAL ARTERY BRANCH OF INTERNAL ILIAC ARTERY
ADDITIONAL - OBTURATOR AND INFERIOR GLUTEAL ARTERIES SUPPLY

*VENOUS DRAINAGE - VESICAL VENOUS PLEXUS—->INTERNAL ILIAC VEIN

*NERVE -

1) PARASYMPATHETIC EFFERENT FIBRES - S2,3,4 → MOTOR TO DETRUSOR MUSCLE
2) SYMPATHETIC EFFERENT FIBRES - T11 TO L2 INHIBITORY TO DETRUSOR MUSCLE

3) SOMATIC PUDENDAL NERVE - SUPPLY-SPHINCTER URETHRAE - VOLUNTARY
4) SENSORY NERVE. PAIN SENSATION CARRIED BY - PARASYMPATHETIC NERVE.

CLINICAL ANATOMY

1.A DISTENDED BLADDER MAY BE RUPTURED BY INJURIES OF THE LOWER ABDOMINAL WALL.
2.CHRONIC OBSTRUCTION TO THE OUT FLOW OF URINE BY AN ENLARGED PROSTATE CAUSES HYPERTROPHY OF BLADDER LEADING TO TRABUCULATED BLADDER.
3.URINARY BLADDER IS ONE OF SITES FOR STONE FORMATION AS CONCENTRATED URINE LIES HERE.

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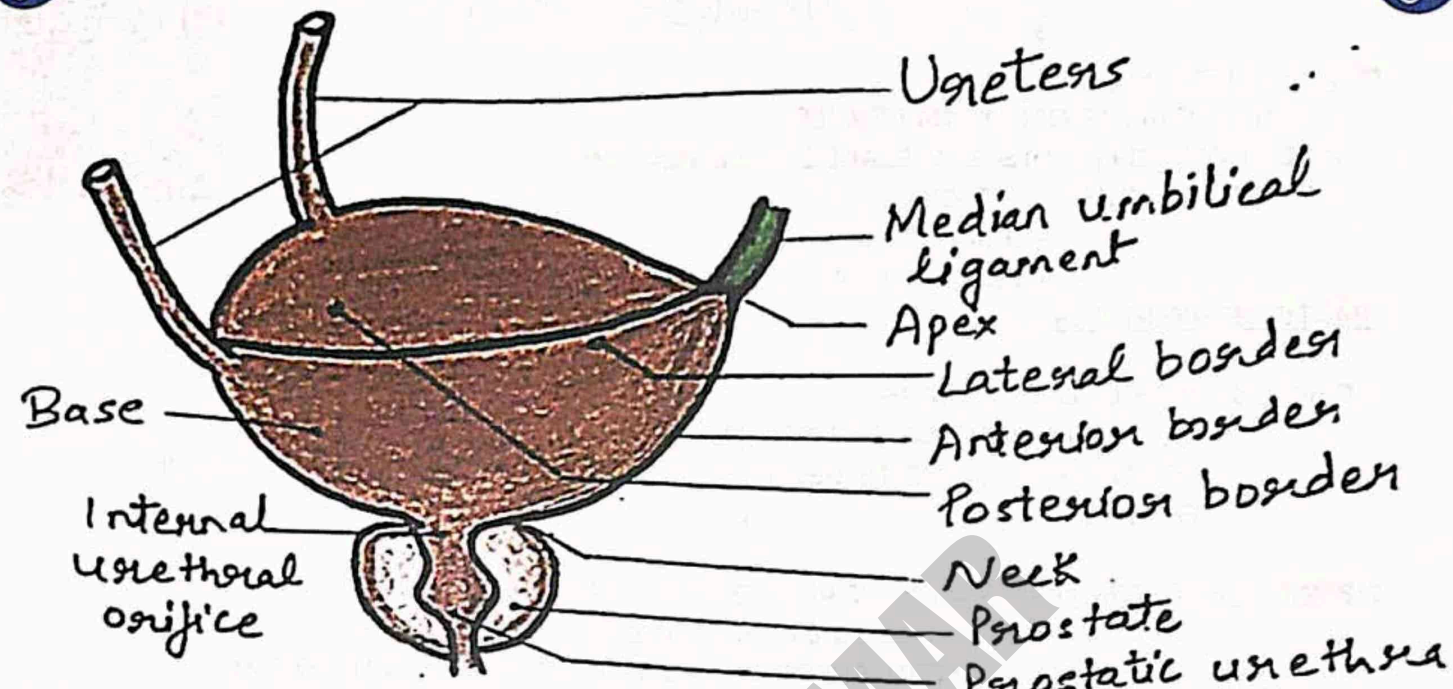
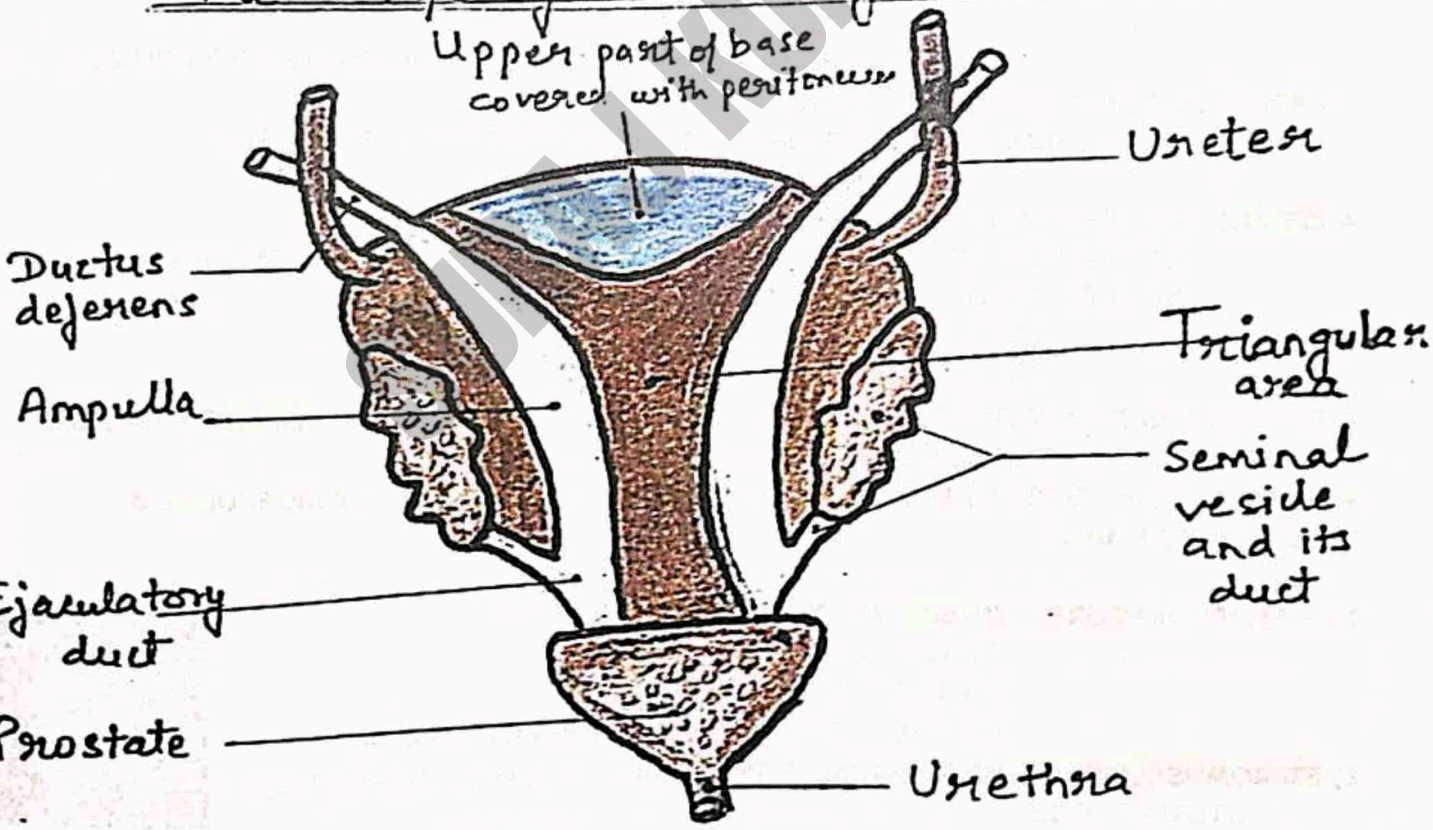


The Shape of the Urinary bladder



URINARY BLADDER

UTERUS

*INTRODUCTION :-
- CHILD BEARING ORGAN IN FEMALE
- SITUATED IN PELVIS B/W BLADDER AND RECTUM
- PYRIFORM SHAPE - 7.5 CM LONG
  - 5 CM BROAD
  - 2.5CM THICK

*PARTS OF UTERUS :-

1.FUNDUS- FREE END OF UTERUS
- LIE ABOUT OPENING OF UTERINE TUBE
- FERTILISED OOCYTE IS USUALLY IMPLANTED IN POSTERIOR WALL OF FUNDUS

2.BODY- ANTERIOR PART - IS FLAT RELATED TO URINARY BLADDER
- COVERED BY PERITONEUM
- FORM SUPERIOR WALL OF UTEROVESICAL POUCH

- POSTERIOR SURFACE - IS CONVEX RELATED TO TERMINAL ILEUM
  → COVERED WITH PERITONEUM
  → FORM ANTERIOR WALL OF RECTOUTERINE POUCH

3.LATERAL BODY - ROUND AND CONVEX
- PROVIDE ATTACHMENT TO BROAD LIGAMENT OF UTERUS

4.CERVIX -THE CERVIX IS LOWER CYLINDRICAL PART OF UTERUS.
- LOWER PART OF CERVIX PROJECTS INTO ANTERIOR WALL OF VAGINA DIVIDED INTO SUPRA VAGINAL AND VAGINAL

SUPPORT OF UTERUS

- THE UTERUS IS A MOBILE ORGAN WHICH UNDERGOES EXTENSIVE CHANGES IN SIZE & SHAPE DURING THE REPRODUCTIVE PERIOD OF LIFE.
- IT IS SUPPORTED BY A NUMBER OF FACTORS WHICH ARE CHIEFLY MUSCULAR & FIBROMUSCULAR :-

1) PRIMARY SUPPORT:- MUSCULAR OR ACTIVE SUPPORT :-
- PELVIC DIAPHRAGM
- PERINEAL BODY
- DISTAL URETHRAL SPHINCTER MECHANISM

2) FIBROMUSCULAR OR MECHANICAL SUPPORT :-
1. UTERINE AXIS
2. PUBOCERVICAL LIGAMENT
3. TRANSVERSE CERVICAL LIGAMENT
4. UTERO SACRAL LIGAMENT
5. ROUND LIGAMENTS OF UTERUS

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3) SECONDARY SUPPORT :-

1. BROAD LIGAMENT
2. VESICOUTERINE POUCH & FOLD OF PERITONEUM
3. RECTOVAGINAL OR RECTOUTERINE POUCH & FOLD OF PERITONEUM

ROLE OF INDIVIDUED SUPPORTS:-

1. PELVIC DIAPHRAGM -
- SUPPORT PELVIC VISCERA & WRIST INCREASE IN INTRAABDOMINAL PRESSURE
- PUBOCOCCYGEUS- INSERT INTO PERINEAL BODY BETWEEN VAGINAL RECTUM
- TORN DURING PARTURITION THE SUPPORT OF UTERUS LOST.

2. PERINEAL BODY- FIBROMUSCULAR NODE TO WHICH TEN MUSCLES ARE ATTACHED.

THE MUSCLE ARE -
- 2 SUPERFICIAL TRANSVERSE PERINEI.
- 2 DEEP TRANSVERSE PERINEI.
- 2 PUBOCOCCYGEUS.
- 2 BULBOSPONGIOUS.
- 1 SINGLE SPHINCTER ANI EXTERNUS.
- LONGITUDINAL MS COATS OF ANAL CANAL.

3. URETHRAL SPHINCTER MECHANISM - UROGENITAL DIAPHRAGM DOES NOT EXIST IN URETHRA IS WITHIN THE WALL OF URETHRA.
- IN ADDITION COMPRESSOR URETHRAE & SPINCTER URETHRAE VAGINALIS INSERT INTO VAGINA SUPPORT UTERUS INDIRECTLY.

4) UTERINE AXIS -
ANTEVERTED POSITION OF UTERUS- PREVENT ORGAN SAGGING DOWN THROUGH VAGINA.

5) PUBOCERVICAL LIGAMENT - CONNECT CERVIX TO POST SURFACE OF PUBIS.
- DERIVED FROM ENDOPELVIC FASICA.

6 TRANSVERSE CERVICAL LIGAMENT - CONNECT LAT. ASPECT OF CERVIX TO LAT. PELVIC WALL
-DERIVED FROM ENDOPELVIC FASICA FAN SHAPE CONDENSATION

7. UTEROSACRAL LIGAMENT -CONNECT CERVIX TO PERIOSTEUM OF SACRUM.
-DERIVED FROM ENDOPELVIC FASICA.
-KEEP CERVIX BACKWARD AGAINST ROUND LIGAMENT.
-2 LIGAMENT COUPLE MAINTAIN UTERINE AXIS

8. ROUND LIGAMENT - 2 FIBROMUSCULAR BAND - 10 TO 12 CM LONG.

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EACH LIG. BEGIN INTO AT LAT. ANGLE OF UTERUS

↓

RUN FORWARD LATERALLY

↓

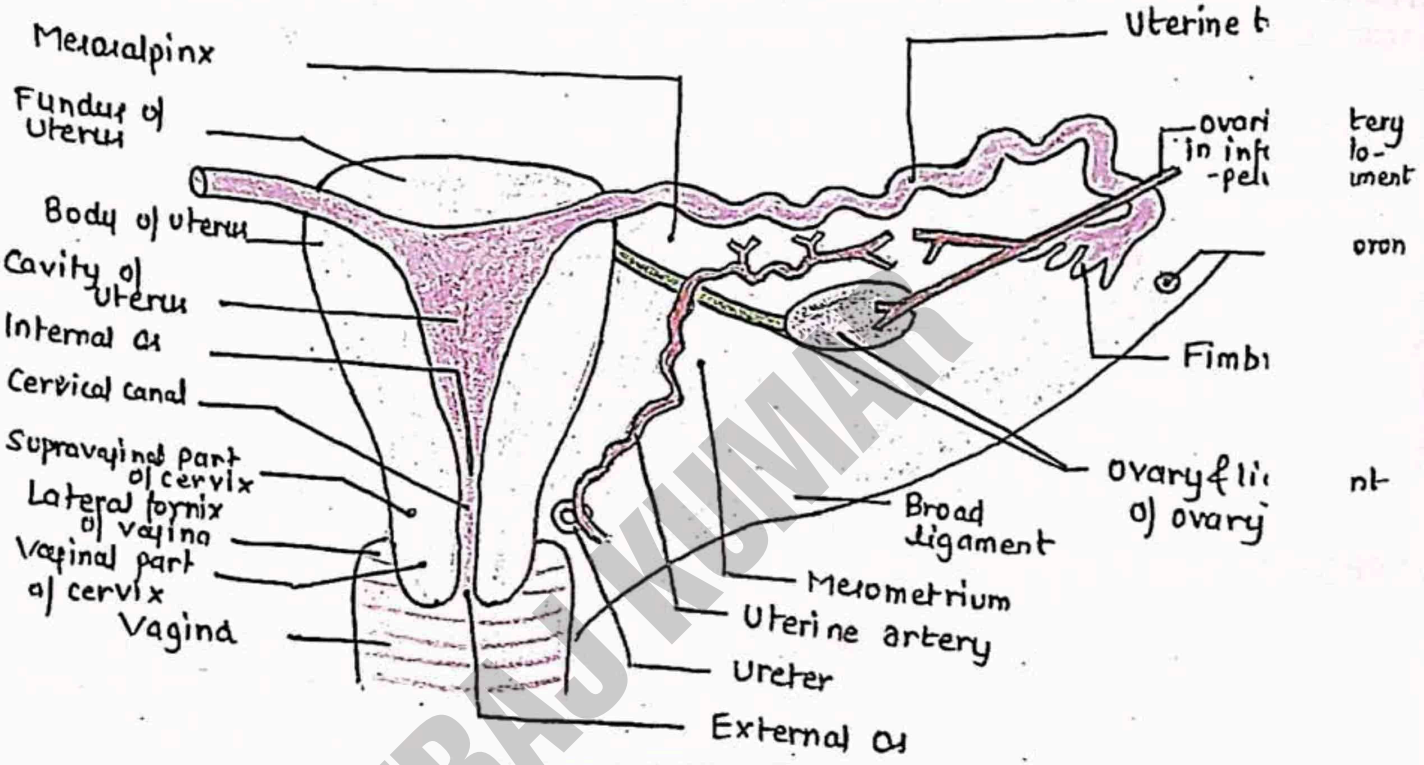
PASS DEEP INGUINAL RING

↓

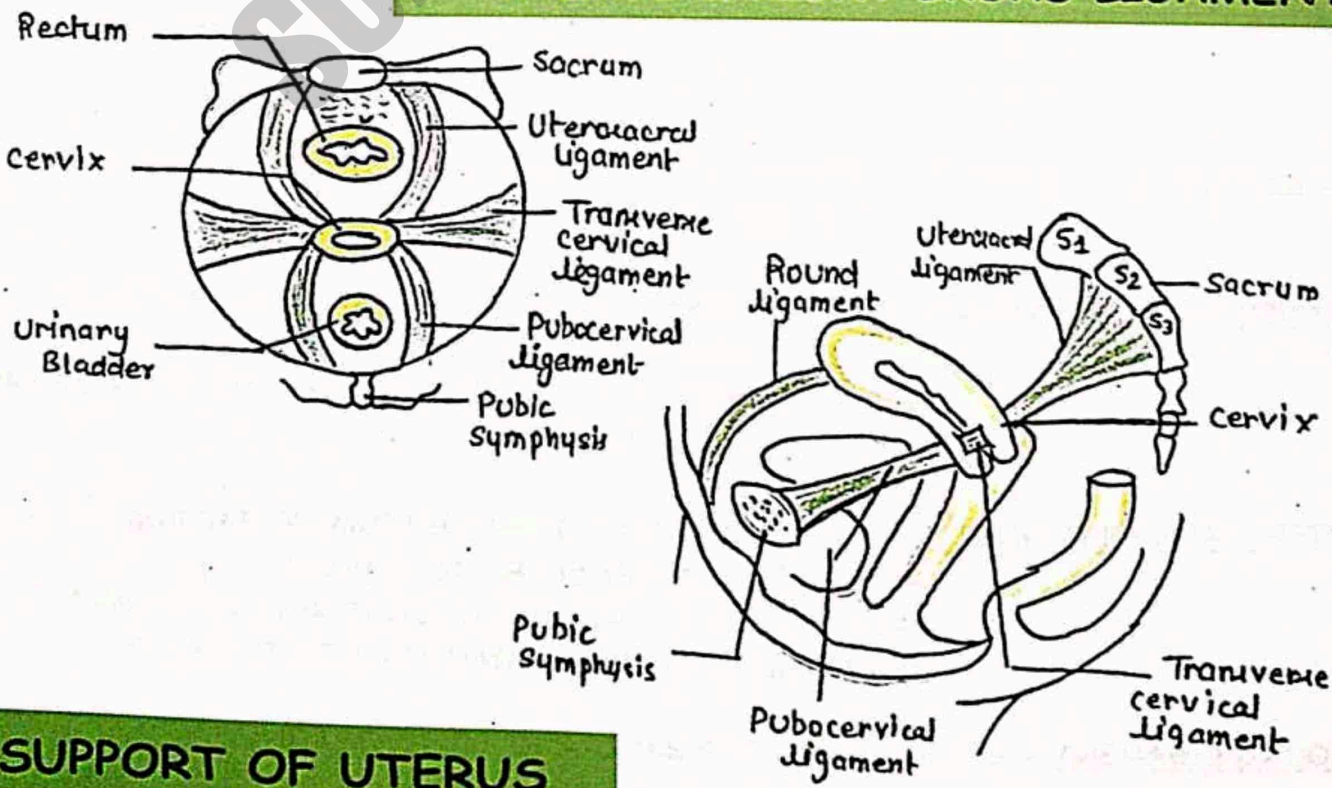
TRAVERSE THE INGUINAL CANAL

↓

MERGE WITH AEROLAR TISSUE TO LABIUM MAJUS.



UTERUS & RIGHT BROAD LIGAMENT



SUPPORT OF UTERUS

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UTERINE TUBES

*DEFINITION- TORTUOUS DUCT WHICH, CONVEY OOCYTE
-SITUATION IN UPPER FREE MARGIN OF BROAD LIGAMENT OF UTERUS
-DIMENSION- UTERINE TUBE IS ABOUT 10CM LONG.

*SUBDIVISION:

1) INFUNDIBULUM - LATERAL END OF UTERINE TUBE IS LIKE FUNNEL SHAPE
- BEAR NUMBER OF FRINGER LIKE PROJECTION - FIMBRIAE

2) AMPULLA - MEDIAL TO THE INFUNDIBULUM
-THIN WALLED DILATED AND TORTUOUS - 4MM DIAMETER

3) ISTHMUS - SUCCEEDS THE AMPULLA
4) INTRAMURAL - INTERSTITIAL PART OF THE TUBE IS ABOUT 1CM LONG

*COURSE:-

1) ISTHMUS - POSTEROLATERALLY IN HORIZONTAL PLANE
2) AMPULLA - ARCHES OVER THE OVARY
3)INFUNDIBULUM - PROJECT BEYOND THE FREE MARGIN OF BROAD LIGAMENT
4) UTERINE TUBE LIE - IN UPPER FREE MARGINE OF THE BROAD LIGAMENT

*BLOOD SUPPLY -

UTERINE ARTERY - SUPPLIES - MEDIAL 2/3 OF TUBE
OVARIAN ARTERY - SUPPLIES- LATERAL 1/3 OF TUBE

*NERVE SUPPLY:-

1) SYMPATHETIC NERVE- T10 TO T_{12} & L1-L2 SEGMENT FROM HYPOGASTRIC PLEXUSES
2) PARASYMPATHETIC NERVE- DERIVED FROM VAGUS - LATERAL HALF

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ANAL CANAL

*INTRODUCTION -

- ANAL CANAL IS TERMINAL PART→ GASTROINTESTINAL TRACK
- LIE - IN ANAL TRIANGLE B/W ISCHIOANAL FOSSA
- EXTEND FROM ANORECTAL JUNCTION TO ANUS
- SURROUNDING - OUTER VOLUNTARY & INNER-INVOLUNTARY SPHINCTER

*RELATION:-

1) ANTERIORLY - BOTH SEX - PERINEAL BODY
- IN MALE - MEMBRANOUS URETHRA AND BULB OF PENIS
- IN FEMALE-LOWER END OF VAGINA

2) POSTERIORLY - ANOCOCCYGEAL LIGAMENT.
- TIP OF COCCYX
3) LATERALLY - ISCHIANAL FOSSA

4) INFERIOR OF ANAL CANAL - DIVIDE INTO 3 PART - UPPER - 15MM
- MIDDLE - 15MM
- LOWER - 8MM

CHARACTERISTIC EPITHELIUM :-

1) UPPER MUCOUS PART:-
- LINE BY COLUMNAR EPITHELIUM (MUCOUS MEMBRANE]
- 6-10 VERTICAL RIDGE - ANAL COLUMNS
- LOWER END OF ANAL COLUMN UNITED BY SHORT TRANSVERSE FOLD CALLED ANAL VALVE
- ABOVE VALUE DEPRESSION- ANAL SINUS
- ANAL VALUE TOGETHER FORM TRANSVERSE LINE

2) MIDDLE PART:-
- NON- KERATINIZING SQUMOUSE EPITHELIUM (ALSO MUCOUS]
- ANAL COLUMNS ARE NOT PRESENT
- MUCOSA BLUISH APPEARANCE -BECAUSE DENSE VENOUS PLEXUS
-THIS REGION REFERRED TO PECTEN OR TRANSITIONAL ZONE
-LOWER LIMIT OF PECTIN OFTEN HAS WHITISH APPEARANCE ALSO REFERRED AS- "WHITE LINE OF HILTON"

3) LOWER PART:-
-LINE BY TRUE SKIN - CONTAINING SEBACEOUS GLAND
(KERATINIZING SQUMOUSE EPITHELIUM)

DEEP PERINEAL SPACE

*DEFINTION - THIN SPACE - - -> UROGENITAL REGION
- SITUATED DEEP TO PERINEAL MEMBRANE

*BOUNDARIES - SUPERFICIAL - PERINEAL MEMBRANE
- ON EACH SIDE - ISCHIOPUBC RAMI

CONTENT-	MALE -	FEMALE-
1) TUBES -	PART OF URETHRA	- URETHRA & VAGINA
2) MUSCLES	- SPHINCTER URETHRA - DEEP TRANSVERSE PERINEI	- SPHINCTER URETHRA - COMPRESSOR URETHRA - SPHINCTER URETHROVAGINALIS
3) NERVE	- DORSAL NERVE OF PENIS - MUSCULAR BRANCHES FROM - PERINEAL NERVE	- DORSAL NERVE OF CLITORIS
4) VESSEL	- ARTERY OF PENIS - ARTERY TO BULB OF PENIS - URETHRA ARTERY - DEEP & DORSAL ATERIES OF PENIS	- ARTERY IF CLITORIS - ARTERY TO BULB OF VESTIBULE - URETHRAL ARTERY

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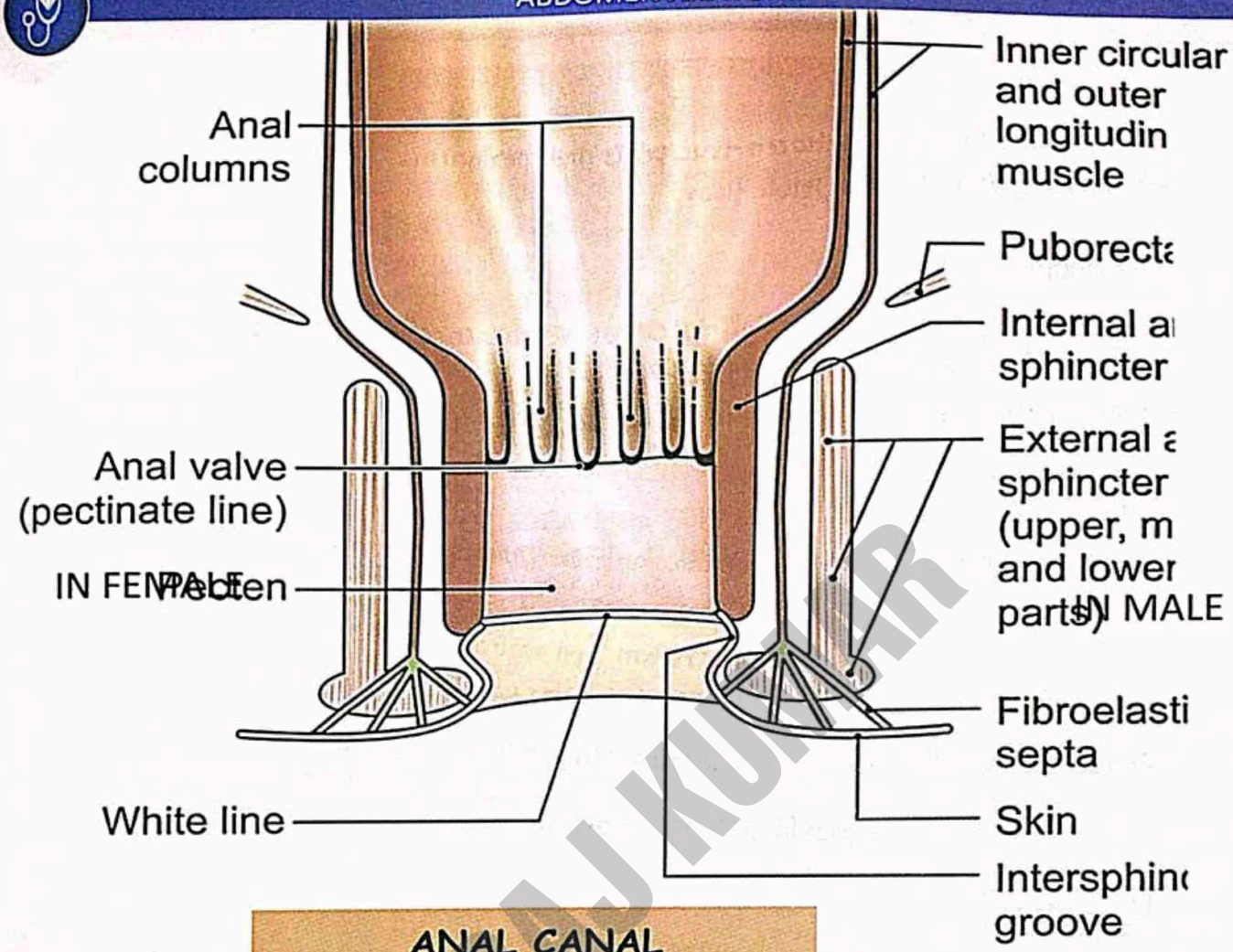
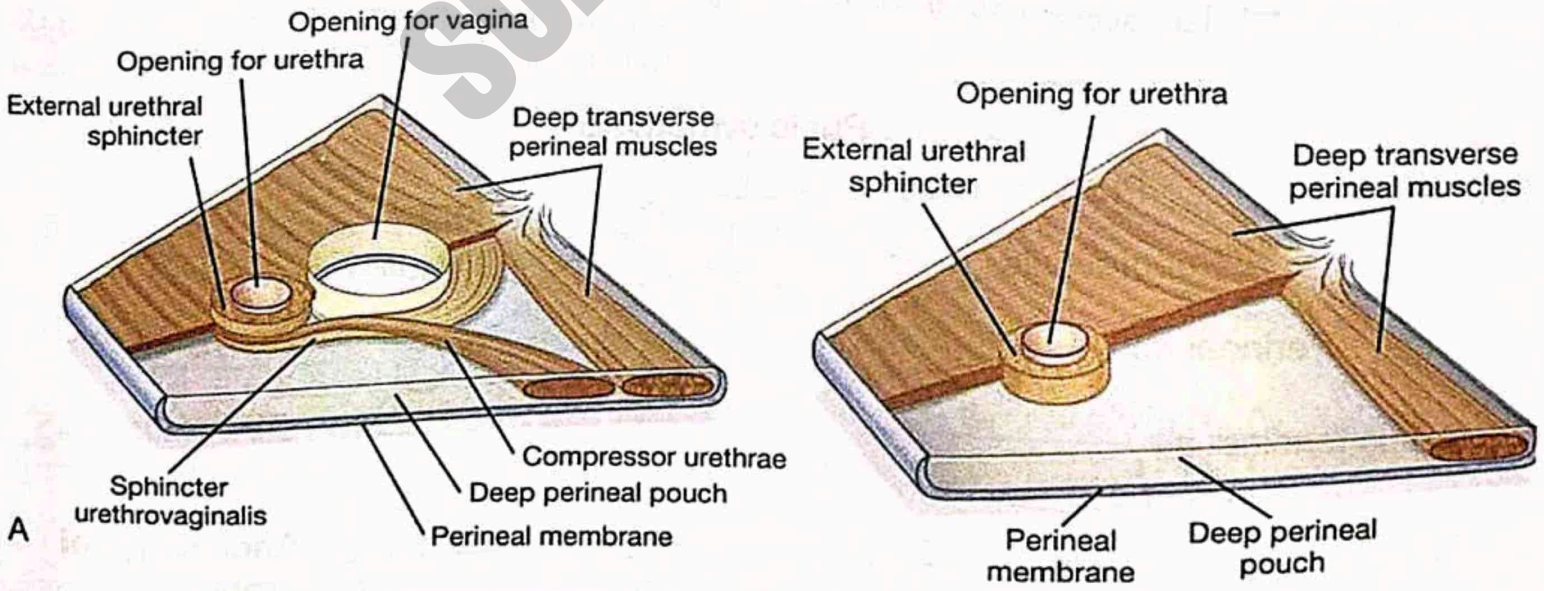


ANAL CANAL



DEEP PERINEAL SPACE

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PELVIC DIAPHRAGM

→ Muscular Partition between true pelvis and perineum.
→ form gutter shaped Pelvic floor.

Formation

→ formed by large levator Ani muscle and small coccygeus muscle on 2 sides & their covering fascia
→ Incomplete anteriorly to allow passage of urethra.

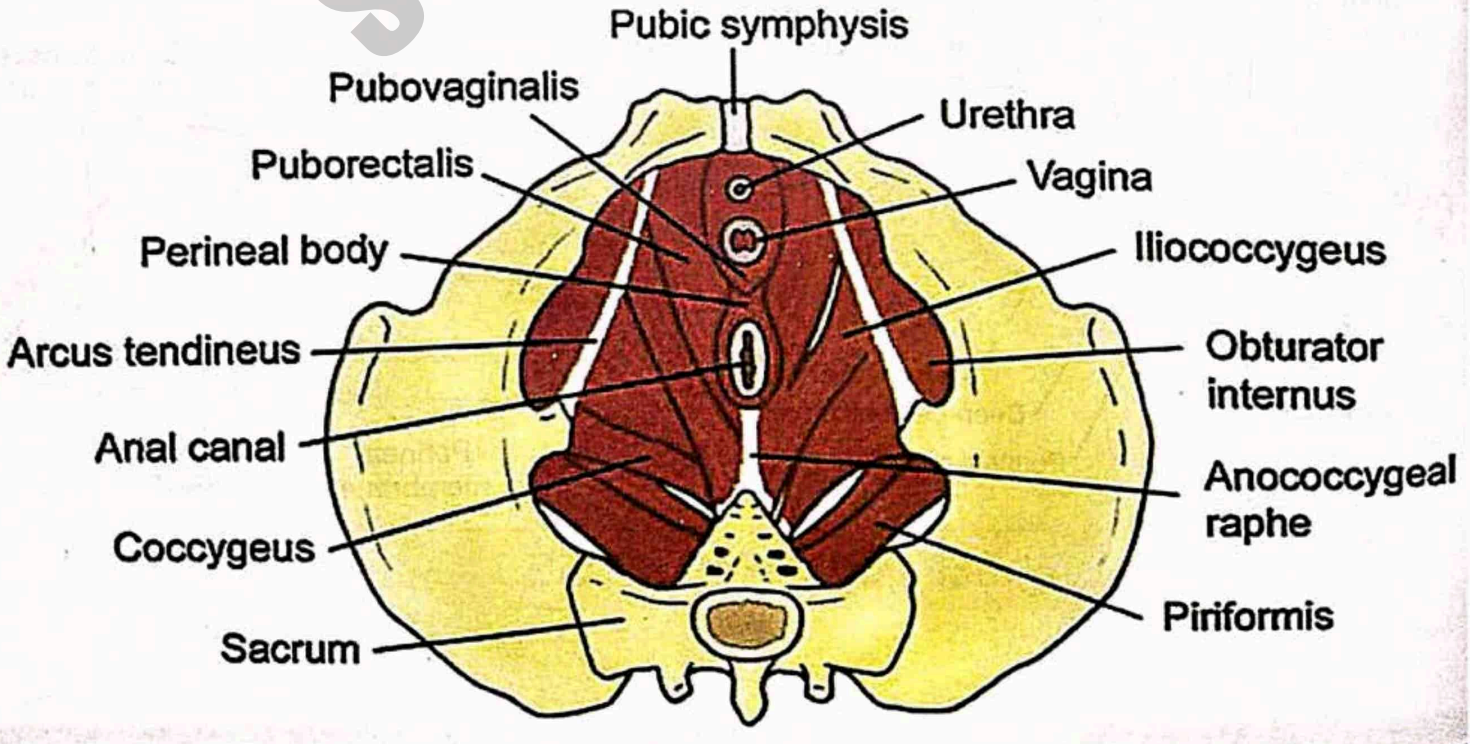
Openings

1) Hiatus Urogenitalis - Δ gap btw anterior fibres of levator Ani muscle
- transmit urethra in male & urethra and vagina in female.
- It closed from below by urogenital diaphragm.

2) Hiatus Rectalis - O(round) opening btw perineal body and anococcygeal raphe.
- provide passage to anorectal junction

Functions

→ provide support to pelvic viscera sphinter action on vagina & rectum.
→ Use abdominal pressure during defecation micturation parturition



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LESSER SAC

*IT IS LARGE RECESS OF PERITONRAL CAVITY BEHIND THE STOAMCH, THE OMENTUM AND CAUDATE LOBE OF LIVER

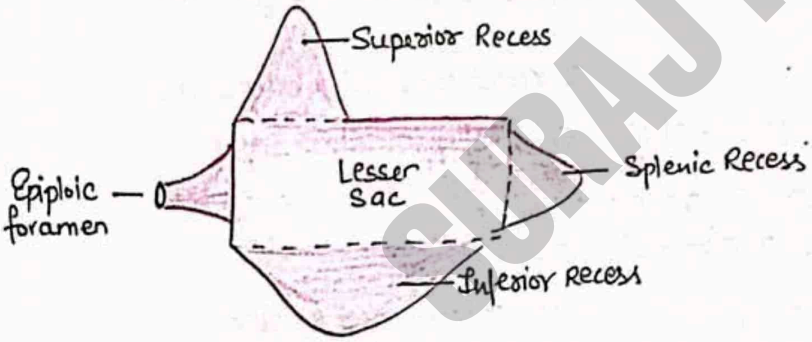
*CLOSE ALL ROUND

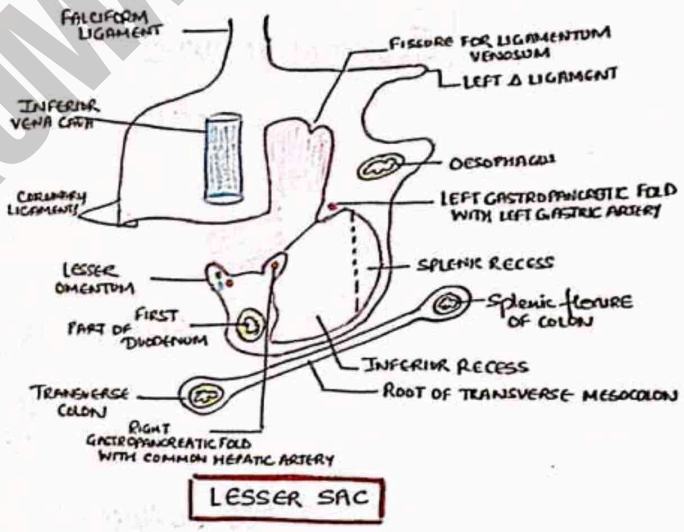
*ECXEPT IN THE UPPER PART OF ITS RIGHT BORDER -> WHERE IT COMMUNICATE WITH GREATER SAC ---> THROUGH --> EPIPLOIC FORAMEN

*BOUNDAIRES

1) ANTERIOR - PERITONEUM COVERING CAUDATE LOBE OF LIVER
   - POSTERIOR LAYER OF LESSER OEMNTUM
   - POSTERIOR SURFACE OF STOMACH
   - 2ND LAYER OF GREATER OMENTUM
2) POSTERIOR - 3RD LAYER OF GREATER OMENTUM
   - TRANSVERSE MESDODERM
   - UPPER LAYER OF TRANSVERSE MESOCOLON

SUB DIVISION OF LESSER SAC

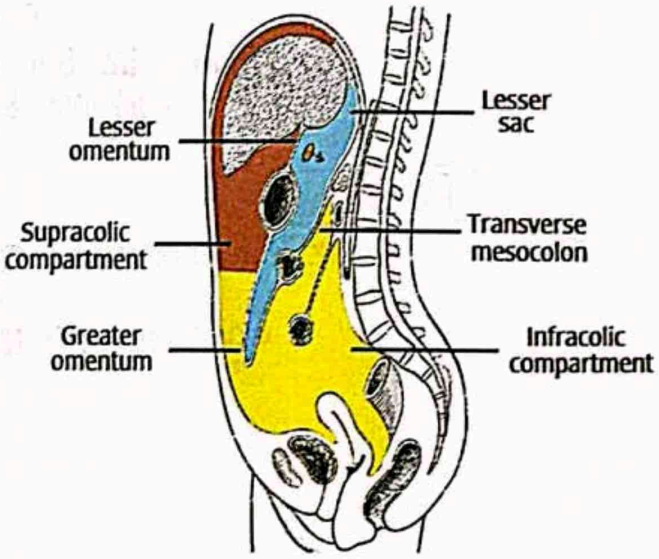




SUPERIOR RECESS - behind the lesser omentum and the liver

INFERIOR RECESS - behind the stomach and within greater omentum

SPLENIC RECESS - between the gastrosplenic and lienorenal ligament



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SPLEEN

Introduction – ∘ lymphatic organ connected to Blood vascular system
∘ act as a filter for blood and plays an important role in immune response for body

Location – ∘ mainly in LEFT and partly in

∘ wedge shaped organ [tetrahedral in shape]

Dimensions – soft, highly vascular and dark purple in colour.
On average – 2.5 - thick
7.5 - Broad
12.5 - long

Position – lie obliquely along the long axis of 10th rib
Directed – downward, forward and laterally

External features –

∘ Two ends - Anterior or lateral end – expanded and is more like border
Directed - downward & forward
Posterior or medial end - round
Directed upward, backward

∘ Three Border - superior - characteristically notched near anterior end
inferior - rounded
intermediate - also rounded

∘ Two surface - Diaphragmatic surface - convex and smooth
visceral surface - concave and irregular

∘ Hilum - lie btw superior and intermediate borders
- pierced by branches and tributaries of splenic

Relations :–

★ Peritoneal relation - spleen is surrounded by peritoneum and suspended by ligament

∘ gastrosplenic ligament – extend hilum of spleen to greater curvature of stomach
contain short gastric vessels

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- Lienorenal ligament - extend from hilum of spleen to anterior surface of left kidney
  - Contain - tail of pancreas
    - Splenic vessels
    - pancreaticosplenic lymph nodes
- Phrenicocolic ligament - not attached to spleen but support its anterior end

* Visceral relation - related to - fundus of stomach
  - anterior surface of left kidney
  - Splenic flexure of colon
  - tail of pancreas
- gastric impression - fundus of stomach
  - btw superior and intermediate border
- Renal impression - Left kidney
  - btw inferior and intermediate border
- Pancreatic impression - tail of pancreas
  - btw hilum and colic impression
- Hilum - lie on inferomedial part of gastric impression
  - Transmits - splenic vessels & nerve
  - provide attachment to gastrosplenic & linorenal ligament
- colic impression - Splenic flexure of colon
  - Occupies Δ area of anterior border

* Diaphragmatic surface -
  - Related to diaphragm
  - which separated the spleen from pleura and lungs.

Arterial supply -
Supplied by splenic artery

Venous supply -
- Drained by splenic vein
- Tributaries
  - → Short gastric vein
  - → left gastroepiploic vein
  - → pancreatic vein
  - → Inferior mesenteric vein

Nerve supply -
Sympathetic fibres - Coeliac plexus

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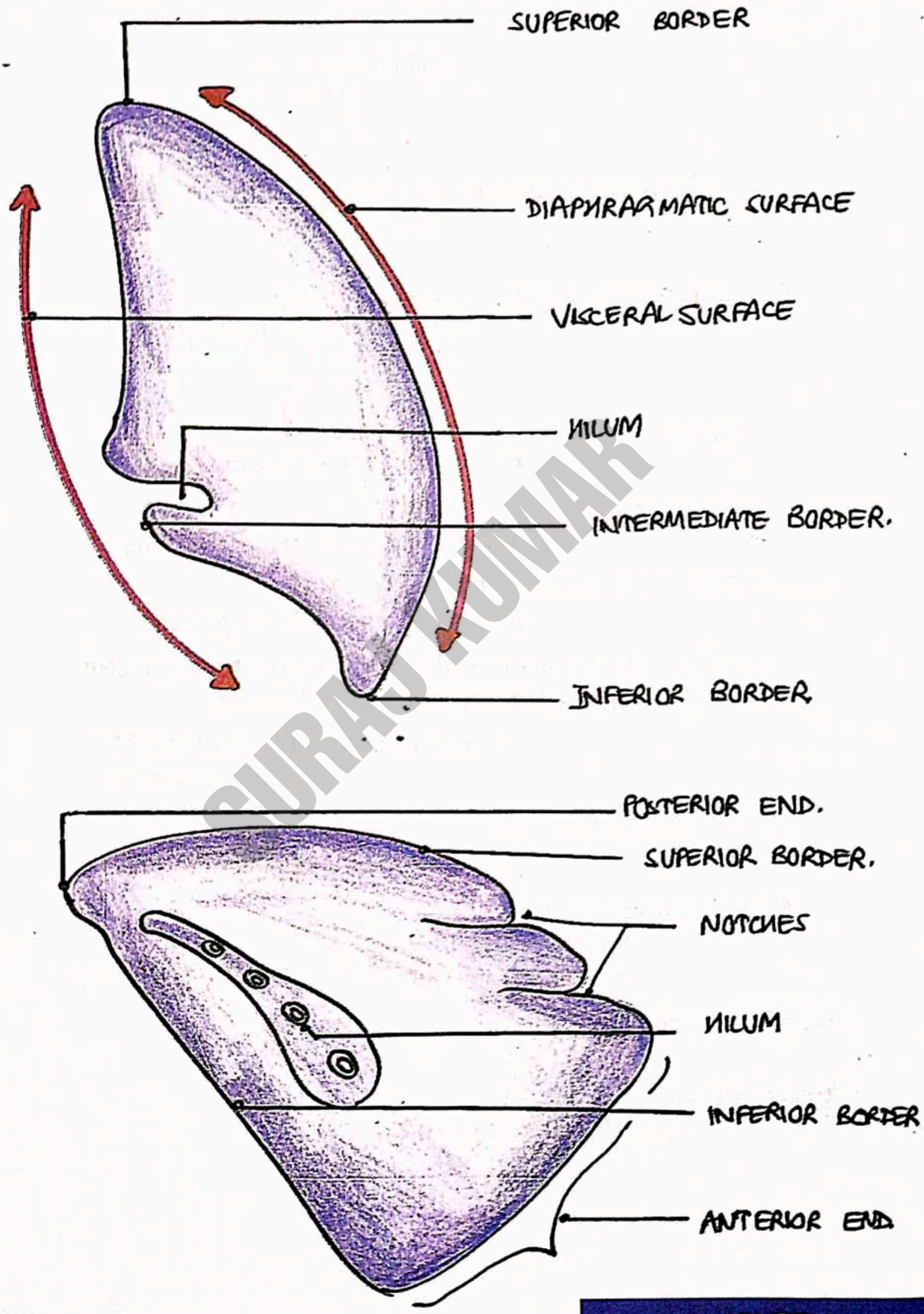

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SPLEEN

STUDY TIPS FOR MAXIMUM MARKS

1.HOW TO WRITE IN UNIVERSITY EXAM AND HOW TO DRAW DIAGRAMS ALREADY MENTION IN UPPER LIMB (PAGE NUMBER - 25 OF VOLUME - 1)

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTION - 20MARKS

1.DESCRIBE INGUINAL CANAL UNDER FOLLOWING HEADINGS -
1.LOCATION & EXTEND 2.BOUNDARY & CONTENT 3.PROTECTIVE MECHANISM
4.APPLIED(HERNIA) 5.SPERMATIC CORD

2.DESCRIBE STOMACH UNDER FOLLOWING HEADINGS -
1.POSTERIOR RELATION 2.ARTERY SUPPLY 3.APPLIED ANATOMY

3.DESCRIBE VERMIFORM APPENDIX UNDER FOLLOWING HEADINGS -
1.POSITION 2.BLOOD SUPPLY & LYPHATIC DRAINAGE 3.APPLIED

4.DESCRIBE EXTERNAL FEATURE, RELATION, LIGAMENT, BLOOD SUPPLY & DEVELOPMENT OF URINARY BLADDER ?

5. DESCRIBE PARTS, RELATION, LIGAMENT, BLOOD SUPPLY & SUPPORT OF UTERUS ?

6.DESCRIBE SECOND PART OF DEUDENUM - UNDER FOLLOWING HEADING -
1.RELATION 2.DEVELOPMENT 3.APPLIED 4.ARTERY SUPPLY

*SHORT NOTES - 5MARKS

1.RECTUS SHEATH
2.DESCEND OF TESTIS
3.LIGAMENT OF LIVER & VISCERAL RELATION
4.PANCREAS
5.DEVELOPMENT OF KIDNEY
6.KIDNEY RELATION
7.EXTRA HAPATIS BILIARY APPARATUS
8.PORTAL VEIN & ITS CLINICAL ANATOMY
9.EPIPLOIC FORAMEN
10.GREATER & LESSER OENTUM
11.ISCHIONAL FOSSA
12.DEVELOPMENT OF ANAL CAN

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ABDOMEN ANATOMY IS COMPLETED !!

YOUR POSITIVE ACTION COMBINED WITHPOSITIVE THINKING RESULTS IN SUCCESS

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

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MCQ ABDOMEN

1) Paralysis of which muscle causes drooping of upper eyelid (ptosis) :
a Occipito-frontalis
b Orbicularis oculi
c levator palpebrae superioris
d superior rectus

2) All of the following muscles have bony attachment except :
a procerus
b Orbicularis oculi
c Risorious
d platysma

3) Which one of the following is motor nerve of scalp:
a lesser occipital
b auriculo temporal
c posterior auricular
d supra orbital

4) Which of the following muscles is an abductor of vocal cords :
a posterior cricoarytenoid
b vocalis
c lateral cricoarytenoid
d cricothyroid

5) Which muscle is supplied by glossopharyngeal nerve
a stylohyoid
b stylopharyngeus
c salpingopharyngeus
d palatopharyngeus

6)Which muscle protrudes the tongue:
a genioglossus
b hyoglossus
c styloglossus
d palatoglossus

7) where does the nasolacrimal duct drains :
a superior meatus
b middle meatus
c inferior meatus
d sphenoethmoidal recess

8) Which muscle tenses the vocal cord
a Thyroarytenoid
b lateral cricoarytenoid
c posterior cricoarytenoid
d cricothyroid

9) Which of the following is present in the suprasternsl space :
a jugular venous arch
b external jugular vein
c supraclavivular nerve
d external jugular vein

10) All of the following pass through the superior orbital fissure except :
a ophthalmic artery
b ophthalmic vein
c occulomotor nerve
d trochlear nerve

11) Which of the following lies deep to the hyoglossus muscle
a Lingual nerve
b lingual artery
c submandibular gland
d submandibular ganglion

12) All of the following are intrinsic muscle of larynx excep
a cricothyroid
b thyrohyoid
c thyroepiglottic
d Thyroarytenoid

13) All of the following are supplied by hypoglossal nerve except:
a genioglossus
b styloglossus
c hyoglossus
d palatoglossus

14) The parotid duct opens into vestibule of mouth at whic level of molar tooth
a upper first
b lower first
c upper second
d lower second

15) Which nerve supplies the superior oblique muscle of t eye ball
a optic
b occulomotor
c trochlear
d abducens

16) Stylopharyngeus muscle recieves its motor innervatio from which nerve
a facial
b glossopharyngea
c vagus
d hypoglossal

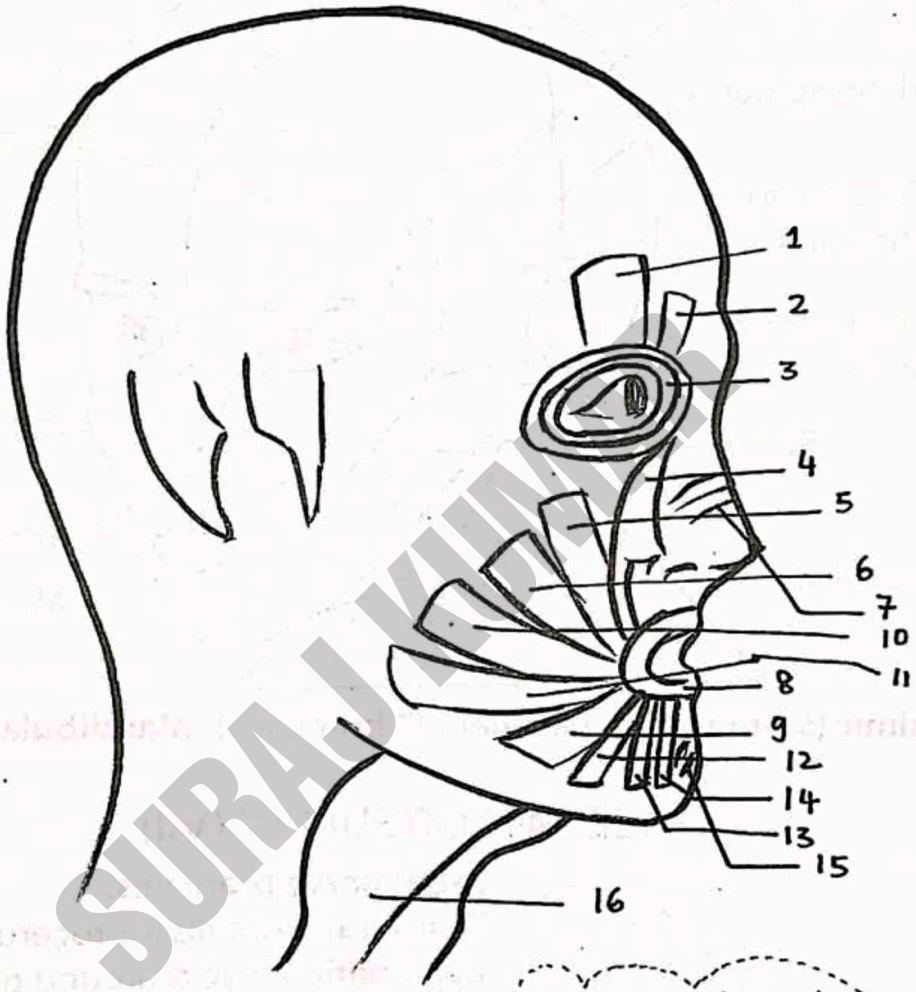
17) Which of the following layer of the scalp is known as dangerous layer
a skin
b connective tissue
c loose areolar tissue
d pericranium

18) The structure that does not pass through the parotid gland
a facial nerve
b external carotid artery
c retromandibular nerve
d auriculo temporal nerve

SHARMA SURAJ KUMAR BBS

1)C 2)D 3)C 4)A 5)B 6)A 7)C 8)D 9)A 10)A 11)B 12)B 13)D 14)C 15)C 16)B 17)C 18)D

FACIAL MUSCLES



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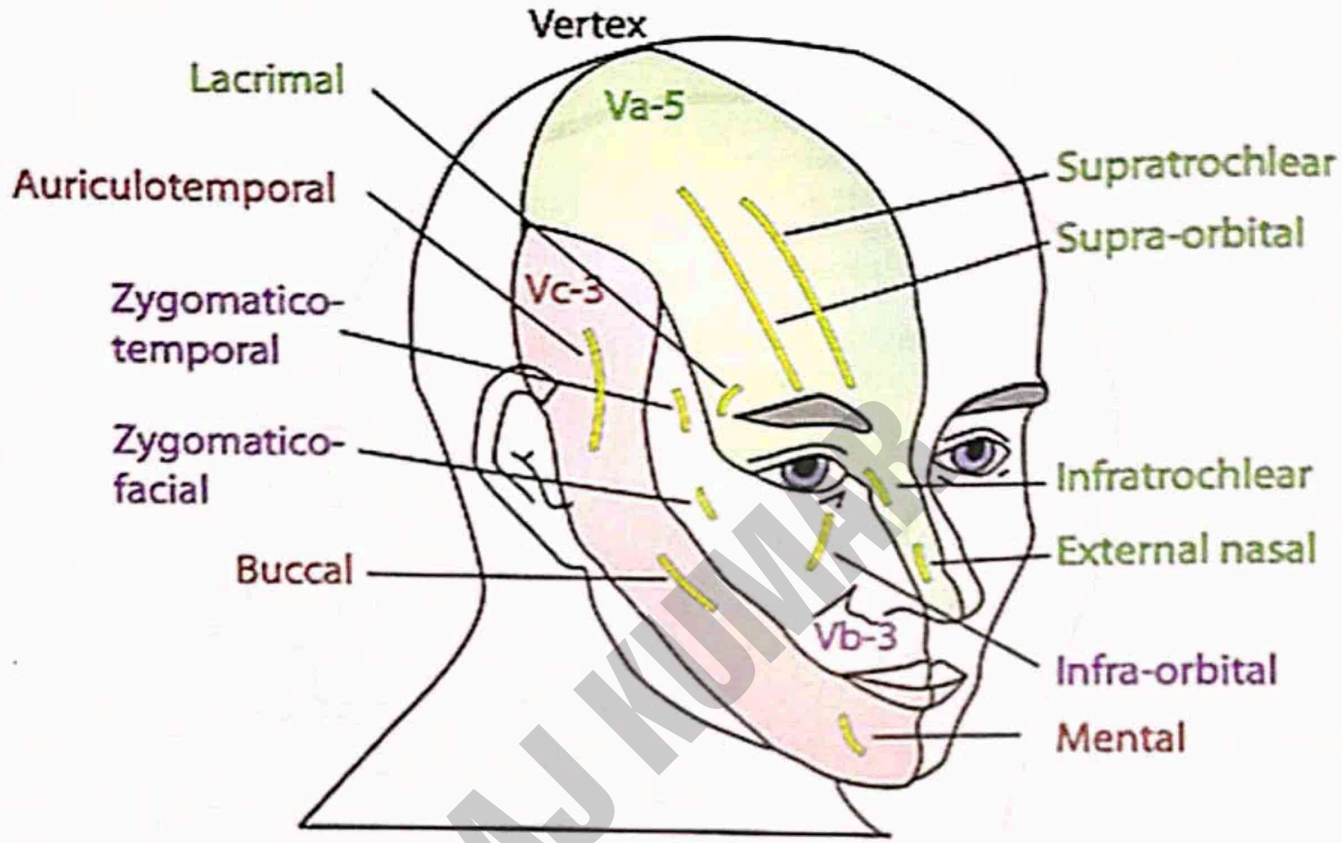
1 - Frontalis
2 - Corrugator supercilli
3 - Orbicularis oculi
4 - Levator labii superioris alaeque nasi
5 - Levator labii superioris
6 - Levator anguli oris
7 - Compressor naris
8 - Orbicularis oris
9 - Buccinator
10 - Zygomaticus minor
11 - Zygomaticor major
12 - Risorus
13 - Depressor labii oris
14 - Depressor labii inferioris
15 - Mentis
16 - Platysma

- Temporal branch
- Zygomatic branch
- Upper buccal branch
- Lower buccal branch
- Marginal Mandibular branch
- Cervical branch

SENSARY INERVATION OF FACE

SENSORY BRANCHES OF TRIGEMINAL N ON FACE



Ophthalmic (5 branches) Maxillary (3 branches) Mandibular (3 branches)

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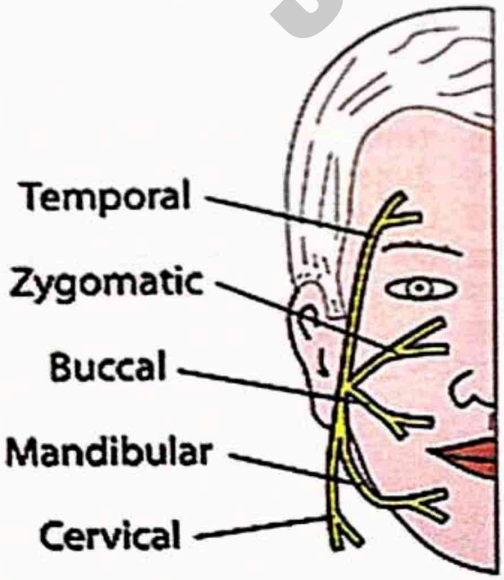
FACE: MOTOR SUPPLY (VII)



Facial nerve branches:
Temporal: frontalis & procerus
Zygomatic 1: eye & around orbit
Zygomatic 2: mid face & smile
Buccal: buccinator & upper lip
Mandibular: lower lip & orbicularis oris
Cervical: platysma
(note: proprioception is supplied by trigeminal)

Mnemonic:
Two
Zebras
Befriended
My
Cat

SCALP ANATOMY

SCAN QR
WITH QRBOT APP
TO DIRECTLY OPEN VIDEO LINK

*SOFT TISSUES COVERING - THE CRANIAL VAULT FROM SCALP

*EXTENT OF SCALP -
1) ANTERIORLY - SUPRAORBITAL MARGIN
2) POSTERIORLY - EXTERNAL OCCIPITAL PROTUBERANCE
- SUPERIOR NUCHAL LINES
3) EACH SIDE - SUPERIOR TEMPORAL LINES

-MADE UP OF FIVE LAYER - SKIN
- SUPERFICIAL FASCIA
- DEEP FASCIA OR GALEA APONEUROTICA
- LOOSE AREOLAR TISSUE
- PERICRANIUM

1) SKIN- THICK AND HAIRY
- IT HAS SWEET GLANDS AND SEBACEOUS GLANDS

2) SUPERFICIAL FASCIA - MORE FIBROUS AND DENSE IN CENTRE
- CONTAIN BLOOD VESSEL
- ITS BINDS THE SKIN TO SUBJACENT APONEUROSIS
- AND PROVIDE PROPER PASSAGE OF VESSEL AND NERVE OF SKIN

-OCCIPITOFRONTALIS MUSCLE:- HAS TWO BELLS OCCIPITALS AND FRONTALIS
- INSERTED-EPICRANIAL APRONEUROSIS

- OCCIPITALIS - SMALL AND SEPARATE
- ARISE - LATERAL 2/3 OF SUPERIOR NUCHAL LINE,
- SUPPLY BY - POSTERIOR AURICULAR BRANCH OF FACIAL NERVE

- FRONTALIS - LARGE AND PARTLY UNITED
- ARISE FROM- SKIN UPPER OF EYELID AND FORE HEAD
- SUPPLY BY- TEMPORAL BRANCH OF FACIAL NERVE

- TEMPARO PARIETALIS- PRESENT ON LATERAL SIDE
-ARISE FROM - TEMPORAL FASCIA AND FUSE WITH EPICRANIAL APONEUROSIS.
-SUPPLY BY TEMPORAL BRANCH OF FACIAL NERVE

3) EPICRANIAL APONEUROSIS-
- FREELY MOVABLE ON PERICRANIUM
- ATTACHMENT- ANTERIORLY - INSERTION-FRONTALIS
- POSTERIORLY - EXTERNAL OCCIPITAL PROTUBERANCE
- HIGHEST NUCHAL LINE B/W OCCIPITAL BELLIES
- SIDE - SUPERIOR TEMPORAL LINE
- EXPANSION - ZYGOMATIC ARCH

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5) LOOSE AREOLAR TISSUE
- EXTEND - ANTERIORLY-EYELIDS
- POSTERIORLY - SUPERIOR NUCHAL LINE.
- EACH SIDE - SUPERIOR TEMPORAL LINE
-GIVE PASSAGE TO EMISSARY VEIN WHICH CONNECT
- EXTRACRANIAL VEIN TO INTRACRANIAL VENOUS SINUSES

6) PERICRANIUM - LOOSELY ATTACHED TO SURFACE OF BONE
- BUT IS FIRMLY ADHERENT TO SUTURE.

*ARTERIAL SUPPLY - IN FRONT - SUPRATROCHLEAR-→BRANCH→INTERNAL CAROTID A.
- SUPRA ORBITAL-→BRANCH-→EXTERNAL CAROTID A.

- BEHIND - POSTERIOR AURICULAR-→EXTERNAL CAROTID ARTERY
- OCCIPITAL----→EXTERNAL CAROTID ARTERY

*VENOUS DRAINAGE -
-SUPRATROCHLEAR & SUPRA ORBITAL VEIN - JOIN EACH OTHER TO FORM ANGULAR VEIN--→ FACIAL VEIN
-SUPERFICIAL TEMPORAL VEIN - DESCEND IN FRONT OF TRAGUS--→TO ENTER PAROTID GLAND
-JOIN MAXILLARY VEIN -→FORM RETROMANDIBULAR VEIN
TERMINATED ANTERIOR & POSTERIOR DIVISION
-POSTERIOR AURICULAR VEIN - IT DESCEND BEHIND AURICLE &
-UNITES THE POSTERIOR DIVISION OF RETROMANDIBULAR VEIN--→FORM EXTERNAL JUGULAR VEIN
-OCCIPITAL VEIN - TERMINATE - SUBOCCIPITAL VENOUS PLEXUS

-VEIN OF SCALP COMMUNICATE WITH INTRACRANIAL DURAL VENOUS SINUSES THROUGH EMISSARY VEIN :
- EMISSARY VEIN - VEIN CONNECTING VEIN OUTSIDE THE CRANIUM WITH INTRACRANIAL VEIN BY PASSING THROUGH FORAMEN IN CRANIUM
- PARIETAL EMISSARY VEIN - COMMUNICATES SUPERIOR SAGGITAL SINUS
----→PASSES PARIETAL FORAMEN
- MASTOID EMISSARY VEIN - SIGMOID SINUS
-----→ PASSES MASTOID FORAMEN

*LYMPHATIC DRAINAGE -
- ANTERIOR PART OF SCALP--→ DRAIN PREAURICULAR OR SUPERFICIAL PAROTID
- POSTERIOR PART OF SCALP--→ POSTERIOR AURICULAR OR MASTOID
---→ OCCIPITAL LYMPH NODE

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*NERVE SUPPLY - SUPPLY BY 10 NERVE

-FRONT OF AURICLE - FIVE NERVE (4, SENSARY & 1 MOTOR)
-SUPRA ORBITAL
-SUPRATROCHLEAR NERVE
-ZYGOMATICOTEMPERAL NERVE
-TEMPORAL BRANCH OF FACIAL NERVE(MOTOR)
-AURICULOTEMPORAL NERVE

-BEHIND AURICLE - FIVE (4 SENSARY & 1 MOTOR)
-GREATER AURICULAR NERVE
-POSTERIOR AURICULAR NERVE(MOTOR)
-LESSER OCCIPITAL NERVE
-GREATER OCCIPITAL NERVE
-THIRD OCCIPITAL NERVE

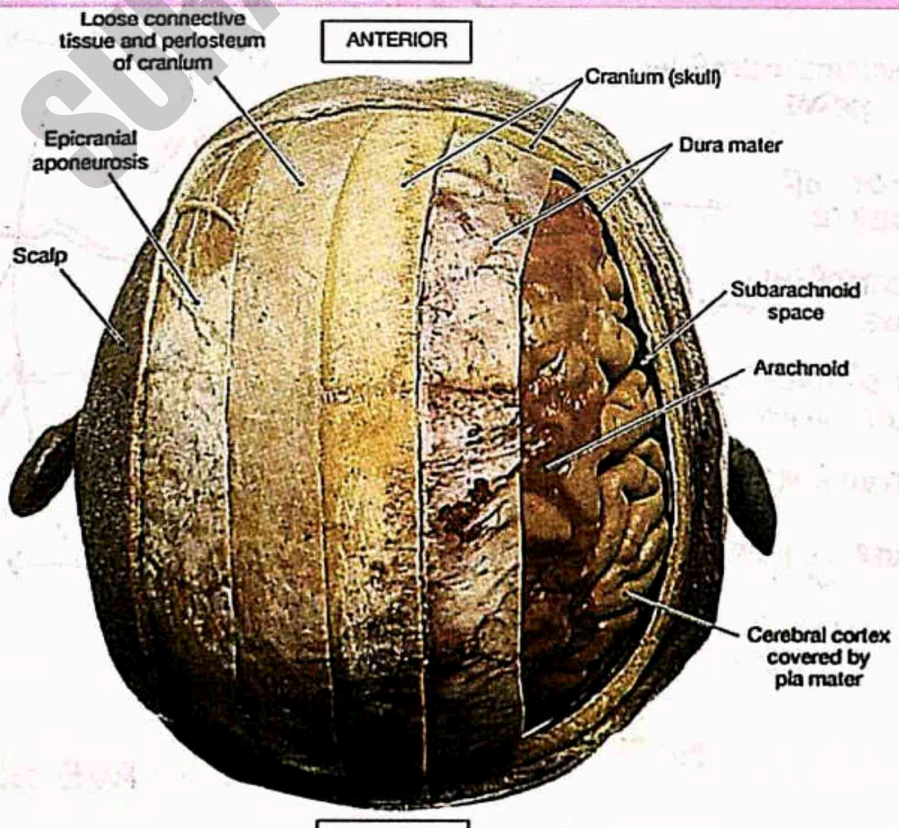
CLINICAL ANATOMY

SCALP

1.THE LAYER OF LOOSE AREOLAR TISSUE IS KNOWN AS THE "DANGEROUS AREA OF THE SCALP" BECAUSE THE EMISSARY VEINS, WHICH COURSE HERE, MAY TRANSMIT INFECTION FROM THE SCALP TO THE CRANIAL VENOUS SINUSES.

2.BECAUSE OF THE ABUNDANCE OF SEBACEOUS GLANDS, THE SCALP IS A COMMON SITE FOR SEBACEOUS CYSTS.

3.COLLECTION OF BLOOD IN THE LAYER OF LOOSE CONNECTIVE TISSUE CAUSES GENERALISED SWELLING OF THE SCALP.



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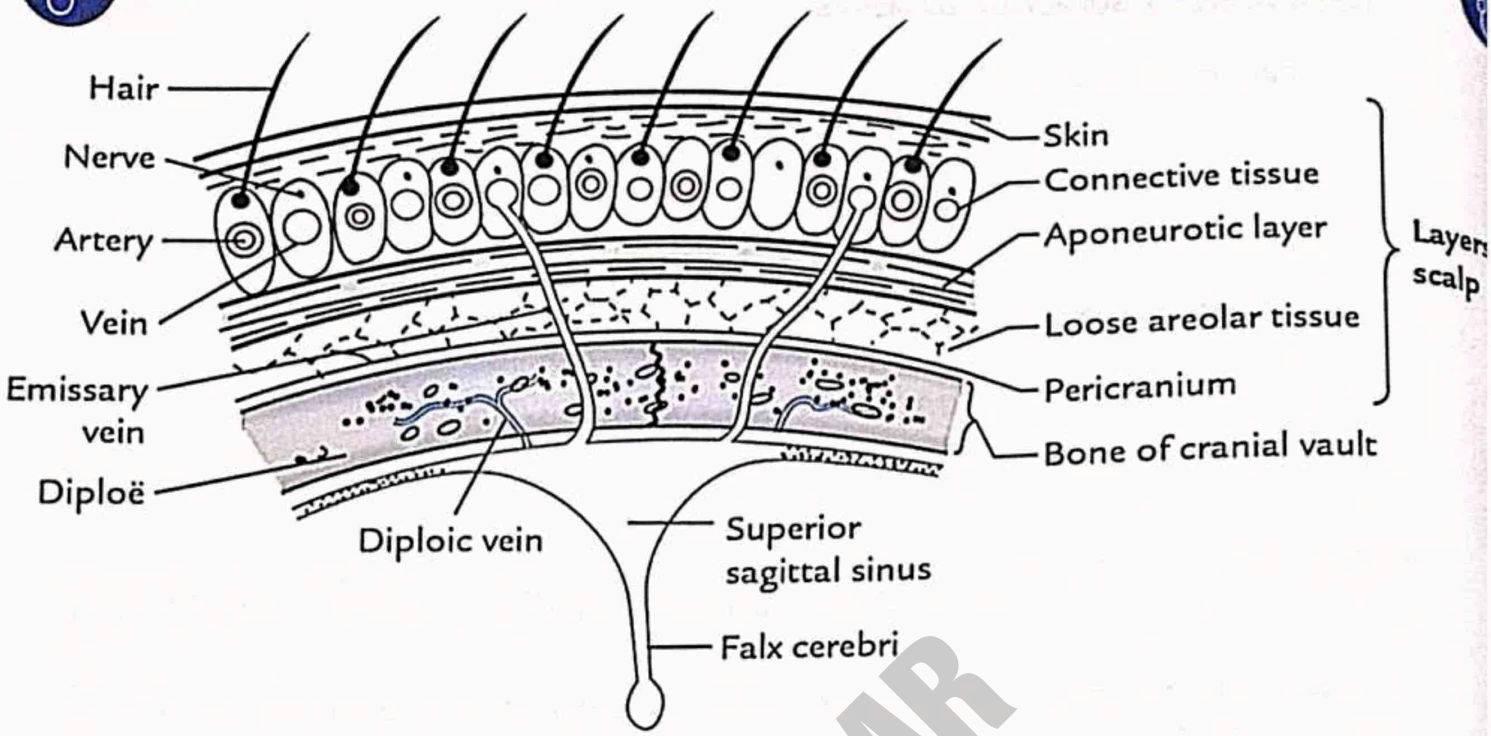

LAYER OF SCALP

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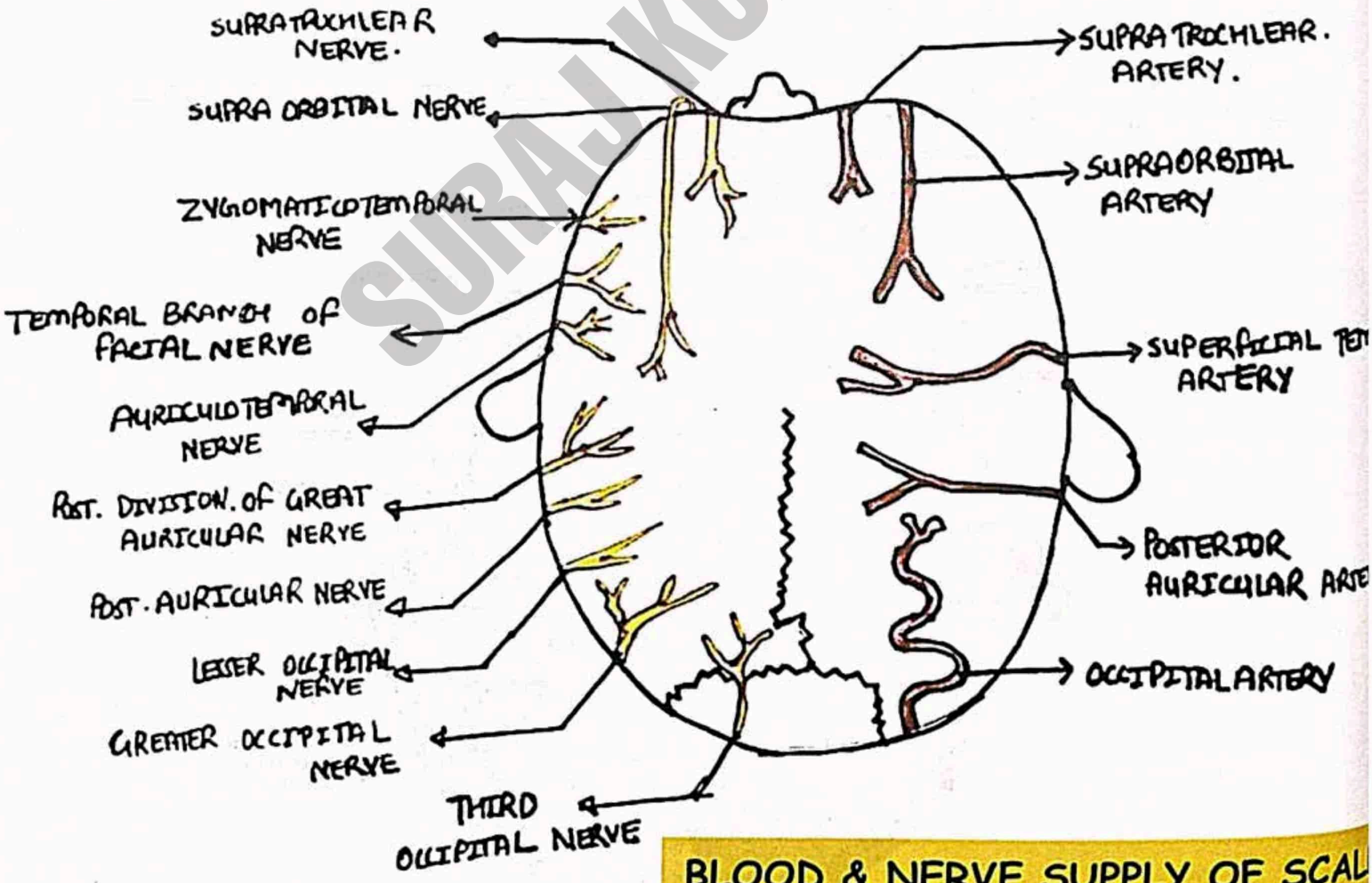

BLOOD & NERVE SUPPLY OF SCAL

LACRIMAL APPARATUS

*COMPONENTS - STRUCTURE CONCERNED WITH SECRETION AND DRAINAGE OF LACRIMAL TEAR FLUID CONSTITUTE THE LACRIMAL APPARATUS

- MADE - LACRIMAL GLAND AND DUCTS
- CONJUNCTIVAL SAC
- LACRIMAL PUNCTA AND LACRIMAL CANALICULI
- LACRIMAL SAC
- NASOLACRIMAL SAC

1) LACRIMAL GLAND - SEROUS GLAND
- SITUATED LACRIMAL FOSSA- ANTEROLATERAL OF ROOF OF BONY ORBIT
- "J" SHAPED GLAND - BEING INDENTED BY - TENDON-OF- LEVATOR PALPEBRAE SUPERIOR
  - ↳ ORBITAL PART - LARGER AND DEEPER
  - ↳ PALPEBRAL PART- SMALLER AND SUPERFICIAL
- DOZEN OF DUCTS PIERCE - CONJUNCTIVA OF UPPER LID AND OPEN INTO CONJUNCTIVAL SAC
- ARTERY - LACRIMAL BRANCH OF OPHTHALMIC ARTERY
- NERVE SUPPLY - LACRIMAL GLANDS NERVE
- LACRIMAL FUILD SECRETED —->LACRIMAL GLAND—->FLOW→ →CONJUNCTIVAL SAC-→ —> PERIODIC BLINKING —>SPREAD FLUID

2) CONJUNCTIVA SAC -
- CONJUNCTIVA LINING-DEEP SURFACE - OF EYELIDS → PALPEBRAL CONJUNCTIVA
  - ↳ FRONT OF EYEBALL →BULBAR CONJUNCTIVA
- POTENTIAL SPACE B/W PALPEBRAL AND BULBAR - CALLED CONJUNCTIVA SAC

3) LACRIMAL PUNCTA & CANALICULI -
- EACH LACRIMAL CANALICULI- BEGINS AT LACRIMAL PUNCTUM, 10MM LONG
- BOTH CANALICULI OPEN- LATERAL WALL OF LACRIMAL SAC
- BEHIND MEDIAL PALPEBRAL LIGAMENT

4) LACRIMAL SAC -
- MEMBRANOUS SAC 12MM LONG AND 5MM WIDE
- SITUATED IN LACRIMAL GROOVE
- BEHIND MEDIAL PALPEBRAL LIGAMENT
- UPPER - END IS BLIND
- LOWER END CONTINOUS WITH NASOLACRIMAL DUCT

5) NASOLACRIMAL DUCT:
- MEMBRANOUS PASSAGE - 18MM LONG
- BEGINS - LOWER END OF LACRIMAL SAC
- RUN DOWNWARD, BACKWARD - OPEN INTO- INFERIOR MEATUS OF NOSE

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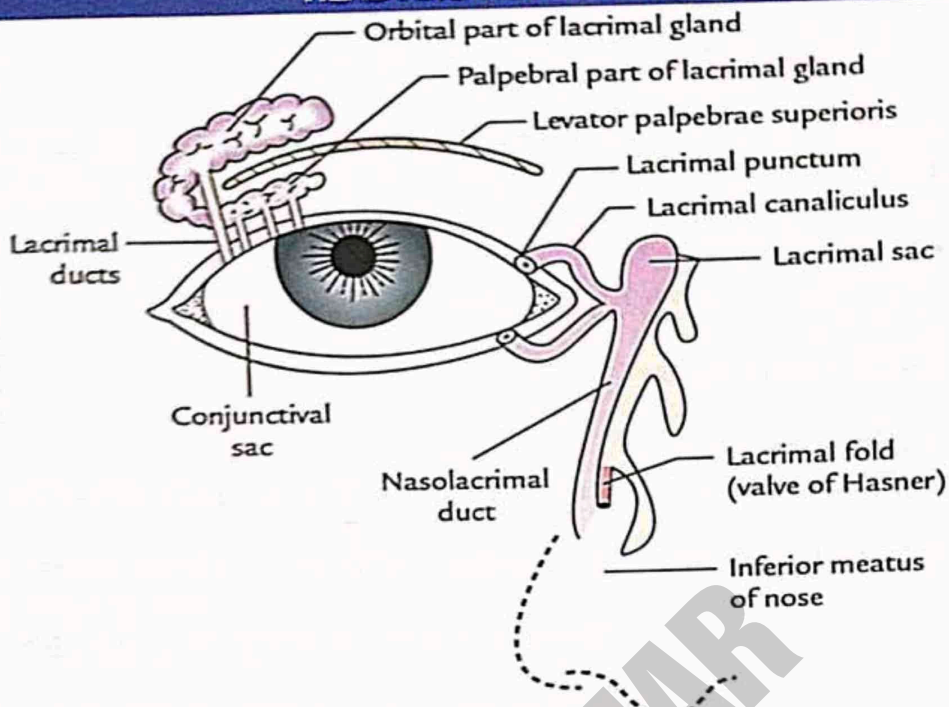
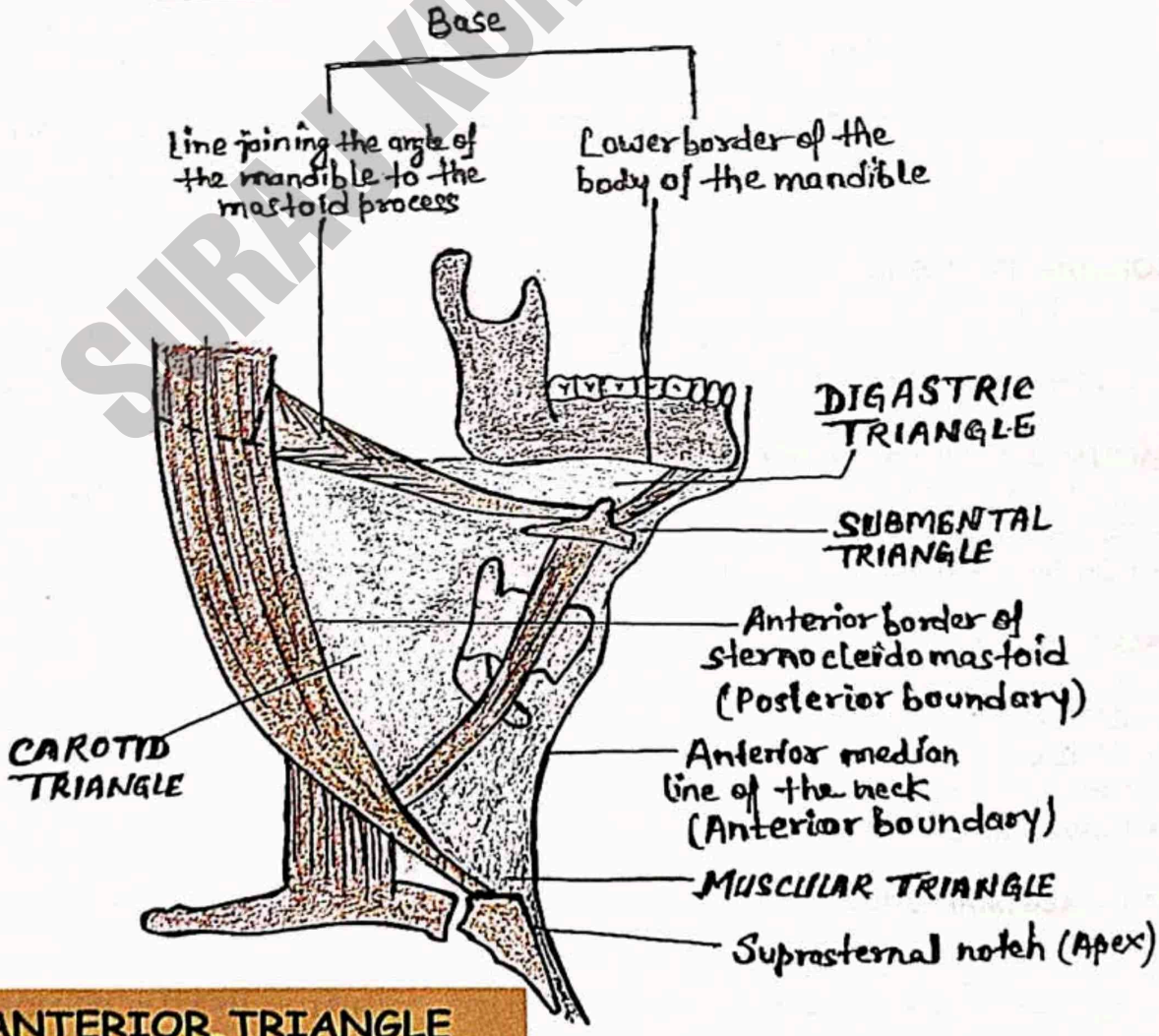


LACRIMAL APPARATUS



ANTERIOR TRIANGLE

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ANTERIOR TRIANGLE

*BOUNDARIES-

1) MEDIALLY-ANTERIOR MEDIAN PLANE OF NECK
2) LATERALLY-STERNOCLEIDOMASTOID
3) BASE -A LINE JOINING THE ANGLE OF MANDIBLE TO MASTOID PROCESS

*SUBDIVISION -
- SUB MENTAL TRIANGLE
- DIGASTRIC TRIANGLE
- CAROTID TRIANGLE
- MUSCULAR TRIANGLE

1) SUBMENTAL TRIANGLE -

*BOUNDARIES - EACH SIDE-ANTERIOR BELLY OF DIGASTRIC MUSCLE
- BASE - HYOID BONE
- FLOOR - RIGHT AND LEFT MYLOHYOID

*CONTENTS
- 4 SUBMENTAL LYMPH NODE
- SMALL SUB-MENTAL VEIN JOIN TO FORM ANTERIOR JUGULAR VEIN

2) DIGASTRIC TRIANGLE

*BOUNDARIES
- ANTERO-INFERIORLY - ANTERIOR BELLY OF DIGASTRIC
- POSTERO-INFERIORLY - POSTERIOR BETLY OF DIGASTRIC
  - STYLOHYOID
- BASE - BASE OF MANDIBLE
- ROOF - SKIN
  - SUPERFICIAL FASCIA→ PLATYSMA
    - CERVICAL BRANCH OF FACIAL NERVE
    - ANTERIOR CUTANEOUS NERVE
  - DEEP FASCIA
- FLOOR - ANTERIORLY - MYLOHYOID MUSDE
  - POSTERIORLY - HYOGLOSSUS MUSCLE

*CONTENTS -
- SALIVARY GLAND
- SUBMANDIBULAR LYMPH NODE
- FACIAL ARTERY AND VEIN
- SUB-MENTAL ARTERY
- HYPOGLOSSAL NERVE
- MYLOHYOID NERVE

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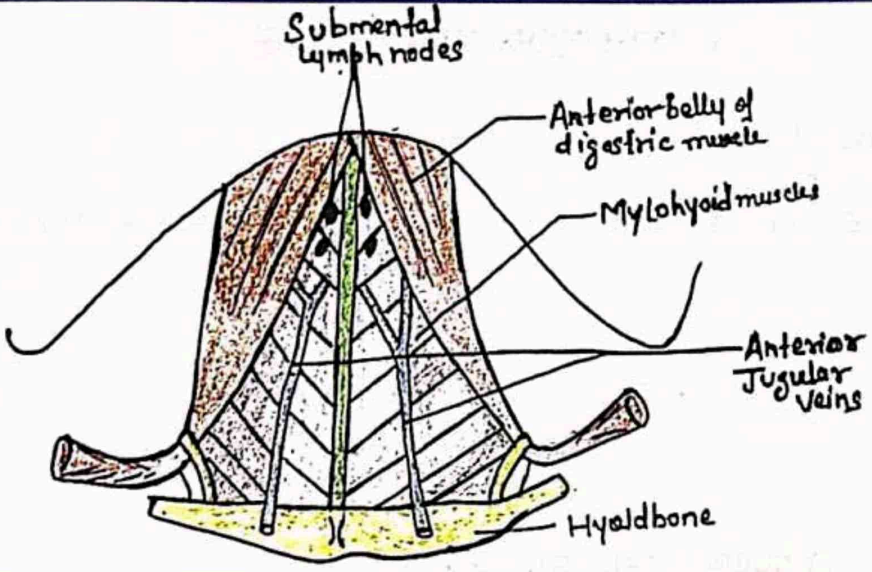
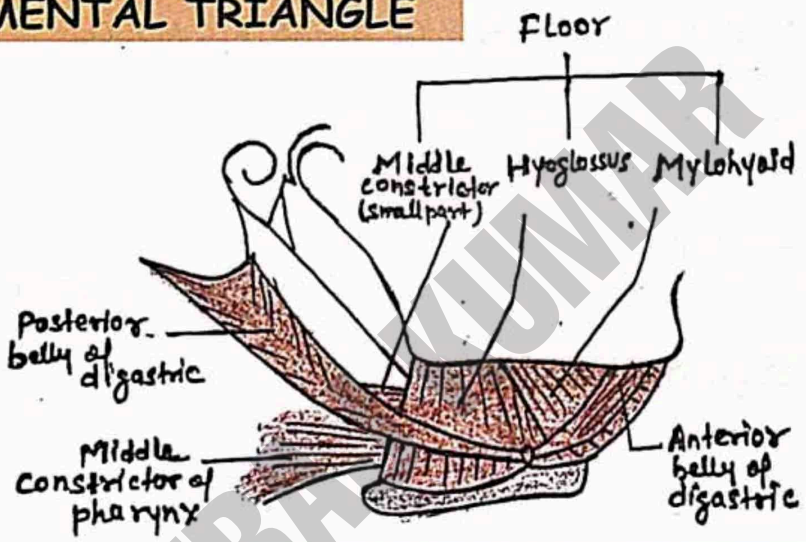
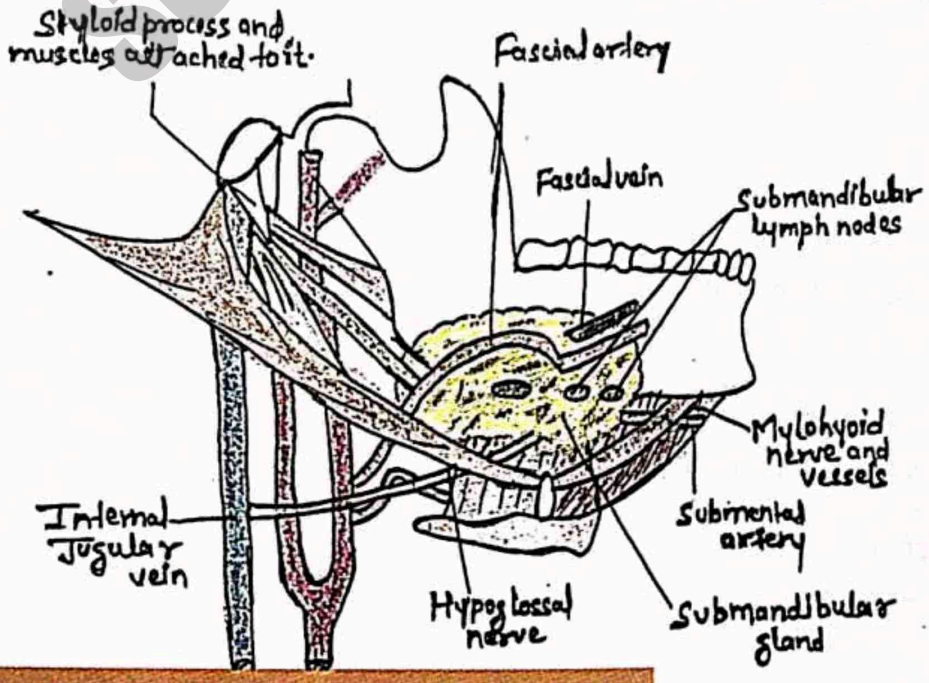


SUB MENTAL TRIANGLE



DIGASTRIC TRIANGLE



CONTENTS OF DIGASTRIC TRIANGLE

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CAROTID TRIANGLE

*INTRODUCTION - SUB-DIVISION OF -ANTERIOR TRIANGLE OF NECK

*BOUNDARIES -

- ANTEROSUPERIORLY - POSTERIOR BELLY OF DIGASTRIC MUSCLE
  - STYLOHYOID
- ANTERO INFERIORLY - SUPERIOR BELLY OF OMOHYOID
- ROOF - SKIN
  -SUPERFICIAL FASCIA CONTAIN - PLASTYSMA
  - CERVICAL BRANCH OF FACIAL NERVE
  -TRANSVERSE CUTANEOUS NERVE
  - DEEP CERVICAL FASCIA
- FLOOR - MIDDLE CONSTRICTOR OF PHARYNX
  - INFERIOR CONSTRICTOR OF PHARYNX
  - THYROHYOID MEMBRANE
- POSTERIORLY - ANTERIOR BORDER OF STERNOCLEIDOMASTOID

*CONTENTS -

*ARTERY

- COMMON CAROTID ARTERY
- INTERNAL CAROTID ARTERY
- EXTERNAL CAROTID ARTERY - SUPERIOR THYROID.
  - LINGUAL
  - FASCIAL
  - ASCENDING PHARYNGEAL ARTERY

*VEIN -

- INTERNAL JUGULAR VEIN
- COMMON FACIAL VEIN
- PHARYNGEAL VEIN
- LINGUAL VEIN

*NERVE-

- VAGUS NERVE
- SPINAL ACCESSARY NERVE
- SUPERIOR LARYNGEAL BRANCH - EXTERNAL LARYNGEAL NERVE
  - INTERNAL LARYNGEAL NERVE
- HYPOGLOSSAL NERVE
- SYMPATHETIC CHAIN NERVE

*LYMPH NODE - DEEP CERVICAL LYMPH NODE -
-JUGULODIGASTRIC NODE
- JUGULO-OMOHYOID NODE

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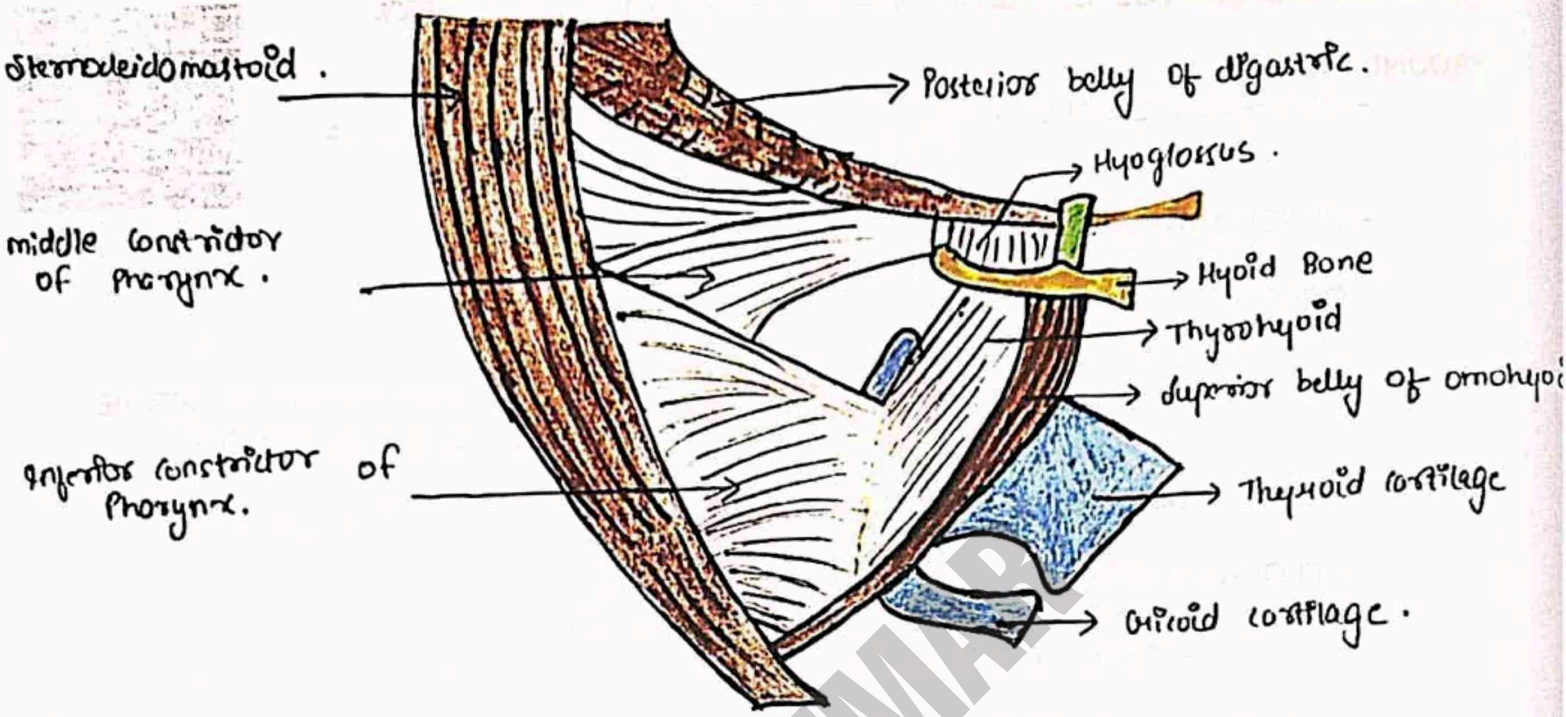
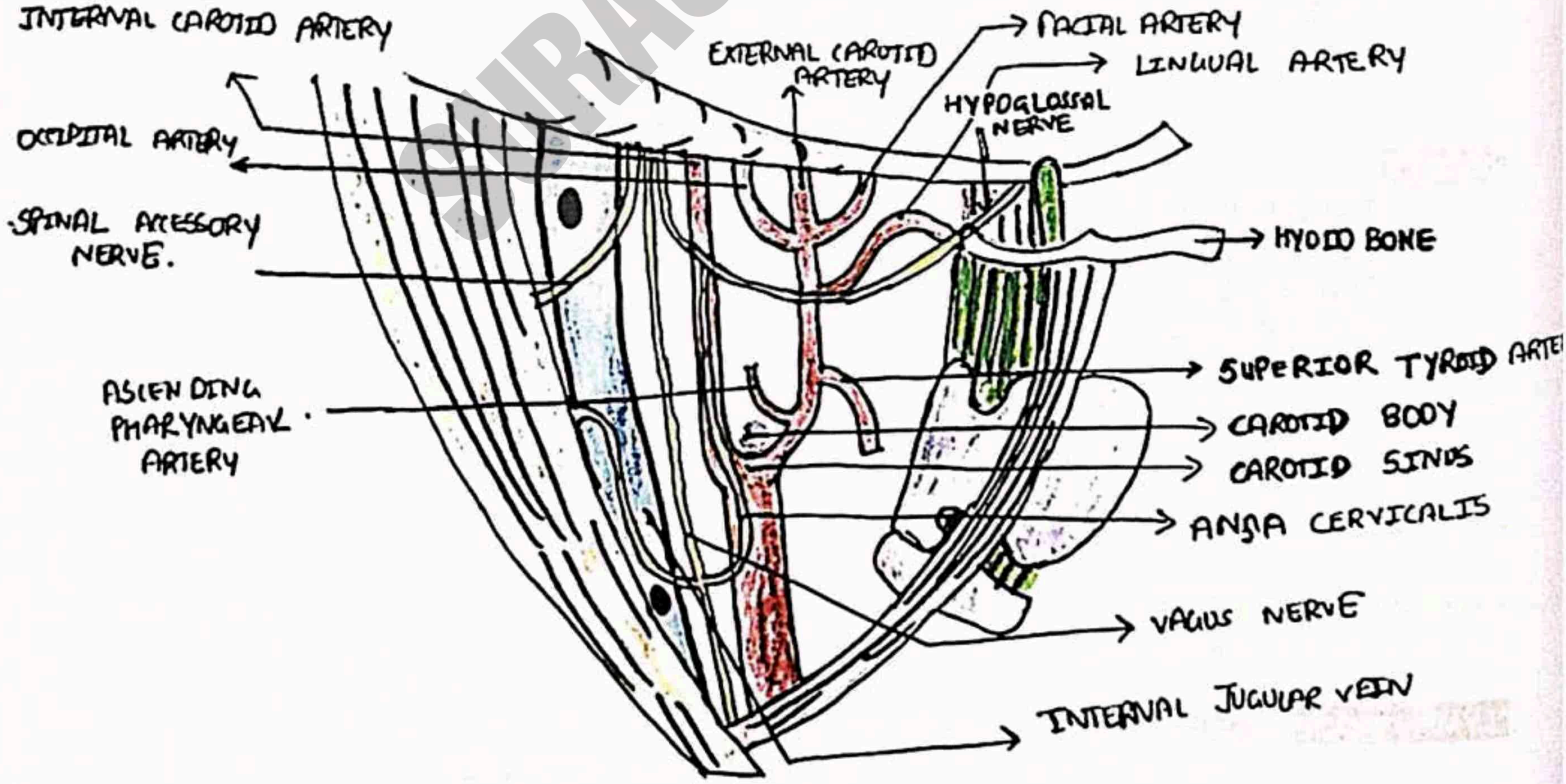


BOUNDARIES OF CAROTID TRIANGLE



CONTENTS OF CAROTID TRIANGLE

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CAROTID SHEATH

-CONDENSATION OF FIBROAREOLAR TISSUE - AROUND MAIN VESSEL OF NECK

*FORMATION: ANTERIOR - PRETRACHEAL FASCIA
POSTERIOR - PREVERTEBRAL FASCIA

*CONTENT-
- INTERNAL CAROTID ARTERIES
- INTERNAL JUGULAR VEIN
- VAGUS NERVE
- IN UPPER PART OF SHEATH - IX, XI, XII NERVE

*RELATION -
- ANTERIOR WALL - ANSA CERVICALIS LIE EMBEDDED
- POSTERIOR WALL - SYMPATHETIC CHAIN LIE BEHIND
- OVERLAPPED BY- STERNOCLEIDOMASTOID MUSCLE
- FUSED TO - DEEP CERVICAL FASCIA

ANSA CERVICALIS

*THIN NERVE LOOP (C1, C2 & C3)
*LIE EMBEDDED - IN ANTERIOR WALL OF CAROTID SHEATH
SUPPLIES - INFRAHYOID MUSCLES-→ THYROHYOID
→ OMOHYOID
→ STERNOHYOID & STERNOHYOID

*FORMATION - FORMED BY - SUPERIOR & INFERIOR ROOT

- SUPERIOR ROOT - FIBRES ARE DERIVED- FROM FIRST CERVICAL NERVE
- CONTINUATION OF DESCENDING BRANCH OF HYPOGLOSSAL NERVE.
- DESCEND OVER - INTERNAL COROTID ARTERY
- COMMON COROTID ARTERY

- INFERIOR ROOT - DERIVED FROM 2ND & 3RD CERVICAL SPINAL NERVE
- DESCENDS - >IT WIND ROUND - INTERNAL JUGULAR VEIN
- CONTINOUS - ANTEROINFERIORLY - TO JOIN SUPERIOR ROOT IN FRONT OF COMMON CAROTID ARTERY

DISTRIBUTION - SUPERIOR ROOT - SUPERIOR BELLY OF OMOHYOID
- INFERIOR ROOT - INFERIOR BELLY OF OMOHYOID
- ANSA CERVICALIS - STERNOTHYROID
- STERNOTHYROID

*THYROHYOID & GENIOHYOID SUPPLIED BY - SEPARATE BRANCHES FROM 1ST CERVICAL NERVE.

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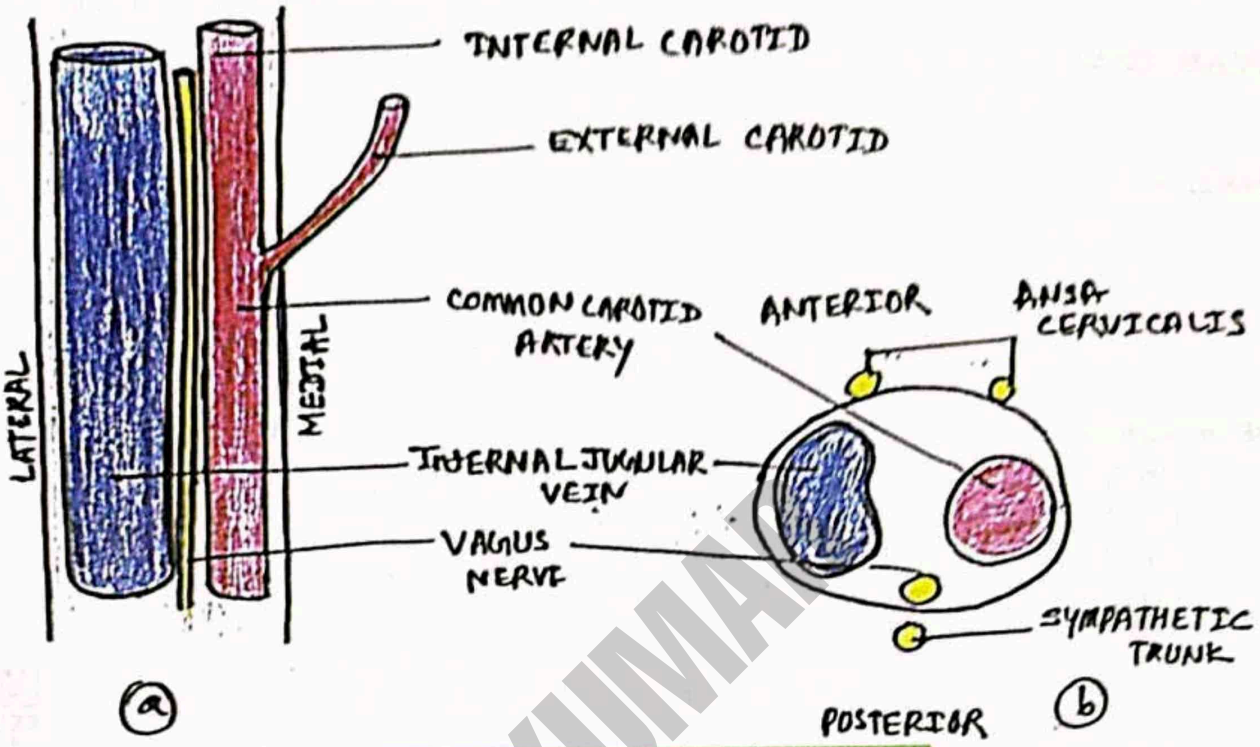
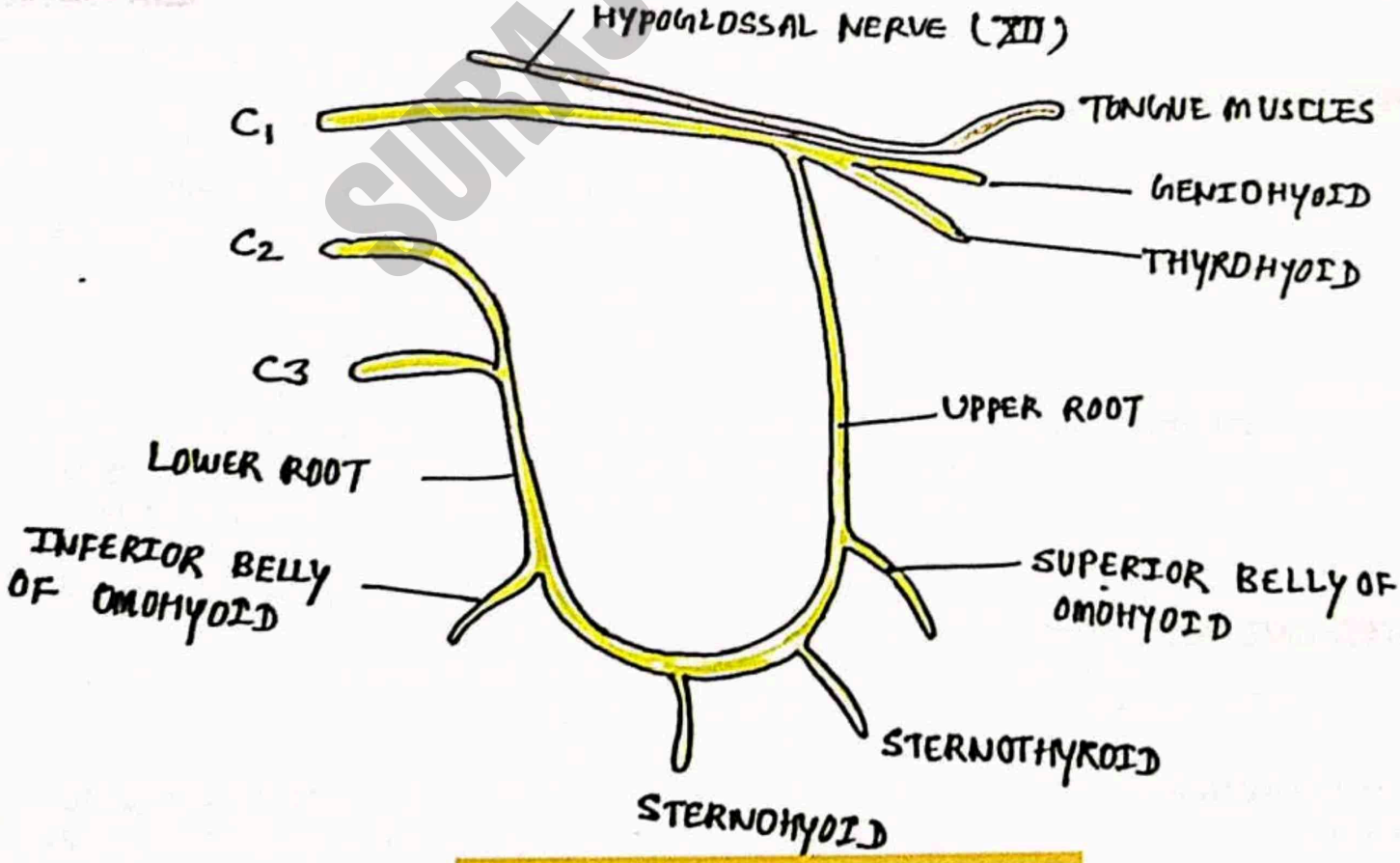


CAROTID SHEATH



ANSA CERVICALIS

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POSTERIOR TRIANGLE

*FEATURES -SPACE ON THE SIDE OF NECK
V- SITUATED BEHIND - STERNOCLEIDOMASTOID MUSCLE

*BOUNDARIES:-
*ANTERIOR - POSTERIOR BORDER OF STERNOCLEIDOMASTOID
*POSTERIOR - ANTERIOR BORDER OF TRAPEZIUS
*BASE - MIDDLE 1/3 OF CLAVICLE
*APEX - LIES ON SUPERIOR NUCHAL LINE
- WHERE TRAPEZIUS AND STERNOCLEIDOMASTOID MEET

*FLOOR - FORMED BY PREVERTEBRAL LAYER OF DEEP CERVICAL FASCIA
- COVERING FOLLOWING MUSCLE
- SPLENIUS CAPITIS
- LEVATOR SCAPULAE
- SCALENUS MEDIUS
- SEMISPINALIS CAPITIS

*ROOF - SKIN
- SUPERFICIAL FASCIA - INVESTING LAYER OF DEEP CERVICAL FASCIA
- CONTAINS - PLATYSMA
-EXTERNAL JUGULAR VEIN
-SUPRACLAVICULAR NERVE
-GREATER AURICULAR
-TRANSVERSE CUTANEOUS
-LESSER OCCIPITAL NERVE
-SUPRASCAPULAR ARTERY
-OCCIPITAL TRANSVERSE CERVICAL ARTERY
-LYMPH VESSEL

*DIVISION OF POSTERIOR TRIANGLE -
- SUB-DIVIDED BY- OMOHYOID MUSCLE
- LARGER UPPER PART - OCCIPITAL TRIANGLE
- SMALL LOWER PART- SUBCLAVIAN TRIANGLE

*CONTENTS -

OCCIPITAL TRIANGLE

1) NERVE
--> SPINAL ACCESSARY NERVE - 4X CUTANEOUS BRANCH
- TRANSVERSE CUTANEOUS NERVE
- SUPRACLAVICULAR NERVE
- GREAT AURICULAR NERVE
- LESSER OCCIPITAL NERVE

-->MUSCULAR BRANCH - 2 LEVATAR SCAPULAE
- 2 TRAPEZIUS
- NERVE TO RHOMBOID

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2) VESSELS - TRANSVERSE CERVICAL ARTERY AND VEIN
- OCCIPITAL ARTERY

3) LYMPH NODE - SUPRA-CLAVICULAR NODE
- OCCIPITAL NODE

SUBCLAVIAN TRIANGLE

1) NERVE -
- ROOT AND TRUNK BRACHIAL PLEXUS
- NERVE TO SUBCLAVIUS
- NERVE TO SERRATUS ANTERIOR
- SUPRASCAPULAR NERVE

2) VESSEL -
- 3RD PART OF SUBCLAVIAN ARTERY AND VEIN
- SUPRASCAPULAR ARTERY
- TRANSVERSE CERVICAL ARTERY
- EXTERNAL JUGULAR VEIN

CLINICAL ANATOMY

ANTTERIOR TRIANGLE

1.THE COMMON ANTERIOR MIDLINE SWELLINGS OF THE NECK ARE:

a.ENLARGED SUBMENTAL LYMPH NODES & SUBLINGUAL DERMOID IN THE SUBMENTAL REGION.
b.THYROGLOSSAL CYST & INFLAMED SUBHYOID BURSA JUST BELOW THE HYOID BONE.
c.GOITRE, CARCINOMA OF LARYNX & ENLARGED LYMPH NODES IN THE SUPRASTERNAL REGION.

POSTERIOR TRIANGLE

*THE MOST COMMON SWELLING IN THE POSTERIOR TRIANGLE IS DUE TO ENLARGEMENT OF THE SUPRACLAVICULAR LYMPH NODES.WHILE DOING BIOPSY OF THE LYMPH NODE, ONE MUST BE CAREFUL IN PRESERVING THE ACCESSORY WHICH MAY GET ENTANGLED AMONGST ENLARGED LYMPH NODES.

CAROTID TRIANGLE

*THE CAROTID SINUS IS RICHLY SUPPLIED BY NERVES.IN SOME PERSONS, THE SINUS MAY BE HYPERSENSITIVE.IN SUCH PERSONS, SUDDEN ROTATION OF THE HEAD MAY CAUSE SLOWING OF HEART.THIS CONDITION IS CALLED "CAROTID SINUS SYNDROME".

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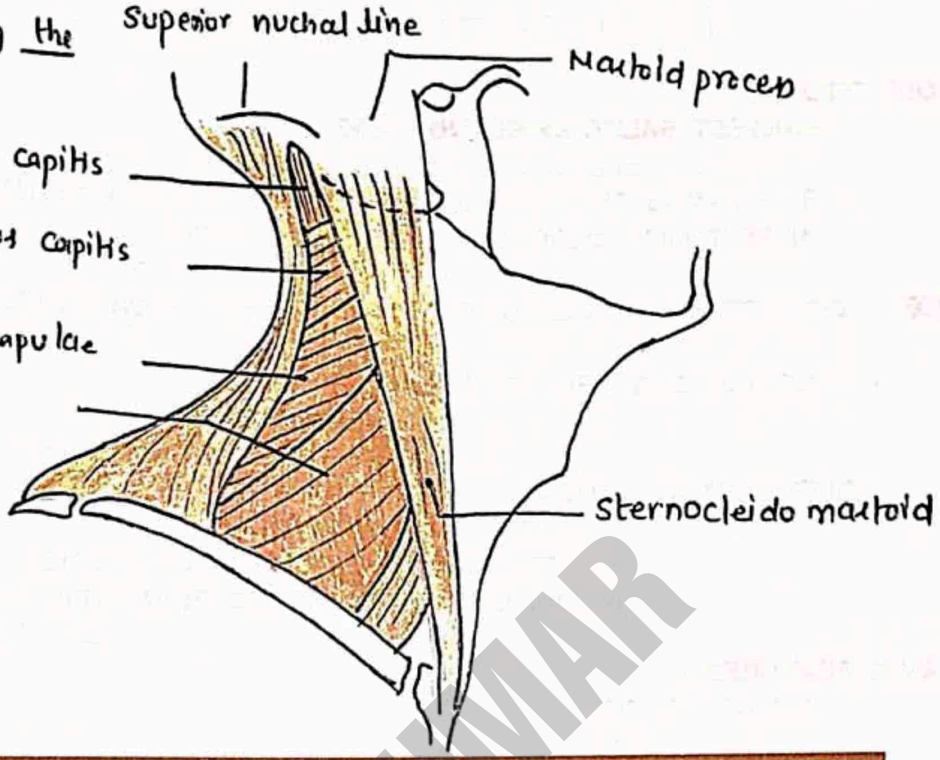
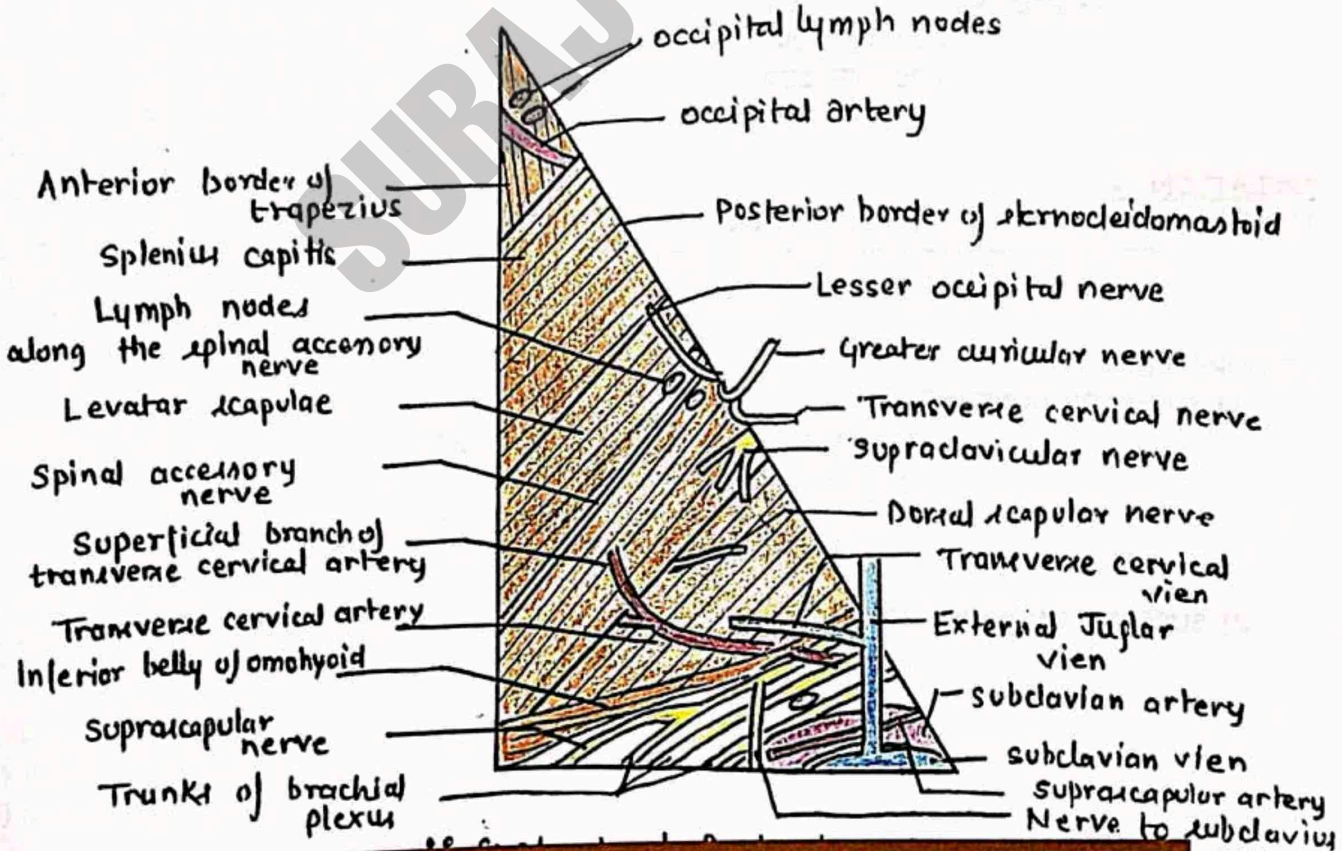
(*) Muscles forming the floor

1) Semispinalis capitis
2) Splenius capitis
3) Levator scapulae
4) Scalenus medius



BOUNDARIES OF POSTERIOR TRIANGLE



CONTENTS OF POSTERIOR TRIANGLE

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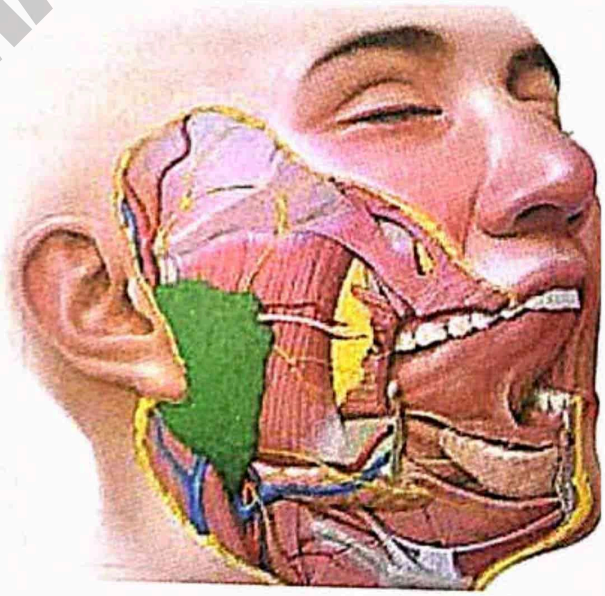
PAROTID GLAND

*INTRODUCTION -
- LARGEST SALIVARY GLAND - 25G
- SITUATED- BELOW EXTERNAL ACOUSTIC MEATUS.
- B/W RAMUS OF MANDIBLE AND STERNOCLEIDOMASTOID
- ANTERIORLY- OVERLAP - MASSETER MUSCLE

CAPSULE .- INVESTING LAYER OF DEEP CERVICAL FASCIA-FORM CAPSULE FOR GLAND
- SUPPLIED-BY AURICULAR NERVE
- FASCIA SPLITS - ENCLOSE GLAND
- SUPERIOR LAMINA- THICK & ADHERENT TO GLAND
  - ATTACHED ABOVE - ZYGOMATIC ARCH
- DEEP LAMINA - THIN
  - ATTACHED - TYMPANI PLATE
  - PORTION OF D.L EXTEND B/W STYLOID PROCESS AND MANDIBLE - IS THICKENED FORM STYLOMANDIBULAR LIGAMENT

*EXTERNAL FEATURES -
- 3 SIDE PYRAMID
- APEX - DOWNWARD
- 4 SURFACE - SUPERIOR
  - SUPERFICIAL
  - ANTEROMEDIAL
  - POSTEROMEDIAL
- 3 BORDER-
  - ANTERIOR
  - POSTERIOR
  - MEDIAL



*RELATION :-
1) APEX.- POSTERIOR BELLY DIGASTRIC
- CERVICAL BRANCH OF FACIAL NERVE
- 2 DIVISION OF RETROMANDIBULAR VEIN

2) SURFACE:-
1) SUPERIOR SURFACE-
- SUPERFICIAL TEMPORAL VESSEL
- AURICULO TEMPORAL NERVE
- EXTERNAL ACOUSTIC MEATUS
- POSTERIOR SURFACE OF TMJ

2) SUPERFICIAL SURFACE → SKIN
- SUPERFICIAL FASCIA
- PAROTID FASCIA
- DEEP PAROTID LYMPH NODE

3) ANTEROMEDIAL SURFACE -

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- RAMUS OF MANDIBLE
- MASSETER
- MEDIAL PTERYGOID
- LATERAL SURFACE OF TMJ

4) POSTEROMEDIAL SURFACE - MASTOID PROCESS
- STYLOID PROCESS

3) BORDER -
1) ANTERIOR BORDER - TERMINAL BRANCH OF FACIAL NERVE
- PAROTID DUCT
- TRANSVERSE FACIAL VESSEL

2) POSTERIOR BORDER SEPARATE — SUPERFICIAL SURFACE TO POSTEROMEDIAL SURFACE

3) MEDIAL BORDER SEPARATE — ANTEROMEDIAL TO POSTEROMEDIAL SURFACE

STRUCTURE WITHIN THE PEROTID GLAND

*ARTERIES - "ECA" ENTER POSTEROMEDIAL SURFACE
- MAXILLARY A. LEAVE —> ANTEROMEDIAL SURFACE
- TFA → ANTERIOR PART - SUPERIOR SURFACE

*VEIN - RETROMANDIBULAR VEIN → FORMED IN GLAND WITH BY STV + MV
- LOWER PART ANTERIOR, POSTERIOR DIVISION —>EMERGE, APEX

*NERVE - FACIAL NERVE - ENTER —> POSTEROMEDIAL SURFACE
- LEAVE —> ANTEROMEDIAL SURFACE

-BRANCH ->
TEMPORAL BRANCH
ZYGOMATIC BRANCH
BUCCAL BRANCH
MARGINAL MANDIBULAR NERVE
CERVICAL BRANCH

*PAROTID DUCT — EMERGS, MIDDLE OF ANTERIOR BORDER
- RUN FORWARD DOWNLOAD ON MASSETER
- PIERCE IN- BUCCINATOR MUSCLE
- FINALLY DUCT OPEN - VESTIBULE OF MOUTH

*BLOOD SUPPLY - ARTERY - EXTERNAL CAROTID ARTERY
VEIN SUPPLY - EJU AND IJU

*NERVE SUPPLY -
1) SENSORY NERVE - AURICULOTEMPORAL NERVE
2) SYMPATHETIC NERVE - VASOMOTAR - PLEXUS AROUND - MIDDLE MENINGEAL ARTERY

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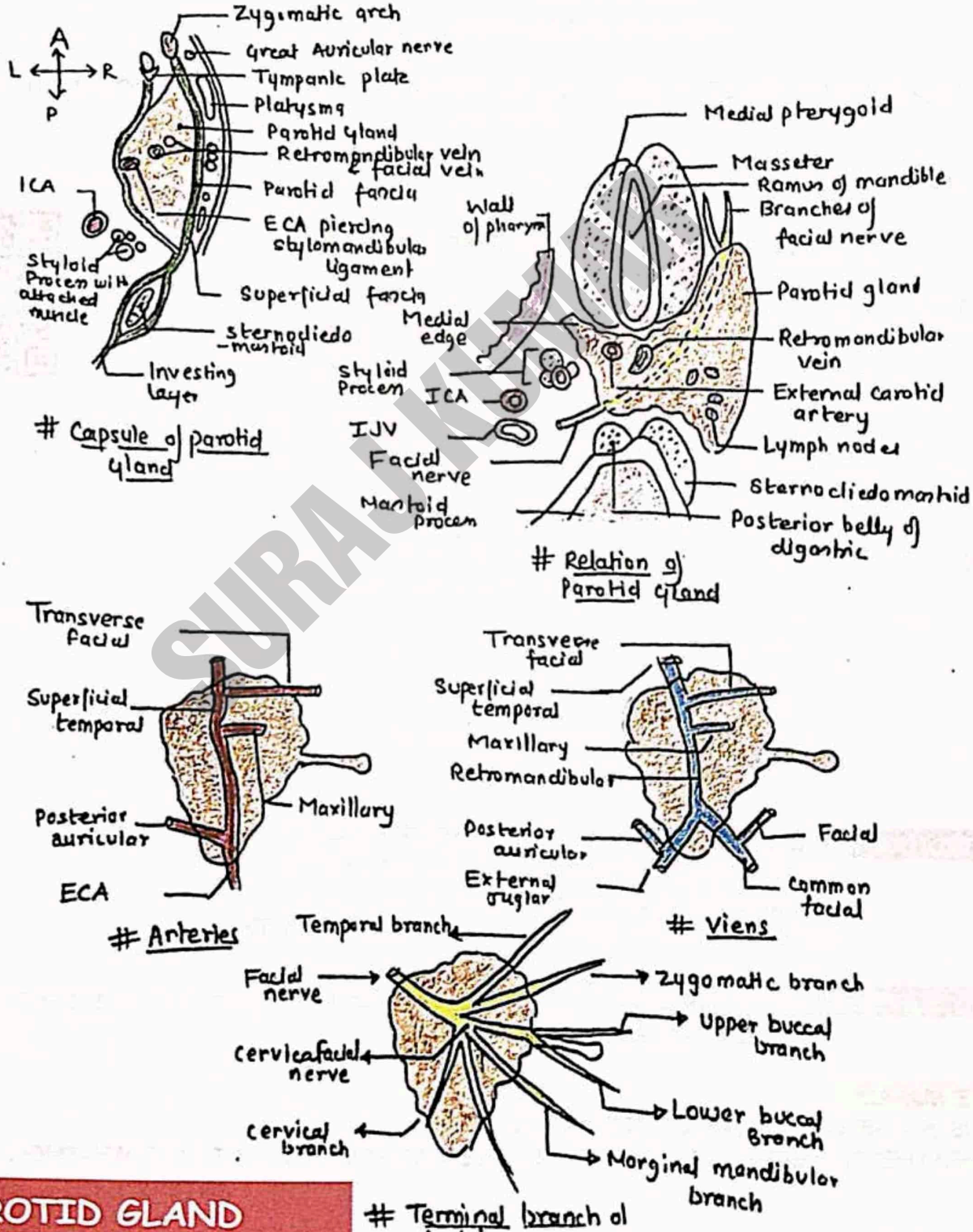
3) PARASYMPATHIC NERVE - SECRETO MOTOR
- INFERIOR SALIVATORY NUCLEUS
- GLOSSOPHARYNGEAL NERVE
- TYMPANIC BRANCH
- TYMPANIC NERVE
- LESSER PETROSAL NERVE
- OTIC GANGLION
- POST-GANGLIONIC FIBRE THROUGH AURICULOTEMPORAL NERVE



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PAROTID GLAND

THYROID GLAND

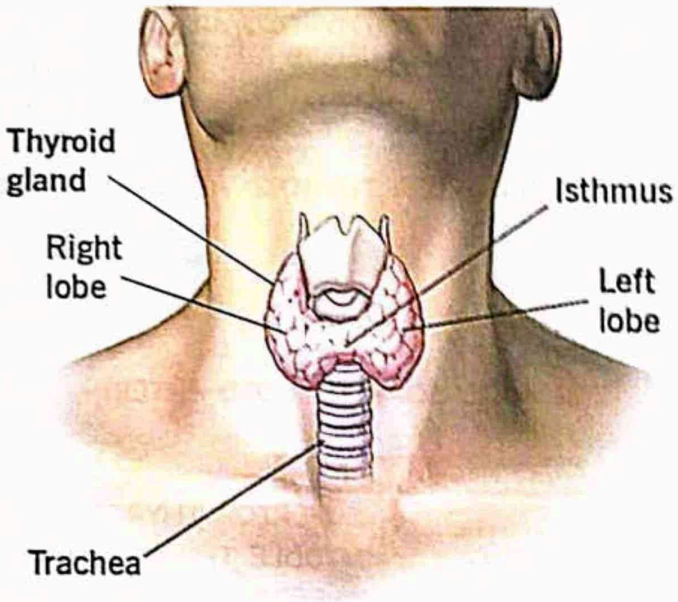
*THYROID GLAND - LARGEST ENDOCRINE GLAND
- LOWER PART OF FRONT AND SIDE OF NECK
- FUNCTION - BASAL METABOLIC RATE
- CALCIUM METABOLISM
- NATURAL IODINE SYNTHESIS

- CONSIST OF - 2 LOBES->RIGHT & LEFT LOBE-> JOINED TO EACH OTHER BY ISTHMUS
- SITUATION - GLAND LIES AGAINST - C5-C7, T1 VERTEBRAE
- EACH LOBE EXTEND→ MIDDLE OF THYROID CARTILAGE TO FIFTH TRACHEAL RING
- ISTHMUS EXTEND→ SECOND TO FOURTH TRACHEAL RING

*CAPSULE OF THYROID -
- TRUE CAPSULE → PERIPHERAL CONDENSATION OF CONNECTIVE TISSUE OF GLAND
- DENSE CAPILLARY PLEXUS IS PRESENT DEEP TO TRUE CAPSULE

- FALSE CAPSULE → DERIVED FROM - PRETRACHEAL LAYER OF DEEP CERVICAL FASCIA
- THIN IN POSTERIOR BORDER OF LOBE
- THICK INNER SURFACE OF GLAND TO FORM SUSPENSARY LIGAMENT (OF BERRY)

*PARTS AND RELATIONS:
1) LOBES - APEX
- BASE
- 3 SURFACES - LATERAL
- MEDIAL
- POSTEROLATERAL
- 2 BORDER - ANTERIOR
- POSTERIOR
2) ISTHMUS - 2 SURFACE - ANTERIOR
- POSTERIOR
- 2 BORDER - SUPERIOR
- INFERIOR



*THYROID GLAND RELATION : -
*LOBES:-
1) APEX- STERNOTHYROID MUSCLE
- SUPERIOR THYROID ARTERY
- EXTERNAL LARYNGEAL NERVE
2) BASE - INFERIOR THYROID ARTERY
- RECURRENT LARYNGEAL NERVE
3) SURFACE:-
1) LATERAL - STERNO THYROID
- STERNO HYOID
- SUPERIOR BELLY OF OMOHYOID
- STERNOCLEIDOMASTOID

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2) MEDIAL - 2X TUBE - TRACHEA
- OESOPHAGUS
-2 MUSCLE - INFERIOR CONSTRICTOR
-CRICOTHYROID
-2 NERVE - EXTERNAL RECURRENT LARYNGEAL NERVE

3) POSTEROLATERAL - CAROTID SHEATH

4) BORDER:-
ANTERIOR BORDER - THIN
- SUPERIOR THYROID ARTERY
POSTERIOR BORDER - THICK
- INFERIOR THYROID ARTERY
- PARATHYROID GLAND
- THORACIC DUCT

*ISTHMUS:-
1) ANTERIOR SURFACE - STERNO THYROID
- STERNO HYOID
2) POSTERIOR SURFACE- 2ND TO 4TH TRACHEAL RINGS
3) UPPER SURFACE- SUPERIOR THYROID ARTERIES
4) LOWER SURFACE - INFERIOR THYROID VEIN

*ARTERIAL SUPPLY :-

1) SUPERIOR THYROID ARTERY - FIRST BRANCH OF ECA
- DOWNWARD, FORWARDS RUN
- RELATION-EXTERNAL LARYNGED NERVE
- APEX OF LOBE
2) INFERIOR THYROID ARTERY - BRANCH OF THYROCERVICAL TRUNK - SUB CLAVIAN ARTERY
- UPWARD, MEDIALLY RUN
- RELATION- RECURRENT LARYNGEAL NERVE
- BASE OF GLAND
3) ACCESSORY THYROID ARTERY - ARISE FROM TRACHEA AND OESOPHAGUS ARTERY

*VENOUS DRAINAGE -
- SUPERIOR THYROID VEIN - DRAIN - INTERNAL JUGULAR VEIN
- MIDDLE THYROID VEIN → INTERNAL JUGULAR VEIN
- INFERIOR THYROID VEIN - LEFT BRACHIOCEPHALIC VEIN

LYMPHATIC :- DEEP CERVICAL LYMPH NODES

*NERVE - DERIVED FROM- MIDDLE CERVICAL GANGLION
PARTLY FROM - SUPERIOR AND INFERIOR CERVICAL GANGLION

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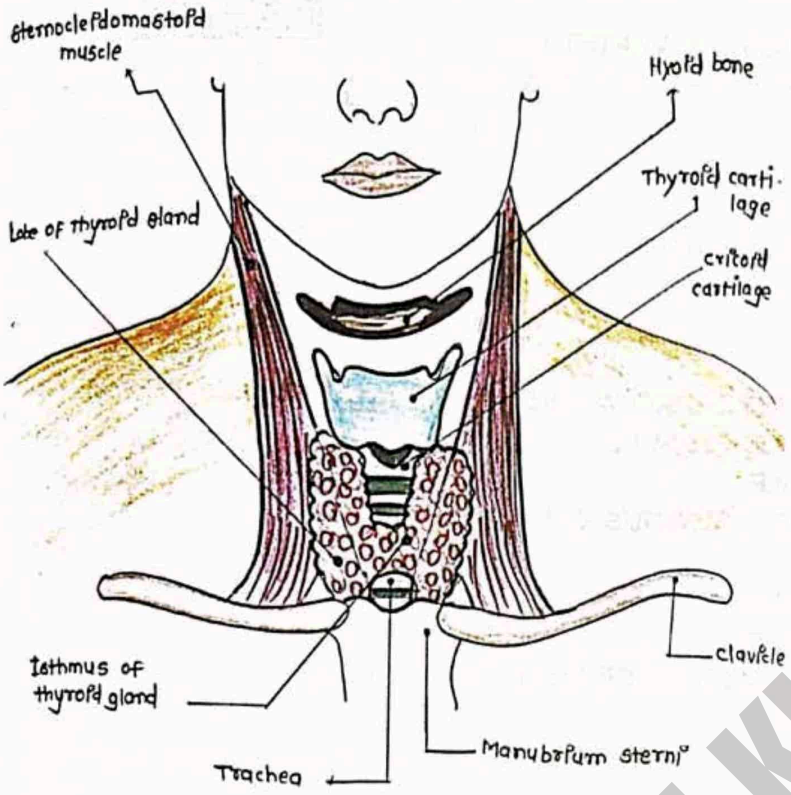
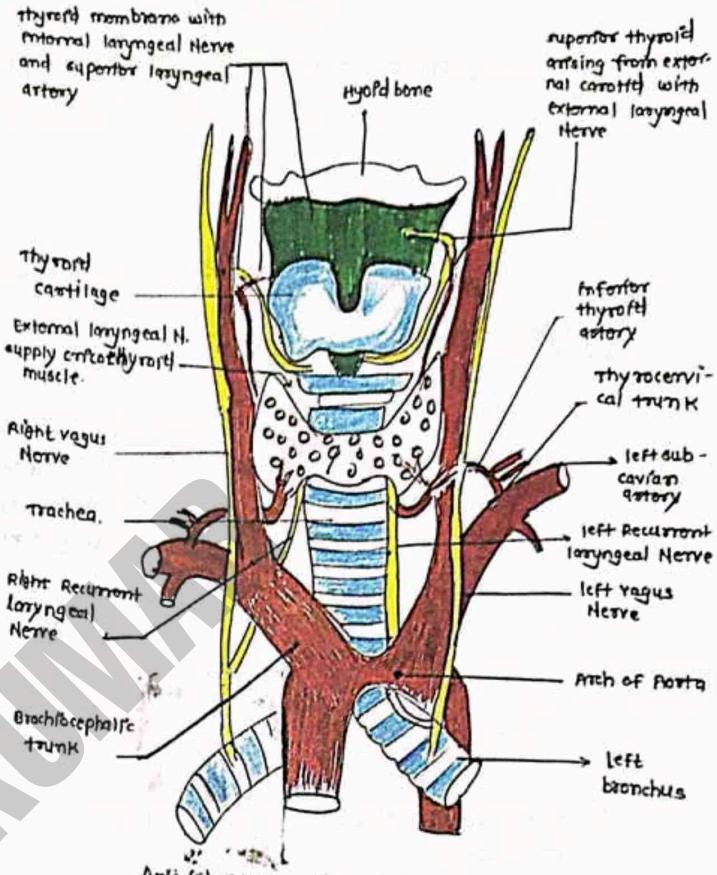
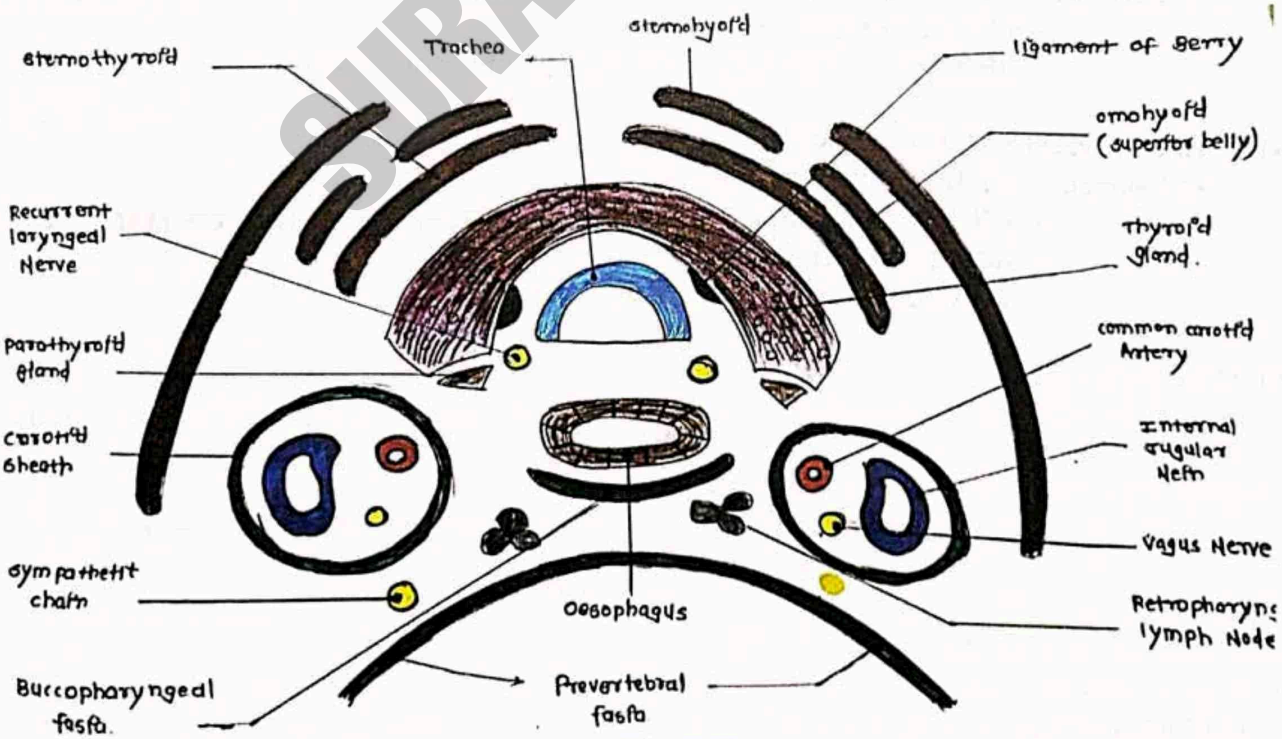


Position of thyroid gland.



Arterial supply of Anterior aspect of thyroid gland



RELATION OF THYROID GLAND

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TEMPAROMANDIBULAR JOINT

TYPE OF JOINT: SYNOVIAL JOINT OF CONDYLAR VARIETY

ARTICULAR SURFACE - UPPER ARTICULAR SURFACE IS FORMED BY - TEMPORAL BONE
- ARTICULAR TUBERCLE
-MANDIBULAR FOSSA
-TYMPANIC PLATE

- INFERIOR ARTICULAR SURFACE → HEAD OF MANDIBLE

LIGAMENTS -

1) FIBROUS CAPSULE -
ATTACH - ABOVE - ARTICULAR TUBERCLE CIRCUMFERENCE OF MANDIBULAR FOSSA
- BEHIND - SQUAMOTYMPANIC FISSURE
- BELOW - NECK OF MANDIBLE
- SYNOVIAL MEMBRANE LINE - FIBROUS CAPSULE

2) LATERAL TEMPOROMANDIBULAR LIGAMENT -
ATTACH - ABOVE - ARTICULAR TUBERCLE
- BELOW - POSTEROLATERAL ASPECT OF NECK OF MANDIBLE
- STRENGTH— LATERAL PART - CAPSULE

3) SPHENOMANDIBULAR LIGAMENT -
ATTACH -
SUPERIORLY - SPINE OF SPHENOID
INFERIARLY - LINGUAL OF MANDIBULAR FORAMEN
-ACCESSARY LIGAMENT
- REMNANT - DORSAL PART OF MECKEL CARTILAGE

4) STYLOMANDIBULAR LIGAMENTS-
ATTACHED - ABOVE- STYLOID PROCESS
- BELOW - ANGLE AND POSTERIOR BORDER OF RAMUS OF MANDIBLE
- ACCESSARY LIGAMENT
- THICKEND - DEEP CERVICAL

5) ARTICULAR DISC-
- OVAL PREDOMINANTLY FIBROUS PLATE JOINT - DIVIDE JOINTS UPPER AND LOWER COMPARTMENT
- COMPOSED OF - ANTERIOR EXTENSION
ANTERIOR BAND - INTERMEDIATE- POSTERIOR BAND BILAMINAR REGION

- DISC REPRESENTS - DEGENERATED PRIMITIVE INSERTION OF LATERAL PTERYGOID
- PREVENTS FRICTION BETWEEN THE ARTICULATING SURFACES
- SHOCK ABSORPTION
- "PROPRIOCEPTIVE FIBRE" HELP REGULATE MOVEMENTS OF JOINT

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RELATION

1) LATERAL - SKIN
- SUPERFICIAL FASCIA
- PAROTID GLAND
- TEMPORAL BRANCH OF FACIAL NERVE

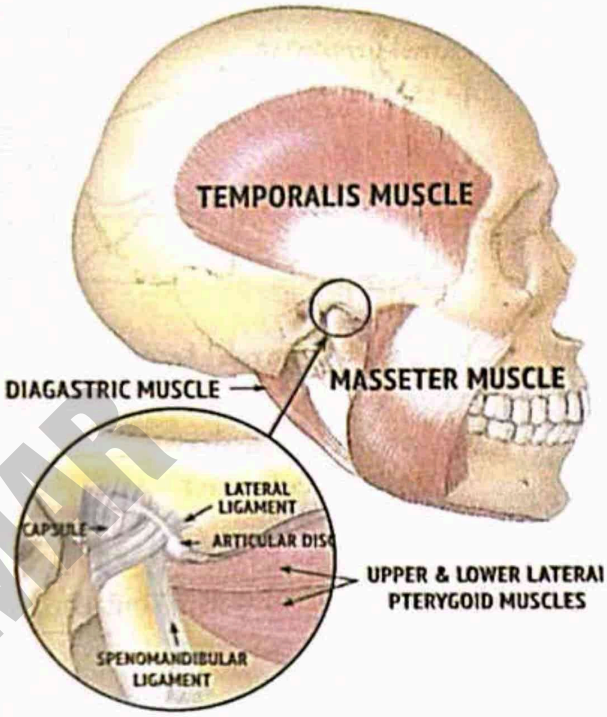
2) MEDIAL - TYMPANIC PLATE
- SPINE OF SPHENOID
- ANTERIOR-LATERAL PTERYGOID
- AURICULO TEMPORAL AND CHORDO-TYMPANI NERVE
- MIDDLE MENINGEAL ARTERY

3) ANTERIOR - LATERAL PTERYGOID
- MASSETERIC NERVE AND ARTERY

4) POSTERIOR - PAROTID GLAND
- SUPERFICIAL TEMPORAL VESSEL
- AURICULO-TEMPORAL NERVE

5) SUPERIOR- MIDDLE CRANIAL FOSSA

6) INFERIOR - MAXILLARY ARTERY



***BLOOD SUPPLY** - SUPERFICIAL TEMPORAL ARTERY AND VEIN
- MAXILLARY ARTERIES AND VEIN

***NERVE SUPPLY** - AURICULO-TEMPORAL NERVE
- MASSETERIC NERUE

***MOVEMENTS -**

1) DEPRESSION COPEN MOUTH) - LATERAL PTERYGOID
-DIGASTRIC AND GENIOHYOID
2) ELEVATION (CLASE MOUTH) → MASSETER AND MEDIAL PTERYGOID
3) PROTRUSION - MEDIAL & LATERAL PTERYGOID.
4) RETRUSION - TEMPORALIS & MASSETER
5) SIDE TO SIDE MOVEMENT DURING CHEWING

CLINICAL ANATOMY

DISLOCATION OF MANDIBLE : DURING EXCESSIVE OPENING OF THE MOUTH, THE HEAD OF THE MANDIBLE OF ONE OR BOTH SIDES MAY SLIP ANTERIORLY INTO THE INFRATEMPORAL FOSSA, AS A RESULT OF WHICH THERE IS INABILITY TO CLOSE THE MOUTH.

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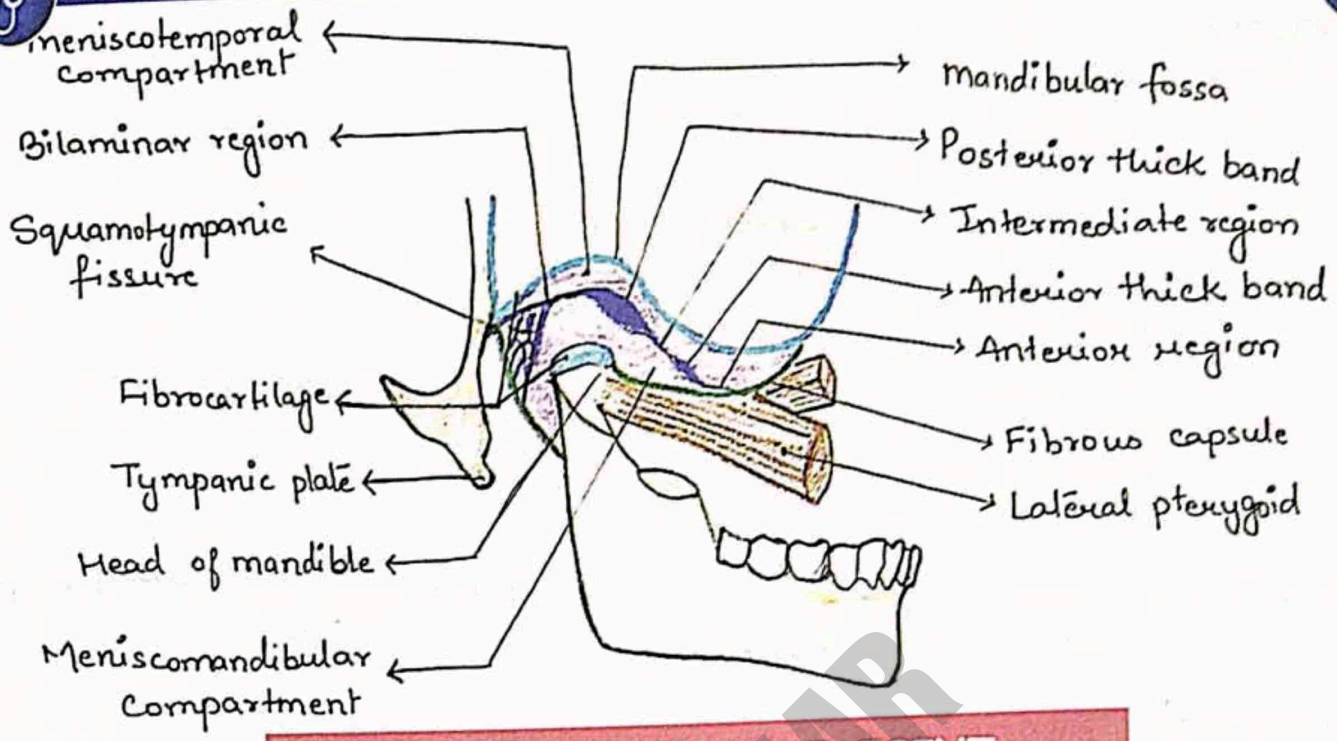
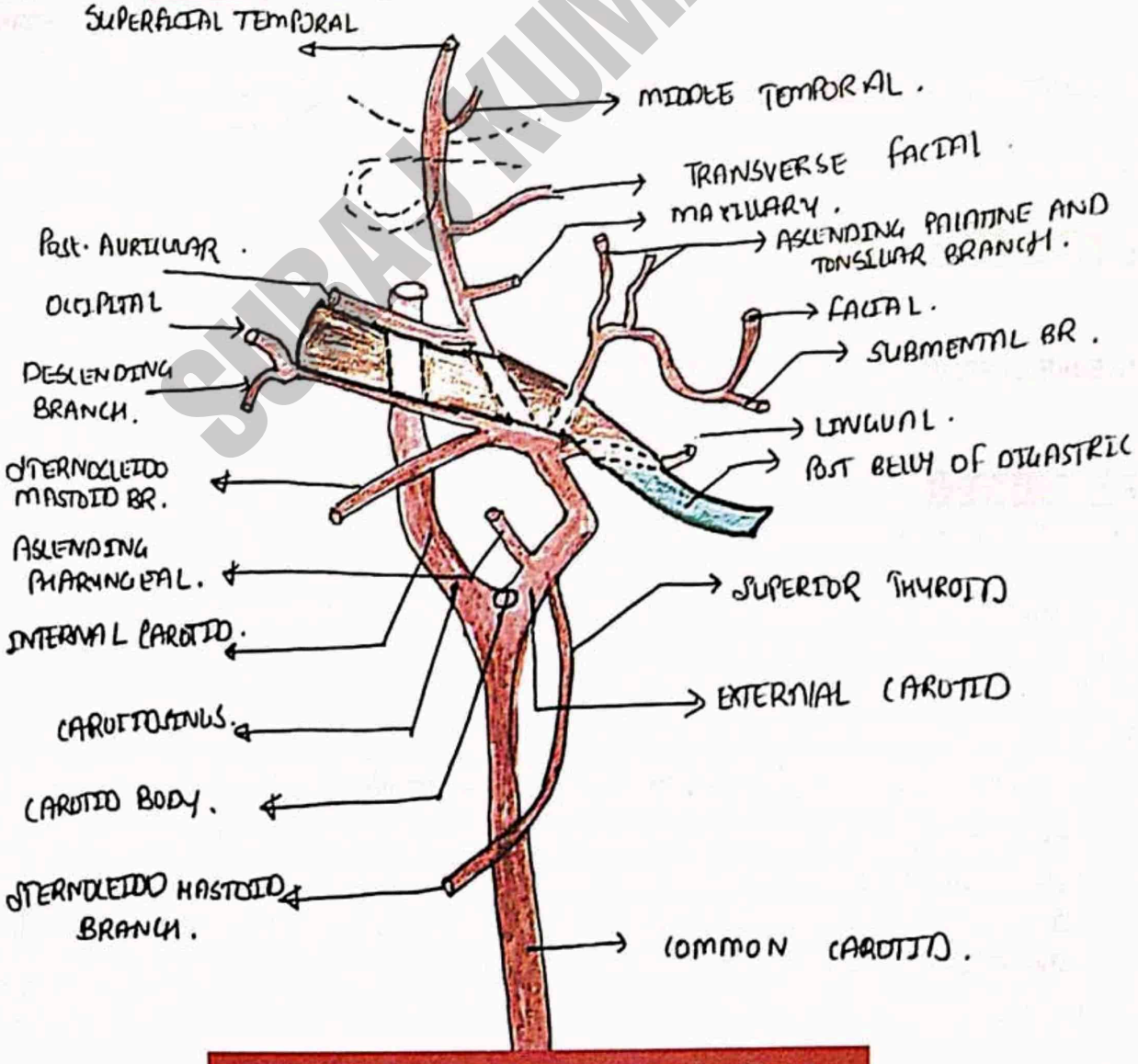


TEMEROMANDIBULAR JOINT



EXTERNAL CAROTID ARTERY

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EXTERNAL CAROTID ARTERY

*ECA IS TERMINAL BRANCH OF COMMON CAROTID ARTERY
*CHIEF ARTERY OF SUPPLY --→ FRONT OF NECK & FACE

*COURSE :-
*BEGIN - IN CAROTID TRIANGLE AT LEVEL OF UPPER BORDER OF THYROID CARTILAGE
- OPPOSITE TO C3 & C4
- RUN UPWARD , BACKWARD AN LATERALLY

*TERMINATE - BEHIND NECK OF MANDIBLE
- DIVIDING - MAXILLARY AND SUPERFICIAL TEMPORAL ARTERY
- SLIGHLY CURVED COURSE - LOWER ANTEROMEDIAL - TO ICA
- UPPER ANTEROLATERAL TO ICA
- IN CAROTID TRIANGLE - IS SUPERFICIAL
- LIE UPPER COVER OF ANTERIOR BORDER OF STERNOCLEIDOMASTOID MUSCLE
- ABOVE CAROTID - LIE - DEEP IN PAROTID GLAND
- RELATED - RETROMANDIBULAR VEIN
- & FACIAL NERVE

*BRANCHES - ANTERIOR - SUPERIOR THYROID A.
- LINGUAL A.
- FACIAL A
- POSTERIOR - OCCIPITAL
- POSTERIOR AUERICULAR A.
- MEDIAL - ASCENDING PHARYNGEAL A.
- TERMINAL - MAXILLARY ARTERY
- SUPERFICIAL TEMPORAL A.

1) SUPERIOR THYROID A. - BRANCH OF ECA - JUST BELOW GERATER CORNUA OF HYOID BONE
-RUN - DOWNWARD, FORWARD PARALLEL TO LATERAL LOBE OF THYROID
- SUPPLY - THYROID GLAND - TERMINAL BRANCH - SUPERIOR LARYNGEAL ARTERY

2) LINGUAL A - ARISE - OPPOSITE TIP OF GREATER CORNUA OF HYOID BONE
- BRANCH - SUPRAHYOID, DOSRAL LINGUAL, SUB- LINGUAL

3) FACIAL A.- SAME AS FACIAL NOTES

4) OCCIPITAL A - OPPOSITE ORIGIN OF FACIAL ARTERY
- CROSSED AT ITS ORIGIN HYPOGLOSSAL NERVE

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5) POSTERIOR ARTICULAR ARTERY - ABOVE POSTERIOR BELLY OF DIGASTRIC
- UPWARD BACKWARD→DEEP PAROTID GLAND & STYLOID PROCESS
- SUPPLY→SKIN-→OVER MASTOID PROCESS & BACK OF SCALP

6) ASCENDING PHARYNGEAL ARTERY - SMALLEST BRANCH
- ARISES→MEDIAL SIDE OF ECA
- VERTICALLY UPWARD->BETWEEN SIDE WALL OF PHARYNX TONSIL & MEDIAL WALL OF MIDDLE EAR

7) MAXILLARY ARTERY - LARGEST BRANCH
- BEHIND NECK OF MANDIBLE UNDER PAROTID GLAND
- BELOW AURICULO-TEMPORAL NERVE

8) SUPERFICIAL TEMPORAL ARTERY - SMALLER TERMINAL BRANCH
- BEHIND NECK OF MANDIBLE
- DIVIDED→ANTERIOR & POSTERIOR BRANCH →SUPPLY TEMPLE & SCALP

FACIAL ARTERY

*CHIEF ARTERY OF FACE
*BRANCH -->OF EXTERNAL CAROTID ARTERY-->GIVE OFF IN CAROTID TRIANGLE-->JUST ABOVE GREATER CORNUA OF HYOID BONE.

*COURSE -

ENTER FACE→WINDING AROUND MANDIBLE-→AND PIERCING→DEEP CERVICAL FASCIA
-FIRST RUN UPWARD FORWARD→1.25CM LATERAL TO ANGLE OF MOUTH
-ASCENDS BY SIDE OF NOSE-→UP TO ANGLE OF EYE
-TERMINATE BY SUPPLYING LACRIMAL SAC →BY ANASTOMOSING WITH DORSAL NASAL BRANCH
-LIE BETWEEN SUPERFICIAL & DEEP MUSCLE OF FACE.

*BRANCHES - ANTERIOR BRANCH - INFERIOR LABIAL
- SUPERIOR LABIAL
- LATERAL NASAL

- POSTERIOR BRANCH - SMALL AND UNNAMED

*ANASTOMOSES - LARGE ANTERIOR→ANASTOMOSES SIMILAR BRANCH OF OPPOSITE SIDE
-SMALLER POSTERIOR-->TRANSVERSE FACIAL INFRA ORBITAL ARTERIES
-TERMINAL BRANCH OF FACIAL ARTERY-→OPHTHALAMIC ARTERY.

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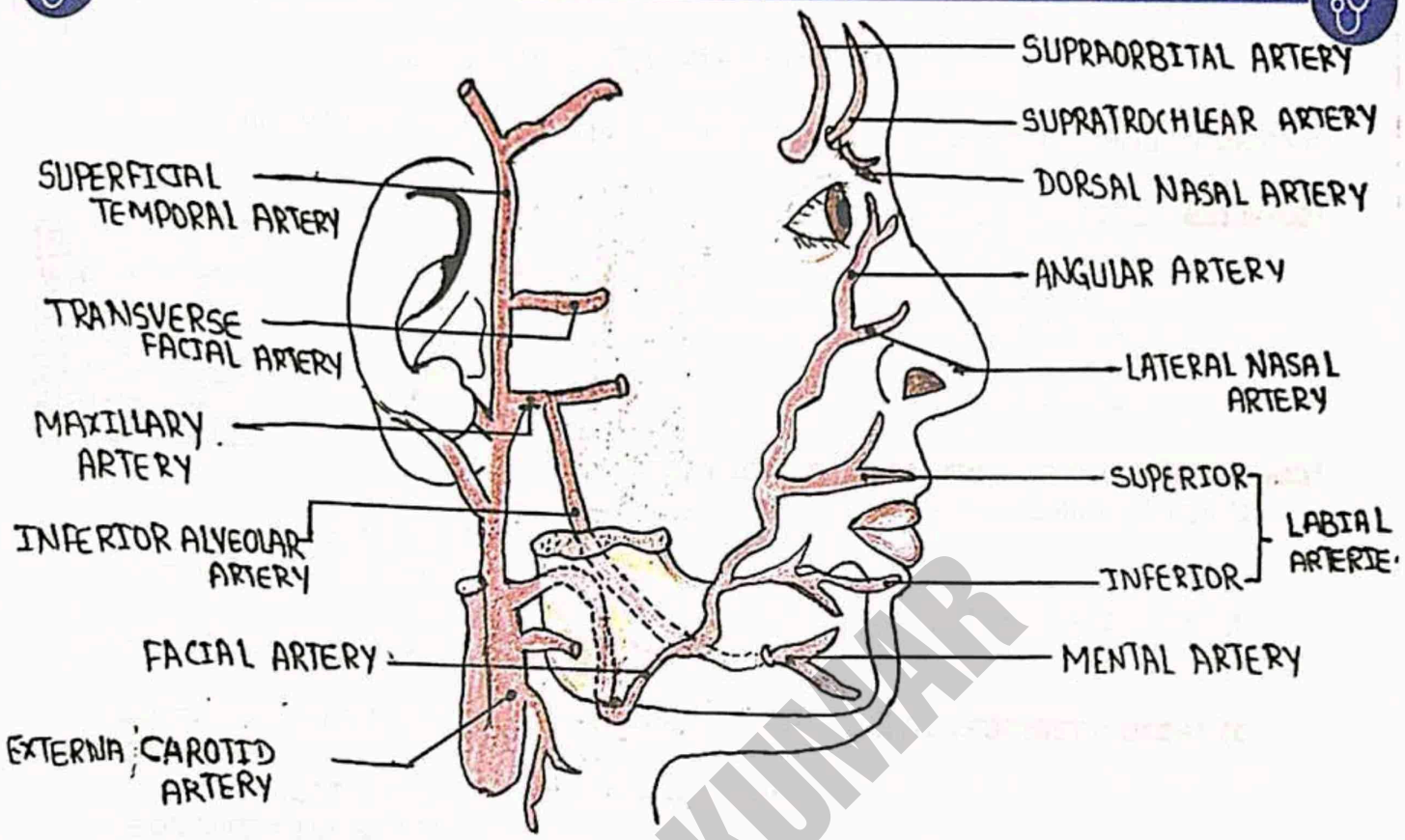
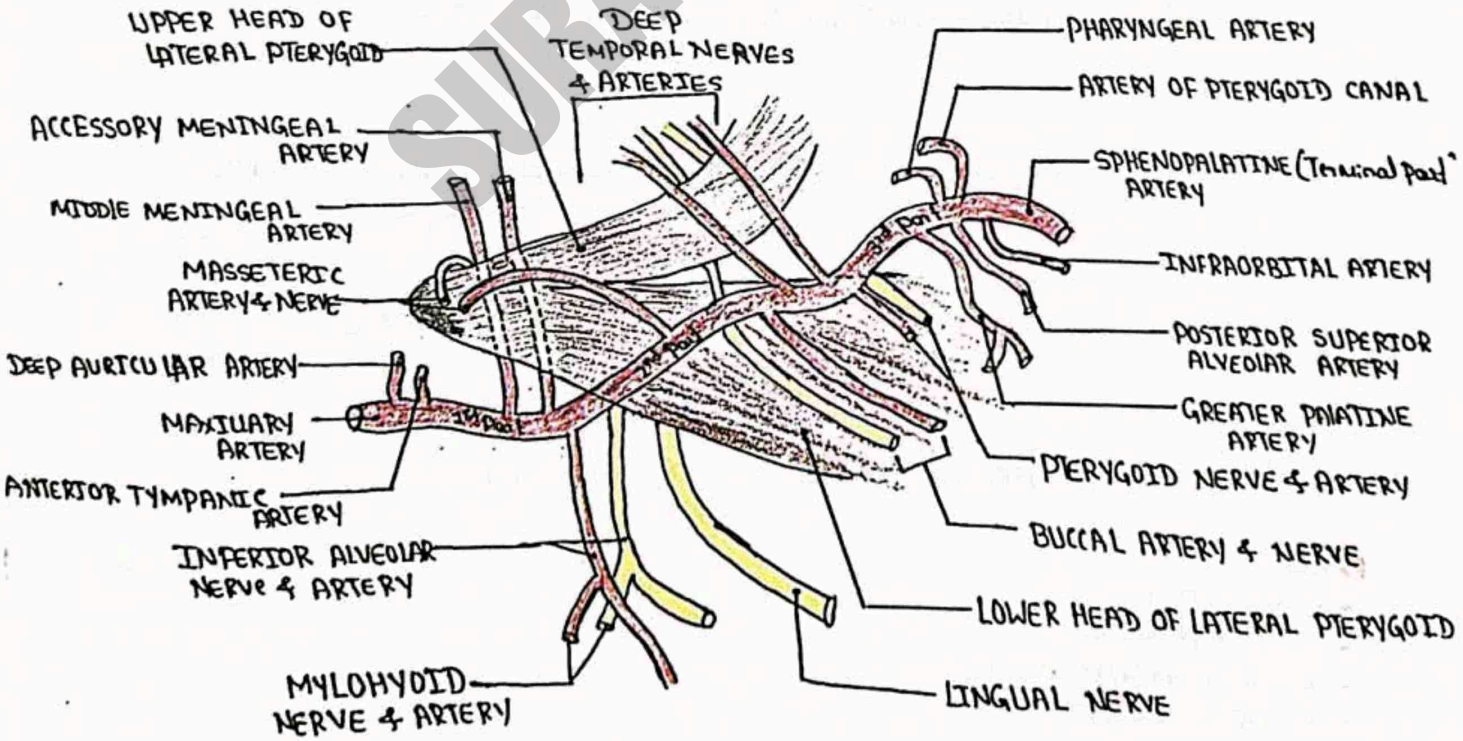


FACIAL ARTERY



MAXILLARY ARTERY

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MAXILLARY ARTERY

*LARGEST TERMINAL BRANCH OF ECA→GIVEN OFF BEHIND NECK OF MANDIBLE

*SUPPLIES - EXTERNAL & MIDDLE EAR
- DURA MATER
- MUSCLE OF TEMPORAL & INFRA TEMPORAL REGION
- NOSE & PARA-NASAL AIR SINUSES
- PALATE
- ROOF OF PHARYNX

*COURSE & RELATION - DIVIDED INTO 3x PARTS -

1) FIRST (MANDIBULAR) PART - RUN HORIZONTALLY FORWARDS
-BETWEEN NECK OF MANDIBLE & SPLENO MANDIBULAR LIGAMENT

2) SECOND (PTERYGOID) PART - RUN UPWARD & FORWARDS
-SUPERFICIAL TO LOWER HEAD OF LATERAL PTERYGOID

3) THIRD (PTERYGOPALATINE) PART - PASSES BETWEEN TWO HEAD OF LATERAL PTERYGOID
-THROUGH PTERYGOMAXILLARY FISSURE→TO
-→ENTER→PTERYGOPALATINE FOSSA

*BRANCHES -PART 1 1) DEEP AURICULAR ARTERY-→SUPPLY EXTERNAL ACOUSTIC MEATUS TYMPANIC MEMBRANE

2) ANTERIOR TYMPANIC→SUPPLY MIDDLE EAR MEDIAL SURFACE OF TYMPANIC MEMBRANE

3) MIDDLE MENINGEAL ARTERY→ENTER FORAMEN SPINOSUM
→REACH MIDDLE CRANIAL FOSSA
-→SUPPLY LARGE FRONTAL BRANCH & SMALL PARIETAL

4) ACCESSARY MENIGEAL ARTERY-→FORAMEN OVALE-→REACH MIDDLE CRANIAL FOSSA
-→SUPPLY INFRA TEMPORAL FOSSA

5) INFERIOR ALVEOLAR ARTERY -→MANDIBULAR FORAMEN
-→SUPPLY MANDIBLE & ROOT OF TOOTH

PART2
- 1) MASSETERIC
- 2) DEEP TEMPORAL
- 3) TO PTERYGOID MUSCLE
- 4) BUCCAL BRANCH

PART-3
1) POSTERIOR SUPERIOR ALVEOLAR ARTERY - ENTER - PTERYGOMAXILLARY FISSURE

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- SUPPLY - MOLAR AND PREMOLAR TEETH
- MAXILLARY AIR SINUS

2) INFRA-ORBITAL ARTERY - ENTER - PTERYGOMAXILLARY FISSURE ->INFERIOR - ORBITAL FISSURE --> ENTER ---> ORBITAL
-SUPPLY - ORBITAL BRANCHES
- MIDDLE SUPERIOR ALEVOLAR BRANCH
- ANTERIOR SUPERIOR ALVEOLAR BRANCH

3) GREATER PALATINE A.- ENTER - GREATER PALATINE FORAMEN ->GREATER PALATINE CANAL
-SUPPLY -> TERMINAL BRANCH ENTER NASAL CAVITY

4) PHARYNGEAL BRANCH - ENTER - PHARYNGEAL CANAL
- SUPPLY - NASOPHARYNX & AUDITORY TUBE

5) ANTERIOR ATERY OF PTERYGOID CANAL - PTERYGOID CANAL
- SUPPLY -> PHARYNX & TYMPANI CAVITY

6) SPHENOPALATINE ARTERY -> SPHENOPALATINE FORAMEN
- POSTEROLATERAL & POSTEROMEDIAL - NASAL BRANCH

MANDIBULAR NERVE

- LARGEST MIXED BRANCH OF TRIGEMINAL NERVE
- OTIC AND SUB-MANDIBULAR GANGLIA - ASSOCIATED WITH THIS NERVE

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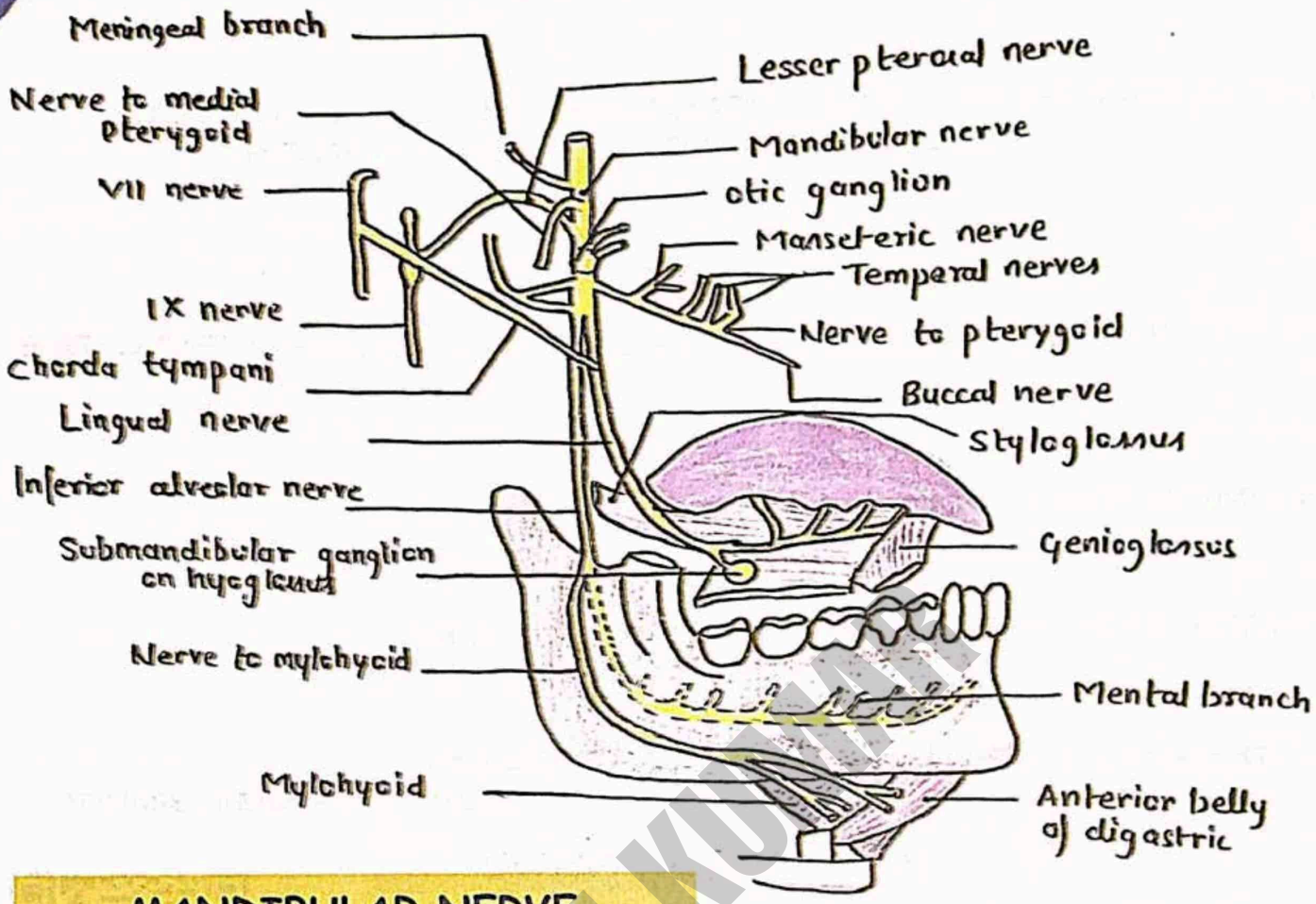
*COURSE-

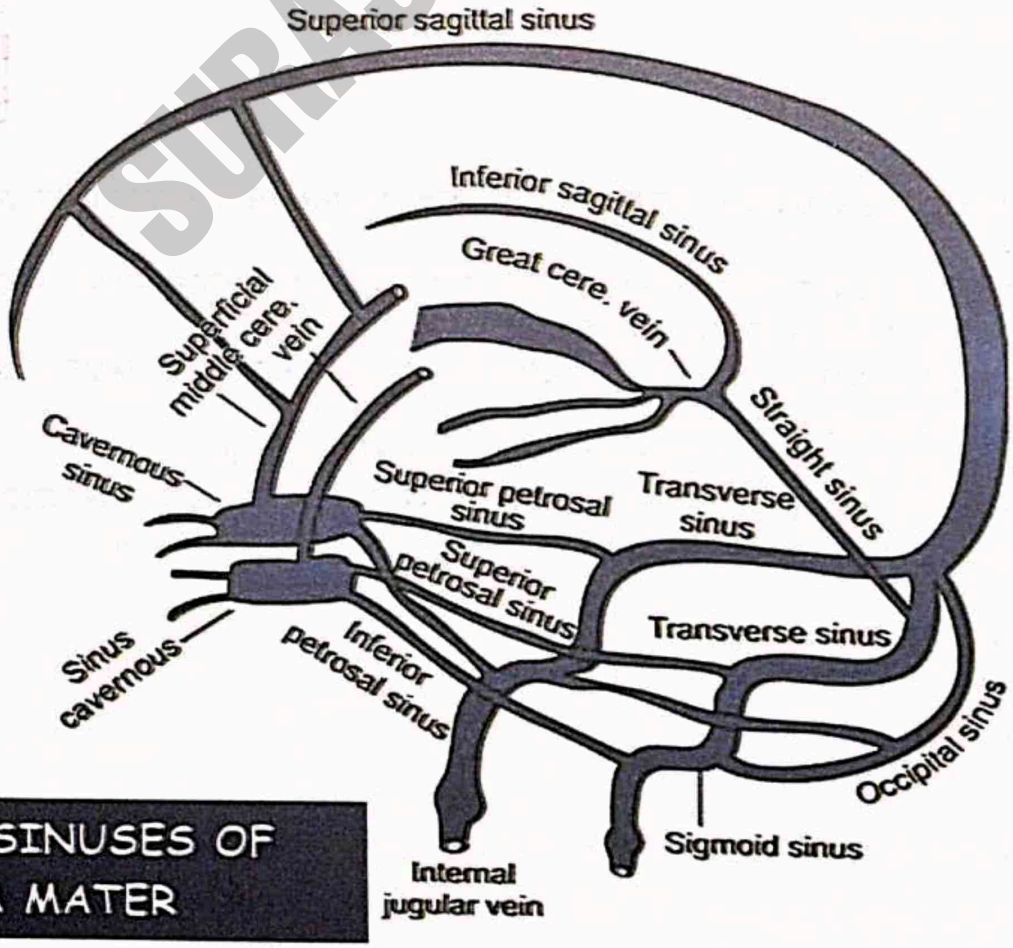
1. BEGIN IN MIDDLE CRANIAL FOSSA THROUGH, LARGE SENSORY ROOT & SMALL MOTOR ROOT
2. LARGE SENSARY ROOT ARISE FROM LATERAL PART OF TRIGEMINAL GANGLION LEAVE CRANIAL CAVITY FORAMEN OVALE
3. SMALL MOTOR ROOT LIE DEEP TO TRIGEMINAL GANGLION - LEAVE - CRANIAL CAVITY ----> FORAMEN OVALE
4. BOTH ROOT JOIN BELOW FORAMEN FORM - MAIN TRUNK
5. MAIN TRUNK IN INFRATEMPORAL FOSSA DEEP TO LATERAL PTERYGOID
6. MAIN TRUNK DIVIDED -> ANTERIOR TRUNK, POSTERIOR TRUNK

*BRANCHES :-

1.MAIN TRUNK - MENINGEAL BRANCH , NERVE TO MEDIAL PTERYGOID
2.ANTERIOR TRUNK - SENSORY BRANCH - BUCCAL NERVE
-MOTOR BRANCH -MASSETERIC, DEEP TEMPORAL NERVE
-NERVE TO LATERAL PTERYGOID
3.POSTERIOR TRUNK - AURICULO TEMPORAL NERVE ,LINGUAL, INFERIOR ALVEOLAR NERVE



MANDIBULAR NERVE



VENOUS SINUSES OF DURA MATER

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VENOUS SINUSES OF DURA MATER

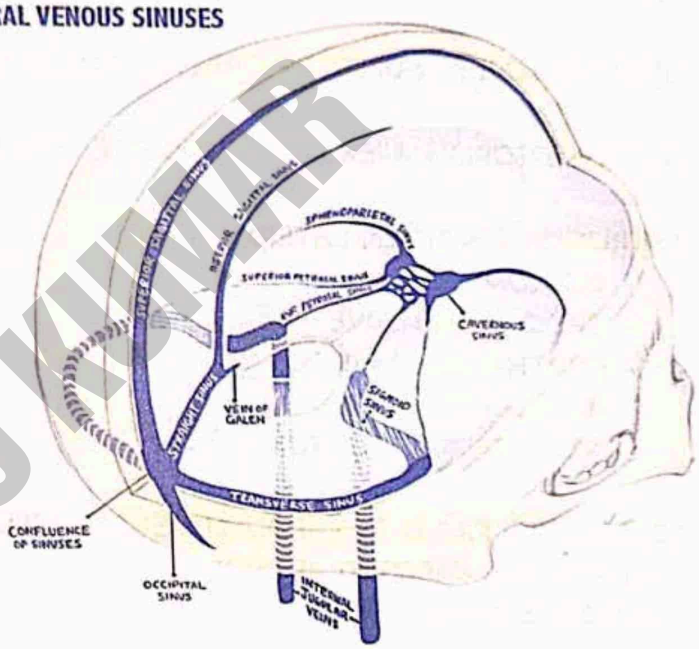
1) VENOUS SPACE IN - WALL WHICH ARE FORMED BY -DURA MATER
l, NO MUSCLE IN THEIR WALL
-THEY HAVE NO VALUE

2) VENOUS SINUSES RECEIVE → VENOUS BLOOD FROM BRAIN , MENINGS BONE OF SKULL
3) CRANIAL VENOUS SINUSES COMMUNICATED → WITH OUTSIDE SKULL THROUGH EMISSARY VEINS
4) COMMUNICATIONS HELP TO KEEP THE PRESSURE OF BLOOD CONSTANT
5) 23 VENOUS SINUSES --→ 8 PAIRED & 7 UNPAIRED

*PAIRED VENOUS SINUSES :-

- CAVERNOUS SINUS
- SUPERIOR PETROSAL SINUS
- INFERIOR PETROSAL SINUS
- TRANSVERSE SINUS
- SIGMOID SINUS
- SPHENO-PARIETAL SINUS
- PETRO-SQUAMOUS SINUS
- MIDDLE MENINGEAL SINUS

DURAL VENOUS SINUSES



*UNPAIRED VENOUS SINUSES:

- SUPERIOR SAGITAL SINUS
- INFERIOR SAGITAL SINUS
- STRAIGHT SINUS
- OCCIPITAL SINUS
- ANTERIOR INTER CAVERNOUS SINUS
- POSTERIOR INTER CAVERNOUS SINUS
- PLEXUS OF VEIN

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CAVERNOUS SINUS

- LARGE VENOUS SPACE- SITUATED IN MIDDLE CRANIAL FOSSA EITHER SIDE, BODY OF SPHENOID

* FLOOR & MEDIAL WALL → ENDOSTEAL DURA MATTER
* ROOF & LATERAL WALL → MENINGEAL DURA MATTER

- EXTEND - ANTERIORILY - SUPERIOR ORBITAL FISSURE
- POSTERIORLY - APEX OF PETROUS TEMPORAL BONE

*RELATION -

1) SUPERIORLY- OPTIC TRACT
- OPTIC CHIASMA
- OLFACTORY TRACT
- INTERNAL CAROTID ARTERY

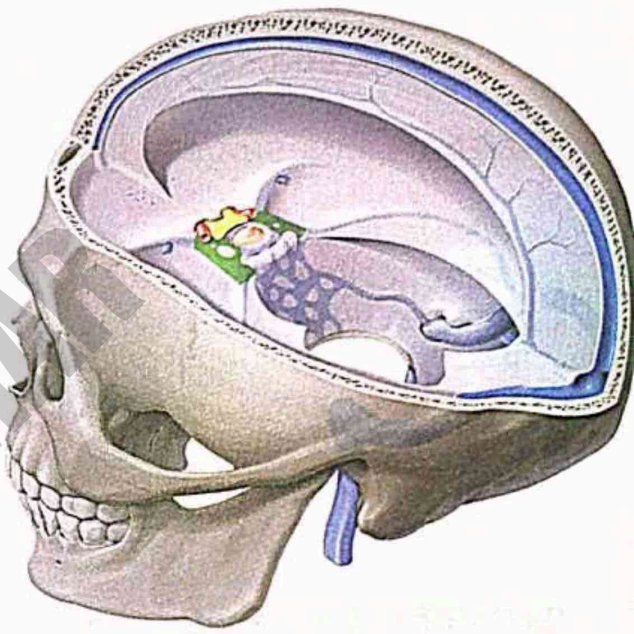
2) INFERIORLY - FORAMEN LACERUM
- GREATER WING OF SPHENOID BONE

3) MEDIALLY - HYPOPHYSIS CEREBRI
- SPHENOIDAL AIR SINUS

4) LATERALLY- TEMPORAL LOBE

5) ANTERIORLY- SUPERIOR ORBITAL FISSURE

6) POSTERIORLY- APEX OF PETROUS TEMPORAL



*STRUCTURE WITHIN LATERAL WALL:
- OCULOMOTOR NERVE
- TROCHLEAR NERVE
- OPHTHALMIC NERVE
- MAXILLARY NERVE
-TRIGEMINAL GANGLION

*STRUCTURE PASSING THROUGH MEDIAL ASPECT OF SINUS :
- INTERNAL CAROTID ARTERY
- ABDUCENT NERVER

*TRIBUTARIES OR INCOMING CHANNEL :
1) FROM ORBIT - SUPERIOR OPHTHALMIC VEIN
- INFERIOR OPHTHALMIC VEIN
- CENTRAL VEIN OF RETINA

2) FROM BRAIN - SUPERFICIAL MIDDLE CEREBRAL VEIN
- INFERIOR CEREBRAL VEIN

3) FROM MENINGES - SPHENO-PARIETAL SINUS
- MIDDLE MENINGEAL WEIN

*DRAINING CHANNEL -
-TRANSVERSE SINUS - THROUGH--→ SUPERIOR PETROSAL SINUS
- INTERNAL JUGULAR VEIN → INTERNAL CAROTID ARTERY
- PTERYGOID PLEXUS - EMISSARY VEIN
- FACIAL VEIN - SUPERIOR OPHTHALMIC VELN

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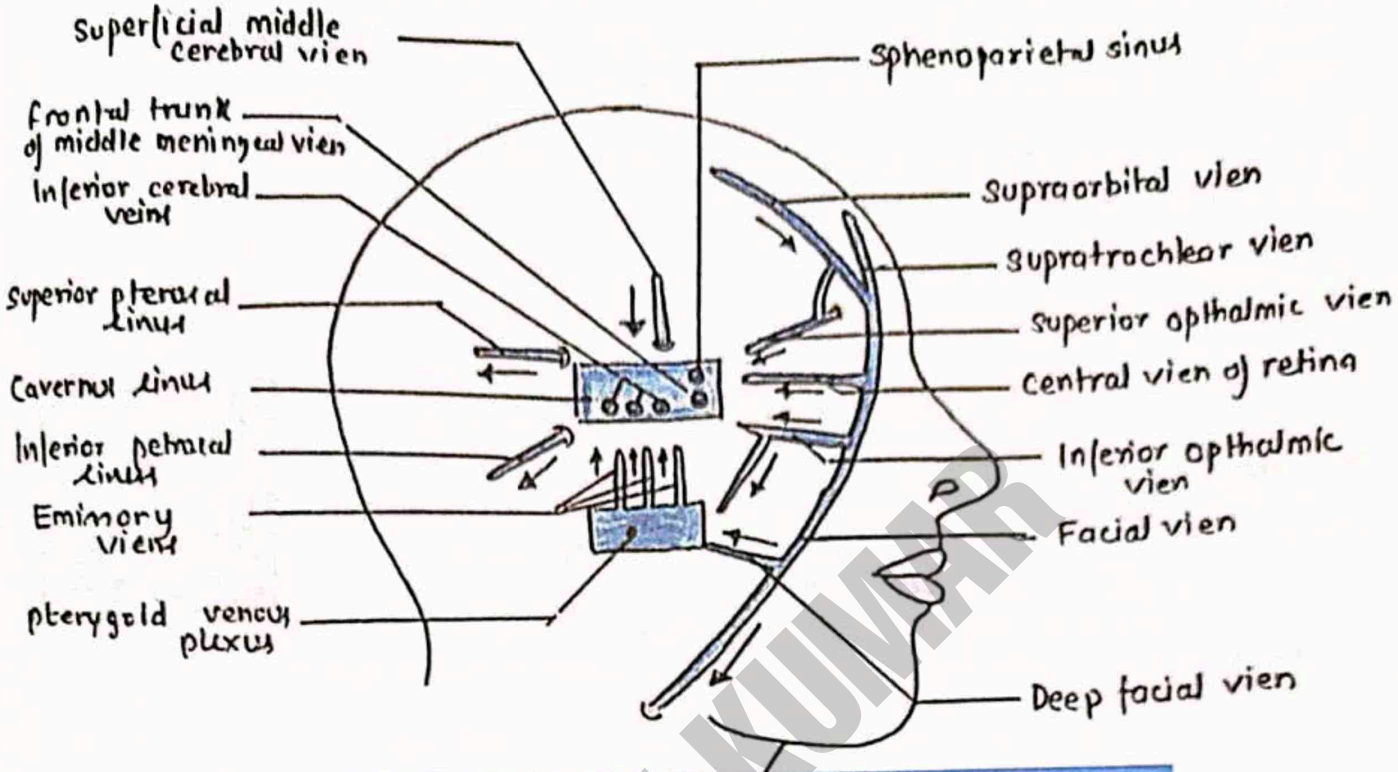
-RIGHT AND CAVERNOUS SINUSES → ANTERIOR & POSTERIOR -→ INTERCAVERNOUS SINUSES



VENOUS SINUSES OF DURA MATER

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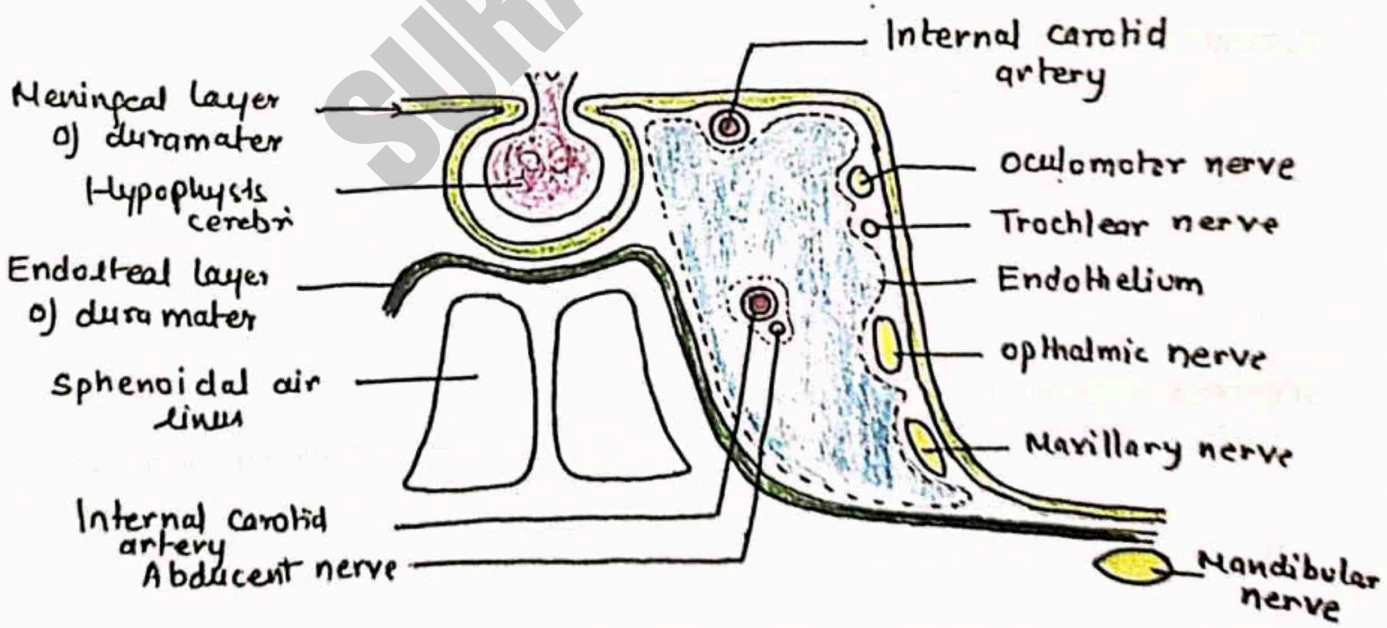

CAVERNOUS SINUS

CLINICAL ANATOMY

*THROMBOSIS OF THE CAVIRNUS SINUS : MAY BE CAUSED BY SEPSIS IN THE DANGEROUS AREA OF THE FACE, IN NASAL CAVITIES, & IN PARANASAL AIR SINUSES.THIS GIVES RISE TO THE FOLLOWING SYMPTOMS :

a.NERVOUS SYMPTOMS:SEVERE PAIN IN THE EYE & FOREHEAD IN THE AREA OF DISTRIBUTION OF OPHTHALMIC NERVE.

b.VENOUS SYMPTOMS : MARKED OEDEMA OF EYELIDS.

MUSCLE OF MASTICATION

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1) MASSETER - QUADRILATERAL

ORIGIN - SUPERFICIAL - ZYGOMATIC PROCESS OF MAXILLA
- ANTERIOR 1/3 ZYGOMATIC ARCH
- DEEP - ZYGOMATIC ARCH
- MIDDLE - 1/3 OF ZYGOMATIC ARCH

INSERTION - SUPERFICIAL - LOWER LATERAL SURFACE OF MANDIBLE
- DEEP - RAMUS OF MANDIBLE
- MIDDLE - CENTRAL PART OF RAMUS OF MANDIBLE

NERVE - MASSETER NERVE OF MANDIBULAR NERVE

ACTION - ELEVATION & PROTRUSION

2) TEMPORALIS - FAN-SHAPED

ORIGIN - TEMPORAL FOSSA EXCLUDING ZYGOMATIC BONE

INSERTION - DEEP SURFACE OF CORONOID PROCESS
- ANTERIOR BORDER OF RAMUL OF MANDIBLE

NERVE - DEEP TEMPORAL BRANCH OF M.N

ACTION -ELEVATION & PROTRUSION

3) LATERAL PTERYGOID:

ORIGIN- UPPER HEAD - INFRA-TEMPORAL SURFACE & INFRA-TEMPORAL CREST OF GREATER WING OF SPHENOID BONE
LOWER HEAD - LATERAL PTERYGOID PLATE

INSERTION- PTERYGOID FOVEA ON ANTERIOR OF NECK OF MANDIBLE

NERVE - ANTERIOR DIVISION OF MANDIBULAR NERVE

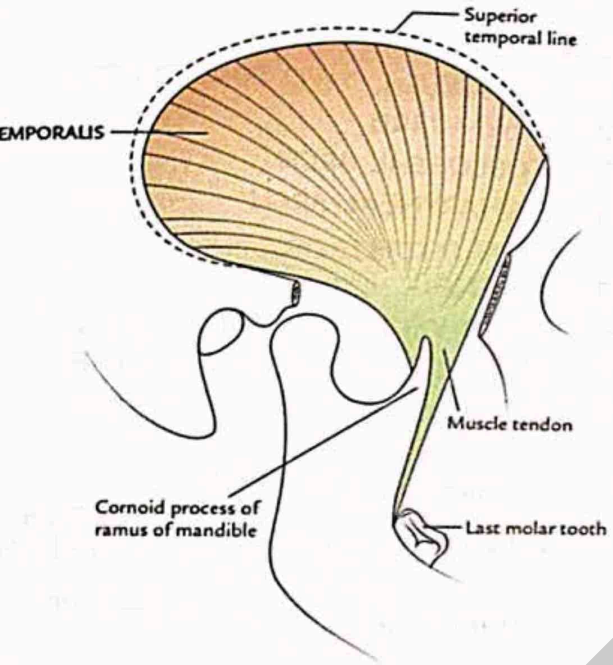
ACTION- DEPRESS & PROTRUSION

4) MEDIAL PTERYGOID - QUADRILATERAL

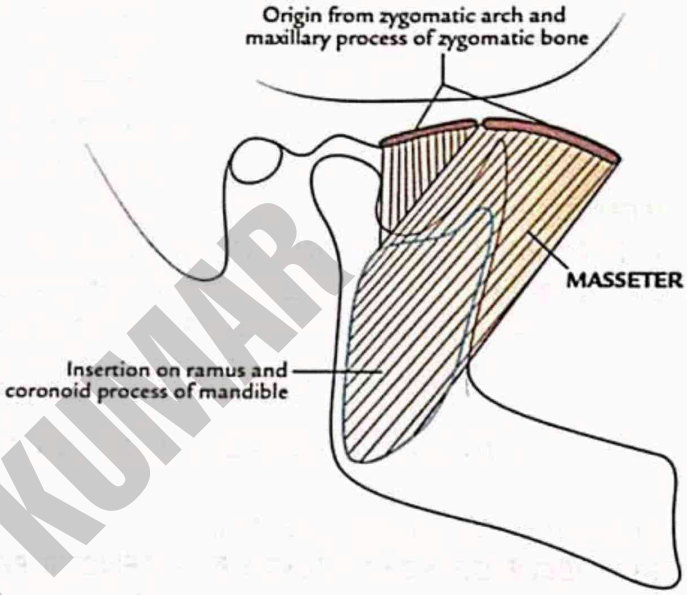
ORIGIN.- SUPERFICIAL HEAD - TUBEROSITY OF MAXILLA
DEEP HEAD - MEDIAL SURFACE OF - LATERAL PTERYGOID PLATE
INSERTION - MEDIAL SURFACE OF ANGLE OF ADJOINING RAMUS OF MANDIBLE

NERVE- NERVE TO MEDIAL PTERYGOID → MANDIBULAR NERVE

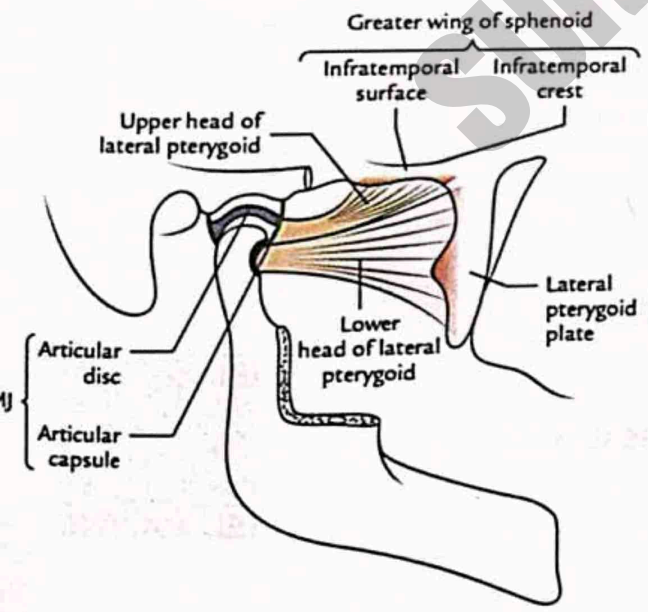
ACTION - ELEVATE & PROTRUDE



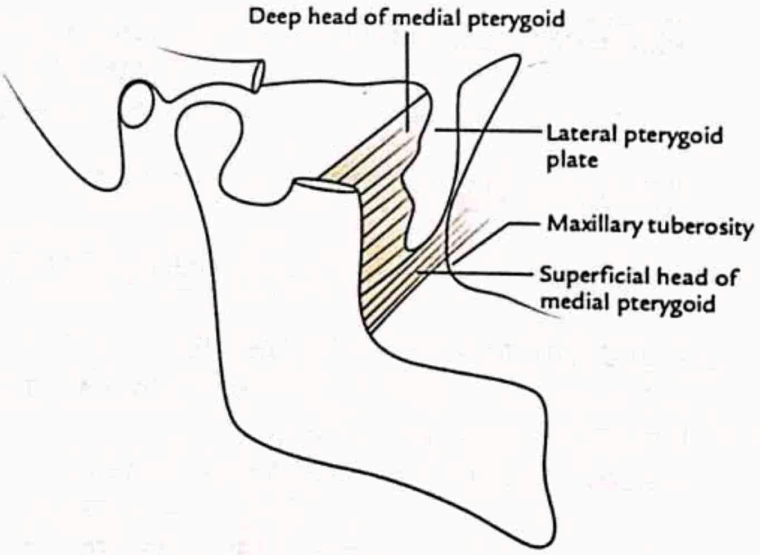
Origin and insertion of the temporalis muscle.



Origin and insertion of the masseter muscle.



Origin and insertion of the lateral pterygoid muscle.



Origin and insertion of the medial pterygoid

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SOFT PALATE

-MOVABLE, MUSCULAR FOLD, SUSPENDED FROM - POSTERIOR BORDER OF HARD PALATE

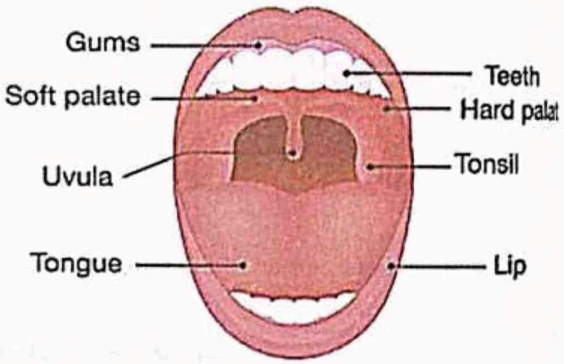
-2 SURFACE →ANTERIOR - CONCAVE
- POSTERIOR- CONVEX FLOOR OF NASAL CAVITY

-2 BORDER → SUPERIOR BORDER ATTACHED, POSTERIOR BORDER OF HARD PALATE
-INFERIOR BORDER - FREE
- CONICAL PROJECTION - "UVULA"
- FROM EACH SIDE BASE OF OVULA, TWO CURVED FOLD EXTEND - LATERALLY DOWNWARD
- ANTERIOR FOLD - PALATOGLOSSAL ARCH
- POSTERIOR FOLD-PALATOPHARYNEAL ARCH

STRUCTURE:-

- SOFT PALATE IS A FOLD OF MUCOUS MEMBRANE CONTAINING:-

1) PALATINE APONEUROSIS - FLATTENED TENDON OF TENSOR VELI PALATINI
- NEAR MEDIAN PLANE - APONEUROSIS SPLITS ENCLOSE - MUSCULUS UVALAE.
- SUPERIOR SURFACE CONTAIN --→ LEVATOR VELI PALATINI
-PALATOGLOSSUS
- INFERIOR SURFACE - PALATOGLOSSUS



2) NUMEROUS MUCOUS GLAND

3) MUSCLE OF SOFT PALATE 5 - TENSOR PALATI
- LEVATOR PALATI
- MUSCULAR UVULAE
- PALATOGLOSSUS
- PALATO PHARYNGEUS

*BLOOD SUPPLY: ARTERIES - GREATER PALATINE BRANCH MARTELY
- ASCENDING PALATINE FACIAL ARTERY
- PALATINE BRANCH OF ASCENDING PHARYNGEAL

*VEIN - TONSILLAR PLEXUSES OF VEIN

PALATINE TONSIL

- SECONDARY LYMPHOID TISSUE
- OCCUPIES- TONSILLAR FOSSA B/W PALATOGLOSSAL AND PALATOPHARYNGEAL ARCHES
- ALMOND-SHAPED - 2 SURFACE - MEDIAL
- LATERAL
- 2 BARDER - ANTERIOR
-POSTERIOR
- 2 POLES - UPPER AND LOWER

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1) SURFACE -

- MEDIAL SURFACE - COVERED BY STRATIFIED SQUAMOUS EPITHELIUM
  - SURFACE HAS - 12-15 CRYPTS
  - LARGEST OF THESE → INTRATONSILLAR CLEFT

- LATERAL SURFACE - COVERED BY SHEET OF FASCIA WHICH FORM HEMICAPSULE OF TONSILE
  - CAPSULE IS AN EXTENSION OF → PHARYNGOBASILAR FASCIA
  - TONSILLAR ARTERY - ENTER ---→ TONSIL
  - PARATONSILLAR VEIN → DESCEND → FROM PALATE IN LOOSE AREALOR TISSUE ON LATERAL SURFACE OF CAPSULE

2) BORDER → ANTERIOR BORDER- PALATOGLOSSAL ARCH
- POSTERIOR BORDER - PALATOPHARYNGEAL ARCH

3) POLE - UPPER POLE - SOFT PALATE
LOWER POLE - TONGUE

*BED OF TONSIL IS FORMED FROM WITHIN OUTWARDS BY :

- PHARYNGOBASILAR FASCIA
- SUPERIOR CONSTRICTOR AND PALATO PHARYNGEUS MUSCLE
- BUCCOPHARYNGEAL FASCIA
- GLOSSOPHARYNGEAL NERVES
- LOWER PART OF STYLOGLOSSUS

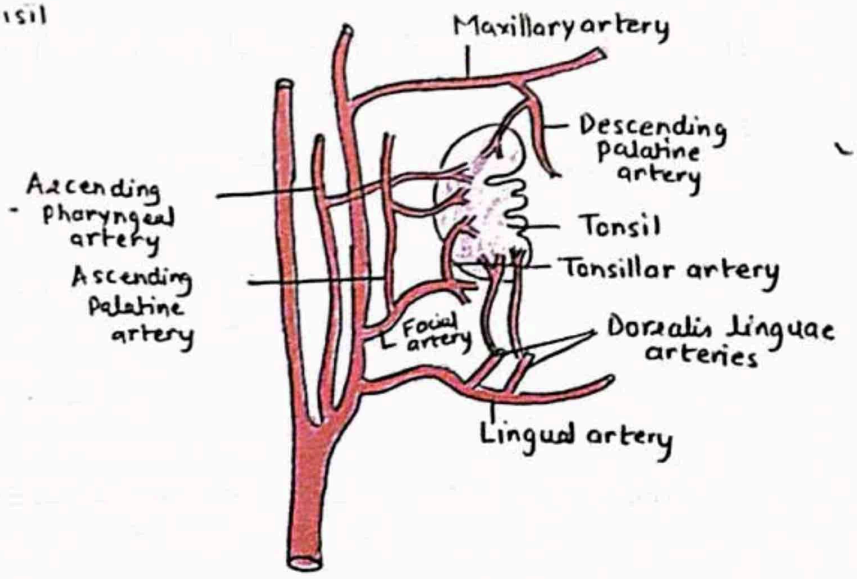
*BLOOD SUPPLY -

- MAIN-TONSILLAR BRANCH OF FACIAL ARTERY(FA)
- ASCENDING PALATINE - FA
- DORSAL LINGUAL BRACH - FA
- ASCENDING PHARYNGEAL BRANCH - ECA
- GREATER PALATINE - MAXILLARY A

*VEIN - 2-3 VEIN LEAVE LOWER PART OF TONSIL JOIN PALATINE,FACIAL VEIN

*NERVE

- GLOSSOPHARYNGEAL NERVE
- LESSER PALATINE NERWE



∴ Arteries supply of tonsil

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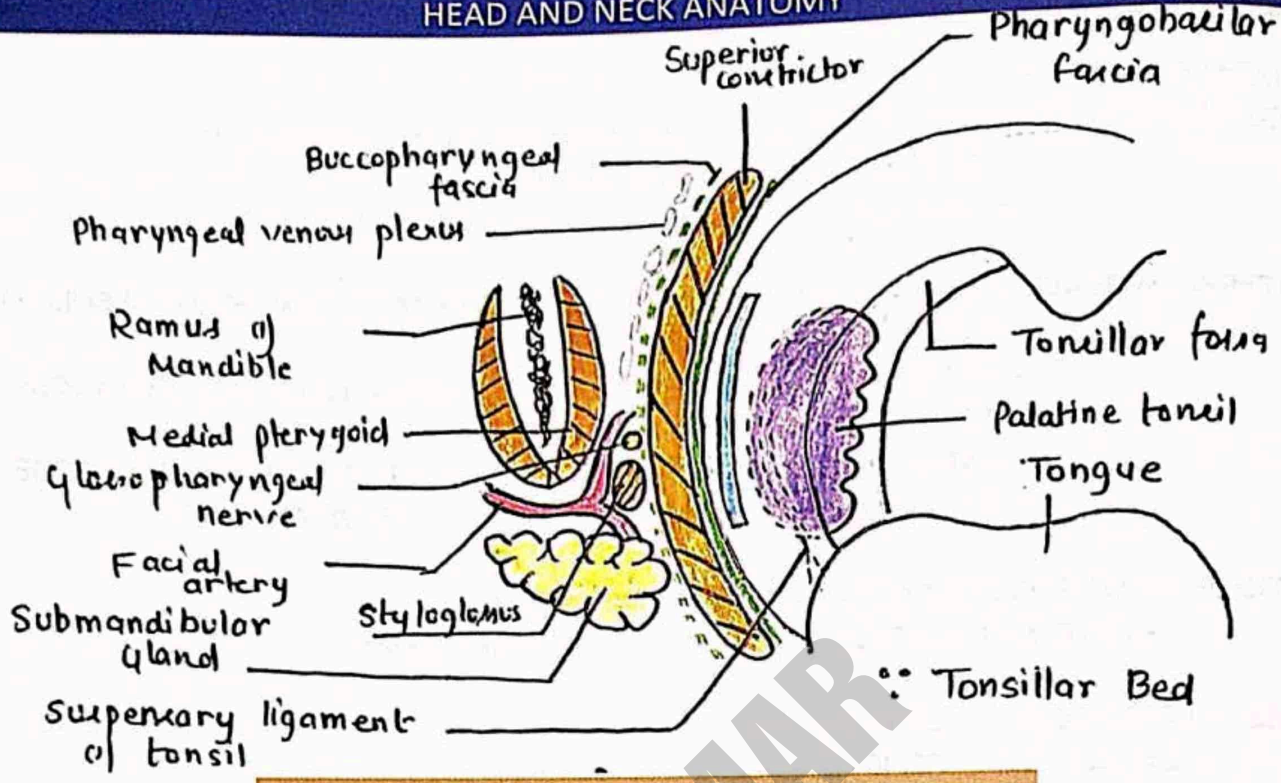

PALATINE TONSIL

TONGUE ANATOMY

*MUSCULAR ORGAN - SITUATED - FLOOR OF MOUTH
*FUNCTION- TASTE, SPEECH, CHEWING AND DEGLUTITION
*EXTERNAL FEATURES :- ROOT
- TIP
- BODY

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

SURAJ NIHAR

1) ROOT - ATTACHED → STYLOID PROCESS & SOFT PALATE - ABOVE
- HYOID BONE - BELOW

2) TIP - ANTERIOR FREE END→ BEHIND UPPER INCISOR TEETH

3) BODY -
DORSUM - CONVEX IN ALL DIRECTION,
- ORAL PART - FLOOR OF MOUTH
- FREE MARGIN -IN CONTACT WITH GUMS AND TEETH
- SUPERIOR SURFACE - COVERED WITH PAPILLAE
- INFERIOR SURFACE - COVERED WITH SMOOTH MEMBRANE
- MEDIAN FOLD- FRENULUM
- LATERAL FOLD - PLICA FRENULUM

- PHARYNGEAL PART - LIE BEHIND PALABOGLOSSAL ARCHES
- BASE OF TONGUE

- MUCOUS MEMBRANE - NO PAPILLAE BUT LYMPHOID FOLLICLE

- POSTERIOR MOST PART - CONNECTED TO EPIGLOTTIS
  - 3 FOLDS OF MUCOSA MEMBRANE
  - 2 LATERAL GLOSSOEPIGLOTTIC FOLD
  - MEDIAN GLOSSOEPIGLOTTIC FOLD

MUSCLE OF TONGUE

*MIDDLE FIBROUS SEPTUM → DIVIDES TONGUE INTO RIGHT & LEFT HALVES
*EACH HALF CONTAIN - 4 INTRINSIC MUSCLES
- 4 EXTRINSIC MUSCLES

*INTERINSIC MUSCLE - OCCUPY UPPER PART OF TONGUE
- ATTACHED- SUBMUCOUS FIBROUS LAYER

1) SUPERIOR LONGITUDINAL - ARISES FROM FIBROUS TISSUE DEEP TO MUCOUS MEMBRANE
- ON DORSUM OF TONGUE
- ACTION - SHORT TONGUE
- MAKE DORSUM CANCAVE

2) INFERIOR LONGITUDINAL - ARISES FROM FIBROUS TISSUE → BENEATH MUCOUS MEMBRANE
- ON TIP OF TONGUE
- ACTION - SHORT TONGUE
- MAKE DORSUM CONVEX

3) TRANSVERSE
- LIE ON EITHER SIDE OF MID LINE
- ORIGIN- FIBROUS MEDIAL SEPTUM TO SUBMUCOUS FIBROUS TISSUE
- MAKE TONGUE NARROW

4) VERTICAL - FOUND AT BORDER OF ANTERIOR PART OF TONGUE
- ACTION-FLATTED

*EXTRINSIC MUSCLE : -

ACTION
1) GENIOGLOSSUS → PROTRUDES
2) HYOGLOSSUS → DEPRESSES
3) STYLO GLOSSUS → RETRACTS
4) PALATOGLOSSUS → ELEVATES

PALATOGLOSSUS - ORIGIN - ORAL SURFACE OF PALATINE APONEUROSIS
-INSERTION- SIDE OF TONGUE
-NERVE - PHARYNGEAL PLEXUS

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HYOGLOSSUS - ORIGIN - LATERAL PART OF HYOID BONE
- INSERTION- SIDE OF TONGUE
- NERVE- REST ALL BE MUSCLE HYPOGLOSSAL NERVE

STYLOGLOSSUS - ORIGIN - TIP OF STYLOID PROCESS
- INSERTION- SIDE OF TONGUE

GENIOGLOSSUS - ORIGIN - UPPER GENITAL TUBERCLE OF MANDIBLE
- INSERTION - UPPER FIBRES INTO TIP OF TONGUE

*BLOOD & NERVE SUPPLY OF TONGUE

*ARTERY → DERIVED FROM → LINGUAL ARTERY BRANCH - ECA
-TIP ALSO SUPPLY BY - TONSILLAR ARTERY - FA

*VENOUS DRAINAGE - DEEP LINGUAL VEIN - LARGEST AND PRINCIPAL VEIN
- INFERIOR SURFACE OF TONGUE
- END IN INTERNAL JUGULAR VEIN THROUGH LINGUAL VEIN

*LYMPHATIC DRAINAGE
- SUBMENTAL NODE (TIP) -----> FINALLY DRAIN OMOHYOID NODE
- SUB-MANDI BULAR NODE (RIGHT AND LEFT HALVES) -----> FINALLY DRAIN OMOHYOID NODE
- DEEP-CERVICAL NODE (POSTERIO MOST) ------> FINALLY DRAIN OMOHYOID NODE

*NERVE SUPPLY :-
- MOTOR - ALL MUSCLE- HYPOGLOSSAL NERVE
EXCEPT - PALATOGLOSSUS - PHARYNGEAL PLEXUS
- SENSORY -ANTERIOR 2/3 - LINGUAL NERVE
- CHORDA - TYMPANI NERVE
POSTERIOR 1/3-GLOSSOPHARYNGEAL NERVE

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CLINICAL ANATOMY

TONGUE

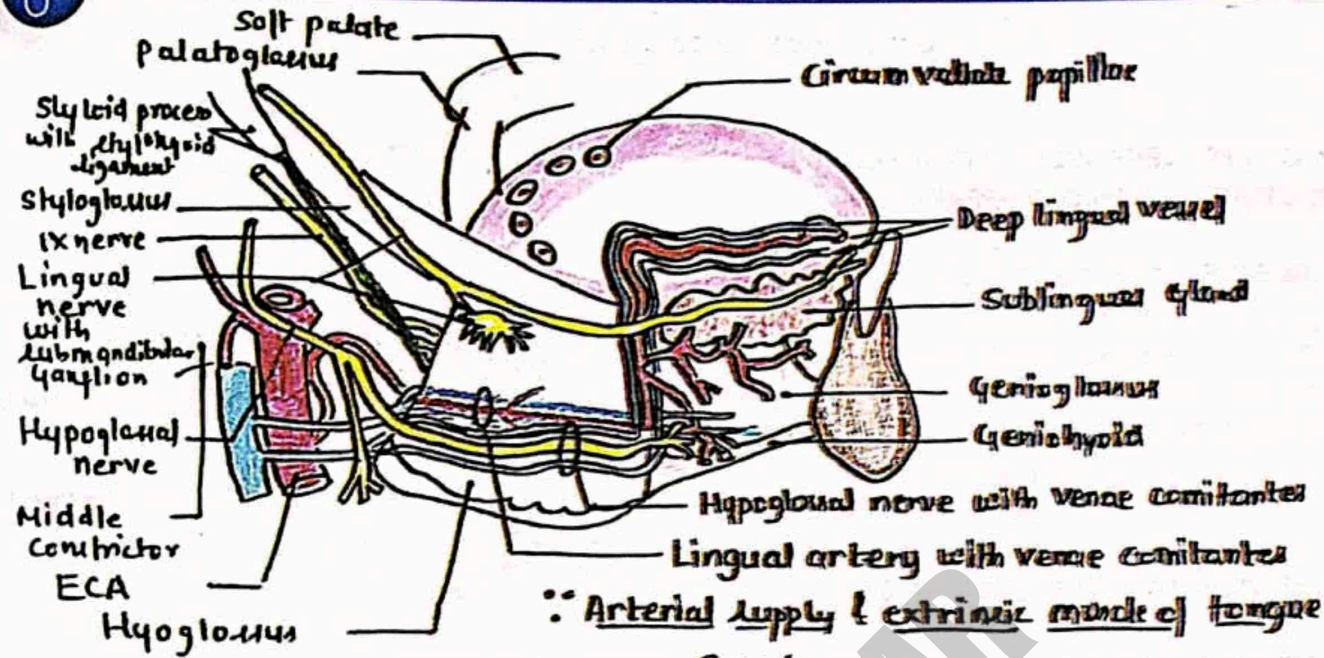
*GLOSSITIS IS USUALLY A PART OF GENERALIZED ULCERATION OF THE MOUTH CAVITY OR STOMATITIS.

*CARCINOMA OF THE TONGUE IS QUITE COMMON.THE AFFECTED SIDE OF THE TONGUE IS REMOVED SURGICALLY.

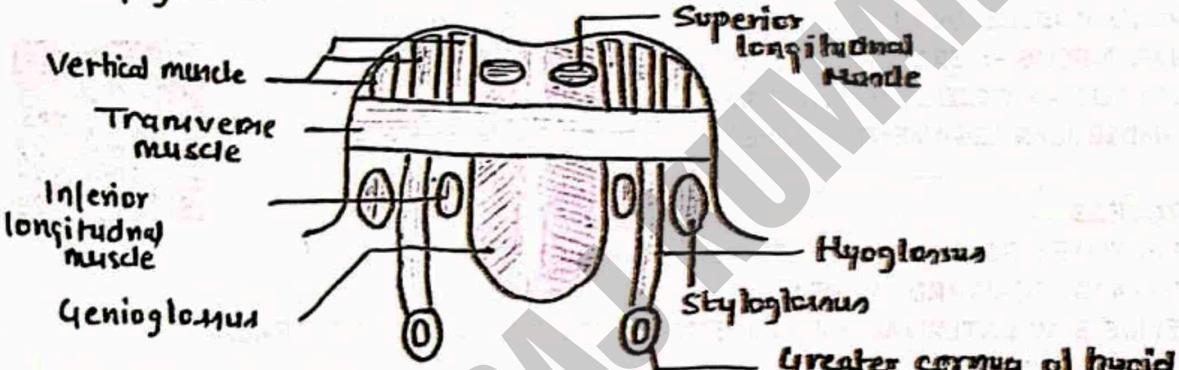
TONSILS

*THE TONSILS ARE LARGE IN CHILDREN.THEY RETROGRESS AFTER PUBERTY.

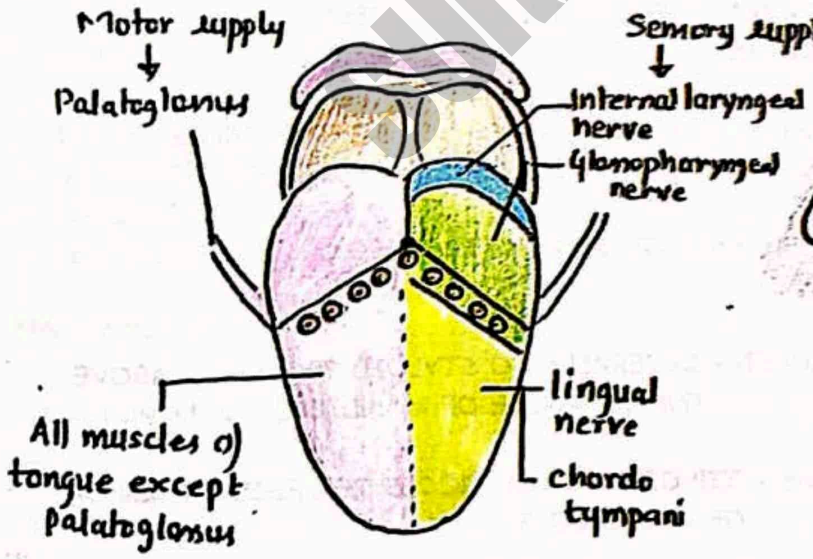
*TONSILLITIS MAY CAUSE REFERRED PAIN IN THE EAR AS GLOSSOPHARYNGEAL NERVE SUPPLIES BOTH THESE AREAS.



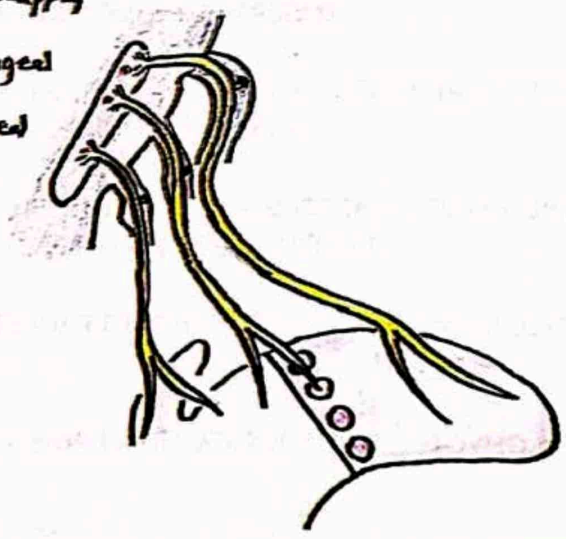
∵ Arterial supply & extrinsic muscle of tongue



∵ Intrinsic muscle



∵ Nerve supply of tongue



∵ Nerve carrying taste sensations

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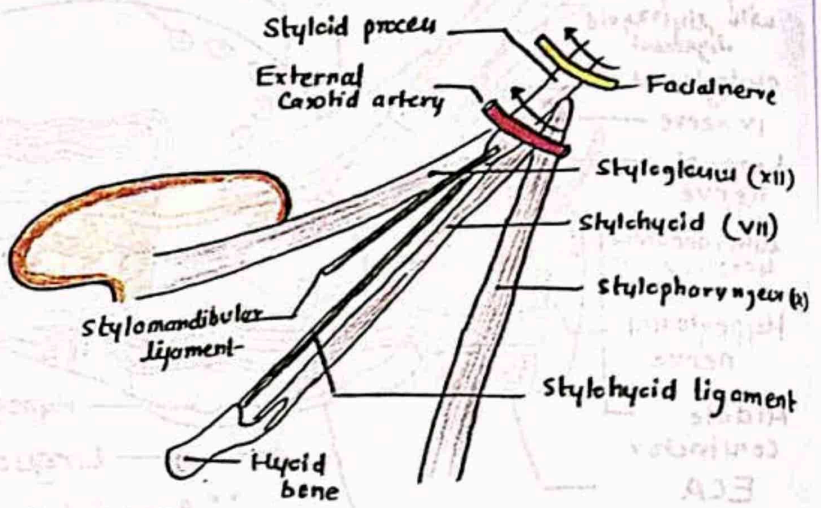
STYLOID APPARATUS

STYLOID PROCESS WITH ATTACHED STRUCTURE → STYLOID APPARATUS

- 3 MUSCLE - STYLOGLOSSUS
  - STYLOHYOID
  - STYLOPHARYNGEUS
- 2 LIGAMENTS - STYLOHYOID
  - STYLO MANDIBULAR



***APPARATUS IS OF DIVERSE ORIGIN :-**

- STYLOHYOID MUSCLE AND LIGAMENT -> 2RD BRANCHIAL ARCH
- STYLOPHARYNGEUS -> 3RD BRANCHIAL ARCH
- STYLOGLOSSUS -> OCCIPITAL MYOTOMES
- STYLOMANDIBULAR LIGAMENT -> DEEP FASCIA OF NECK

***STYLOID PROCESS :-**

- LONG POINTED BONY PROCESS OF TEMPORAL BONE
- DOWNWARD, FORWARD, MEDIALLY
- DESCENDS B/W EXTERNAL AND INTERNAL CAROTID ARTERY TO REACH SIDE OF PHARYNX

STYLOGLOSSUS - ARISES - ANTERIOR SURFACE OF STYLOID PROCESS
-INSERT SIDE OF TONGUE

STYLOPHARYNGEUS - ARISES - MEDIAL SURFACE OF BASE OF STYLOID PROCESS.
-INSERT - POSTERIOR BORDER OF THYROID CARTILAGE

STYLOHYOID - ARISES - POSTERIOR SURFACE OF STYLOID PROCESS
-INSERT - HYOID BONE

STYLOMANDIBULAR LIGAMENT ATTACHED - LATERALLY TO STYLOID PROCESS - ABOVE
-INSERT ANGLE OF MANDIBLE - BELOW

STYLOHYOID LIGAMENT EXTEND FROM - TIP OF STYLOID PROCES TO LESSER CORNUA OF HYOID BONE

***FEATURES -**

*EXTERNAL CAROTID ARTERY→ TIP OF STYLOID PROCESS
*FACIAL NERVE CROSSES -→ BASE OF STYLOID PROCES

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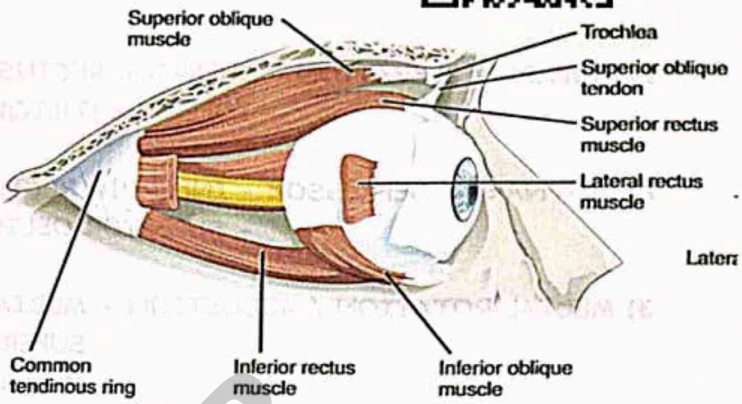
EXTRAOCULAR MUSCLE

*VOLUNTARY MUSCLE :-

4X RECT - SUPERIOR RECTUS
- INFERIOR RECTUS
- MEDIAL RECTUS
- LATERAL RECTUS

2X OBLIQUE - SUPERIOR OBLIQUE
- INFERIOR OBLIQUE

LEVATAR PALPEBRAE SUPERIORIS



Lateral view of the right eye

*INVOLUNTARY MUSCLE -

1) SUPERIOR TARSAL MUSCLE -DEEPER PORTION OF LPS
- INSERTED-UPPER MARGIN OF SUPERIOR TARSUS
- ELEVATE UPPER EYELID

2) INFERIOR TARSAL MUSCLE - EXTEND- FASCIAL SHEATH OF INFERIOR RECTUS
- DEPRESSES THE LOWER EYELID

3) ORBITALIS BRIDGE - THE INFERIOR ORBITAL FISSURE

VOLUNTARY MUSCLE -

*ORIGIN :-

-4 RECTI - COMMON ANNULAR TENDON - RING ATTACHED - SUPERIOR ORBITAL FISSURE
- LATERAL RECTUS ADDITIONAL ORIGIN -→ GREATER WING OF SPHENOID BONE
- SUPERIOR OBLIQUE ARISES --→ LESSER WING OF SPHENOID BONE
- INFERIOR OBLIQUE ARISES -→ ORBITAL SURFACE OF MAXILLA
- LEVATOR PALPEBRAE SUPERIOR -→ LESSER WING OF SPHENOID BONE

*INSERTION :-

- RECTI INSERTED → SCLERA (POSTERIOR TO LIMBUS]

- SUPERIOR OBLIQUE - PASSES THROUGH→ FIBRO-CARTILAGE PULLY - ATTACHED TO TROCHLEAR FOSSA OF FRONTAL BONE
-INSERTION - SCLERA BEHIND EQUATOR OF EYEBALL

- INFERIOR OBLIQUE MUSCLE INSERTED-→ CLOSE TO SO.

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NERVE - RECTUS X 3 = CN=3/OCCULOMOTAR NERVE
- SUPERIOR OBLIQUE = CN=4 / TROCHLEAR NERVE
- LATERAL RECTUS - CN=6 ABDUCENT NERVE

ACTION :-

1) UPWARD / ELEVATION - SUPERIOR RECTUS
- INFERIOR OBLIQUE

2) DOWNARD / DEPRESSOR - INFERIOR RECTUS
- SUPERIOR OBLIQUE

3) MEDIAL ROTATION / ADDUCTION - MEDIAL RECTUS
SUPERIOR RECTUS
INFERIOR RECTUS

4) LATERAL ROTATION / ABDUCTION - LATERAL RECTUS
SUPERIOR OBLIQUE
INFERIOR OBLIQUE

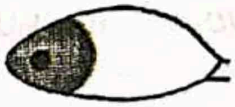
5) TORSION - SUPERIOR OBLIQUE
- SUPERIOR RECTUS

6) EXTORSION - INFERIOR OBLIQUE
- INFERIOR RECTUS

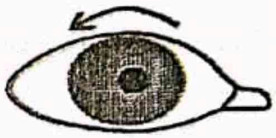
Supraduction

Infraduction

Adduction


Abduction

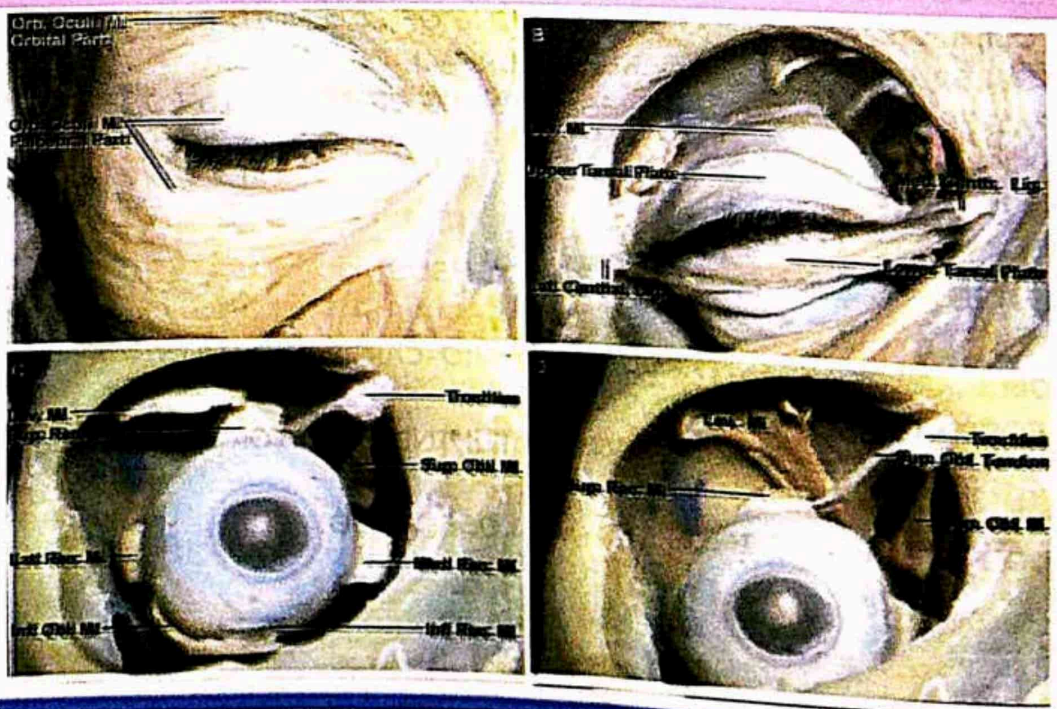
Intorsion


Extorsion

CLINICAL ANATOMY

*WEAKNESS OR PARALYSIS OF A MUSCLE CAUSES SQUINT OR STRABISMUS, WHICH MAY BE CONCOMITANT OR PARALITIC.

*MYSTAGMUS IS CHARACTERIZED BY INVOLUNTARY RYTHMICAL OSCILLATORY MOVEMENTS OF THE EYES.



GOPAL KUMAR

LATERAL WALL OF NOSE

*LATERAL WALL OF NOSE → IRREGULAR OWING → PRESENCE OF 3X SHELF LIKE BONE PROJECTION CALLED CONCHAE
*CONCHAE - INCREASE SURFACE AREA OF NOSE
- FOR EFFECTIVE AIR-CONDITIONING OF INSPIRED AIR

*SEPARATE THE NOSE - FROM ORBIT ABOUE
- MAXILLARY SINUS BELOW
- FROM LACRIMAL SAC AND NASOLACRIMAL DUCT FRONT

*SUB-DIVIDED INTO 3X PART - ANTERIOR - VESTIBULE
MIDDLE - ATRIUM OF MIDDLE MEATUS
POSTERIOR - MEATUSES

*SKELETON OF LATERAL WALL IS PARTLY BONY, PARTLY CARTILAGINOUS AND PARTLY MADE UP OF SOFT TISSUE
-BONY PART - NASAL
- FRONTAL PROCESS OF MAXILLA
- LACRIMAL
- LABYRINTH OF ETHMOID
- INFERIOR NASAL CONCHA
- PERPENDICULAR PLATE OF PALATINE BONE
- MEDIAL PTERYGOID PLATE

- CARTILAGINOUS PART - SUPERIOR NASAL CARTILAGE
- INFERIOR NASAL CARTILAGE
- 3-4 SMALL CARTILAGES OF ALA

- CUTICULAR LOWER PART -→ FORMED BY FIBROFATTY →COVERED WITH SKIN

CONCHAE AND MEATUSES

NASAL CONCHAE → CURVED BONE PROJECTION
- DIRECTED DOWNWARD AND MEDIALLY
- INFERIOR CONCHA - INDEPENDENT BONE
- MIDDLE CONCHA - MEDIAL SURFACE OF ETHMOIDAL LABYRINTH
- SUPERIOR CONCHA - MEDIAL SURFACE OF ETHMOIDAL LABYRINTH

MEATUSE OF NOSE - PASSAGE BENEATH THE OVERHANGING CONCHAE

1) INFERIOR MEATUS - LARGEST OF THREE MEATUSES

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- NASOLACRIMAL DUCT OPEN INTO JUNCTION OF POSTERIOR 2/3 & ANTERIOR 1/3

2) MIDDLE MEATUS - ETHMOIDAL BULLA - ROUND ELEVATION
- HIATUS SEMILUNARIS - DEEP SEMICIRCULAR SULCUS
- OPENING OF FRONTAL AIR SINUS → ANTERIOR PART OF HS
- OPENING OF ETHMOIDAL AIR SINUS → BEHIND F.A
- OPENING OF MAXILLARY AIR SINUS → POSTERIOR PART OF HS

3) SUPERIOR MEATUS - SHORTEST AND SHALLOWEST OF MEATUSES
- OPENING OF POSTERIOR ETHMOIDAL AIR SINUSES

SPHENO ETHMOIDAL RECESS - TRIANGLE FOSSA
- JUST ABOVE - SUPERIOR CONCHA
- RECEIVES - OPENING OF SPHENOIDAL AIR SINUS

***ARTERIAL SUPPLY -**
1) ANTERIOR POSTERIOR ETHMOIDAL A. - ANTEROSUPERIOR QUADRANT
2) SPHENOPALATINE A. - POSTEROSUPERIOR QUADRANT
3) FACIAL A. - ANTEROINFERIOR QUADRANT
4) GREATER PALATINE A. - POSTEROINFERIOR QUADRANT

***VENOUS DRAINAGE** - VEIN FROM A PLEXUS WHICH DRAINS -
- ANTERIORLY - FACIAL VEIN
- MIDDLE - PTERYGOID PLEXUS VEIN
- POSTERIORLY - PTERYGOID PLEXUS VEIN

***NERVE SUPPLY -**
1) ANTERIOR ETHMOID NERVE - ANTEROSUPERIOR QUADRANT
2) LATERAL POSTERIOR NASAL BRANCH NERVE - POSTEROSUPERIOR QUADRANT
3) ANTERIOR SUPERIOR ALVEOLAR NERVE - ANTEROINFERIOR QUADRANT
4) ANTERIOR PALATINE NERVE - POSTEROINFERIOR QUADRANT

CLINICAL ANATOMY

*HYPERTROPHY OF THE MUCOSA OVER THE INFERIOR NASAL CONCHA IS A COMMON FEATURE OF ALLERGIC RHINITIS, WHICH IS CHARACTERISED BY SNEEZING, NASAL BOLCKAGE AND EXCESSIVE WATERY DISCHARGE FROM NOSE.

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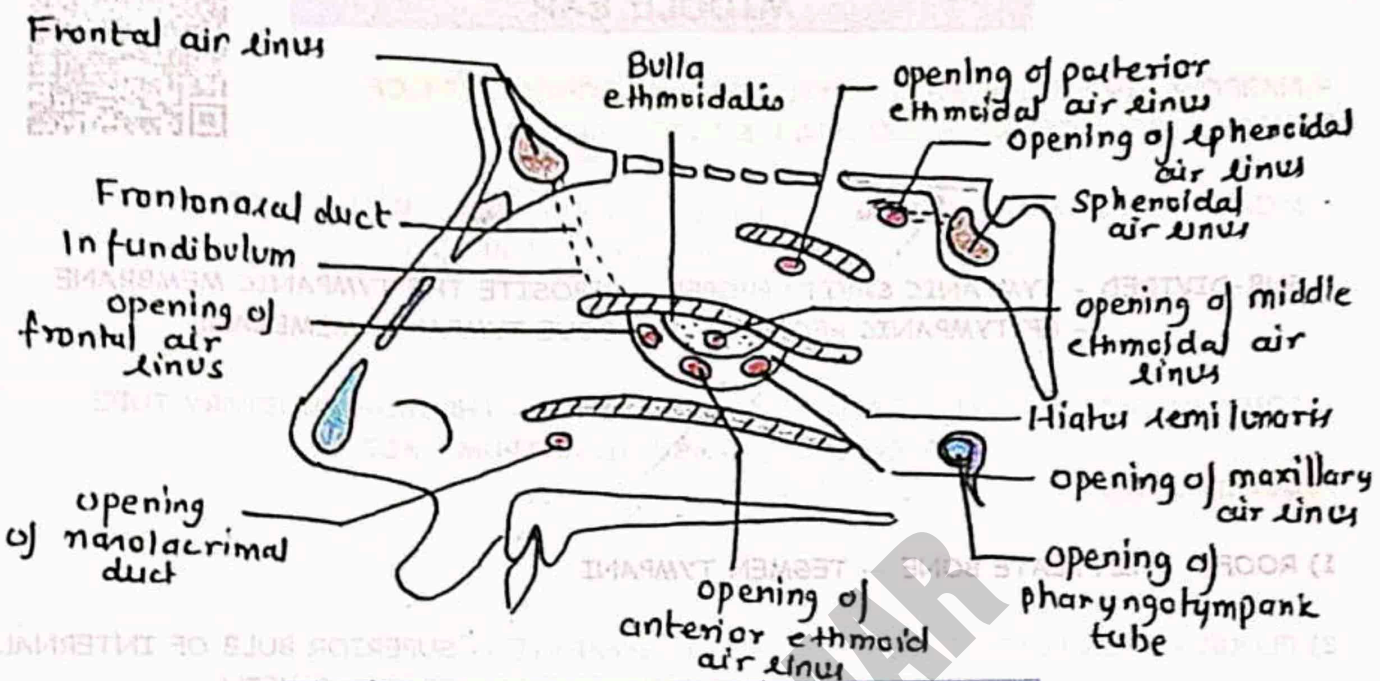
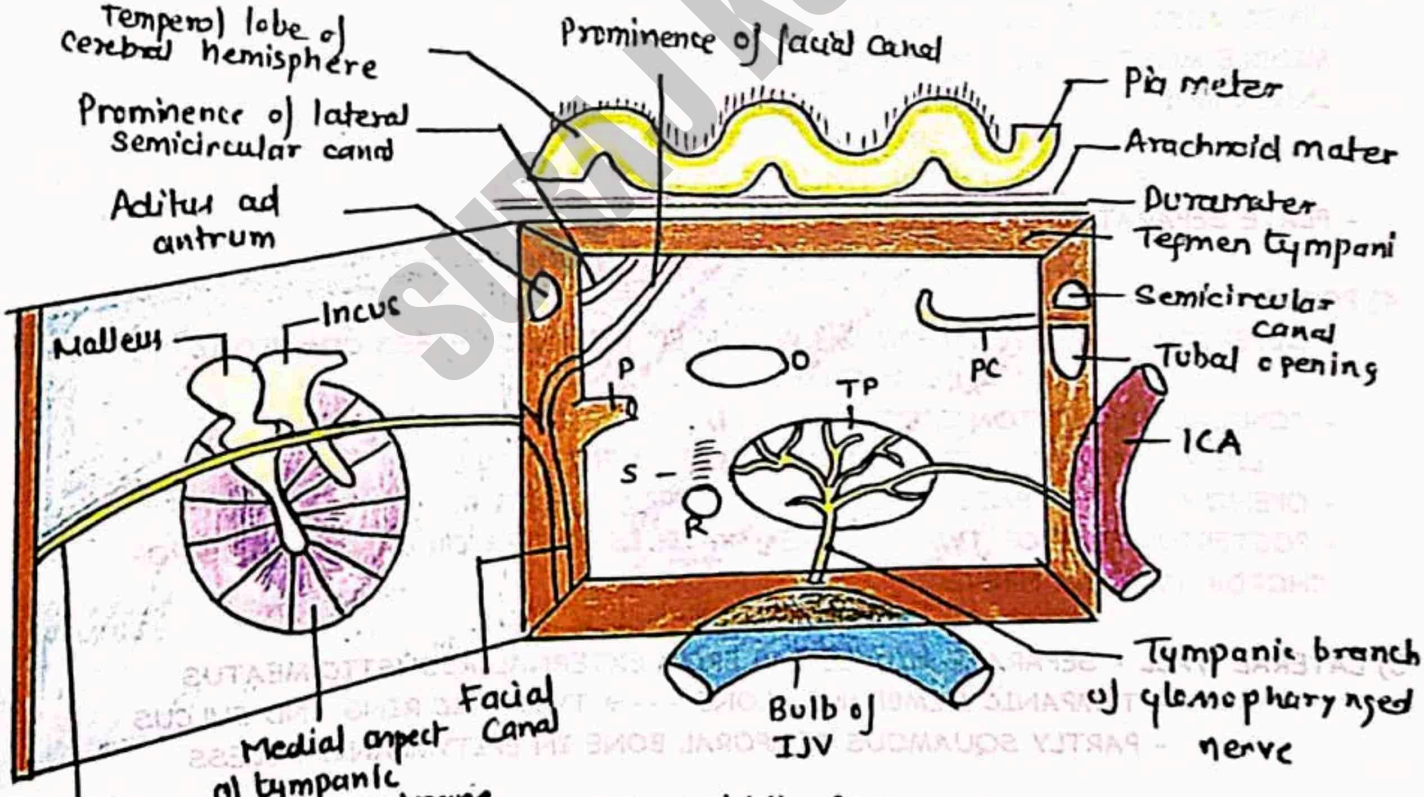


LATERAL WALL OF NOSE



Middle Ear

[P → Pyramid, O → oval window, PC → Procesus cochleariformis, PM → Promontory, R → Round window, S → Sinus tympani, TP → Tympanic Plexus]

MIDDLE EAR

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MIDDLE EAR

- NARROW AIR FILLED SPACE - SITUATED IN PETROUS PART OF TEMPORAL BONE BETWEEN EXTERNAL & INTERNAL EAR

- MIDDLE EAR SHAPED, LIKE CUBE - LATERAL AND MEDIAL WALL - LARGE
  - OTHER WALL - NARROW
- SUB-DIVIDED - TYMPANIC CAVITY PROPER - OPPOSITE THE TYMPANIC MEMBRANE
  - EPITYMPANIC RECESS - LIE ABOUE TYMPANIC MEMBRANE

- COMMUNICATION → ANTERIORLY - NASOPHARYNX - THROUGH AUDITARY TUBE
  POSTERIORLY - MASTOID ANTRUM - ADITUS

*BOUNDARIES -

1) ROOF - THIN PLATE BONE → TEGMEN TYMPANI

2) FLOOR - THIN PLATE BONE OF TEMPORAL SEPARATE → SUPERIOR BULB OF INTERNAL JUGULAR VEIN
   - GLOSSOPHARYNGEAL NERVE

3) ANTERIOR WALL -
   UPPER MOST - TENSOR TYMPANI
   MIDDLE MOST - AUDITORY TUBE
   INNER MOST - THIN PLATE OF BONE
   ↓ WHICH
   FORM POSTERIOR WALL OF CAROTID CANAL
   - PLATE SEPARATED INTERNAL CAROTID ARTERY

4) POSTERIOR WALL -
   - SUPERIORLY - ADITUS THROUGH WHICH EPITYMANIC RECESS COMMUNICATE TYMPANIC ANTRUM
   - CONICAL PROJECTION CALLED PYRAMID
     LIE - NEAR JUNCTION OF POSTERIOR & MEDIA WALL
   - OPENING -→ FOR PASSAGE OF TENDON OF" STAPEDIUS"
   - POSTERIOR EDGE OF TYMPANIC MEMBRANE IS POSTERIOR CANALICULUS FOR CHORDA TYMPANI NERVE

5) LATERAL WALL - SEPARATE MIDDLE EAR FROM EXTERNAL ACOUSTIC MEATUS
   - TYMPANIC MEMBRANE ALONG ---→ TYMPANIC RING AND SULCUS
   - PARTLY SQUAMOUS TEMPORAL BONE IN EPITYMPANIC RECESS

6) MEDIAL WALL - SEPARATE MIDDLE EAR FROM INTERNAL EAR
   - PROMONTARY - FIRST TURN OF COCHLEA

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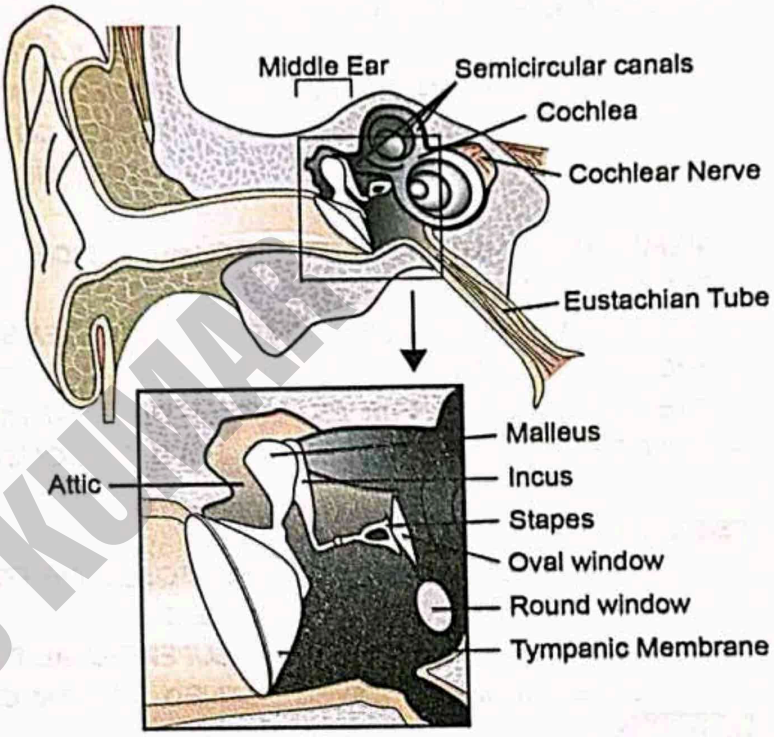
-FENESTRA COCHLEAE - ROUND OPENING
- POSTERO INFERIOR → PROMONTORY

-FENESTRA VESTIBULI - OVAL OPENING
- POSTERO SUPERION → PROMONTORY
-SEMICIRCULAR CANAL → ABOVE FACIAL CANAL
-FACIAL CANAL - ABOVE FENESTRA VESTIBULI



*CONTENTS :-
- 3X SMALL BONE' - MALLEUS
INCUS
STAPE

- LIGAMENT OF EAR OSSICLE

- 2X MUSCLE - TENSAR TYMPANI
- STAPEDIUS

- VESSELS SUPPLY AND DRAINING THE MIDDLE EAR

- NERVE - CHORDA TYMPANI
- TYMPANIC PIEXUS

*ARTERIAL SUPPLY -

- ANTERIOR TYMPANIC BRANCH - MAXILLARY ARTERY
- POSTERIOR TYMPANIC BRANCH POSTERIOR AURICULAR ARTERY
- PETROSAL AND SUPERIOR TYMPANIC BRANCH- MIDDLE MENINGEAL ARTERY
- BRANCHES OF ASCENDING PHARYNGEAL ARTERY
- TYMPANIC BRANCHES - ICA

*VENOUS DRAINAGE
- DRAIN --→ INTO SUPERIOR PETROSAL SINUS & PTERYGOID PLEXUS

*LYMPHATIC DRAINAGE
- PREAURICULAR & RETROPHARYNGEAL LYMPH NODE

*NERVE SUPPLY
- DERIVED FROM - TYMPANIC PLEXUS --→WHICH LIE OVER PROMOTONARY

- PLEXUS FORMED BY - 1) TYMPANIC BRANCH OF GLOSSOPHARYNGEAL NERVE
2) SUP. & INF. CAROTICOTYMPANIC NERVE-→ARISE FROM SYMPATHETIC PLEXUS

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TYMPANIC MEMBRANE

*THIN, TRANSLUCENT PARTITION BETWEEN EXTERNAL ACOUSTIC MEATUS AND MIDDLE EAR

*OVAL IN SHAPE

*PLACED - OBLIQUELY AT AN ANGLE OF 55 DEGREE WITH FLOOR OF MEATUS

*FACE - DOWNWARD, FORWARD AND LATERALLY

- 2 SURFACE ----→OUTER SURFACE - LINED BY THIN SKIN
  - CONCAVE
  ---→ INNER SURFACE - PROVIDE ATTACHMENT TO HANDLE OF MALLEUS
  - INNER CONVEX
  - POINT OF MAX. CONVEXITY LIE-→AT TIP OF HANDLE OF MALLEUS CALLED UMBO
- MEMBRANE THICKENED AT CIRCUMFERENCE →WHICH FIXED TYMPANIC SULCUS OF TEMPORAL PLATE
- SUPERIOR SULCUS DEFICIENT -→ HERE MEMBRANE ATTACHED TO →TYMPANIC NOTCH
- FROM END OF NOTCH - 2 BAND - ANT. & POST. MALLEOLAR FOLD PROLONGED
- GREATER PART OF T.M IS -→TIGHTLY STRETCHED ---→ PARS TENSA
- PART BETWEEEN 2 MALLEOLAR FOLD OS LOOSE ---→ PARS FLACCIDA

*STRUCTURE

- TYMPANIC MEMBRNE IS COMPOSED OF FOLLOWING 3 LAYER
- OUTER CUTICULAR LAYER - SKIN
- MIDDLE FIBROUS LAYER - SUPERFICIAL RADIATING FIBRES
- INNER MUCOUS LAYER - LINED BY LOW CILIATED COLUMNAR EPITHELIUM

*BLOOD-

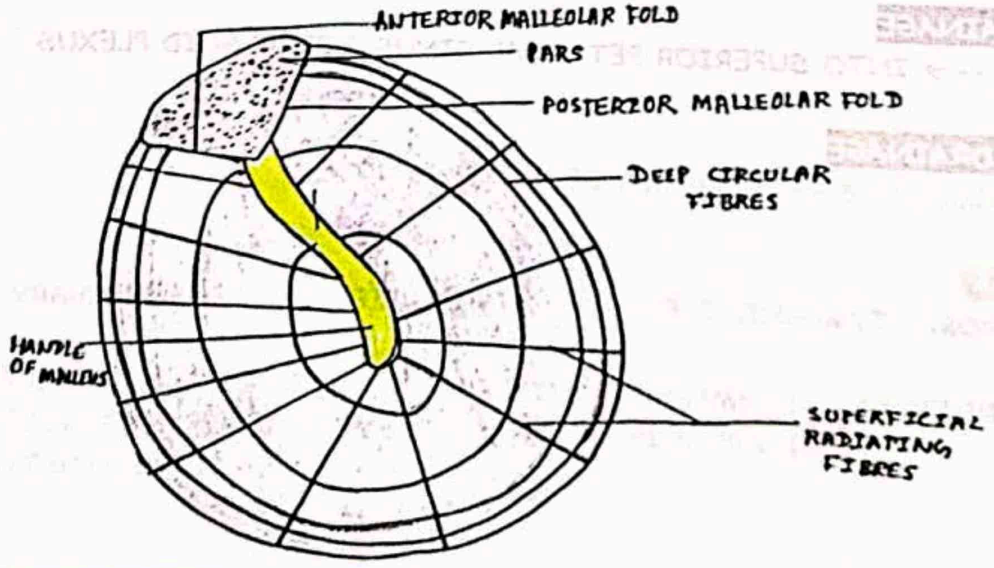
OUTER SYRFACE - DEEP AURICULAR A. BRANCH OF MAXILLARY ARTERY

INNER SURFACE - ANTERIOR TYMPANIC BRANCH - MAXILLARY A.
- POSTERIOR TYMPANI BRANCH - POSTERIOR AURICULAR A.

*NERVE-

OUTER SYRFACE - AURICLUOTEMPORAL NERVE

INNER SURFACE - TYMPANIC BRANCH →GLOSSOPHARYNGEAL NERVE



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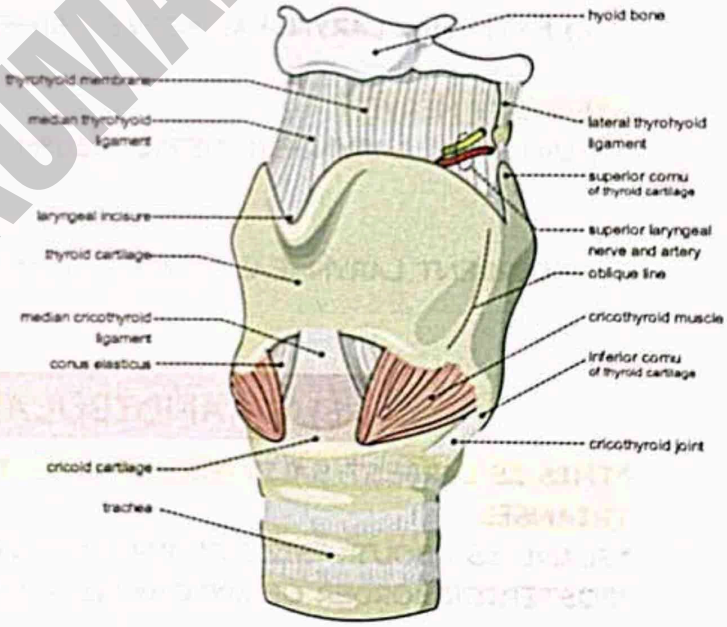
LARYNX ANATOMY

*LARYNX - IS ORGAN FOR PRODUTION OF VOICE OR PHONATION
*IT IS NALSO AN AIR PASSAGE AND ACT AS SPHINCTER AT INLET OF LOWER RESPIRATORY PASSAGES

*SITUATION-
LIE IN -→ ANTERIOR MIFLINE OF NECK
EXTENDING FROM - ROOT OF TONGUE TO TRACHEA
IN ADULT MALE - C3-C6 VERTEBRAE
IN ADULT FEMALE - C1 - C4

*SIZE - LENGTH -→ 44MM IN MALE & 36MM IN FEMALE

*CARTILAGES OF LARYNX -
-THE LARYNX CONTAINS NINE CARTILAGES, OF WHICH THREE ARE UNPAIRED AND THREE ARE PAIRED.



UNPAIRED CARTILAGES
1 THYROID (GREEK SHIELD-LIKE)
2 CRICOID (GREEK RING-LIKE)
3 EPIGLOTTIS (GREEK LEAF-LIKE)

PAIRED CARTILAGES
1 ARYTENOID (GREEK CUP-SHAPED)
2 CORNICULATE (LATIN HORN-SHAPED)
3 CUNEIFORM (LATIN WEDGE-SHAPED)

* INTRINSIC MUSCLES OF LARYNX
1. CRICOTHYROID
2. POSTERIOR CRICOARYTENOID TRIANGULAR
3. LATERAL CRICOARYTENOID
4. TRANSVERSE ARYTENOID
5,6. OBLIQUE ARYTENOID AND ARYEPIGLOTTICAN
7,8. THYROARYTENOID AND THYROEPIGLOTTIC
9. VOCALIS

*MUSCLES ACTING ON THE LARYNX

MOVEMENT	MUSCLE
1. ELEVATION OF LARYNX	- THYROHYOID, MYLOHYOID
2. DEPRESSION OF LARYNX	- STERNOTHYROID, STERNOHYOID
3. OPENING INLET OF LARYNX	- THYROEPIGLOTTIC
4. CLOSING INLET OF LARYNX	- ARYEPIGLOTTIC
5. ABDUCTOR OF VOCAL CORDS	- POSTERIOR CRICOARYTENOID ONLY
6. ADDUCTOR OF VOCAL CORDS	- LATERAL CRICOARYTENOID,

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TRANS-VERSE AND OBLIQUE ARYTENOIDS

7. TENSOR OF VOCAL CORDS AND MODULATION OF VOICE - CRICOTHYROID
8. RELAXOR OF VOCAL CORDS - THYROARYTENOID AND VOCALIS

*BLOOS SUPPLY

- UP TO VOCAL FOLD - BY SUPERIOR LARYNGEAL A.
  -VEIN - DRAIN BY SUPERIOR LARYNGEAL VEIN

- BELOW THE VOCAL FOLD - BY INFERIORLARYNGEAL A.
  - VEIN - DRAIN BY INFERIOR LARYNGEAL VEIN

*NERVE SUPPLY -

-MOTOR NERVE-

1) RECURRENT LARYNGEAL NERVE - SUPPLIS ALL INTRINSIC MUSCLE
   - EXCPET - CRICOTHYROID

2) EXTERNAL LARYNGEAL NERVE - SUPPLIES CRICOTHYROID MUSCLE

-SENSORY NERVE -

1) INTERNAL LARYNGEAL NERVE - SUPPLIES - MUCOUS MEMBRANE UP TO LEVEL OF VOVAL FOLD

2) RECURRENT LARYNGEAL NERVE - SUPPLIES - MUCOUS MEMBRANE BELOW LEVEL OF VOVAL FOLD

SUBMANDIBULAR SALIVARY GLAND

*THIS IS LARGEST SALIVARY GLAND, SITUATED IN ANTERIOR PART OF DIGASTRIC TRIANGEL

*GLAND IS ABOUT - SIZE OF WALNUT WEIGHING ABOUT - 15 TO 20G

*POSTERIOR BORDER OF MYLOHYOID - DIVIDE - INTO LARGER PART -SUPERFICIAL TO MUSCLE
- SMALL PART - DEEP TO MUSCLE

*COVERINGS -

1) GLAND PARTIALLY ENCLOSED BETWEEN TWO LAYERS OF DEEP CERVICAL FASCIA
2) SUPERFICIAL LAYER OF FASCIA - COVER - INFERIOR SURFACE OF SURFACE & ATTACHED TO BASE OF MANDIBLE
3) DEEP LAYER - COVER - MEDIAL SURFACE OF GLAND & SUPERIORLY TO MYLOHYOID LINE OF MANDIBLE

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*SUPERFICIAL PART -

- THIS PART OF GLAND FILLS - DIGASTRIC TRIANGLE
- EXTENDS - SUPERIORLY DEEP TO MANDIBLE UP TO MYLOHYOID LINE

*DEEP PART -

- THIS PART IS SMALL IN SIZE
- LIE DEEP TO MYLOHYOID

*BLOOS SUPPLY & LYMPHATIC DRAINAGE :-

*ARTERY -FACIAL ARTERY

*VEIN DRAIN - FACIAL VEIN

*NERVE SUPPLY -

1) SENSORY FIBRES - LINGUAL NERVE

2) VASOMOTOR SYMPATHETIC FIBRES FROM - PLEXUS ON FACIAL ARTERY

3) SECRETOMOTOR FIBRES - SUPERIOR SALIVATORY NUCLEUS
↓
7TH CRANIAL NERVE
↓
CHORDA TYMPANIC BRANCH
↓
JOINS LINGUAL NERVE
↓
SUSPENDS SUBMANDIBULAR GANGLION
↓
SUBMANDIBULAR G

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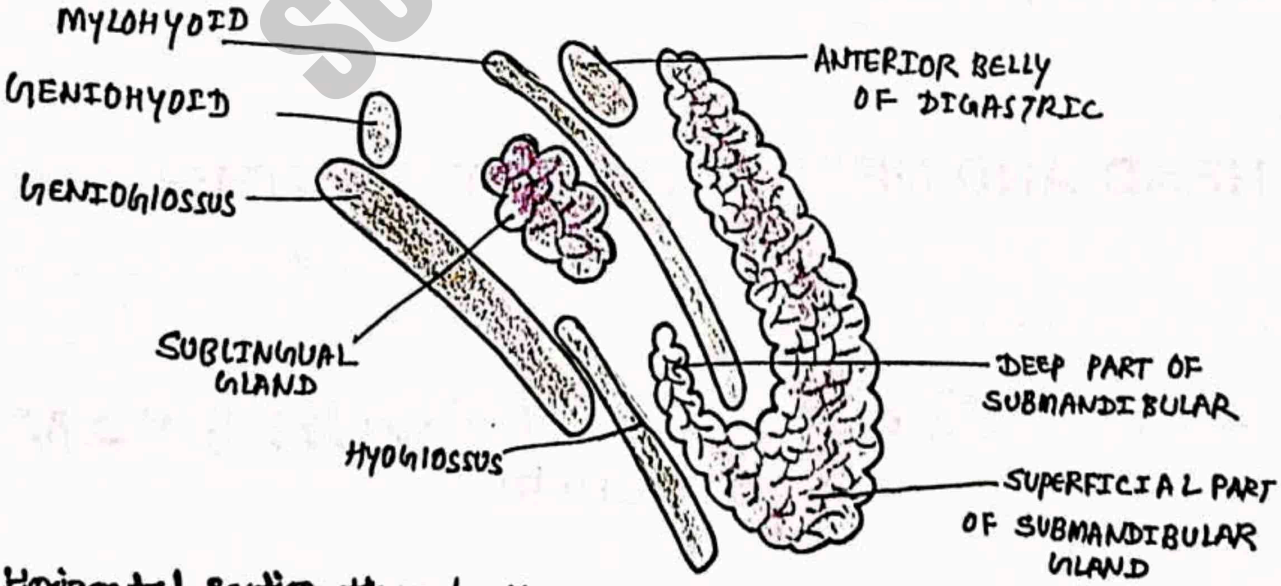

Horizontal section through the submandibular region showing the location of the submandibular and sublingual glands.

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20MARKS

1. CLASSIFY DURAL VENOUS SINUS & DESCRIBE CAVERNOUS SINUS IN DETAIL
2. DESCRIBE MIDDLE EAR UNDER FOLLOWING HEADINGS -
   1.LOCATION 2.CONTENT 3.BOUNDARY & NERVE SUPPLY 4.APPLIED
3. DESCRIBE TONGUE UNDER THE FOLLOWING HEADINGS -
   1.LOCATION & PARTS 2.NERVE SUPPLY & LYMPHATIC DRAINAGE 3.DEVELOPMENT
4. DESCRIBE ANTERIOR TRIANGLE & CAROTID TRIANGLE
5. DESCRIBE PALATINE TONSIL UNDER FOLLOWING HEADINGS -
   1.EXTERNAL FEATURES 2.BLOOD SUPPLY 3.HISTOLOGY
6. DESCRIBE THYROID GLAND OR PAROTID GLAND UNDER FOLLOWING HEADINGS -
   1.LOCATION 2.EXTERNAL FEATURE & RELATION 3.BLOOD & NERVE SUPPLY
7. DESCRIBE TEMPEROMADIBULAR JOINT

*SHORT QUESTIONS - 5MARKS

1. SCALP
2. LACRIMAL APPARATUS
3. POSTERIOR TRIANGLE
4. EXTERNAL CAROTID ARTERY
5. MAXILLARY ARTERY & MAXILLARY NERVE
6. MUSCLE MUSTIGATION
7. DEVELOPMENT OF SOF PALATES
8. HISTOLOGY DIAGRAM OF PALATINE TONSIL
9. EXTRA OCULAR MUSCLE
10. LARYNX
11 OTIC GANGLION
12. SUBMANDICULAR GLAND
13. LATERAL WALL OF NOSE

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HEAD AND NECK ANATOMY IS COMPLETED !!

IT ALWAYS SEEMS IMPOSSIBLE UNTIL IT'S DONE

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

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MCQ HEAD AND NECK

1) Paralysis of which muscle causes drooping of upper eyelid (ptosis) :
a Occipito-frontalis
b Orbicularis oculi
c levator palpebrae superioris
d superior rectus

2) Which one of the following is motor nerve of scalp:
a lesser occipital
b auriculo temporal
c posterior auricular
d supra orbital

3) Which of the following muscles is an abductor of vocal cords :
a posterior cricoarytenoid
b vocalis
c lateral cricoarytenoid
d cricothyroid

4) Which muscle is supplied by glossopharyngeal nerve
a stylohyoid
b stylopharyngeus
c salpingopharyngeus
d palatopharyngeus

5) Which muscle protrudes the tongue:
a genioglossus
b hyoglossus
c styloglossus
d palatoglossus

6) where does the nasolacrimal duct drains :
a superior meatus
b middle meatus
c inferior meatus
d sphenoethmoidal recess

7) Which muscle tenses the vocal cord
a Thyroarytenoid
b lateral cricoarytenoid
c posterior cricoarytenoid
d cricothyroid

8)All of the following pass through the superior orbital fissure except :
a ophthalmic artery
b ophthalmic vein
c occulomotor nerve
d trochlear nerve

9) Which of the following lies deep to the hyoglossus muscle
a Lingual nerve
b lingual artery
c submandibular gland
d submandibular ganglion

10) All of the following are intrinsic muscle of larynx except
a cricothyroid
b thyrohyoid
c thyroepiglottic
d Thyroarytenoid

11) All of the following are supplied by hypoglossal nerve except:
a genioglossus
b styloglossus
c hyoglossus
d palatoglossus

12) The parotid duct opens into vestibule of mouth at which level of molar tooth
a upper first
b lower first
c upper second
d lower second

13) Which nerve supplies the superior oblique muscle of the eye ball
a optic
b occulomotor
c trochlear
d abducens

14)stylopharyngeus muscle recieves its motor innervations from which nerve
a facial
b glossopharyngea
c vagus
d hypoglossal

15) Which of the following layer of the scalp is known as dangerous layer
a skin
b connective tissue
c loose areolar tissue
d pericranium

16) The structure that does not pass through the parotid gland is:
a facial nerve
b external carotid artery
c retromandibular nerve
d auriculo temporal nerve

17) All the following nerves traverses carotid triangle except
a glossopharyngeal
b vagus
c spinal part of acessory
d hypoglossal

18) Muscle responsible for depression of mandible
a masseter
b temporalis
c medial pterygoid
d lateral pterygoid

1)C 2)C 3)A 4)B 5)A 6)C 7)D 8)A 9)B 10)B 11)D 12)C 13)C 14)B 15)C 16)D 17)A 18)D

SURAJ KUMAR BBS

NEUROANATOMY BASICS

NEURON - PARTS OF NEURON:
- CELL BODY
- AXON

NUCLEUS - AGGREGATION OF CELL BODIES IN CENTRAL NERVOUS SYSTEM (CNS)
-DERIVED FROM NEUROECTODERM

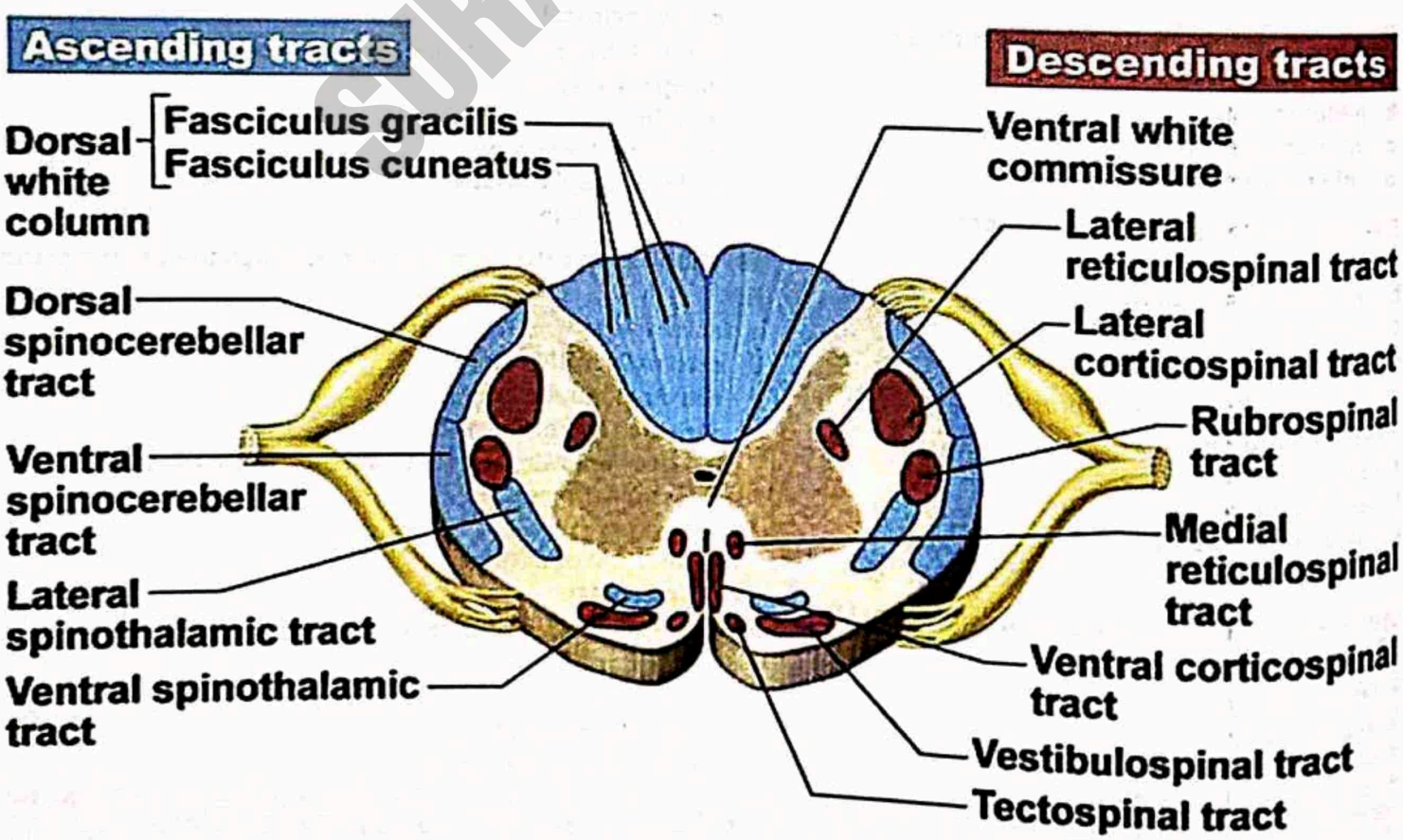
GANGLION - AGGREGATION OF CELL BODIES IN PERIPHERAL NERVOUS SYSTEM (PNS)
-DERIVED FROM NEURAL CREST CELLS

*THE NUCLEUS IS ALSO CALLED GREY MATTER.
*THE AXONS ARE ALSO CALLED WHITE MATTER/FASCICULUS/LEMNISCUS.
*THE COLLECTION OF AXONS IN CNS IS CALLED THE TRACT.
*THE COLLECTION OF AXONS IN PNS IS CALLED THE NERVE.

ASCENDING & DESCENDING TRACT

1) ASCENDING TRACTS ARE SENSORY PATHWAYS THAT BEGIN AT THE SPINAL CORD AND STRETCH ALL THE WAY UP TO THE CEREBRAL CORTEX.

2) DESCENDING TRACTS ARE THE PATHWAYS BY WHICH MOTOR SIGNALS ARE SENT FROM THE BRAIN TO THE SPINAL CORD. THEY ARE ALSO CALLED MOTOR TRACTS, DUE TO THEIR INVOLVEMENT IN MOVEMENT COORDINATION.



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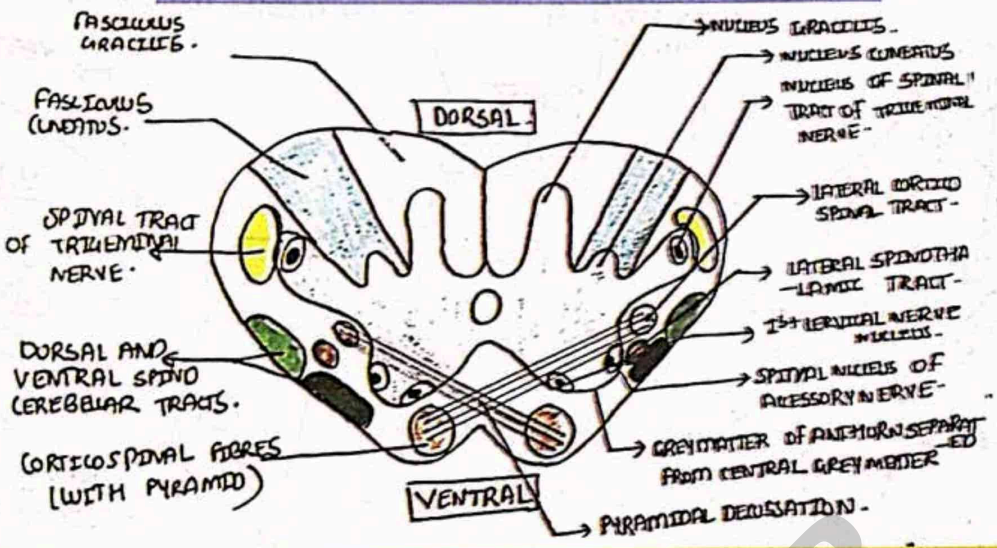
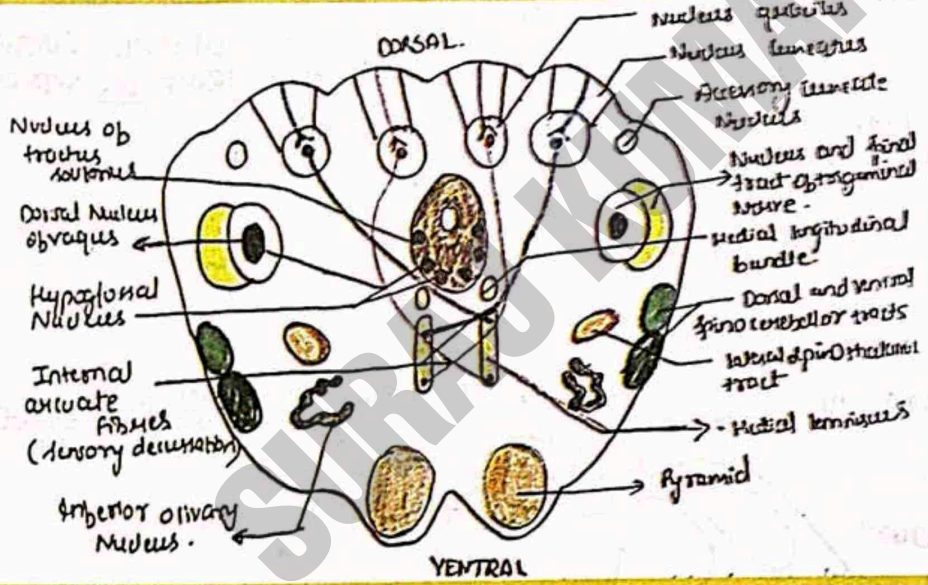
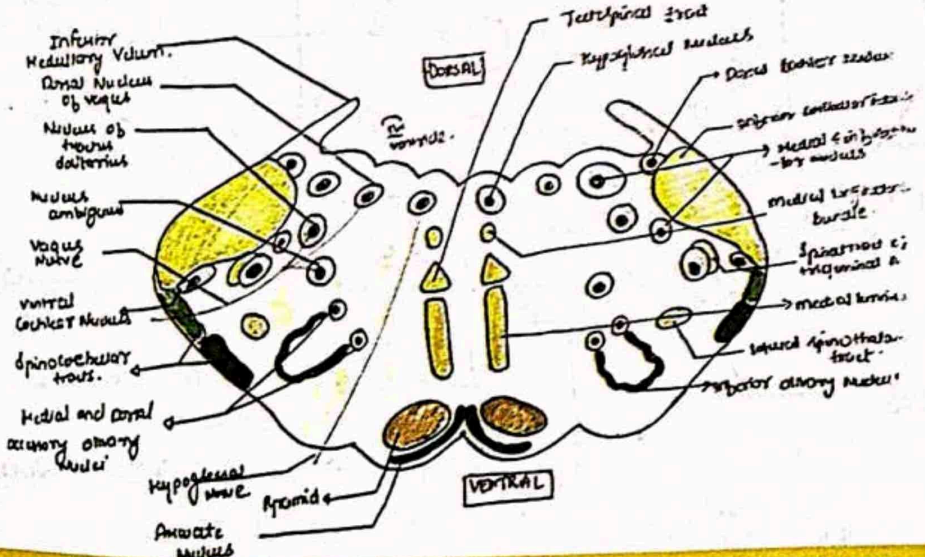
BRAIN STEM



TS OF MEDULLA AT LEVEL OF PYRAMIDAL DECUSSATION



TS OF MEDULLA AT LEVEL OF SENSORY DECUSSATION



TS OF MEDULLA AT LEVEL OF OLIVARY NUCLEUS

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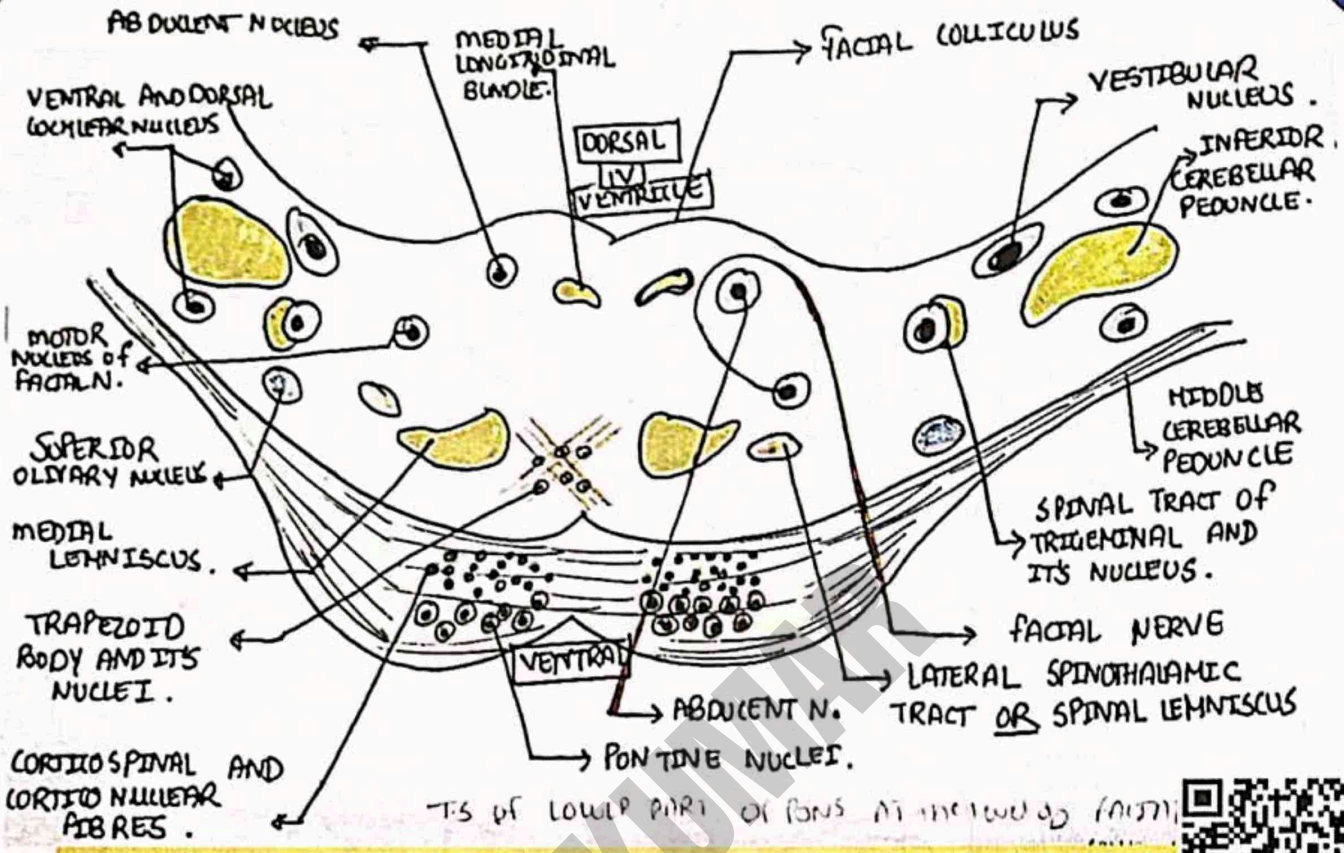
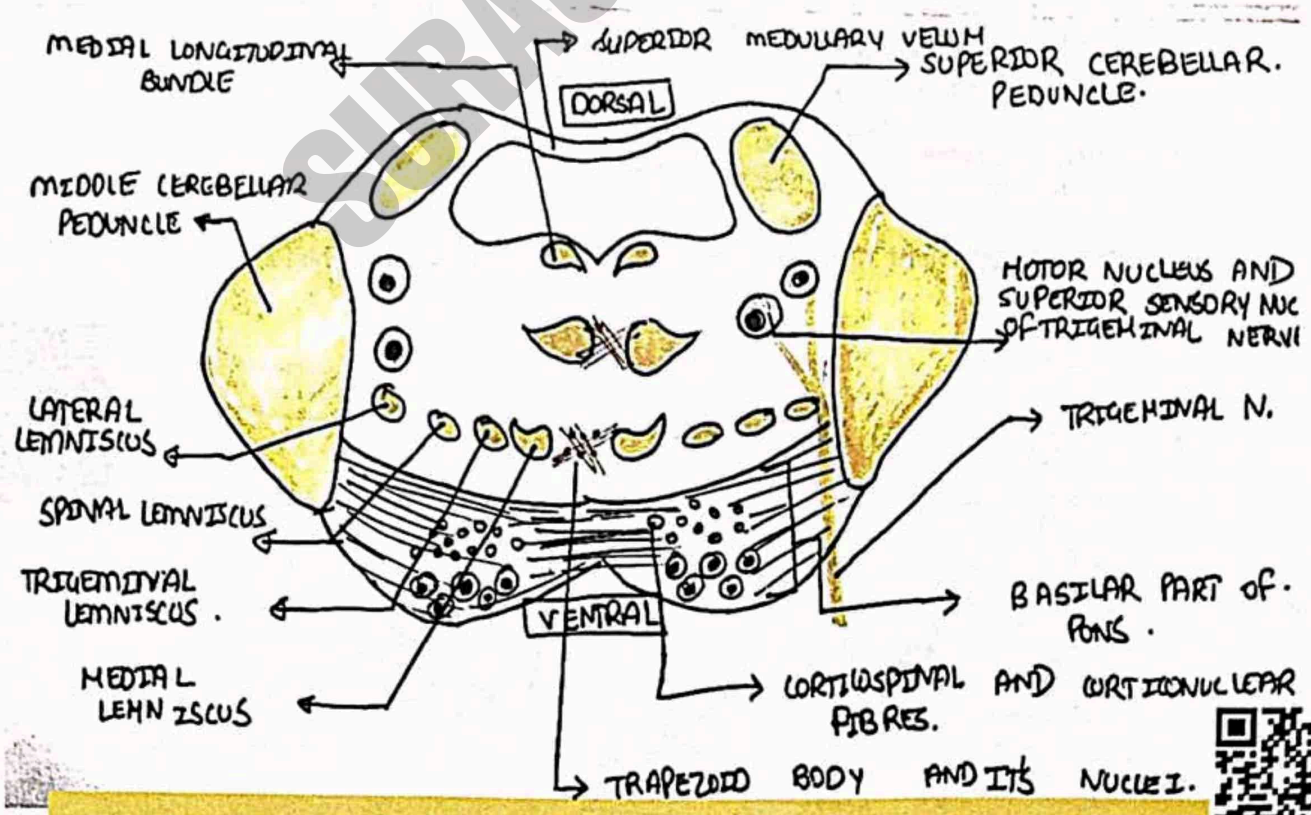


TS OF OF PONS AT FACIAL COLLICULUS



TS OF UPPER PART OF PONS

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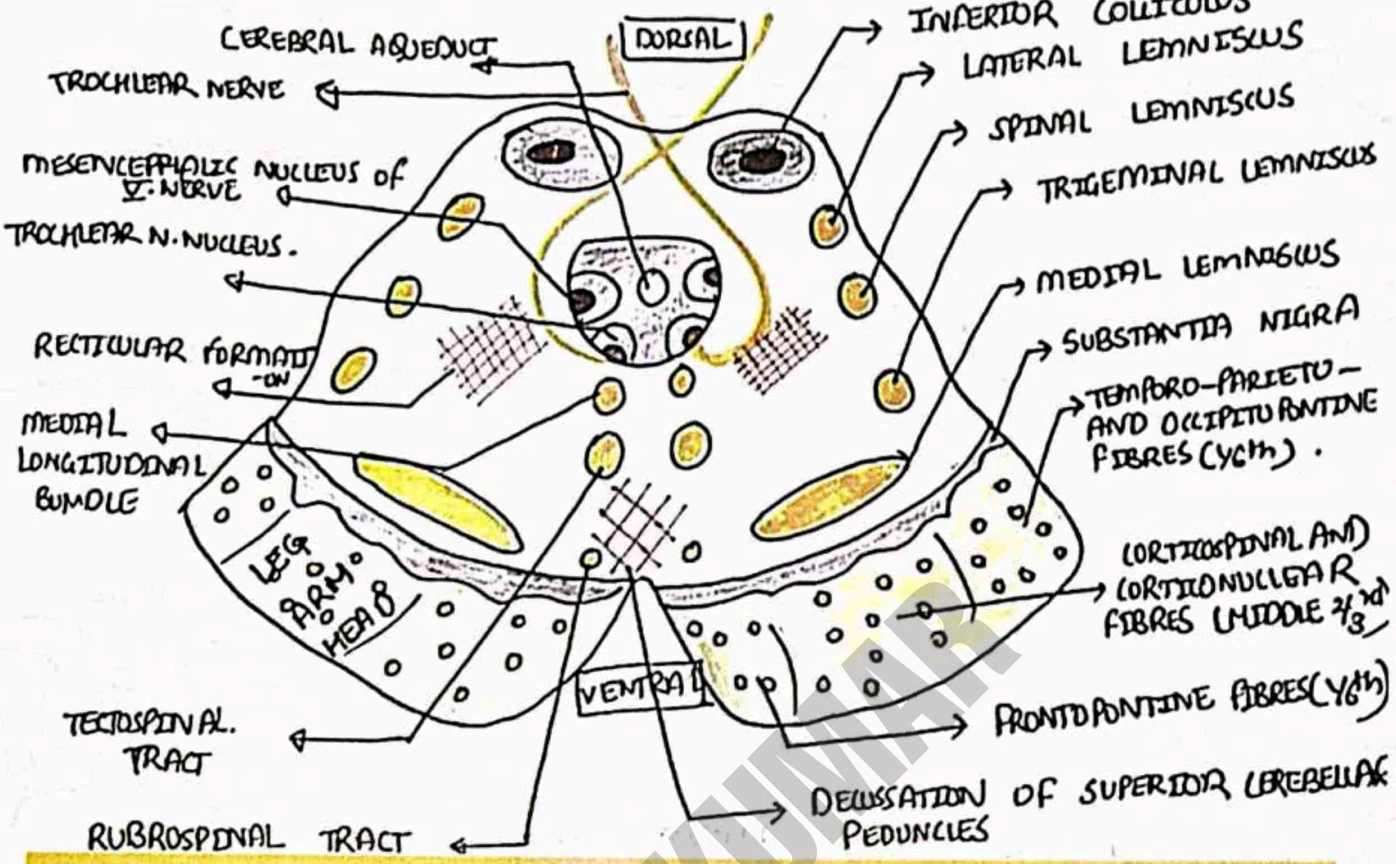
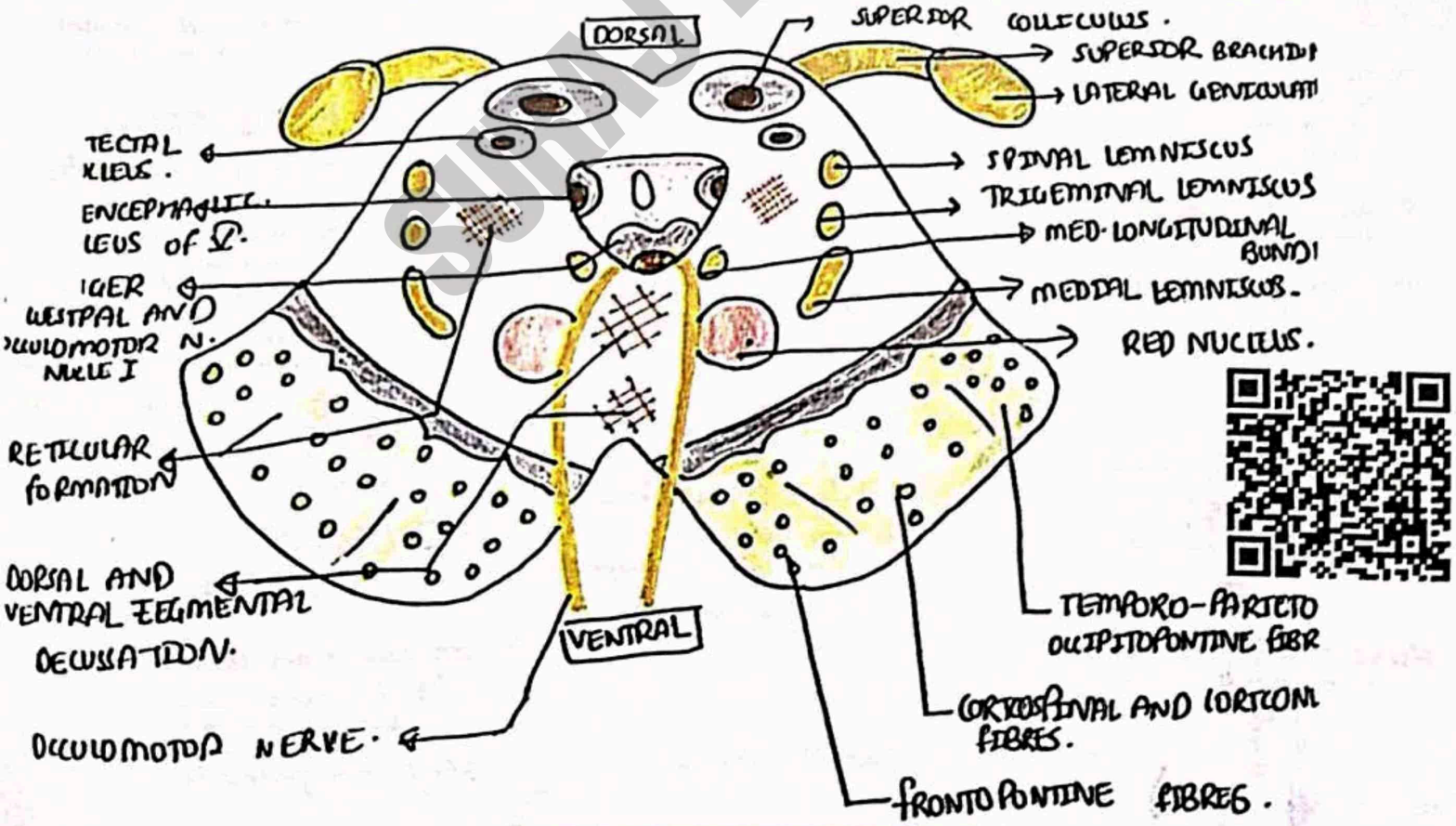


TS OF MIDBRAIN AT LEVEL OF INFERIOR COLLICLUS



TS OF MIDBRAIN AT LEVEL OF SUPERIOR COLLICLUS

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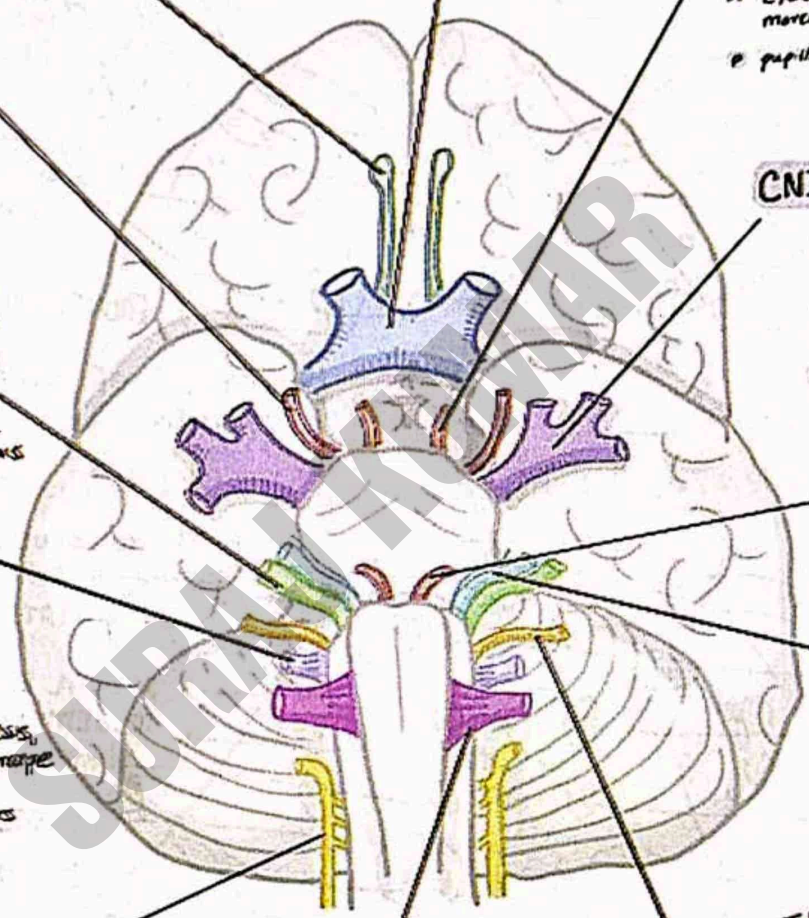
CRANIAL NERVES

MAI SIDWALT

ANATOMY & FUNCTIONS

FUNCTIONS
—Legend—
- S Sensory
- M Motor
- P Parasympathetic
- SM Sympathetic



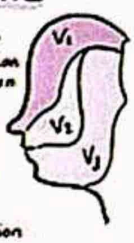
CN I: OLFACTORY
S smell

CN II: OPTIC
S sight

CN III: OCULOMOTOR
M eyeball movement: MR, IR, SR, IO
movement of upper eyelid: LP
P pupillary light reflex

CN IV: TROCHLEAR
M eyeball movement: superior oblique

CN V: TRIGEMINAL
V_1: OPTHALMIC
S facial sensation
SM pupillary dilation
V_2: MAXILLARY
S facial sensation
V_3: MANDIBULAR
M chewing
S facial sensation



CN VIII: VESTIBULO-COCHLEAR
S Hearing: cochlea
Balance: vestibular apparatus

CN VI: ABDUCENS
M eyeball movement
-lateral rectus

CN X: VAGUS
M swallowing, elevation of soft palate, talking, coughing, gag reflex
P heart rate, GIT peristalsis, sweating, vasoangel synapse
SM peripheral chemoreceptors
-hyperemia

CN VII: FACIAL
S taste: anterior 2/3 of tongue
M facial expression
eyelid closing
stapedius muscle: dampens sound
P lacrimation
salivation – submandibular + sublingual glands

CN XI: ACCESSORY
M head turning: SCM
shrugging: trapezius

CN XII: HYPOGLOSSAL
M most muscles of tongue (except palatoglossus)
→tongue movement

CN IX: GLOSSO-PHARYNGEAL
S sensation from posterior 1/3 of tongue, tonsils, middle ear, pharynx
taste from posterior 1/3 of tongue
M stylopharyngeus muscle - elevates pharynx + larynx
P salivation – parotid gland

FUNCTION MNEMONIC
S-sensory M-motor B-both

Some Say Marry Money But My Brother
Cranial Nerve: I II III IV V VI VII

Says Big Brains Matter More
VIII IX X XI XII

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YouTube

OLFACTORY NERVE

- 1st cranial nerve

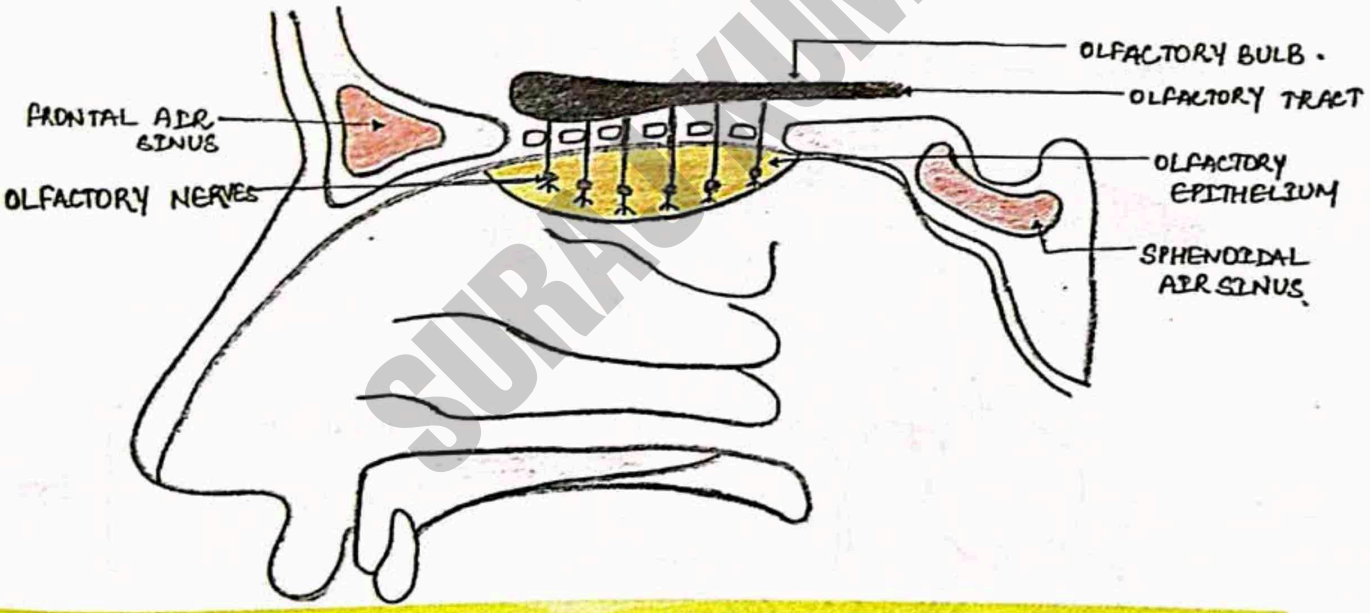
NEURONS

- Receptors & 1st neuron — olfactory cells (in olfactory part of nasal mucosa)
  olfactory neurons (central processes of olfactory cells
- 2nd neuron — located in primary olfactory cortex
- 3rd neuron — located in primary olfactory cortex
- 4th neuron — fibers goes to secondary olfactory cortex

COURSE

- Fibers from primary receptor neurons pass through → foramina in cribriform plate of ethmoid → enter → Anterior cranial fo
  olfactory bulb ← terminate

- Nerves distributes — olfactory region of nasal cavity.



OLFACTORY NERVE

OPTIC NERVE

- 2nd cranial nerve
- "Nerve of sight"

COURSE

- Fibers arise from ganglionic cells of retina of eyeball converge → towards optic disc
  optic disc ← pierce ← outer layer of retina
  eyeball ← leaves ← . outer layer of retina
  . choroid
  . sclera

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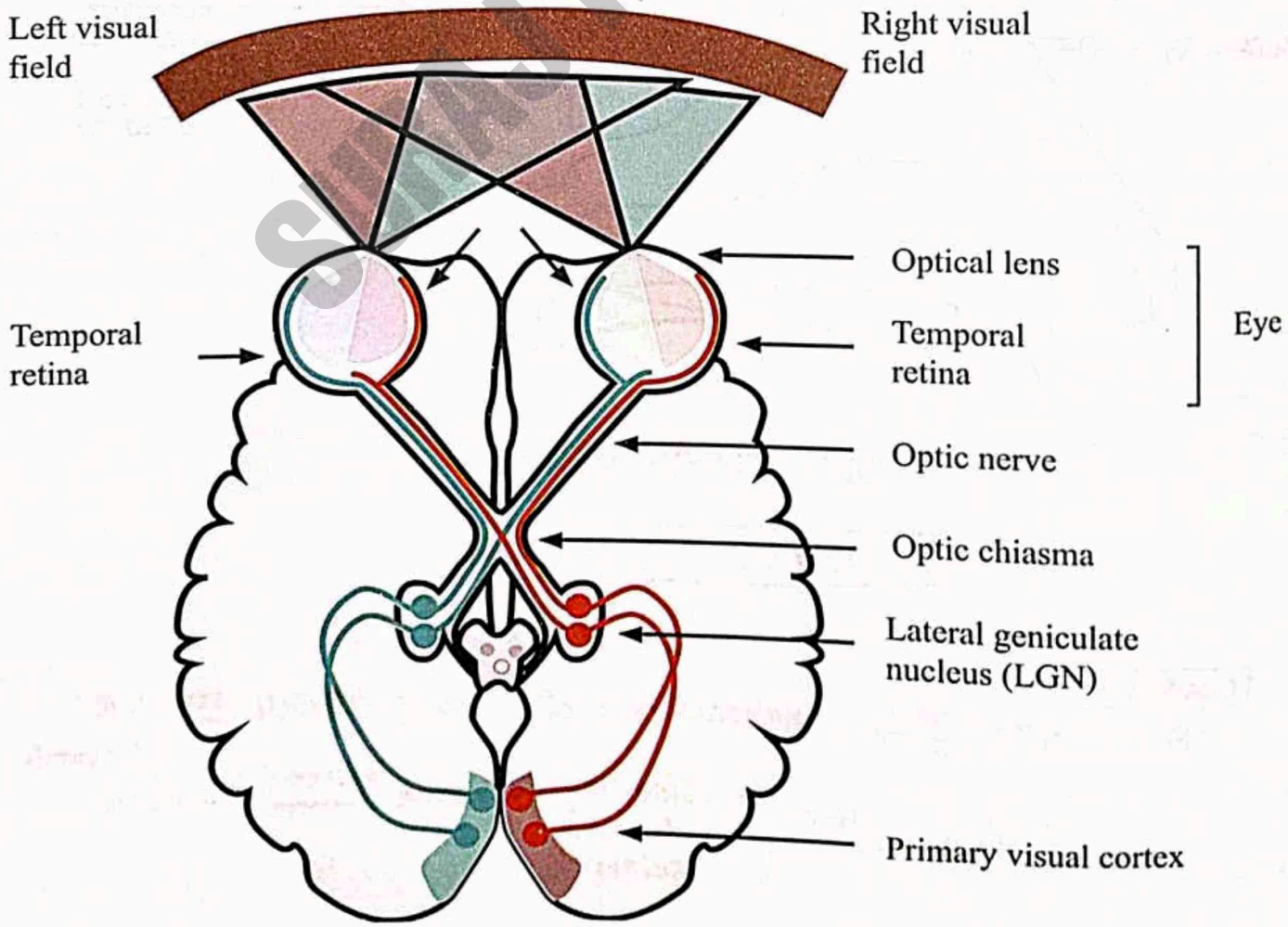
- Fibers unite & form → OPTIC NERVE → runs posteromedially ↓ through posterior half of orbit ↓ enters middle cranial fossa, through optic canal. ↓ unites to form → OPTIC CHIASMA

- Optic nerve divided into → intra-orbital part, canalicular part, intra-cranial part
- Enclosed by 3 meninges of brain

RELATIONS

- At apex ⟶ Recti muscles, ciliary ganglion
- Superior ⟶ ophthalmic artery, nasociliary nerve, superior ophthalmic vein
- Inferior ⟶ nerve to medial rectus.



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OCCULOMOTOR NERVE

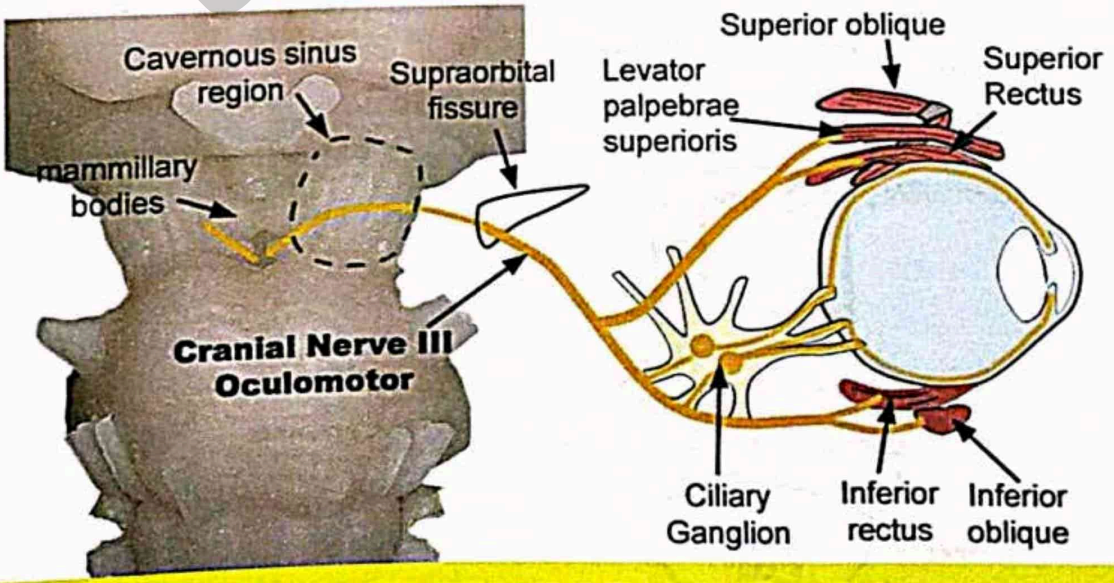
- 3rd cranial nerve
- Distributed to extraocular & intraocula muscles.

NUCLEUS
- Occulomotor nucleus

COURSE

Intraneural Course
- Nerve pass → ventrally through → tegmentum, red nucleus & substantia nigra
- Nerve attached → occulomotor sulcus on medial side of crus cerebri
- Nerve passes between → Superior cerebellar & posterior cerebellar arteries
  - runs forward in interpeduncular cistern, lateral side to posterior communicating artery
  - reach → cavernous sinus
  - descends → lateral wall of sinus, & lies above trochlear nerve
  - Anterior part of sinus → nerve divides → Upper division, Lower division
- Divisions enter → orbit through → middle part of superior orbital fissure

In orbit
- → Smaller upper division → supplies → Superior rectus, Levator palpebrae superioris
- → Larger lower division → divides into → Medial rectus, Inferior rectus, Inferior oblique (↑ longest)
  - supplies → inferior oblique muscle.



COURSE OF TROCHLEAR NERVE

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TROCHLEAR NERVE

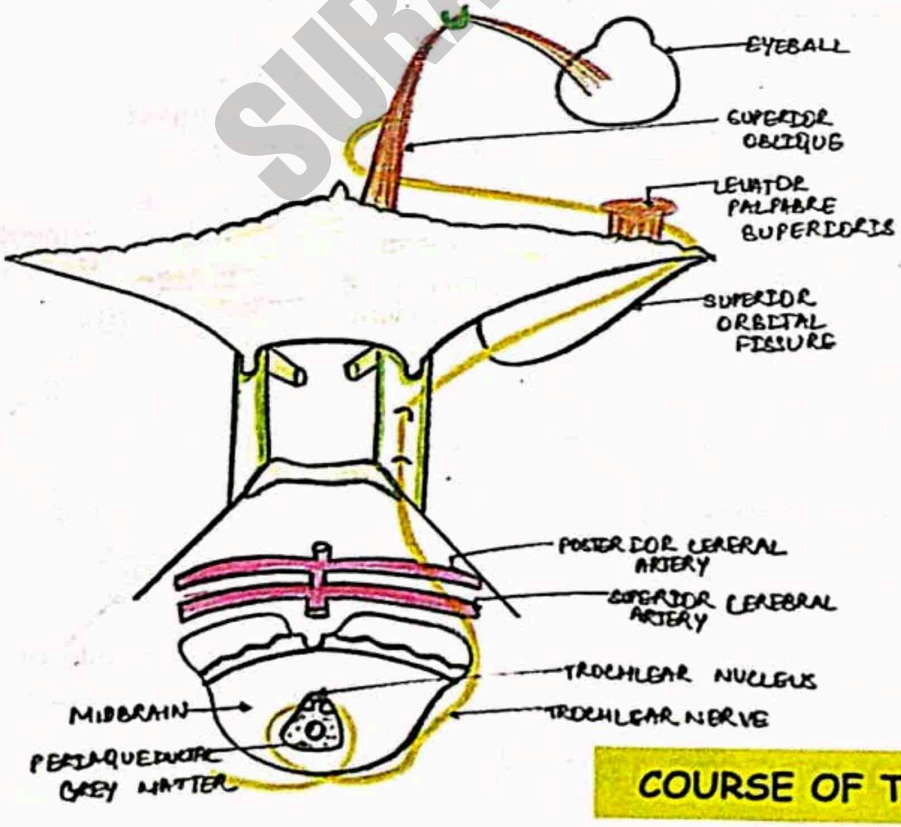
- 4th cranial nerve
- Supplies only superior oblique muscle of eyeball

NUCLEUS - Trochlear nucleus

COURSE

Intraneural Course

- Nerve runs dorsally round central grey matter → reach upper part of superior & anterior medullary velum
- Nerve attached each side – superior medullary velum.
  - ★ Only nerve to emerge on dorsal aspect of brainstem.
- Nerve enters cavernous sinus piercing posterior corner of roof → runs forward in lateral wall of cavernous sinus, between occulomotor & ophthalmic nerves
  - ★ It has longest intracranial course.
- Nerve enters orbit through lateral part of superior orbital fissure
- In orbit runs medially, above origin of levator palpebrae superioris → supply superior oblique muscle on its orbital surface



COURSE OF TROCHLEAR NERVE

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TRIGEMINAL NERVE

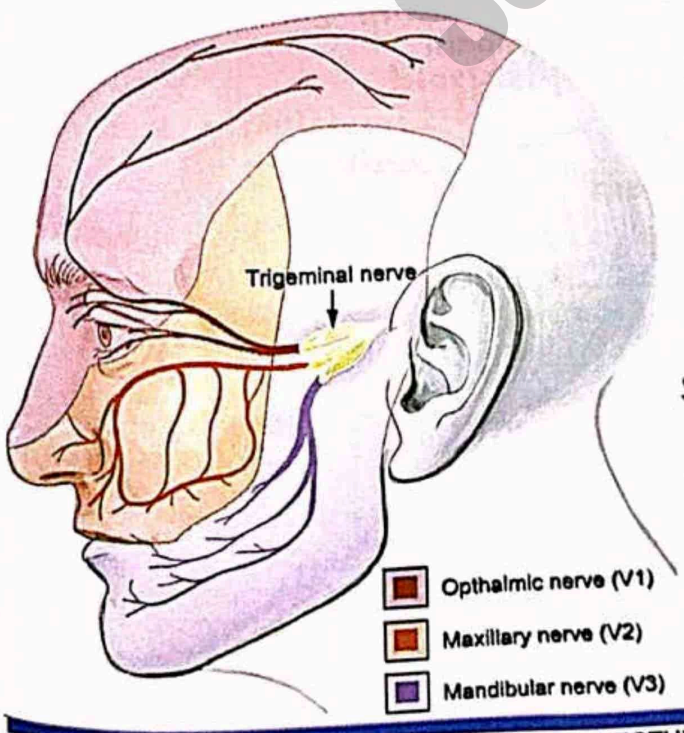
- 5th cranial nerve · Largest · Nerve of 1st brachial arch
- Consists of — Ophthalmic, Maxillary, Mandibular

NUCLEUS

1. General somatic afferent column
   a. Spinal nucleus of V nerve
   b. Principal sensory nucleus of V nerve
   c. Mesencephalic nucleus
2. Branchial efferent column

COURSE

- Trigeminal Nerve
  - larger lateral sensory root → runs forward & lateral → enters middle cranial fa → Trigeminal (Gasserian) ganglia in shallow fossa of petrous temporal bon
    - Ophthalmic Nerve → enters lateral wall of cavernous sinus, below trochlear nerve → branches: nasociliary, Lacrimal, frontal → enter orbit → through superior orbital fissure
    - Maxillary Nerve → enters lateral wall of cavernous sinus, lowest position → enter pterygopalatine fossa → through foramen rotundum
    - Mandibular Nerve → enter infratempora fossa → throu foramen ovale
  - smaller medial motor root → runs forward & laterally → enters infratemporal fossa, through foramen ovale → joins Mandibular nerve



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BRANCHES

OPHTHALMIC NERVE (SENSORY)

- FRONTAL
  - Supratrochlear → Upper eyelid, Conjunctiva, lower part of forehead
  - Supra orbital → Upper eyelid, Frontal air sinus, forehead, scalp
- NASOCILIARY
  - long ciliary → eyeball
  - Ciliary ganglion
  - Ethmoid
    - Anterior
    - posterior
  - Infratrochlear → Both eyelid, lacrimal sac
- LACRIMAL → lacrimal gland

MAXILLARY NERVE (SENSORY)

- MIDDLE CRANIAL FOSSA — Meningeal branch
- PTERYGOPALATINE FOSSA
  - Ganglionic branch
  - Zygomatic
    - → Zygomaticotemporal
    - → Zygomaticofacial
  - Posterior superior alveolar
- INFRAORBITAL CANAL
  - Middle superior alveolar
  - Anterior superior alveolar
- FACE — Infraorbital
  - → Palpebral
  - → Labial
  - → Nasal

MANDIBULAR NERVE (SENSORY + MOTOR)

- TRUNK
  - Meningeal
  - Medial pterygoid
    - → tensor veli palatini
    - → tensor tympani
    - → medial pterygoid
- ANTERIOR DIVISION
  - Deep temporal
  - Lateral pterygoid
  - Masseteric
  - Buccal
- POSTERIOR DIVISION
  - Auriculotemporal
  - Lingual
  - Inferior alveolar

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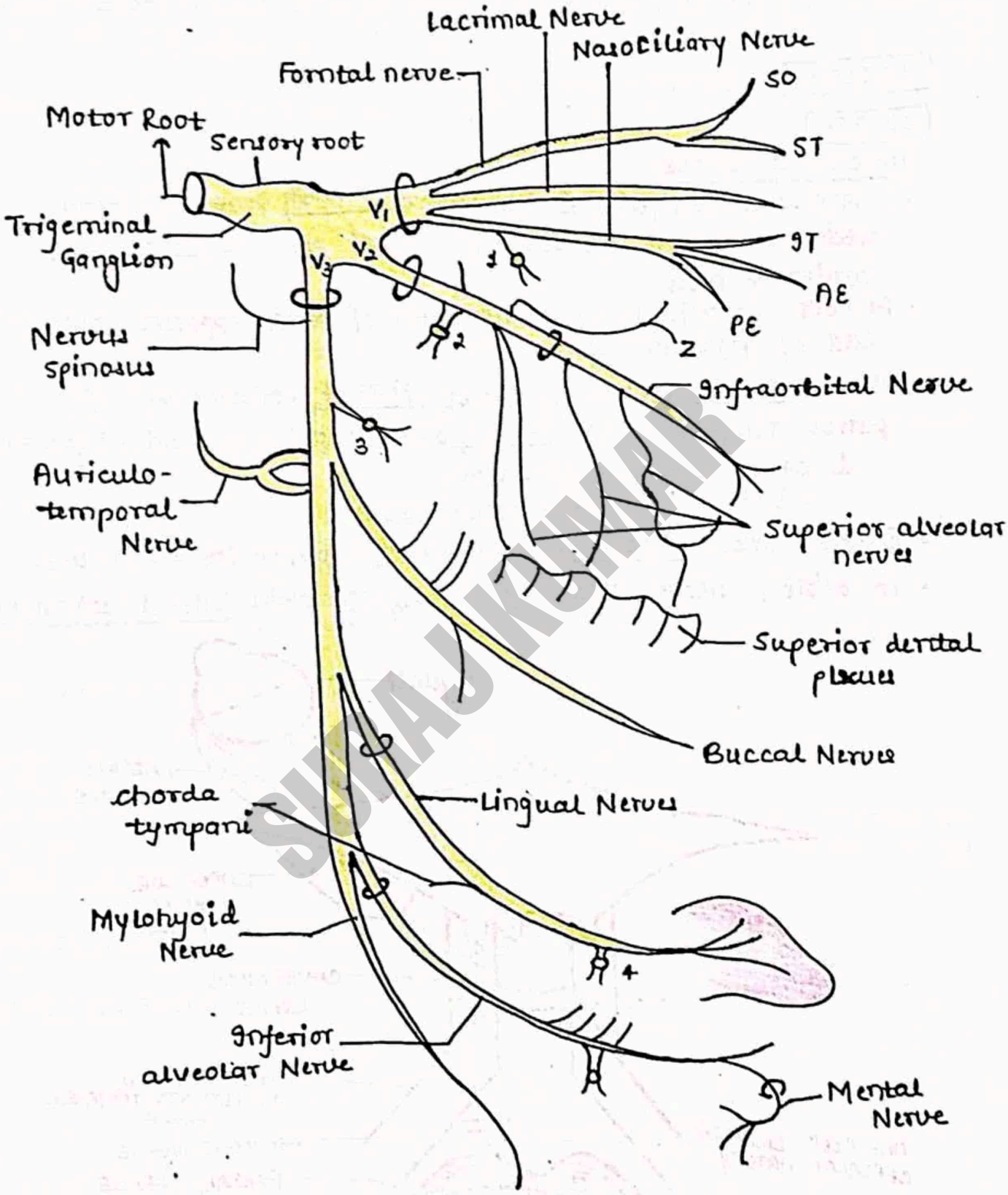

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BRANCHES OF TRIGEMINAL NERVE

ABDUCENT NERVE

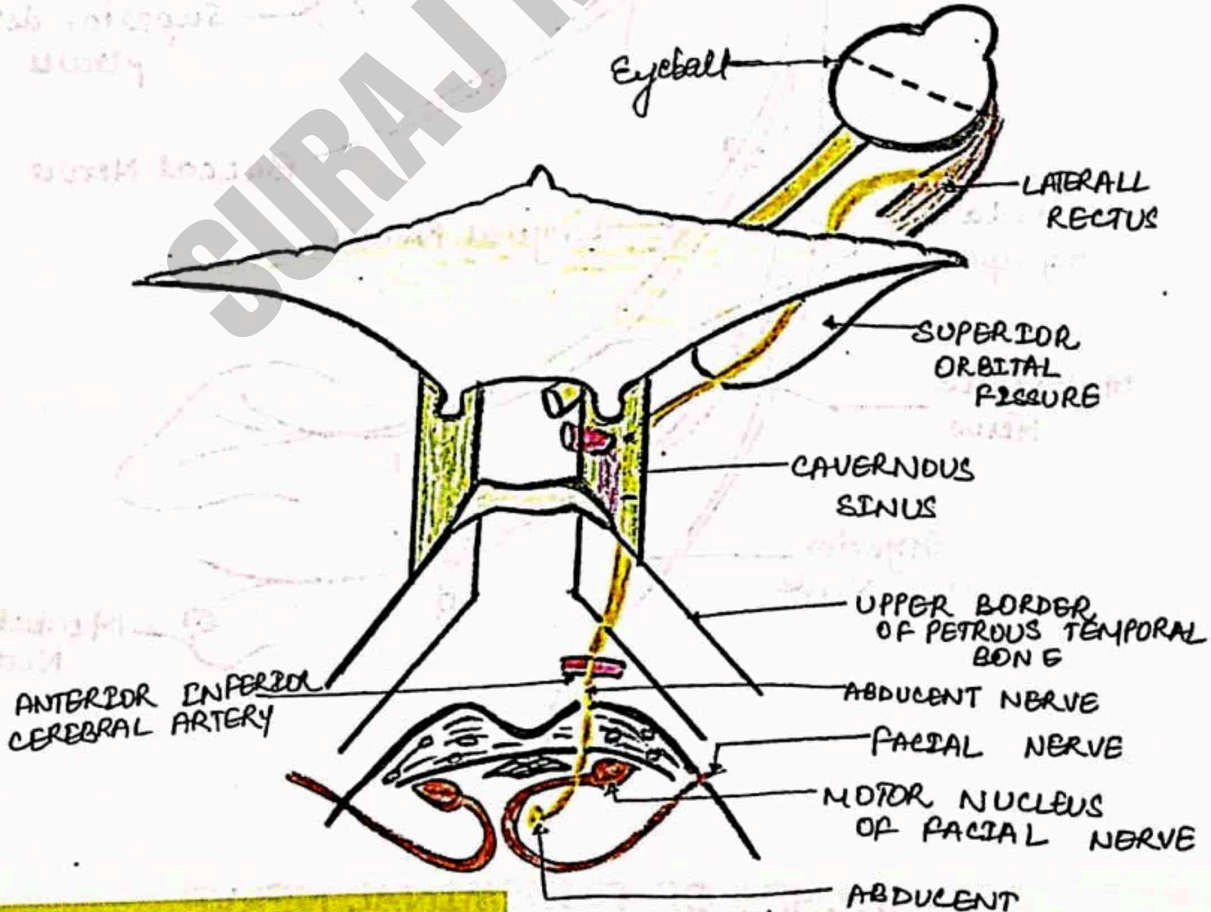
- It is the 6th cranial nerve and supplies lateral rectus muscle of eyeball.

NUCLEUS Abducent nucleus.

COURSE

Intraneural Course

- Fibers run ventrally & downward through trapezoid body, medial lemniscus & basilar part of pons to reach lower border of pons.
- Nerves attached to lower end of pons, opposite upper end of pyramid of medulla
- Nerve enters cavernous sinus piercing posterior wall of petrous temporal bone passing beneath petrosphenoidal ligament & bends sharply forwards.
  - It has long intracranial course
- Enters orbit through middle part of superior orbital fissure.
- In orbit, nerve ends by supplying only Lateral rectus musc



ABDUCENT NERVE

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FACIAL NERVE

- FACIAL NERVE IS NERVE OF SECOND BRANCHIAL ARCH

*NUCLEI : FIBRES OF NERVE ARE CONNECTED TO 4X NUCLEI→ SITUATED IN LOWER PONS
- MOTOR NUCLEUS
- SUPERIOR SALIVATORY NUCLEUS OR PARASYMPATHETIC
- LACRIMATORY NUCLEI - PARA SYMPATHETIC
- NUCLEUS OF TRACTUS SOLITARIUS

*COURSE :-
- ATTACHED TO BRAINSTEM BY 2 ROOTS, MOTOR AND SENSORY
- TWO ROOTS OF FACIAL NERVE ARE ATTACHED TO LATERAL PART OF PONS
- TWO ROOTS RUN LATERALLY AND FORWARD WITH, 8TH CRANIAL NERVE TO REACH---→INTERNAL ACOUSTIC MEATUS

*IN MEATUS → 2X ROOTS, SENSORY AND MOTOR FUSE
- 1ST PART --→ DIRECTLY LATERALLY ABOVE VESTIBULE
- 2RD PART ---→ RELATED TO MEDIAL WALL ABOVE - PROMONTORY.
- 3RD PART→ DIRECTLY VERTICALLY DOWNWARD BEHIND PROMONTORY

-FACIAL NERVE LEAVE SKULL BY PASSING THROUGH - STYLOMASTOID FORAMEN

*EXTRACRANIAL COURSE :-

- FACIAL NERVE CROSSES THE -> LATERAL SIDE OF BASE OF STYLOID PROCESS
- IT ENTERS - POSTERMEDIAL SURFACE OF PAROTID GLAND
- RUNS FORWARD THROUGH THE GLAND CROSSING -→ RETROMANDIBULAR VEIN & ECA
- BEHIND THE NECK OF MANDIBLE - IT DIVIDE INTO 5X TERMINAL BRANCHES EMERGE ALONG ANTERIAR BORDER OF PAROTID GLAND

*BRANCHES AND DISTRIBUTION-

1 WITHIN THE FACIAL CANAL: A. GREATER PETROSAL NERVE
B. THE NERVE TO THE STAPEDIUS
C. THE CHORDATYMPANI

2 AT ITS EXIT FROM THE STYLOMASTOID FORAMEN: A. POSTERIOR AURICULAR
B. DIGASTRIC
C. STYLOHYOID.

3 TERMINAL BRANCHES WITHIN THE PAROTID GLAND: A. TEMPORAL
B. ZYGOMATIC
C. BUCCAL
D. MARGINAL MANDIBULAR
E. CERVICAL.

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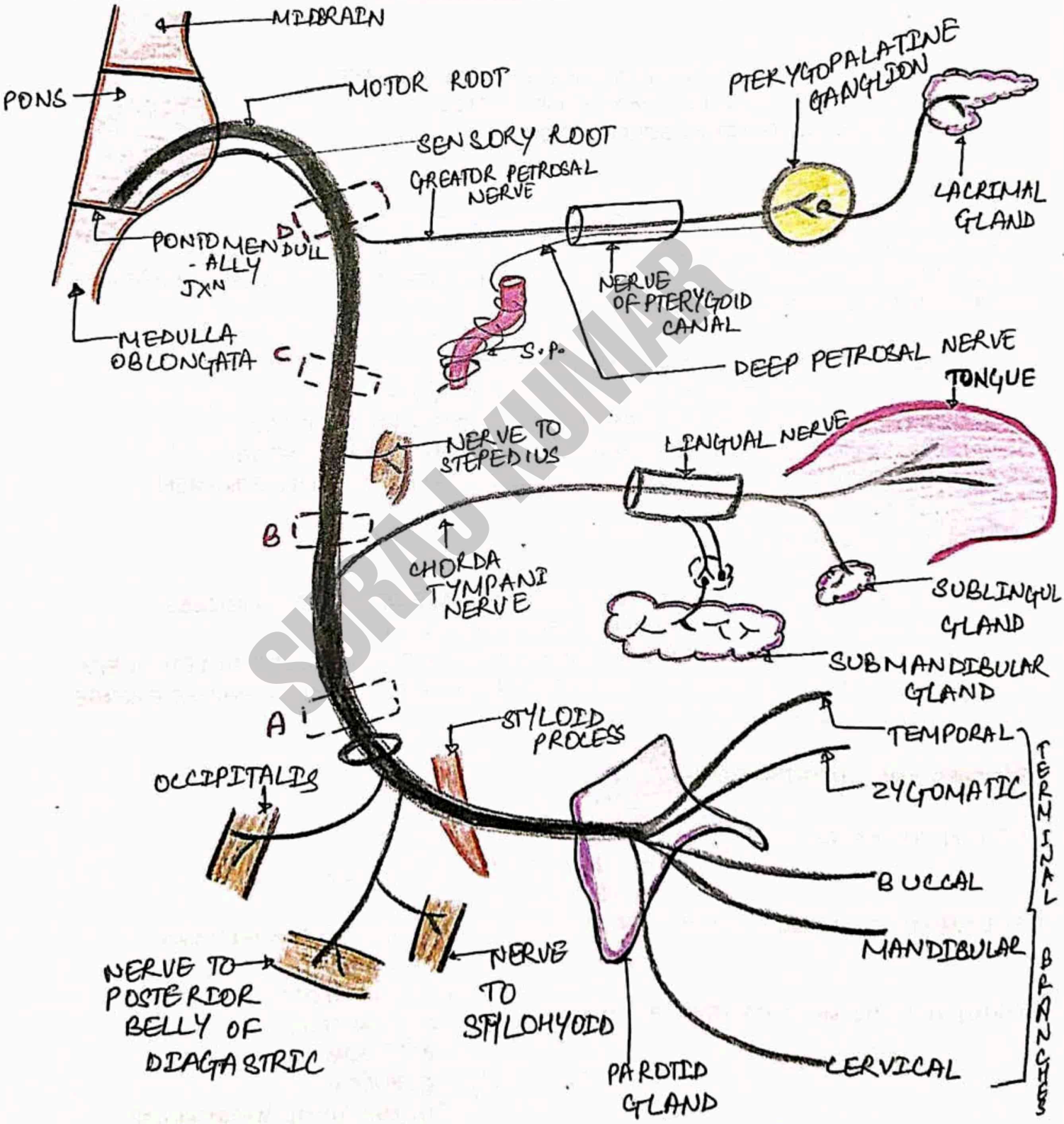
CLINICAL ANATOMY

*BELLS PALSY : SUDDEN PARALYSIS OF NERVE AT THE STYLOMASTOID FORAMEN RESULT IN ASYMMETRY OF CORNER OF MONTH , INABILITY TO CLOSE THE EYE . OSS OF WRINKLING OF SKIN OF FOREHEAD ON THE SAME SIDE



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COURSE OF FACIAL NERVE

VESTIBULO-COCHLEAR NERVE

- It is the 8th cranial nerve.
- It comprises of — hearing & vestibular parts.

PATHWAY OF HEARING

1st neuron — in spiral ganglion, bipolar
- Peripheral → organ of Corti
  Central → Cochlear nerve
- Terminates in dorsal & ventral cochlear nuclei.

2nd neuron — in dorsal & ventral cochlear nuclei

3rd neuron — in superior olivary nucleus
- Axons form lateral lemniscus terminates inferior colliculus

4th neuron — in inferior colliculus
- Axon pass through inferior brachium — reach medial geniculate body.

5th neuron — in medial geniculate body
- Axon form auditory radiation, passes through sublentiform part of internal capsule, to reach auditory area in temporal lobe

VESTIBULAR PATHWAY

- Receptors are maculae of saccule & utricle (static balance), & crista of ampullaris of semicircular ducts (Kinetic balance)
- Fibres of crista of posterior semicircular canal lie → foramen singulare
- Most of fibres from maculae of utricle & saccule lie → in inferior vestibular area.
- Impulses can influence movements of — eyes, head, neck, trunk

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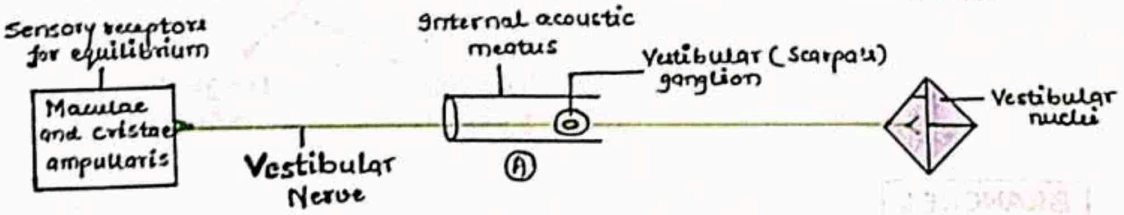

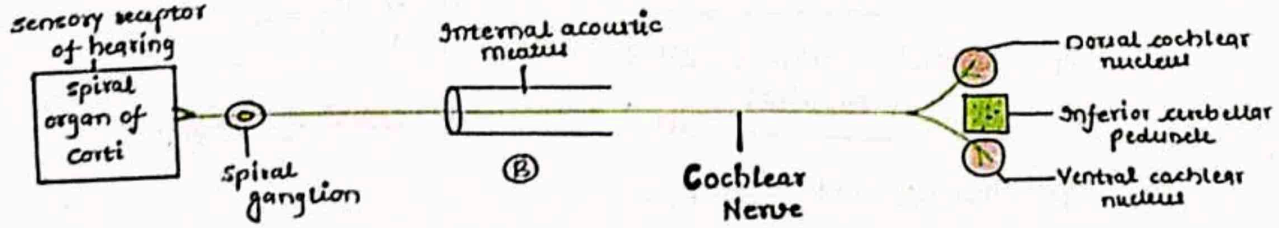

A Vestibular Nerve B Cochlear Nerve

GLOSSOPHARYNGEAL NERVE

- It is the 9th cranial nerve and nerve of third branchial arch.

NUCLEI The 3 nuclei in upper part of medulla are :—

(1) Nucleus Ambigus
(2) Inferior salivatory Nucleus
(3) Nucleus of tractus solitarius

COURSE

Intraneural Course

- Fibers pass forward & laterally, between olivary nucleus & inferior cerebellar peduncle, through reticular formation of medulla.
- At base of brain — Just above rootlets of Vagus nerve, nerve is attached by 3 to 4 filaments to upper part of posterolateral sulcus of medulla.

Intracranial course

- Filaments unite → forms single trunk → passes forward & laterally → JUGULAR FORAMEN
- Nerve is lodged in deep groove → leads to → COCHLEAR CANALICULUS
- Nerve is separated from vagus & Accessory nerves, by inferior petrosal sinus.

Extracranial Course Nerve descends

- Between Internal Jugular vein & Internal Carotid Artery (deep to styloid process & muscles attached to it
- then turns forwards — winding stylopharyngeus & supplies pharynx to give pharyngeal branches
- Enters submandibular region — by passing deep to hyoglossus → breaks into → tonsillar branch, lingular branch.

BRANCHES

(1) Tympanic branch (Jacobson's nerve) → gives off → lesser petrosal nerve
(2) Carotid nerve (nerve of Herring) → supplies → carotid sinus & carotid body.
(3) Pharyngeal branches
(4) Muscular branches → supply → stylopharyngeus

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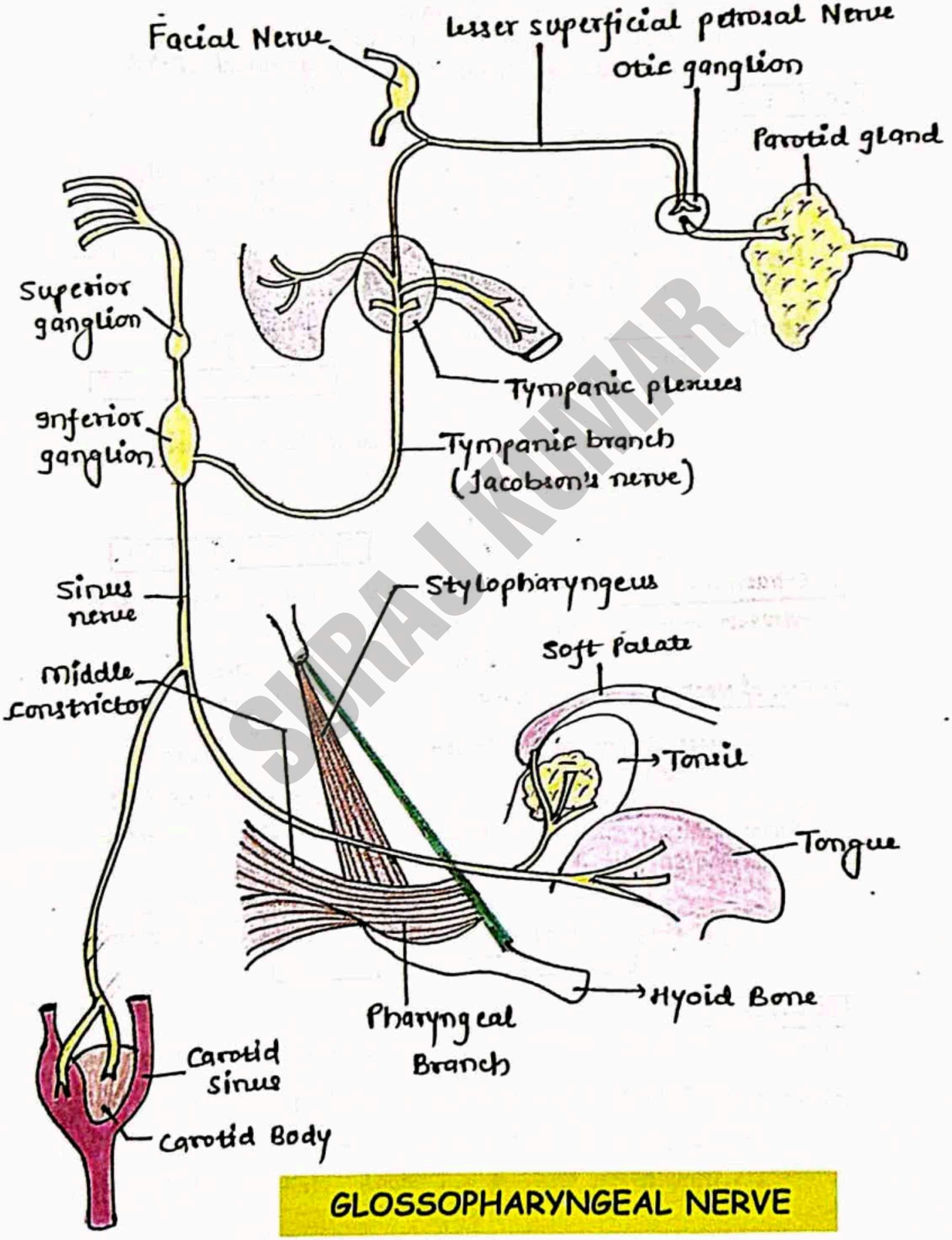
(5) Tonsillar branches <u>supply</u> → tonsil, fauces & palate

(6) Lingual branches <u>supply</u> → posterior 1/3 tongue, vallate papillae



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GLOSSOPHARYNGEAL NERVE

VAGUS NERVE

- 10th cranial nerve
- Named - so for its extensive (vague) course

NUCLEI

(1) Nucleus Ambigus
(2) Dorsal nucleus of vagus
(3) Nucleus of tractus solitarius
(4) Nucleus of spinal tract of trigeminal nerve

COURSE

In Intracranial Course

- Fibers run forward & laterally, between olivary nucleus & inferior cerebellar peduncle; through reticular formation of medulla.
- Nerve attached by about 10 rootlets to posterolateral sulcus of medulla.
- Rootlets → (unites & forms) → large trunk → passes laterally across → **JUGULAR TUBERCLE** with Glossopharyngeal & Accessory nerves → reaches → **JUGULAR FORAMEN**

In Extracranial Course

- Nerve descends within Carotid sheath, between
  - lateral : Internal Jugular vein
  - medial : Common carotid Arteries

At Root of Neck

- Right Vagus → crosses 1st part of subclavian artery → then inclinig medially, behind brachiocephalic vessels → reach Right side of trachea → **THORAX**
- Left Vagus → passing between Left common carotid & Left subclavian arteries behind Internal Jugular & Brachiocephalic veins → **THORAX**

BRANCHES

(1) Meningeal branch → supply → dura mater of posterior cranial fossa

(2) Auricular branch (Arnold's nerve or Alderman's nerve) → supply → floor of meatus & tympanic membrane
- It's stimulation → ear cough, vomiting, cardiac arrest

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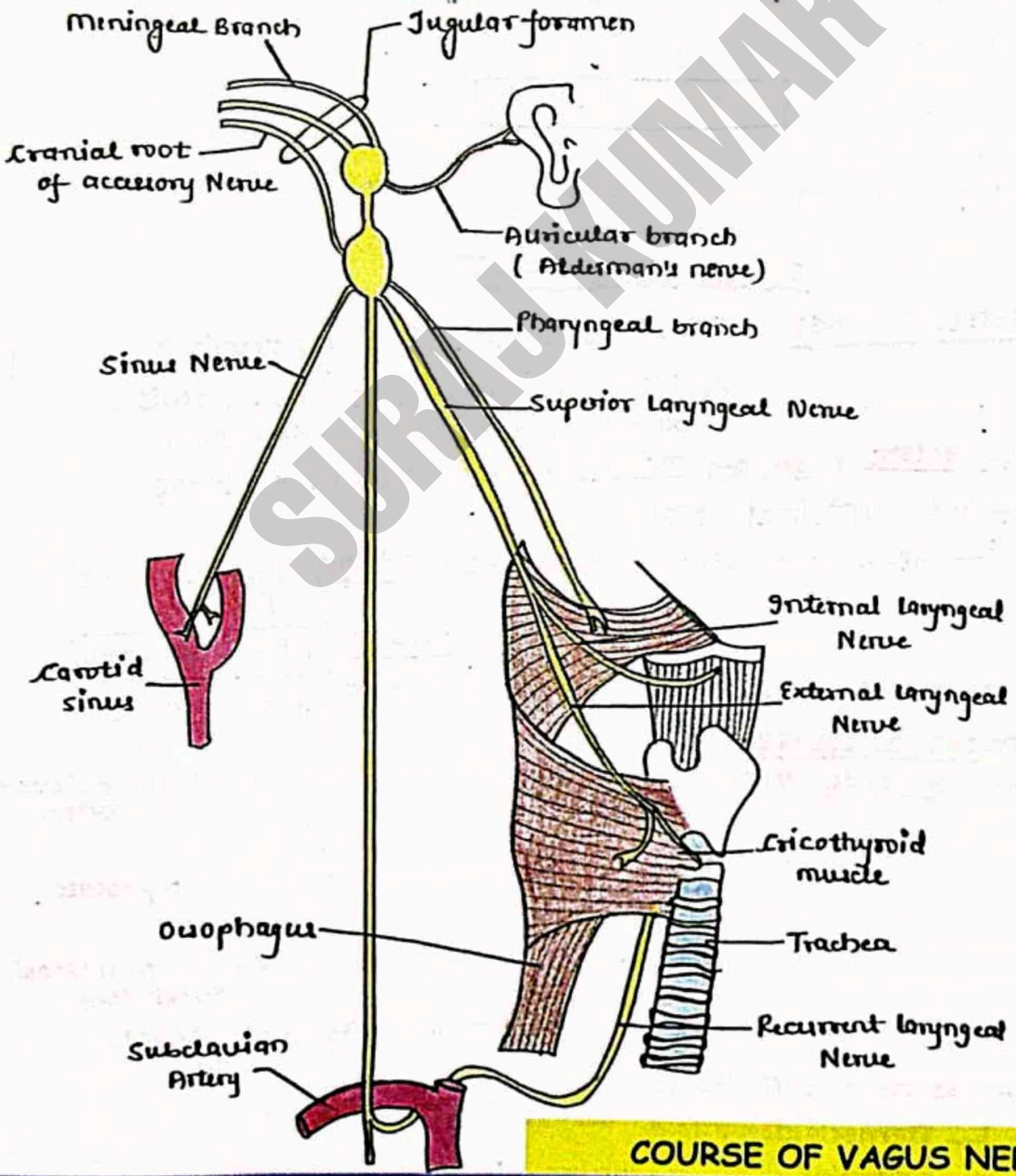
(3) Pharyngeal branch supplies
- → all muscles of pharynx, except – Stylopharyngeus ↑ supplied by Glossopharyngeal nerve
- → all muscles of soft palate, except – tensor palati ↑ supplied by mandibular nerve

(4) Branches to carotid body

(5) Superior laryngeal nerve
- Motor – external laryngeal nerve
- Sensory – internal laryngeal nerve

(6) Superior & inferior cervical cardiac branches

(7) Recurrent laryngeal nerve (nerve of 6th arch).



COURSE OF VAGUS NERVE

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ACCESSORY NERVE

- It is the 11th cranial nerve & has 2 roots → cranial, → spinal

NUCLEI

CRANIAL ROOT — Nucleus Ambiguus
SPINAL ROOT — Long Spinal nucleus

COURSE

CRANIAL ROOT

- Emerges in form of 4 to 5 rootlets — attached to posterolateral sulcus of medulla.
  ↓ joins to form
  single trunk
- Runs laterally with Glossopharyngeal, Vagus & Spinal Accessory
  ↓ crosses
  Jugular tubercle
  ↓ reaches
  Jugular foramen.
- Finally fuses with Vagus — at inferior ganglion, and distributes to muscles of palate, pharynx, larynx & heart.

SPINAL ROOT

In vertebral canal

- Filaments unite → form single trunk → ascends in front of dorsal nerve roots, behind ligamentum denticulatum.
- Nerve enters → cranium through → foramen magnum, lying behind vertebral artery.
  ← nerve runs upwards & laterally → crossing Jugular tubercle & reaches **Jugular foramen**
- Nerve leaves skull through → middle part of Jugular foramen.

In extracranial Course

- Nerve descends, vertically, between internal Jugular vein & Internal carotid artery
  ↓ reaches between
  angle of mandible & mastoid process
  ↓ runs
  downward & backward — superficial to internal jugular vein, surrounded by lymph nodes.
- Nerve is then distributed to
  - Sternocleidomastoid — chin turning muscle
  - trapezius — shrugging muscle.

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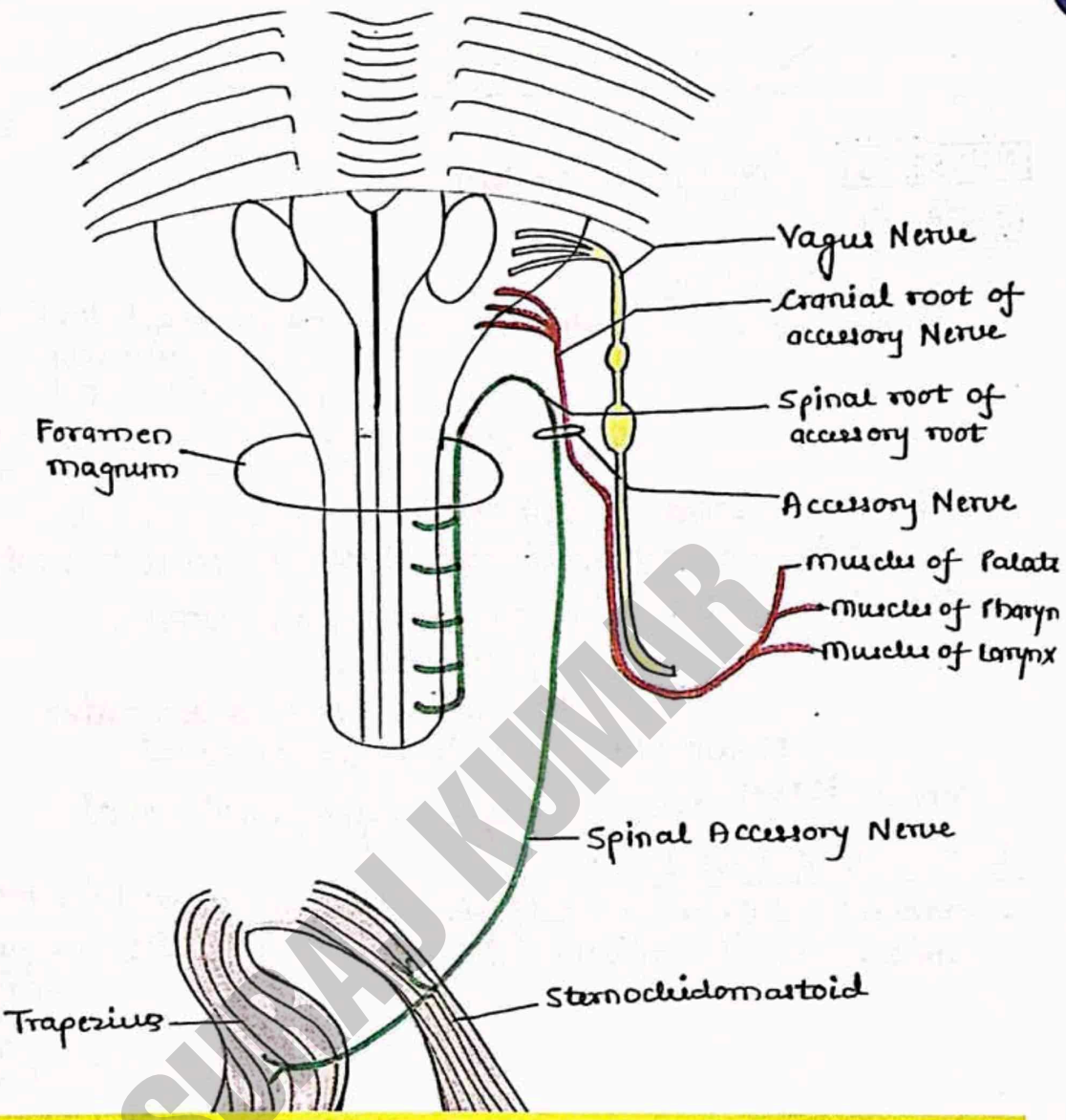

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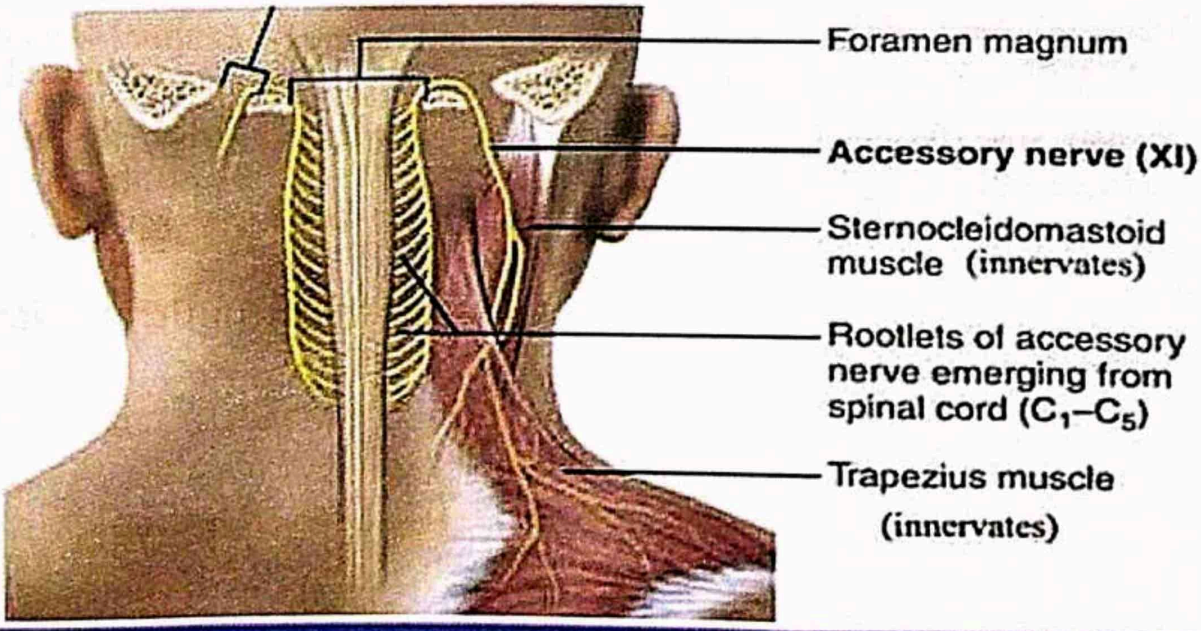
COURSE OF ACCESSORY NERVE



HYPOGLOSSAL NERVE

- It is the 12th nerve and supplies muscles of tongue

NUCLEUS

- Hypoglossal nucleus

COURSE

Intraneural Course

- Fibers runs forwards – lateral → medial longitudinal bundle, medial leminiscus, pyramidal tract
  medial → reticular formation and olivary nucleus.
- Nerve attached to anterolateral sulcus of medulla, between pyramid and olive by 10 to 15 rootlets.
- Rootlets run laterally behind vertebral artery joins to form 2 bundles — pierce dura mater separately near hypoglossal canal.
- Nerve leaves skull through hypoglossal canal

Extracranial Course

- Nerve lies deep to internal jugular vein → inclines b/w internal jugular vein & internal carotid artery crosses → Vagus (laterally in front
  ↓
  deep to parotid gland, styloid process, posterior belly of digastric
- At lower border of posterior belly of digastric – curves forward
  ↓ crosses
  Internal & External carotid Arteries & loop of lingual Artery ← passes deep to ← posterior belly of digastric again ← enters ← Submandibular region
- Nerves runs forward on genioglossus & hyoglossus, deep to submandibular gland & mylohyoid
  enters → tongue to supply — all intrinsic muscles & most extrinsic muscles.

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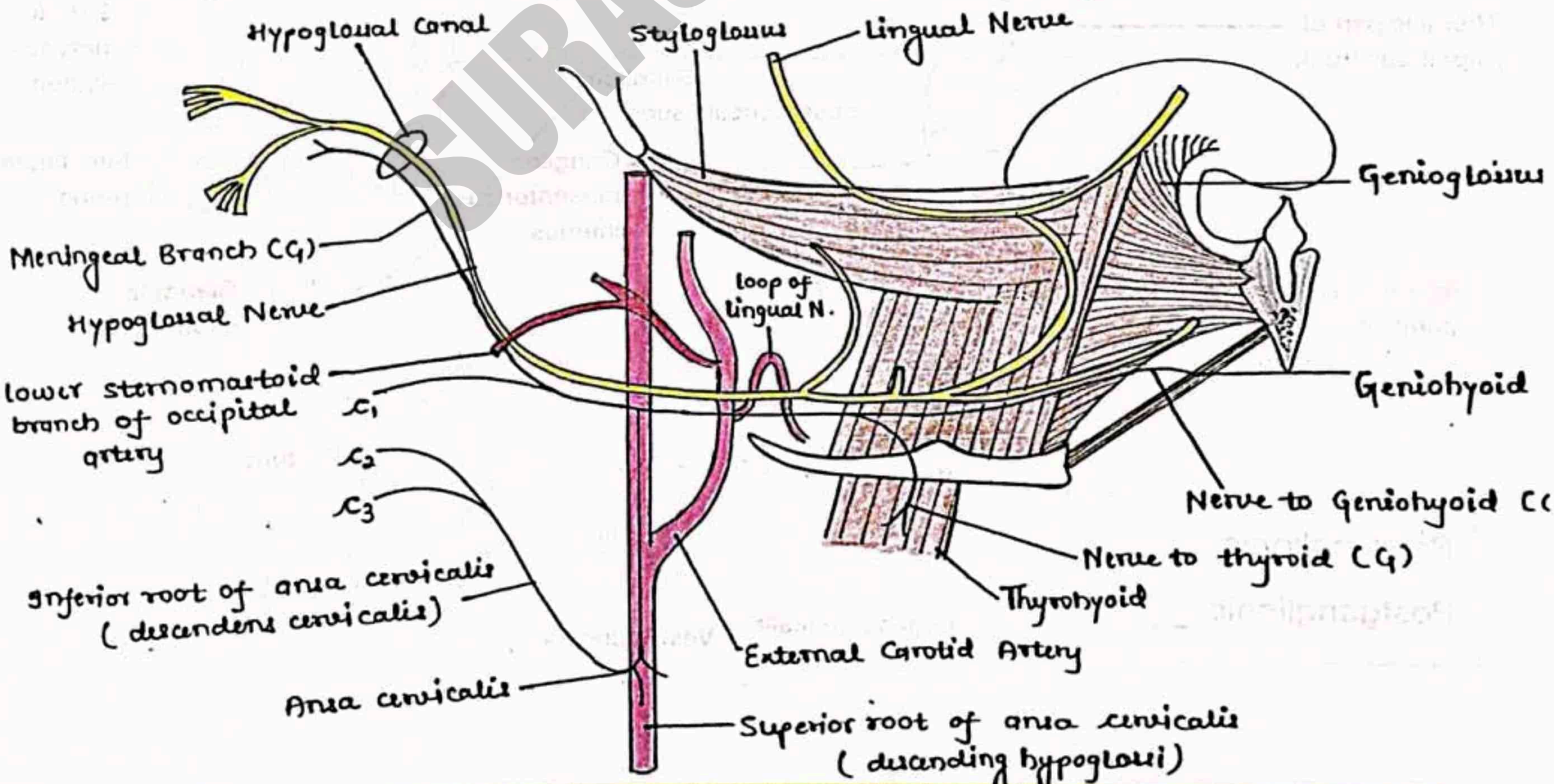
BRANCHES & DISTRIBUTION

- **Branches containing fibres of hypoglossal nerve proper**
  - Supplies all intrinsic muscles & extrinsic muscles
    - Superior longitudinal
    - Inferior longitudinal
    - Transverse & Vertical muscles
    - Styloglossus
    - Genioglossus
    - Hyoglossus
  - ★ Exception : Palatoglossus supplied by → Cranial Accessory Nerve

- **Branches of hypoglossal nerve containing fibers of nerve C**
  - (1) Meningeal branch supplies → bones & meninges in anterior part of Posterior Cranial fossa
  - (2) Descending branch continues → Descendens hypoglossi in ansa cervicalis
  - (3) Branches also supplies → thyrohyoid muscle, Geniohyoid muscle.



COURSE OF HYPOGLOSSAL NERVE

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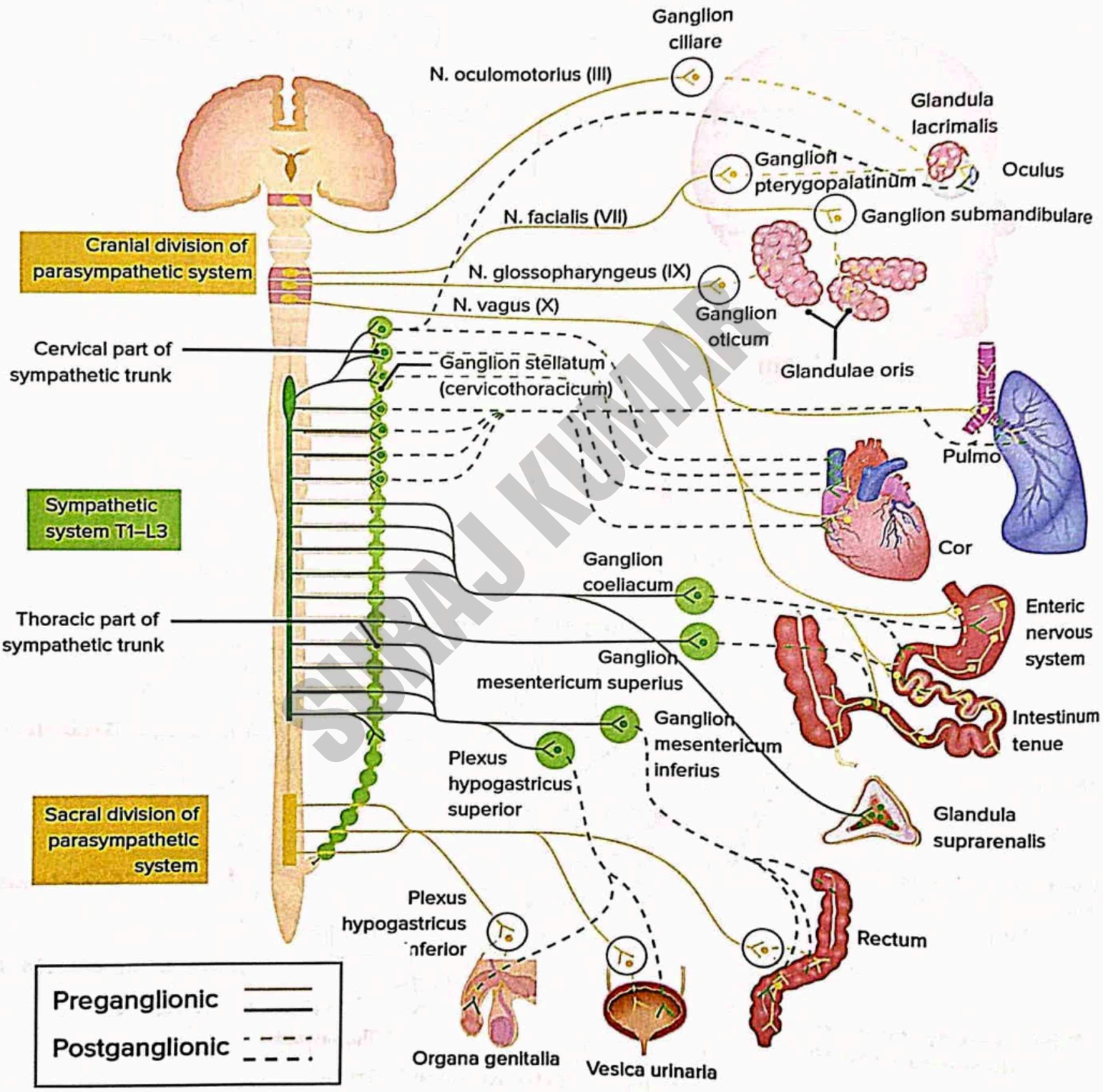
PARASYMPATHETIC & SYMPATHETIC GANGLIONS



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SUBMANDIBULAR ganglia

- It is a parasympathetic ganglion, which serves as a relay station for secretomotor fibres supplying submandibular and sub-lingual salivary glands.
- Topographically connected to :- Trigeminal Nerve
  Functionally connected to :- Facial Nerve

Location

→ Size → Pin-head
→ Situation → on outer surface of hyoglossal muscle.

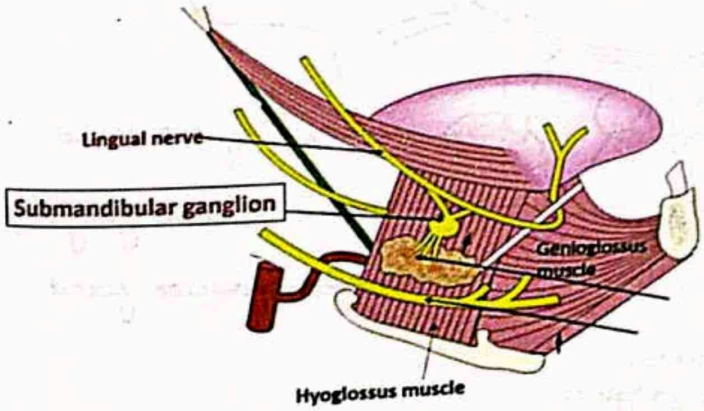
Relations

Above → Lingual Nerve
Below → Deep part of submandibular gland
Medial → Hyoglossus muscle
Lateral → Submandibular gland

Roots

→ Has 3 roots :-

1. Parasympathetic Root :- Superior Salivatory Nucleus
   ↓
   Nervus Intermedius
   ↓
   Facial Nerve
   ↓
   Chorda tympani Nerve
   ↓
   Joins Lingual Nerve
   ↓
   Submandibular Ganglion
   ↓ Relay
   Postganglionic Fibres
   ↓
   Submandibular and Sublingual glands



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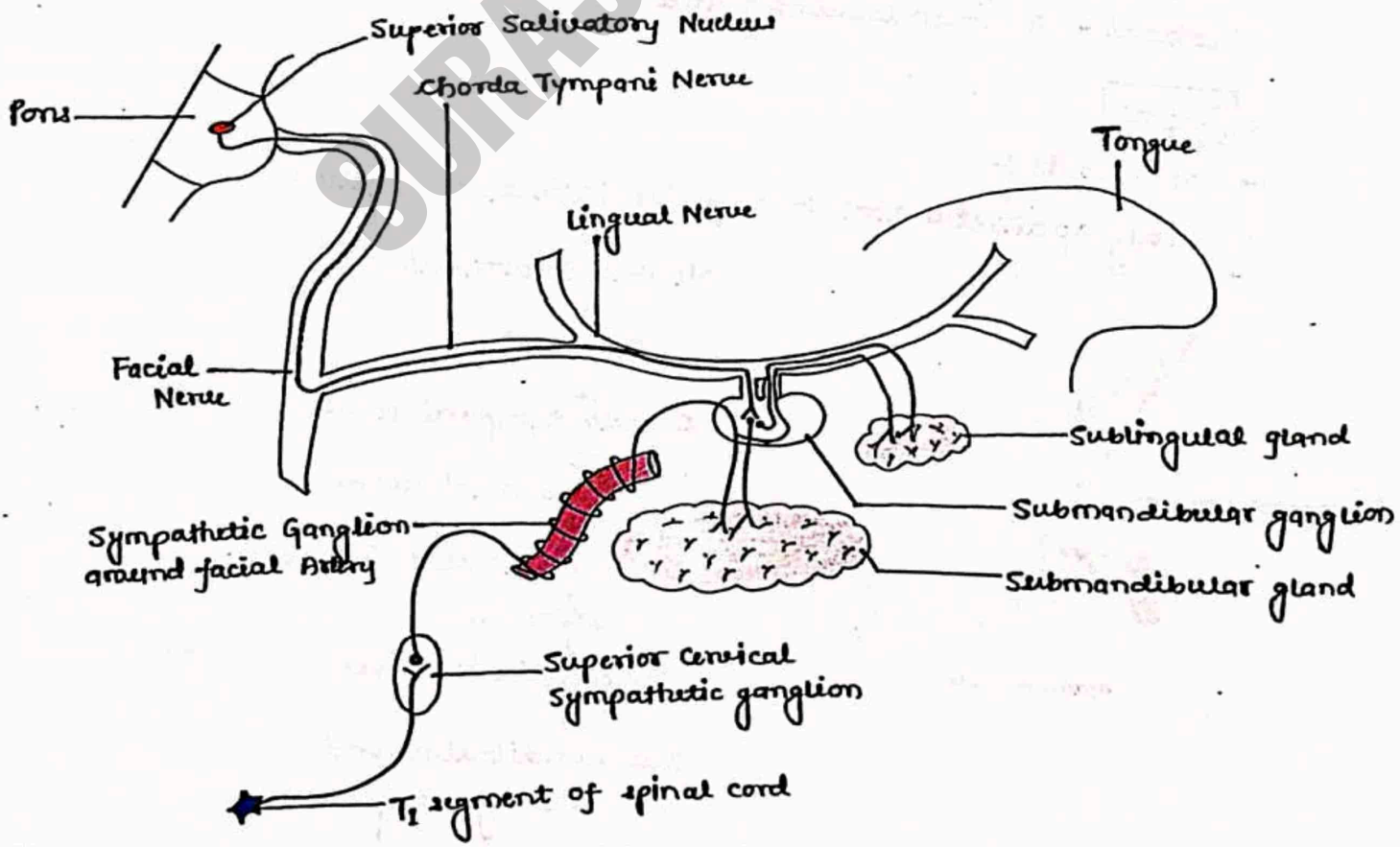
2. Sympathetic Root :- Sympathetic Plexus around facial Artery
↓
First thoracic spinal segment (T_1)
(pre ganglionic fibres)
↓
Superior cervical Sympathetic Ganglion
(post ganglionic fibres)
↓
Plexus around facial Artery
↓
Pass through ganglion without relay
↓
Supply Submandibular and Sublingual
Salivary glands

3. Sensory Root → Derived from lingual Nerve

- **Branches**

→ 5 to 6 :- to supply submandibular gland
→ others :- join sublingual Nerve to supply sublingual & anterior lingual gland.



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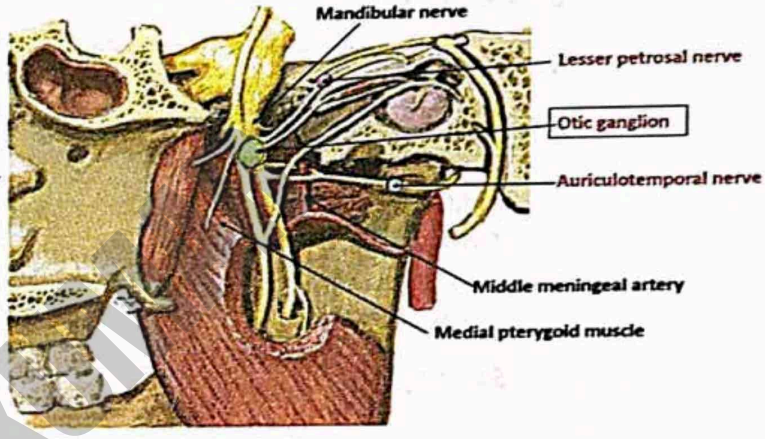
OTIC GANGLIA

- Small parasympathetic ganglion connected to mandibular div. of Trigeminal N.
- Provide relay station to secretomotor fibres to parotid gland.
- Topographically connected to :- Mandibular Nerve
- Functionally connected to :- Glossopharyngeal Nerve



- **Size**

→ Pin-head

- **Location**

→ Infratemporal fossa

- **Relation**

Lateral → Mandibular Nerve
Medial → Tensor Palati muscle
Posterior → Middle Meningeal Artery
Anterior → Medial Pterygoid Muscle

- **Roots**

1. Parasympathetic Root :- Inferior Salivatory Nucleus
↓
Glossopharyngeal N. (IX)
↓
Tympanic Br. of IX N. (Jacobson's N.)
↓
Tympanic Plexus
↓
Lesser Petrosal Nerve
↓
Otic Ganglion
↑ Relay
↓
Postganglionic fibres through auriculo-temporal N.
↓
Supply Parotid gland

2. Sympathetic Root :- Sympathetic plexus around Middle Meningeal Artery
↓
T_1 & T_2 spinal segment
(pre ganglionic fibres)
↓
Superior cervical sympathetic Ganglion

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↕

Plexus around Middle Meningeal Artery

↓

Pass through ganglion without relay

↓

Supply parotid gland (vasomotor in nature & cause thick salivary secretions)

3. Sensory Root :- From auriculotemporal Nerve

4. Somatic Motor :- Nerve to medial pterygoid

- **Branches**

Postganglionic Parasympathetic
Postganglionic Sympathetic
Sensory

} All supply parotid gland through auriculotemporal Nerve

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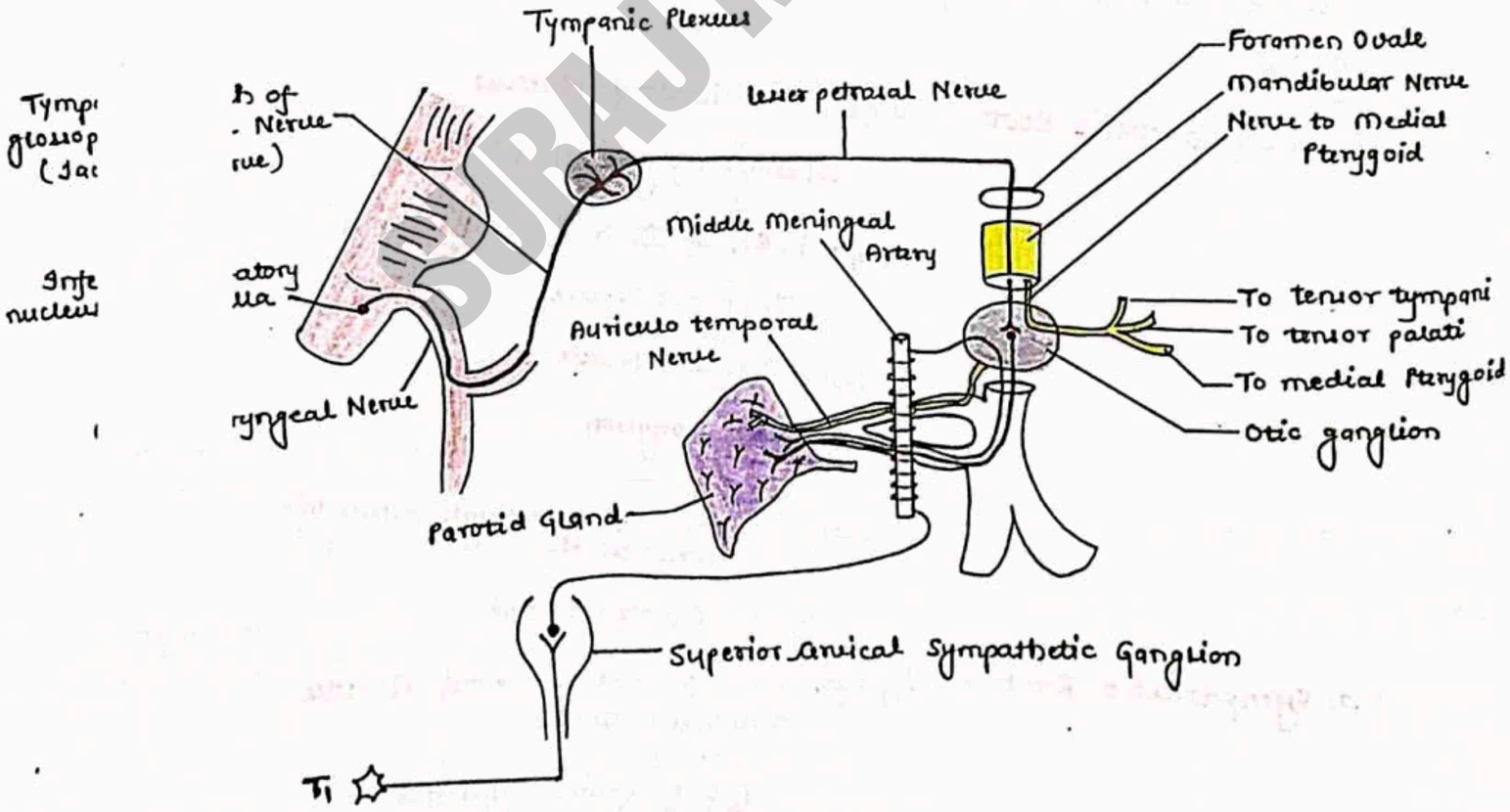

ROOTS & DISTRIBUTION OF OTIC GANAGLION

CILIARY GANGLIA

- Peripheral parasympathetic ganglion —connected with→ Nasociliary N.

- Topographically Connected to :- Nasociliary Nerve
- Functionally Connected to :- Oculomotor Nerve

- **Location**

→ Minute body lying near apex of orbit between optic and lateral rectus muscle.

- **Roots**

1. Parasympathetic Root → Edinger Westphal Nucleus
↓
III Nerve
↓
Nerve to Inferior oblique
↓
Branch to ciliary ganglion
↓
Short ciliary nerve supplies ciliaris & constrictor pupillae muscle.

2. Sympathetic Root :- Plexus around Internal Carotid Artery
↓
Superior Cervical Sympathetic Ganglion
(Post ganglionic fibres)
↓
Passes through ganglion without relay
↓
Short ciliary Nerve
↓
Supply dilator pupillae and Blood vessels of Eyeball

3. Sensory Root :- Derived from Nasociliary Nerve

- **Branches**

→ Branches are :- Short Ciliary Nerves (8-10)

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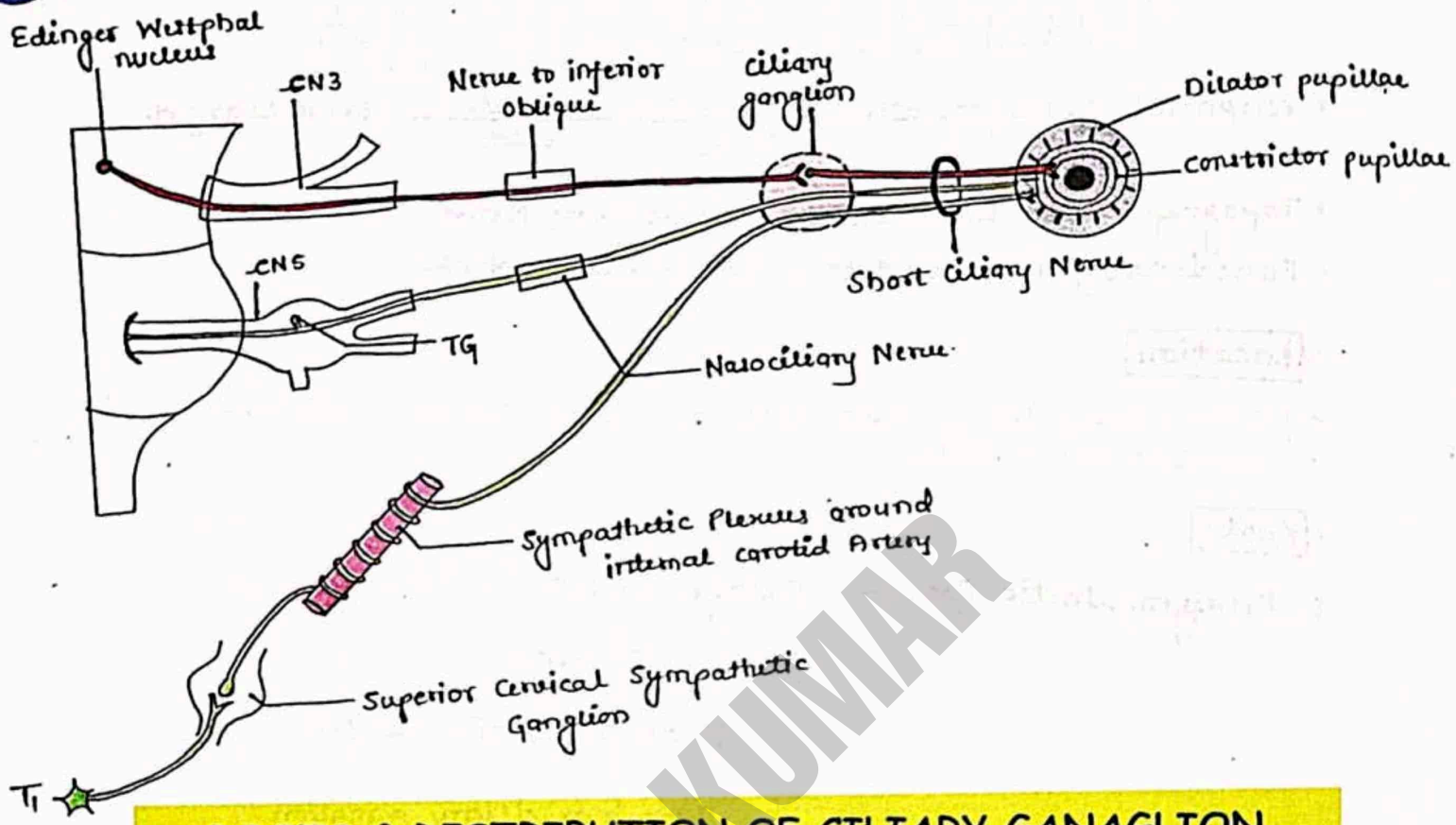

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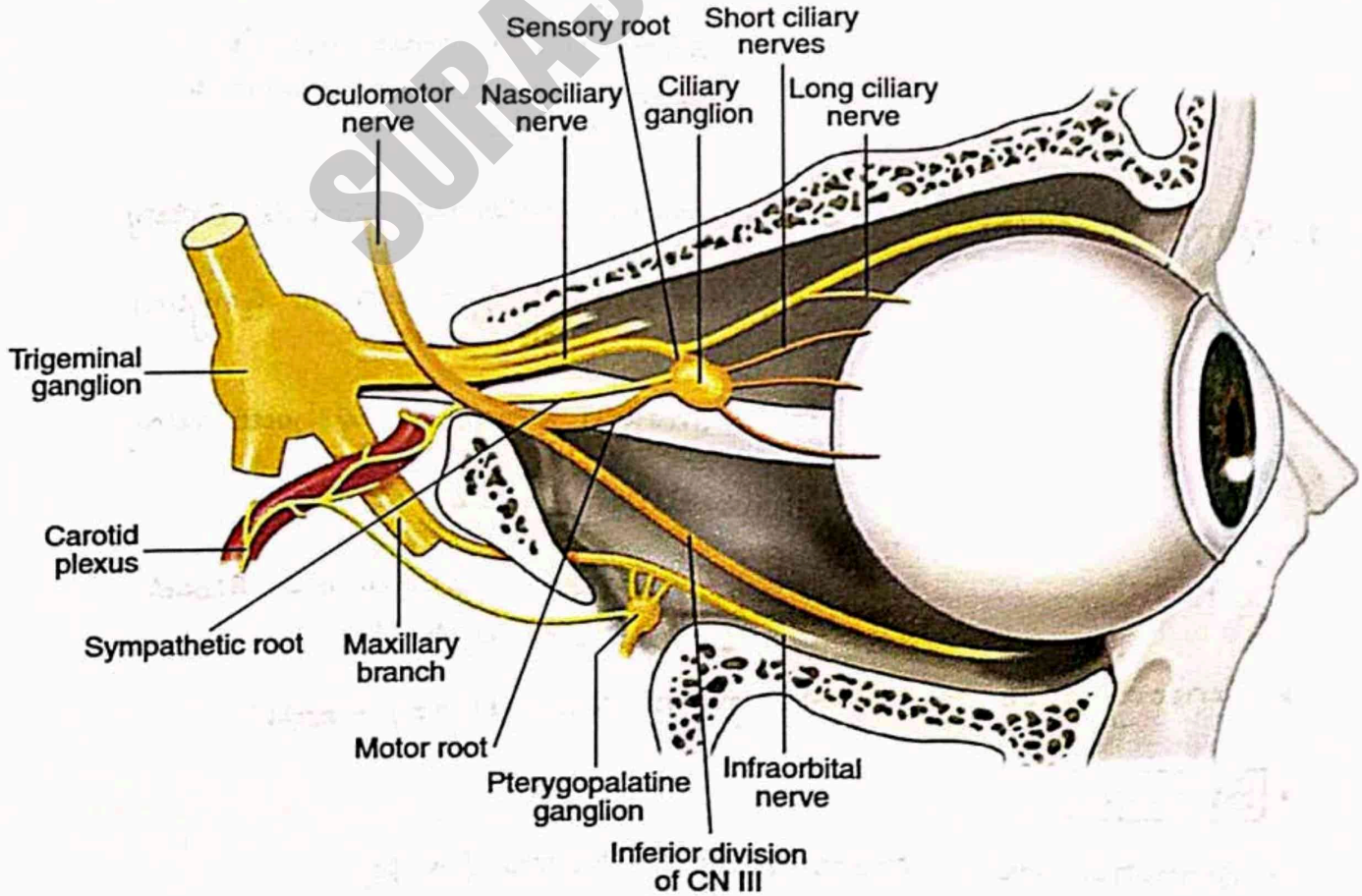
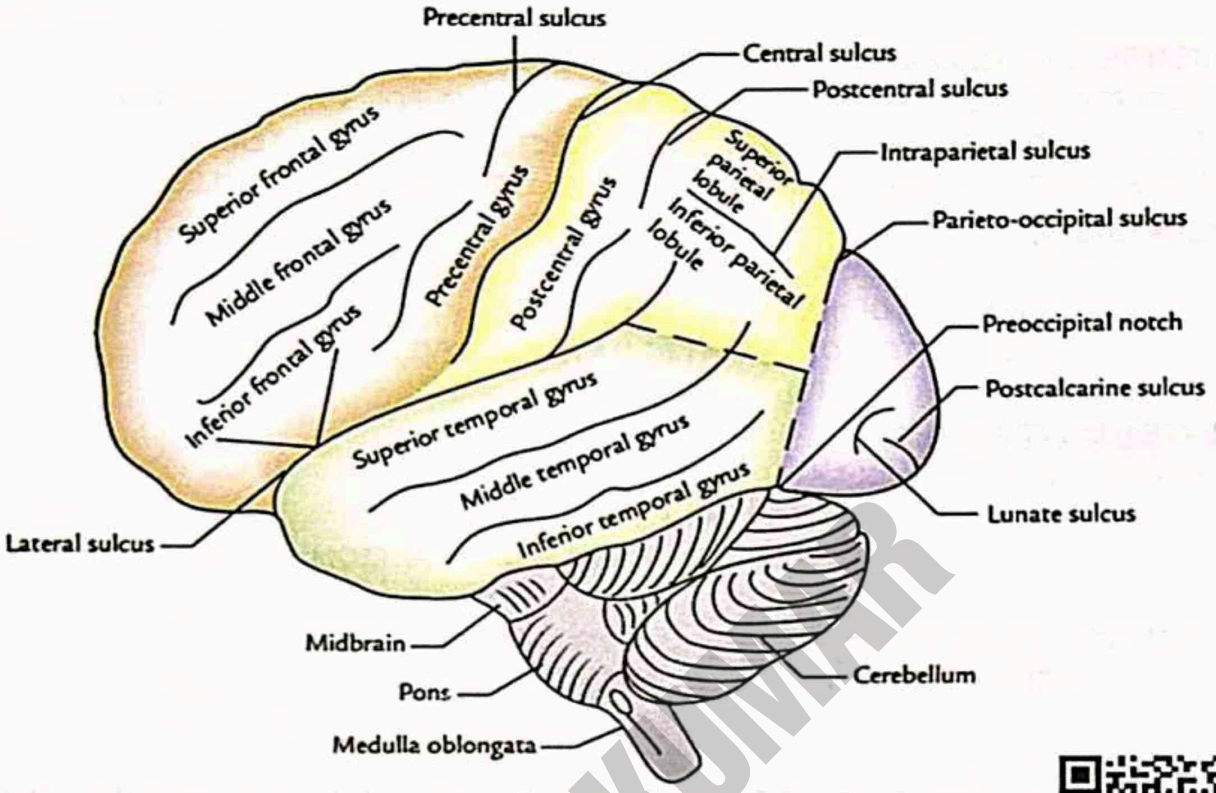
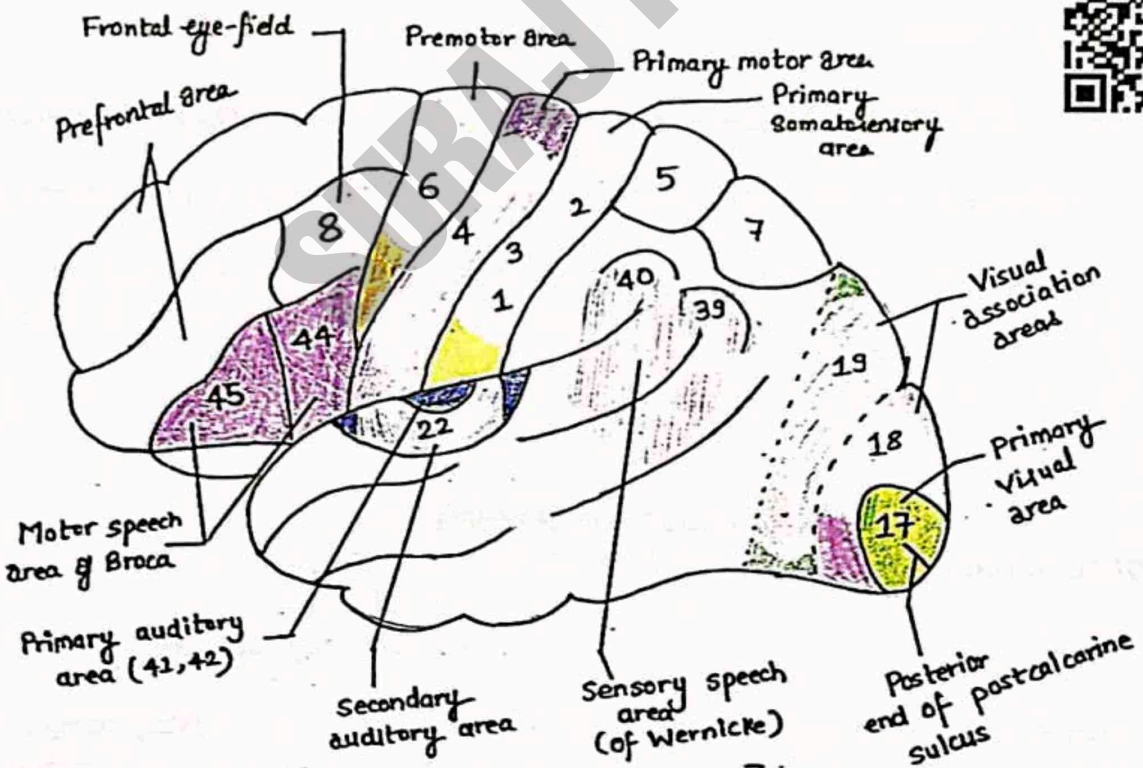
ROOTS & DISTRIBUTION OF CILIARY GANAGLION



SUPEROLATERAL SURFACE OF SURFACE





[∵ Functional cortical areas on superolateral surface of cerebral hemisphere.]

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CEREBELLUM ANATOMY

- CEREBELLUM IS LARGEST PART OF HIND BRAIN
- LOCATED IN POSTORIOR CRANIAL FOSSA BEHIND PONS AND MEDULLA
- EACH HEMISPHERE OF CEREBELLUM CONNECTED TO BRAINSTEM BY 3 PAIR OF 19 LARGE FIBRE TRACT CALLED CEREBULLAR PEDUNCLES

FUNCTION:-

- MAINTANACE OF EQUILIBRIUM
- REGULATION OF MUSCLE TONE
- COORDINATION OF SOMATIC MATAR ACTIVITIES

***EXTERNAL FEATURES**

- PARTS - 3X PART - 2X LATERAL HEMISPHERICAL LOBES
  -2X CEREBELLAR HEMISPHERE
  -NARROW MEDIAN WARN LIKE PARTION - VERMIS

-SURFACE - SURPERIOR SURFACE
-INFERIOR

*NOTCHES :-
-ANTERIOR ASPECT OF CEREBELLUM MARKED BY WIDE SHALLOW ANTERIOR CEREBELLAR NOTCH
- POSTERIOR CEREBELLAR NOTCH IS, DEEP AND NARROW

*FISSURE. - HORIZONTAL FISSURE → ALONG LATERAL AND POSTERIAR MARGIN OF CEREBELLAM
*MARKS JUNCTION B/W - SUPERIOR AND INFERIOR SURFACE OF CEREBELLUM.

-POSTEROLATERAL SURFACE → INFERIOR SURFACE OF CEREBELLUM
- SEPARATE - FLOCCULONODULAR LOBE FROM REST OF CEREBELLUM

-V-SHAPED FISSURE PRIMA - DIVIDE CEREBELLUM INTO ANTERIOR AND POSTERIOR LOBES

SUB-DIVISION OF CEREBELLUM:

ANATOMICAL SUB-DIVISIONS:- INTO 3X LOBES - ANTERIOR
- POSTERIAR
- FLOCCULONODULAR

1) ANTERIOR LOBE- LIE ON SUPERIOR SURFACE ANTERIOR TO FISSURE PRIMA
2) POSTERIOR/MIDDLE LOBE:- BETWEEN FISSURA PRIMA AND POSTEROLATERAL
3) FLOCCULONODULAR LOBE :- SMALLEST OF ALL
- LIES ON INFERIOR SUFACE

MORPHOLOGICAL SUB-DIVISIONS :-

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*BASED ON PHYLOGENETIC AND FUNCTIONAL CRITERIA

1) ARCHICEREBULLUM (VESTIBULAR CEREBELLUM)

- OLDEST PART OF CEREBELLUM
- CONSIST OF FLOCCULO NODULAR AND LINGULA
- FUNCTION- MAINTANANCE OF EQUILIBRIUM, TONE

2) PALEO CEREBELLUM (SPINAL CERE BELLUM) -

- PHYLOGENETICALLY, IT IS NEXT PART OF CEREBELLUM TO APPEAR
- CONSIST OF ANTERIOR LOBE (EXCEPT LINGULA) AND PYRAMID AND UVALA OF INFERIOR VERMIS
- FUNCTION- CONCERNED WITH TONE, POSTURE AND CRUDE MOVEMENT OF LIMB

3) NEOCEREBELLUM (CEREBRAL CEREBALLUM) -

-PHYLOGENETICALLY-MOST RECENT PART OF CEREBELLUM TO DEVELOP
-MADE UP OF- MIDDLE LOBE, LARGE PART OF CEREBELLUM
-CHIEFLY CORTICO PONTOCEREBELLAR IN CONNECTION
-CONCERNED WITH SMOOTH PERFORMANCE OF SILKED VOLUNTARY MOVEMENT

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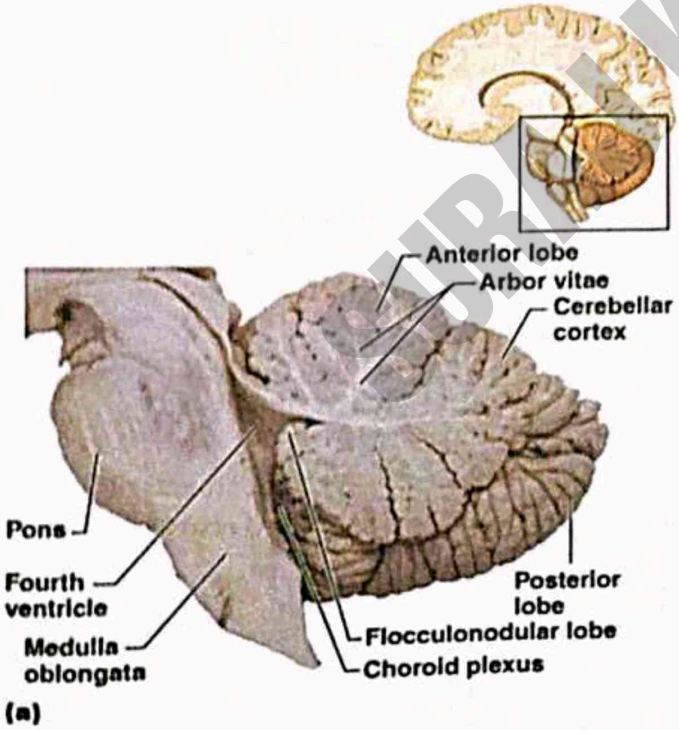

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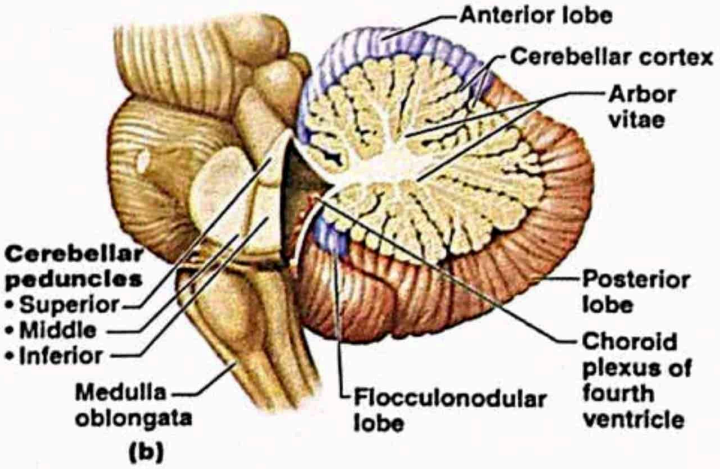

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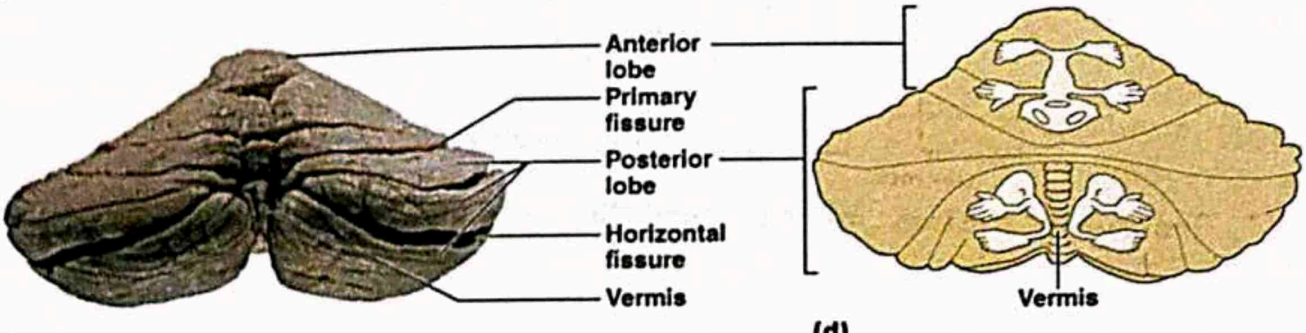

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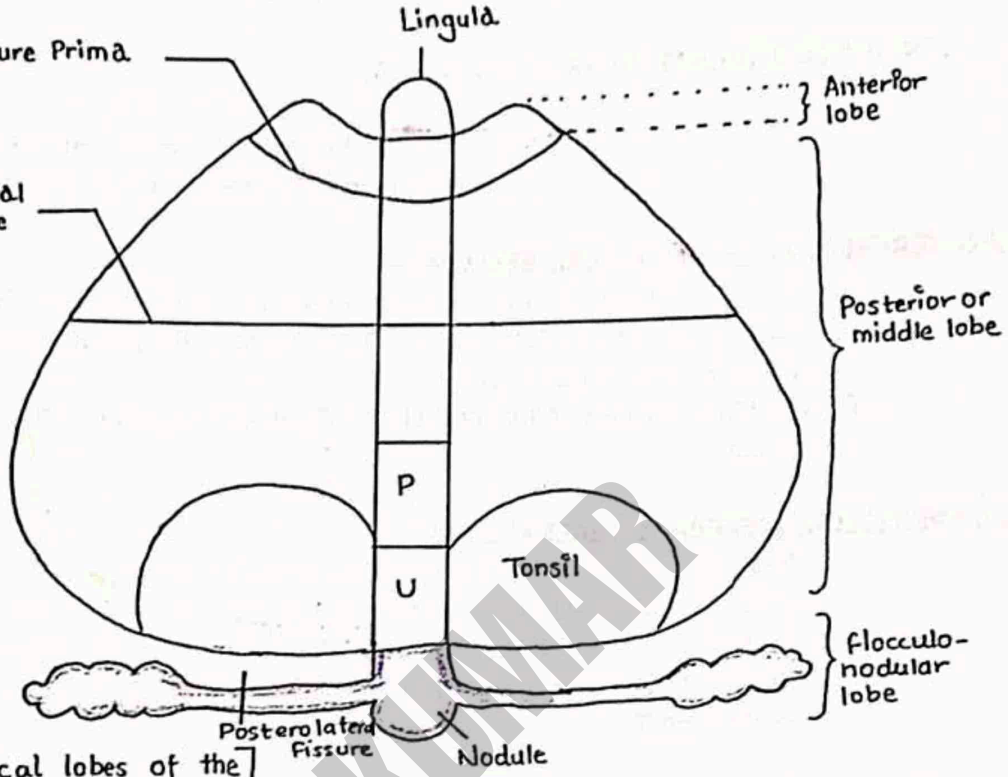
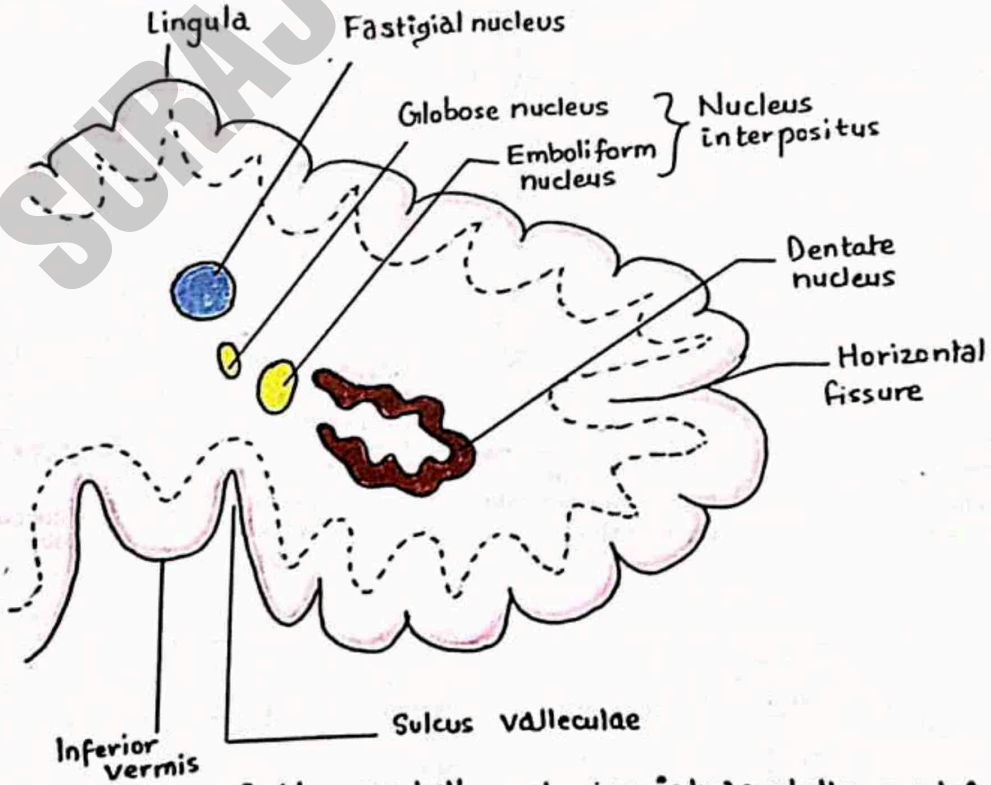


[∵ Anatomical lobes of the Cerebellum]

CEREBELLUM ANATOMY



∵ Transverse section of the cerebellum showing intracerebellar nuclei.

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LATERAL VENTRCILE

- TWO IRREGULAR CAVITIES IN CEREBRAL HEMISPHERE
- COMMUNICATION THIRD VENTRIDE THROUGH - INTERVENTRICULAR FORAMEN (FORAMEN OF MONRO]
- DIVIDE - CENTRAL PART AND 3 HORNS - ANT,POST,INF. HORN

*CENTRAL PART

-EXTEND INTERVENTRICULAR FORAMEN TO SPLENIUM OF CORPUS CALLOSUM

- BOUNDARIES :- ROOF - CORPUS CALLOSUM
  FLOOR - BODY OF CAUDATE NUCLEUS
  - THALAMOSTRIATE VEIN & THALAMUS
  MEDIAL WALL - SEPTUM PELLUCIDUM
  - BODY OF FORNIX

1) ANTERIOR HORN :-

- EXTEND - FRONT OF INTER VENTRICULAR FORAMEN EXTEND INTO FRONTAL LOBE

*BOUNDARIES - ANTERIOR→ POSTERIOR SURFACE OF GENU OF CORPUS CALLOSUM
ROSTRUM OF CORPUS CALLOSUM
- ROOF - CORPUS CALLOSUM
- FLOOR - CALLDATE NUCLEUS
- MEDIAL - SEPTUM PELLUCIDUM ,COLUMN OF FORNIX

2) POSTERIOR HORN

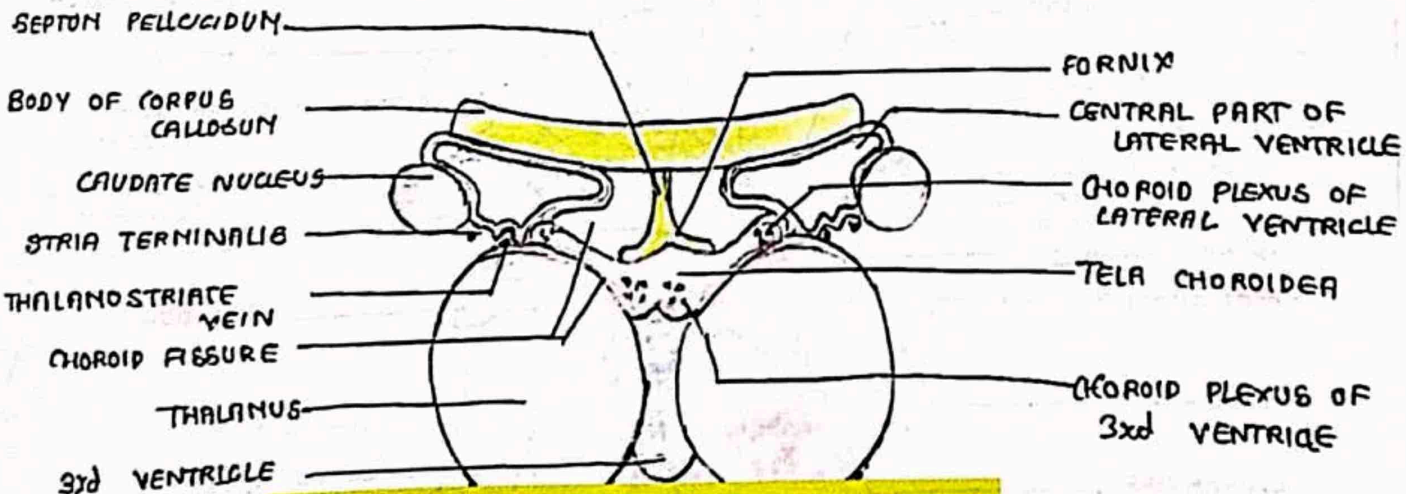
- LIE BEHIND SPLENIUM OF CARPUS CALLOSUM AND EXTEND INTO OCCIPITAL LOBES

*BOUNDARIES - FLOOR AND MEDIAL WALL → FORCEP MAJOR
- ROOF AND LATERAL WALL → TAPETUM OF OPTIC RADIATION

3)I NFERIOR HORNS

- BEGINS AT JUNCTION OF CENTRAL PART WITH POSTERIOR HORN INTO TEMPORAL LOBE

*BOUNDARIES - ROOF AND LATERAL WALL - TAIL OF CAUDATE NUCLEUS, AMYGDALOID
FLOOR → HIPPOCAMPUS



THIRD VENTRCILE

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THIRD VENTRCILE

-DIENCEPHALON CAVITY IS A -→ MEDIAN CLEFT B/W TWO THALAMI
-THIRD AND LATERAL VENTRICLE SECRETE CEREBRO-SPINAL FLUID

COMMUNICATIONS -

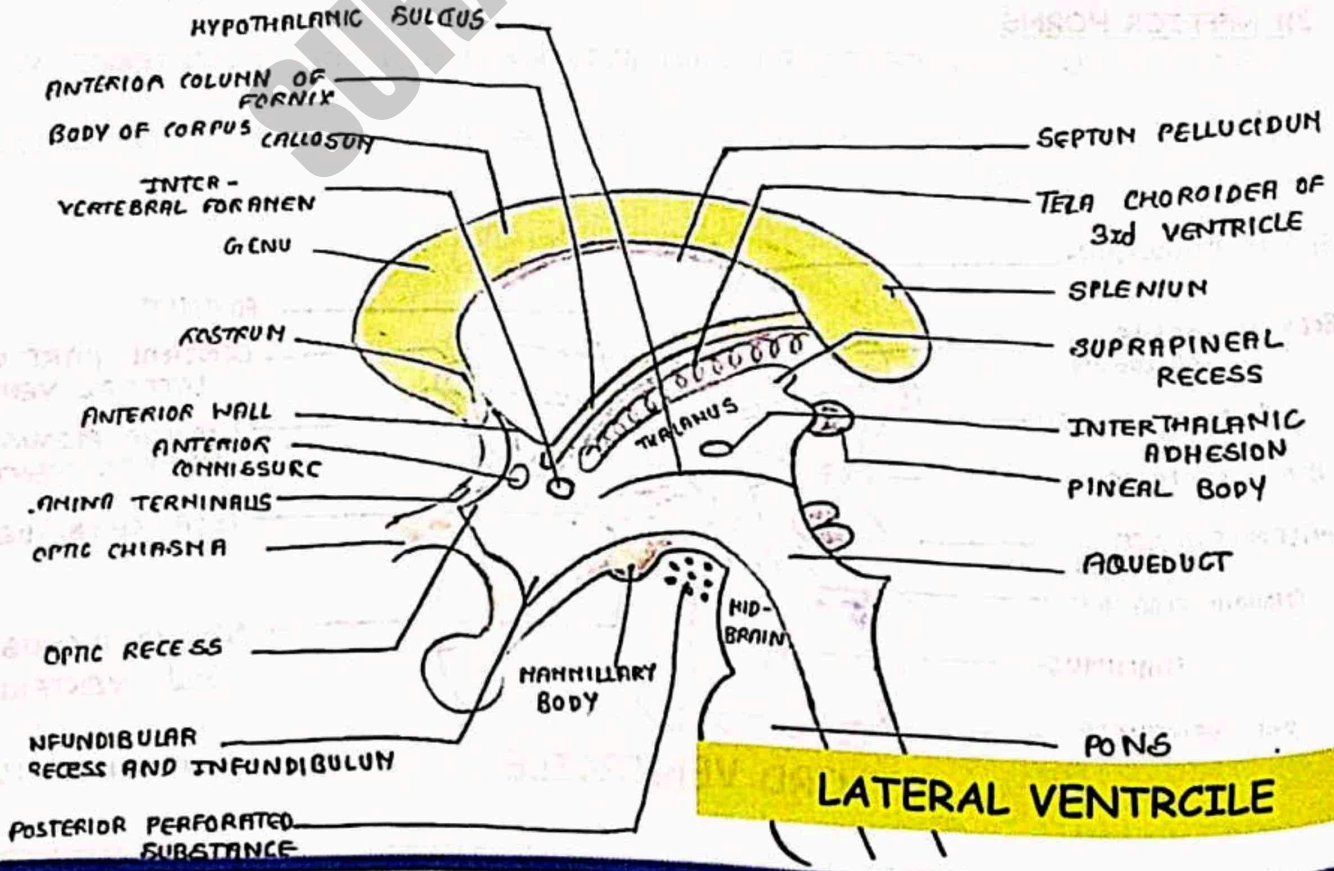
* ANTEROSUPERIORLY - LATERAL VENTRICLE THROUGH, FORAMEN OF MONRO.
* POSTEROINFERIORLY - FOURTH VENTRICLE THROUGH, CEREBRAL AQUEDUCT

BOUNDARIES

* ANTERIOR WALL - LAMINA TERMINALIS
  - ANTERIOR COMMISSURE
* POSTERIAR WALL - PINEAL BODY
  - POSTERIOR COMMISSURE
  - CEREBRAL AQUEDUCT
* ROOF - BODY OF FORNIX
  - TELA CHOROIDEA
* FLOOR - MAMMILLARY BODIES
  - INFUNDIBULUM (PITUITARY STALK)
  - OPTIC CHIASMA
  - TEGMENTUM OF MIDBRAIN
* LATERAL WALL - MEDIAL SURFACE OF THALAMUS
  - HYPOTHALAMUS

***RECESSES:** RECESSES ARE EXTENSIONS OF CAVITY

-SUPRAPINEAL
-PINEAL
-INFUNDIBULAR
-OPTIC



LATERAL VENTRCILE

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YouTube

FOURTH VENTRICLE

-CAVITY OF HIND BRAIN

-TENT-SHAPED SPACE BETWEEN PONS AND UPPER PART OF MEDULLA OBLONGATA

*STRUCTURAL LAYERS: FLOOR IS LINED BY - EPENDYMA
- THIN LAYER OF NEUROGLIA
- LAYER OF GREY MATTER

*BOUNDARIES:-

1) SUPEROLATERALLY -> SUPERIOR CEREBELLAR PEDUNCLE

2) INFEROLATERALLY -> INFERIOR CEREBELLAR PEDUNCLE
- GRACILE TUBERCLES
- CUNEATE TUBERCLE

3) FLOOR-->ALSO CALLED RHOMBOID FOSSA
- FORMED BY - DORSAL SURFACE OF LOWER PONS
- DORSAL UPPER OF MEDULLA OBLONGATA

-PART - IT IS DIVISIBLE INTO
- UPPER TRIANGULAR PART FORMED BY → DORSAL SURFACE OF PONS
- LOWER TRIANGULAR PART FORMED BY → DORSAL SURFACE OF MEDULLA
- INTERMEDIATE PART →JUNCTION OF PONS AND MEDULLA
- PART IS MARKED BY TRANSVERSELY RUNNING FIBRE → STRIA MEDULLARIS

*FEATURE OF FLOOR -
- DORSAL MEDIAN SULCUS - DIVIDE FLOOR → INTO 2 SYMMETRICAL HALVES
- SULCUS LIMITANS - DIVIDE EACH HALF - MEDIAN & LATERAL EMINENCE
- SULCUS LIMITANS PRESENTS DEPRESSION- AT CRANIAL END CALLED - SUPERIOR FOVEA
-AT CAUDAL END CALLED - INFERIOR FOVEA
- MEDIAL EMINENCE → FACIAL COLLICULUS - DEEP TO F.C IS GENU OF FACIAL NERVE
- UPPERMOST PART OF SULCUS LIMITANS → BLUISH IN COLOUR → CALLED LOCUS COERULEUS
- DESCENDING FROM INFERIOR FOVEA SULCUS RUN OBLIQUE - TOWARD →MIDLINE DIVIDES MEDIAL EMINENCE INTO 2 TRIANGLE - HYPOGLOSSAL & VAGAL TRIANGLE

4) ROOF - DIAMOND SHAPE & DIVIDE INTO SUPERIOR , INFERIOR PART

*SUPERIOR/CRANIAL PART - FORM BY - SUPERIOR CEREBELLAR PEDUNCLES
- SUPERIOR MEDULLARY VELUM
-SCP EMERGING -> WHITE MATTER OF CEREBELLUM REACH, INFERIOR COLLICULI
-SMU MADE UP OF -> NERVOUS TISSUE B/W SCP
*INFERIOR / CAUDAL PART FORM BY - VENTRICULAR EPENDYMA AND
- DOUBLE FLOD OF PIA MATTER

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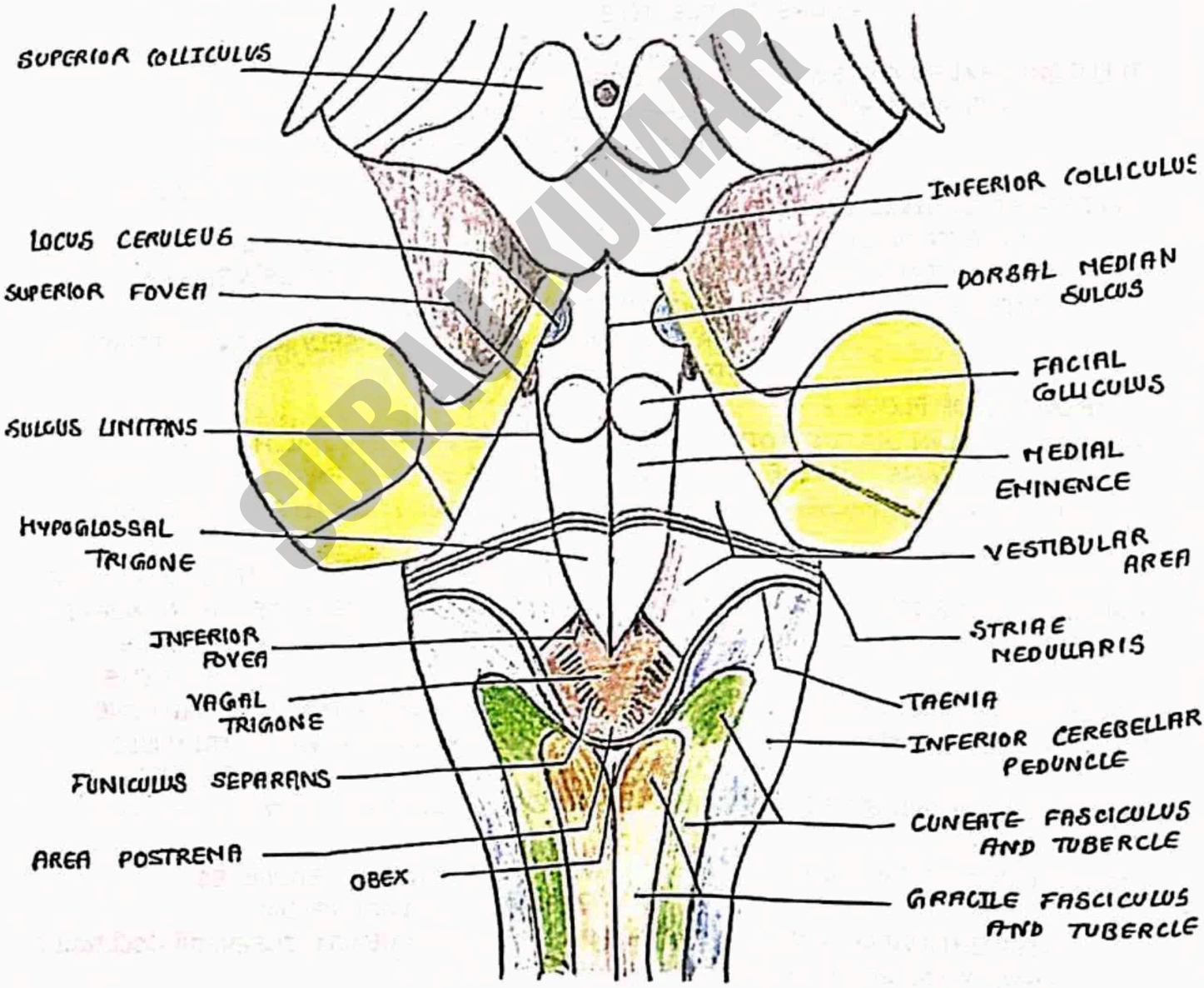
*COMMUNICATION: SUPERIORLY - CEREBRAL AQUEDUCT
INFERIORLY - CENTRAL CANAL

*OPENINGS IN ROOF :- 3X OPENING - 2X- LATERIAL - FORAMEN OF MAGENDIE
- 1X MEDIAN - FORAMINA OF LUSCHKA

*RECESSES OF FOURTH VENTRICLE:-
- 2X LATERAL RECESSES
- 1X MEDIAN DARSAL RECESSES
- 2X LATERAL DORSAL RECESSES



FOURTH VENTRICLE

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THALAMUS

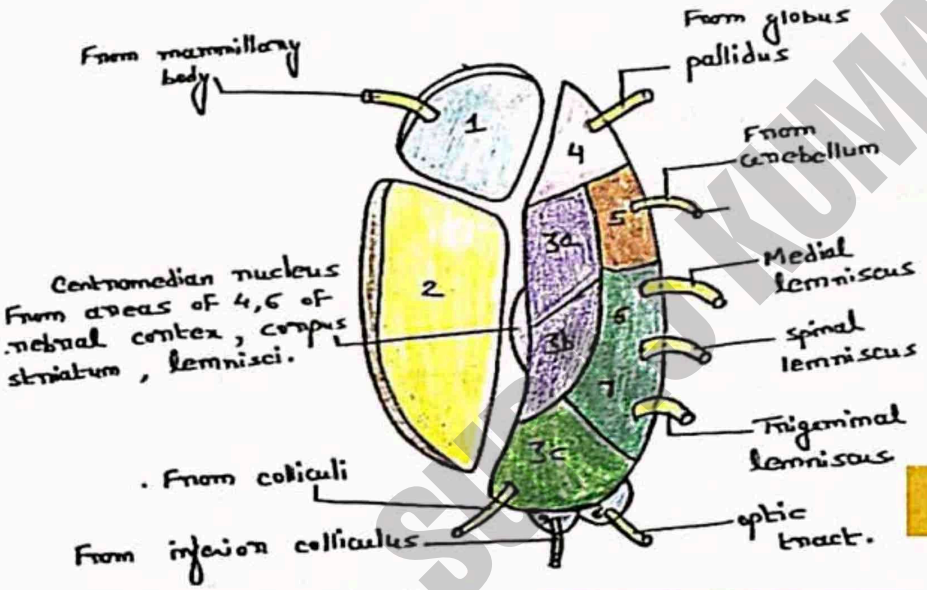
*THALAMUS-→LARGE MASS OF GREY MATTER
*4CM IN EACH - TRANSVERSE
- VERTICAL
- ANTERO-POSTERIOR DIAMETER

*SITUATED-→IN LATERAL WALL OF 3RD VENTRICLE & FLOOR OF CENTRAL PART OF →LATERAL VENTRICLE

STRUCTURE & NUCLEI OF THALAMUS -

1.WHITE MATTER-→EXTERNAL MEDULLARY LAMINA--→COVER LATERAL SURFACE
-INTERNAL MEDULLARY LAMINA--→DIVIDE THALAMUS--→IN 3x PARTS
-ANTERIOR
-MEDIAL
-LATERAL

2.GREY MATTER - DIVIDED INTO 7x NUCLEI



1. Anterior Nucleus.
2. Medial Nucleus.
3. Lateral nuclei
   a. Lateral dorsal.
   b. Lateral posterior.
   c. Pulvinar.

* Ventral	Nuclei
4. Ventral	Anterior
5. Ventral	lateral
6. Ventral	posterior lateral
7. Ventral	posterior medial
8. Lateral	geniculate body
9. Medial	geniculate body

THALAMUS

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CORPUS CALLOSUM

- LARGEST COMMISSURE OF THE BRAIN.
- CONNECT 2 CEREBRAL HEMISPHERE.

PART OF BRAIN CONNECTED -
- CONNECTS ALL PART OF CEREBRAL CORTEX OF 2X SIDE EXCEPT, LOWER AND ANTERIOR PART OF TEMPORAL LOBES - WHICH CONNECT BY ANTERIOR COMMISSURE

*PART OF CARPUS CALLOSUM-
1) GENU - 4CM BEHIND THE FRONTAL POLE
RELATION - ANTERIORLY - ANTERIOR CEREBRAL ARTERIES
POSTERIORLY - ANTERIOR HORN OF LATERAL VENTRICLE

2) ROSTRUM - DIRECTED- DOWNWARD AND BACKWARD FROM GENU

- END BY JOINING → LAMINA TERMINALIS IN FRONT
ANTERIOR COMMISSURE
- RELATION - SUPERIOR TO ANTERIOR HORN

3) TRUNK - MIDDLE PART, BETWEEN GENU AND SPLENIUM
OVERLAPPED BY → CINGULATE GYRUS.
4)SPLENIUM -6CM FRONT OF OCCIPITAL POLE
- RELATION - INFERIOR SURFACE - TELA CHOROIDEA
- SUPERIOR SURFACE- INFERIOR SAGITTAL SINUS

*FIBRES OF CORPUS CALLOSUM-
*ROSTRUM CONNECT - ORBITAL SURFACE OF 2X FRONTAL LOBES
*FORCEPS MINOR CONNECT - 2X FRONTAL LOBES (MADE UP OF FIBRES OF GENU)
*FORCEPS MAJOR CONNECT - 2 X OCCIPITAL LOBES (MADE UP OF FIBRES OF SPLENIUM
*TAPETUM →(FORMED BY SOME FIBRES FROM TRUNK & SPLENIUM OF CORPUS)
IT'S FORM - ROOF AND LATERAL WALL OF POSTERIOR HORN
→ LATERAL WALL OF INFERIOR HORN

WHITE MATTER OF CEREBRUM

- WHITE MATTER OF CEREBRUM CONSIST OF MYELINATED FIBRES

- WHICH CONNECT VARIOUS PART OF CORTEX TO ONE ANOTHER

- FIBRES ARE CLASSIFIED INTO 3X - ASSOCIATION FIBRES (ARCUATE)
- COMMISSURAL FIBRES
- PROJECTION FIBRES
- CONNECT DIFFERENT CORTICAL AREA OF SAME HEMISPHERE TO ONE ANOTHER
- SUBDIVIDED INTO 2X GROUP-

1) SHORT ASSOCIATION FIBRES - CONNECT, ADJACENT GYRI
2) LONG ASSOCIATION FIBRES - CONNECT, MORE WIDELY SEPARATED GYRI
*UNCINATE FASCICULUS- CONNECT TEMPORAL LOBES TO MOTOR SPEECH AREA AND ORBITAL CORTEX

*CINGULUM -> CONNECT CINGULATE GYRUS TO PARAHIPPOCAMPAL GYRUS

*SUPERIOR LONGITUDINAL FASCICULUS - CONNECT -> FRONTAL LOBES TO OCCIPITAL AND TEMPORAL LOBES

*INFERIOR LONGITUDINAL FASCICULUS CONNECT -> OCCIPITAL LOBE TO TEMPORAL LOBES
3) COMMISSURE FIBRES - CONNECT - CORRESPONDING PARTS 2 HEMISPHERES
4) CORPUS CALLOSUM - CONNECTING CEREBRAL CORTEX OF TWO SIDES
5) ANTERIOR COMMISSURE - ARCHIPALLIA OF BOTH SIDES
6) POSTERIOR COMMISSURE - SUPERIOR COLLICULI
7) COMMISSURE OF FORNIX → CONNECTING THE CRURA OF FORNIX

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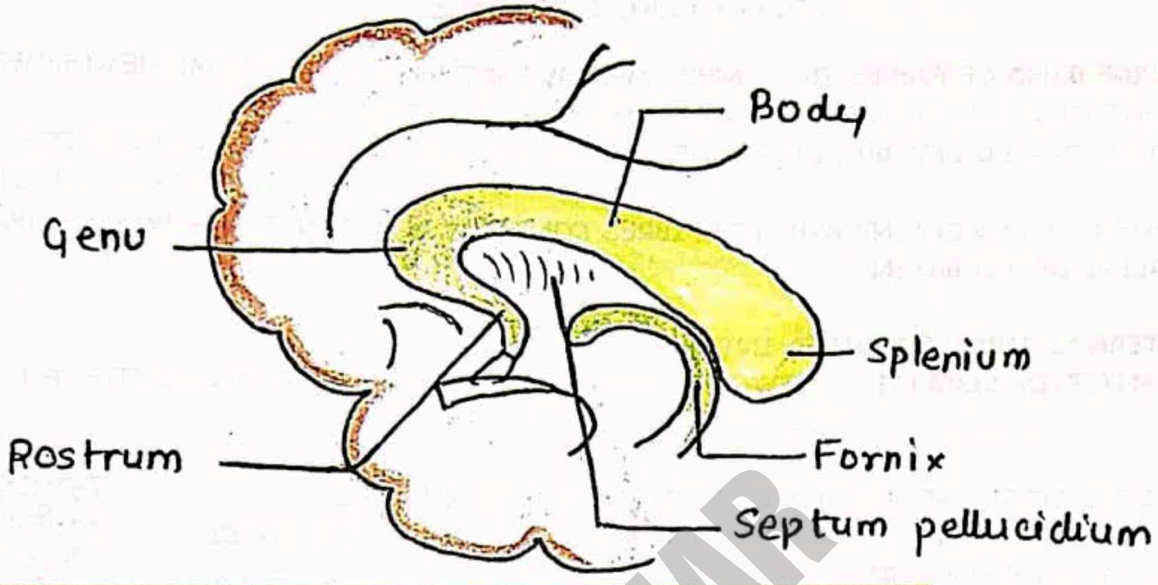
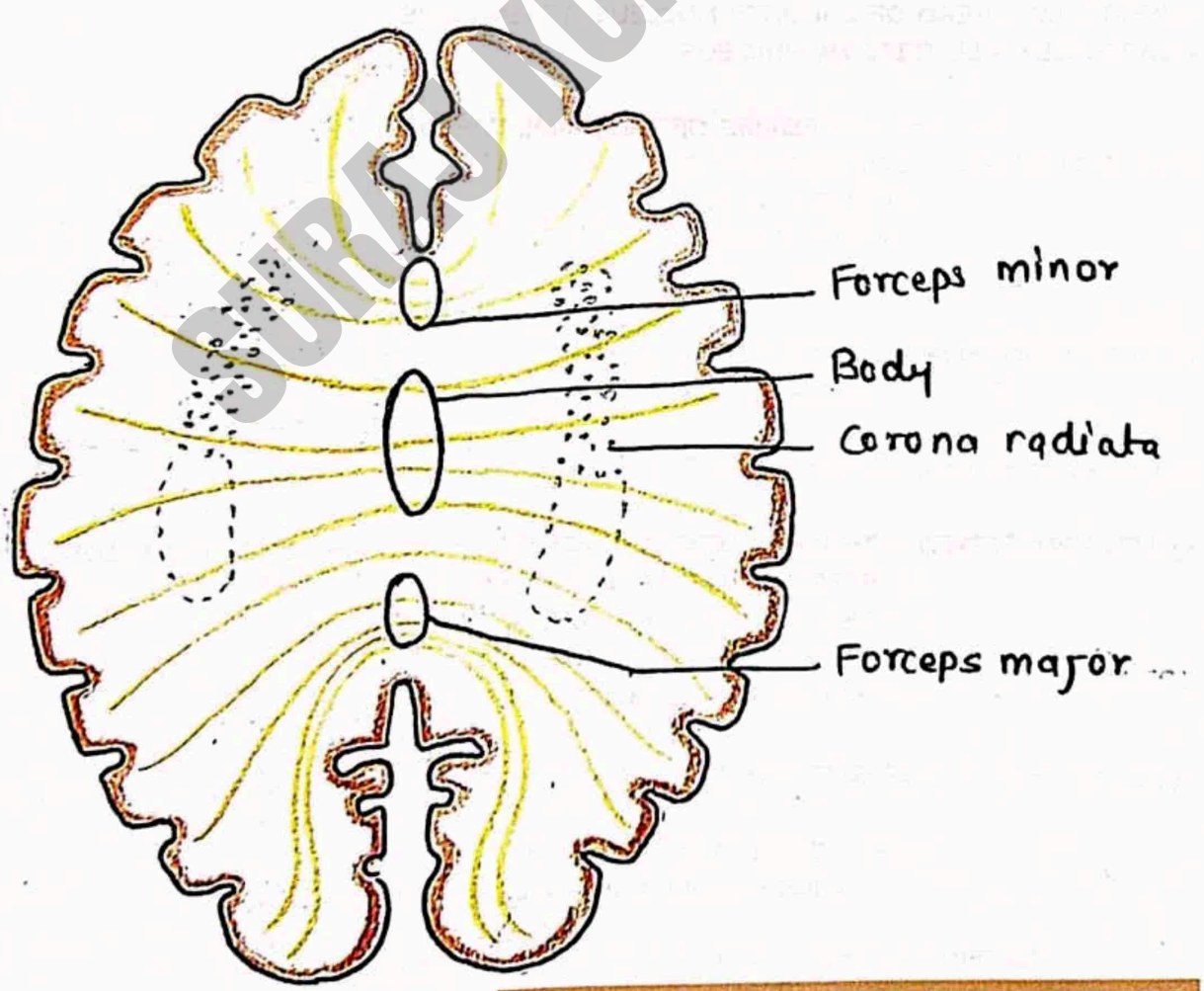


CORPUS CALLOSUM



WHITE MATTER OF CEREBRUM

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INTERNAL CAPSULE

-LARGE BAND OF FIBRES IN - INFEROMEDIAL PART OF EACH CEREBRAL HEMISPHER
-CONTAINS FIBRES GOING TO AND COMING FROM CEREBRAL CORTEX
-WHEN TRACED UPWARD, FIBRES OF CAPSULE DIVERGE AND CONTINUOUS WITH CORONA RADIATA

-WHEN TRACED DOWNWARD, ITS FIBRES CONVERGE AND CONTINUE- WITH - CRUS CEREBRI OF MIDBRAIN

INTERNAL CAPSULE DIVIDED INTO :-
1) ANTERIOR LIMB LIE - BETWEEN HEAD OF CAUDATE NUCLEUS AND LENTIFORM NUCLEUS

2) GENU BEND - BETWEEN ANTERIOR AND POSTERIOR LIMB
3) POSTERIOR LIMB → BETWEEN THALAMUS AND LENTIFORM NUDEI
4) SUB LENIFARM PART - LIE BELOW LENTI FARM NUDEUS
5) RETROLENTIFORM PART - LIE BEHIND LENTIFORM NUCLEUS

*RELATION :-
- MEDIALLY - HEAD OF CAUDATE NUCLEUS & THALAMUS
- LATERALLY - LENTIFORM NUCLEUS

FIBRES OF INTERNAL CAPSULE :-

1) MOTOR FIBRES - CORTICOPONTINE
FRONTOPONTINE
PARIETOPONTINE
TEMPOROPONTINE

2)-PYRAMIDAL FIBRES → CORTICONUCLEAR
→ CORTICOSPINAL

3) EXTRAPYRAMIDAL FIBRES - CORTICORUBRAL FIBRE & CORTICOSTRIATE

4) SENSORY FIBRES - THALAMOCORTICAL FIBRES FROM THALAMIC RADIATION
- ANTERIOR THALAMIC RADIATION
- SUPERIOR THALAMIC RADIATION
- INFERIOR THALAMIC RADIATION
- POSTERIOR THALAMIC RADIATION

*BLOOD SUPPLY - ANTERIOR LIMB - BRANCH OF ANTERIOR CEREBRAL
---→ ARTERY FROM RECCURENT BRANCH
- GENU - BRANCH OF INTERNAL CAROTID
- BRANCH OF POSTERIOR COMMUNICATING

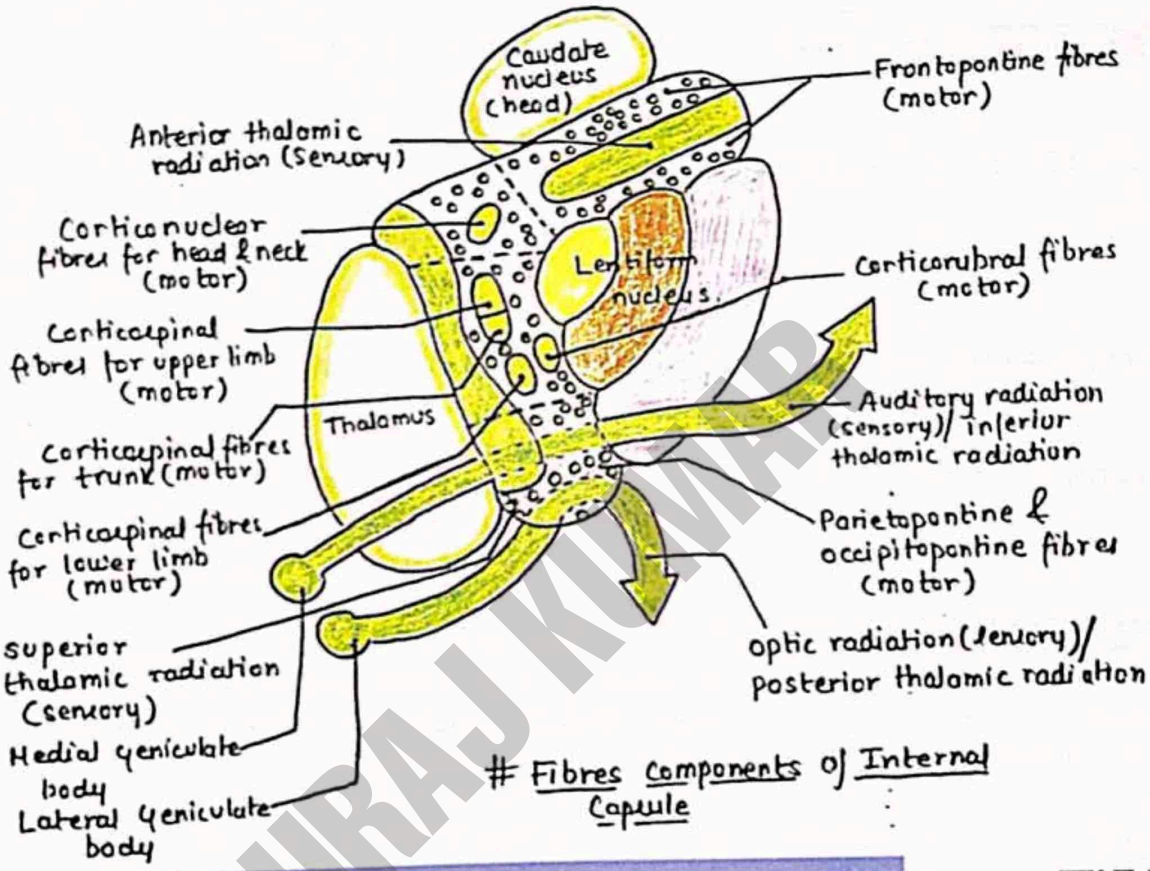
POSTERIOR LIMB → BRANCH OF ANTERIOR CHOROIDAL

SUBAL KUMAR

- BRANCH OF POSTERIOR COMMUNICATING
- BRANCH OF LATERAL AND MEDIAL STRIATE ARTERIES

SUBLENTIFORM → BRANCH OF ANTERIOR CHOROIDAL
-BRANCH OF POSTERIOR CEREBRAL

RETROLENTI FORM → BRANCH OF POSTERIOR CEREBRAL



Fibres components of Internal Capsule

CIRCLE OF WILLIS

*HEXAGONAL ARTERIAL CIRCLE
* SITUATED -->BASE OF BRAIN→IN INTERPEDUNCULAR FOSSA
* FORMED BY - ANTERIOR CEREBRAL-->BRANCH INTERNAL CAROTID
- TERMINAL PART OF INTERNAL CAROTID ARTERIES
- POSTERIOR CEREBRAL-→BRANCH BASILAR ARTERY

-TWO ANTERIOR CEREBRAL ARTERIES-->CONNECTED BY ANTERIOR COMMUNICATING ARTERY.

-INTERNAL CAROTID & POSTERIOR CEREBRAL ARTERY OF SAME SIDE-->UNITED BY POSTERIOR COMMUNICATING ARTERY.

*FORMATION - ANTERIORLY - ANTERIOR COMMUNICATED ARTERY
- ANTEROLATERALLY - ANTERIOR CEREBRAL ARTERY
- LATERALLY - INTERNAL CAROTID ARTERIES
- POSTEROLATERALLY - POSTERIOR COMMUNICATING ARTERY
- POSTERIORLY - POSTERIOR CEREBRAL ARTERIES

SURAJ KUMAR

*BRANCHES - CORTICAL BRANCHES-->SUPPLY CEREBRUM(ANASTOMOSE FREELY)
- CENTRAL BRANCHES-->THALAMUS
-CARPUS STRIATUM (DOESN'T ANASTOMOSE)
-INTERNAL CAPSULE
- CHOROIDAL BRANCH-->CHOROID PLEXUSES OF VENTRICLE

1.CORTICAL BRANCHES -->ARE - ANTERIOR CEREBRAL ARTERIES
- MIDDLE CEREBRAL ARTERIES
- POSTERIOR CEREBRAL ARTERIES

SUPPLY - CEREBRUM --- >SUPEROLATERAL SURFACE--->MAINLY BY MIDDLE CEREBRAL ARTERY
*EXCEPT - OCCIPITAL LOBE - PCA
- TEMPORAL LOBE - PCA

→ MEDIAL SURFACE → MAINLY BY ANTERIOR CEREBRAL ARTERY
*EXCEPT - OCCIPITAL LOBE------- >PCA
- TEMPORAL LOBE ------- >PCA
→ INFERIOR SURFACE----> BY POSTERIOR CEREBRAL ARTERY
*EXCEPT - MEDIAL 1/3 ORBITAL LOBE-ACA
- LATERAL 2/3 TEMPORAL LOBE-MCA

2.CENTRAL BRANCHES - ANTEROMEDIAL---->ARISES ANTERIOR CEREBRAL ARTERY
---->ENTER MEDIAL MOST PART OF ANTERIOR PERFORATED SUBSTANCE
-ANTEROLATERAL-->ARISES MIDDLE CEREBRAL ARTERY
->SUPPLY - MEDIATE STRIATE → CAUDATE NUCLEUS
→ INTERNAL CAPSULE
- LATERAL STRIATE → ENTER INTERNAL CAPSULE
- POSTEROMEDIAL → ARISE POSTERIOR CEREBRAL ARTERY
-POSTEROLATERAL--->ARISES POSTERIOR CEREBRAL ARTERIES
3.CHOROIDAL BRANCH - ANTERIOR CHOROIDAL--->BRANCH INTERNAL CAROTID A.
----->SUPPLIES INFERIOR HORN OF LATERAL VENTRICLE

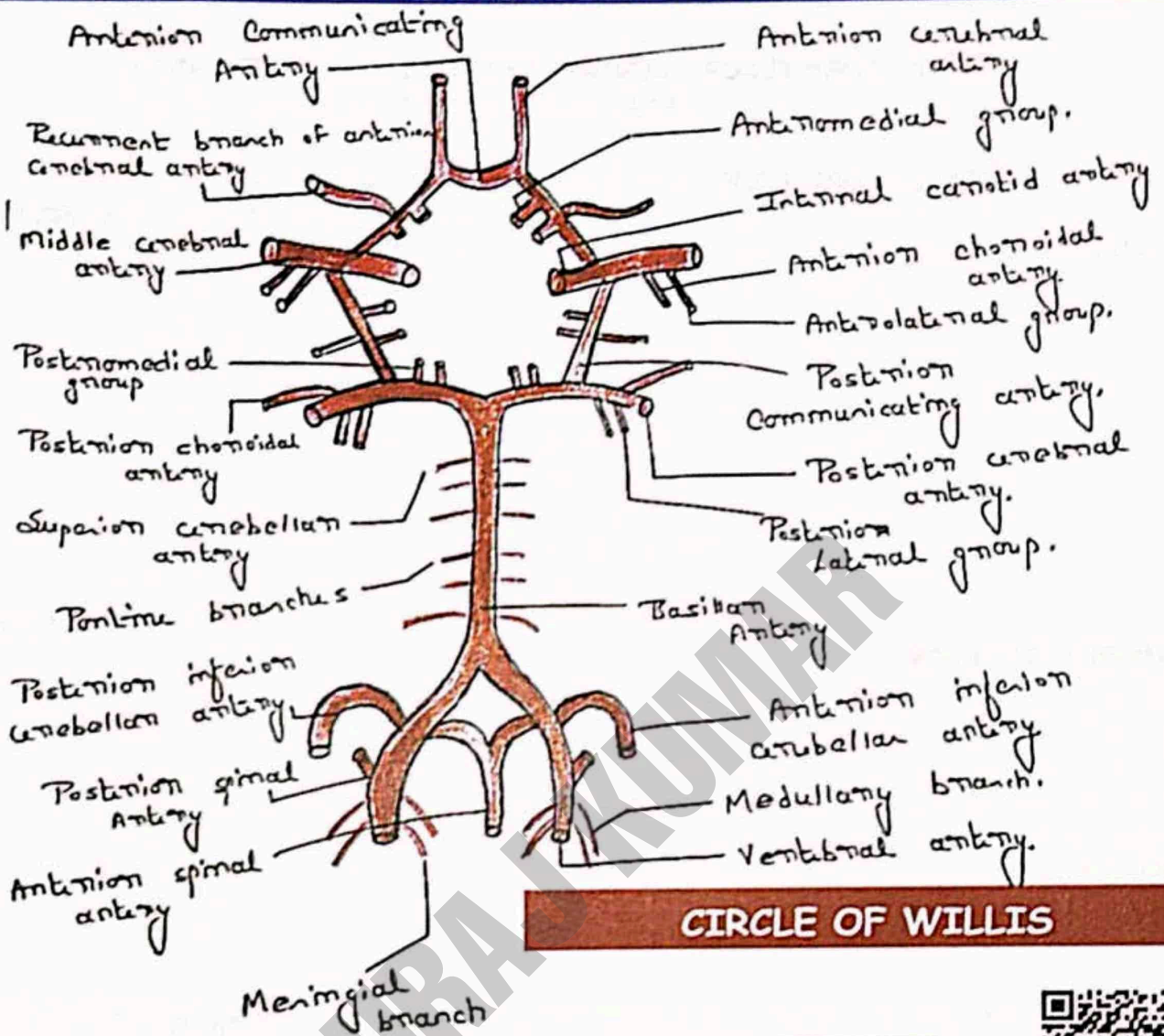
- POSTERIOR CHOROIDAL---->BRANCH POSTERIOR CEREBRAL
------->SUPPLIES REST PART OF CHOROID PLEXUS OF LATERAL VENTRICLE

CLINICAL ANATOMY

*THROMBOSIS OF THE PARACENTRAL ARTERY (TERMINAL CORTICAL BRANCH OF THE ANTERIOR CEREBRAL ARTERY) CAUSES CONTRALATERAL LOWER LIMB MONOPLEGIA.

*THROMBOSIS OF LATERAL STRIATE BRANCHES OF MIDDLE CEREBRAL ARTERY CAUSES MOTOR AND SENSORY LOSS TO MOST OF THE OPPOSITE SIDE OF BODY EXCEPT LOWER LIMB.

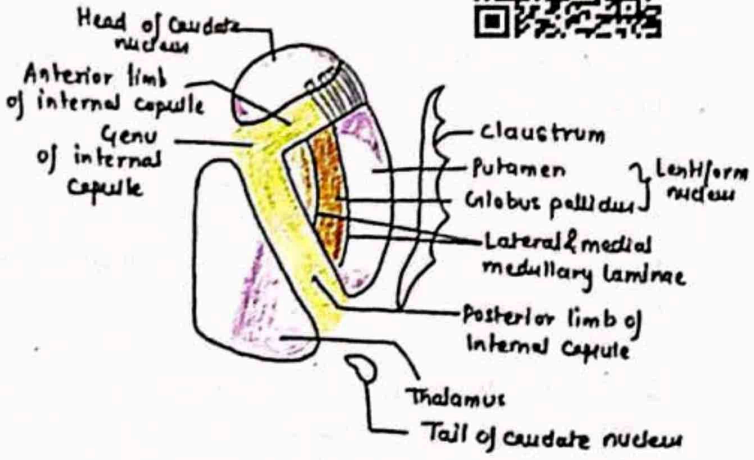
SUBAJ KUMAR



CIRCLE OF WILLIS

SUBA KUMAR

BASAL NUCLEI



- BASAL NUCLEI ARE SUBCORTICAL, INTRACEREBRAL MASSES OF GREY MATTER
- INCLUDES 1) CORPUS STRIATUM -
  - *CAUDATE NUCLEUS
  - *TENTIFORM NUCLEUS
    - GLOBUS
    - PALLIDUS
  2) AMYGDALOID BODY
  3) CLAUSTRUM

1) CARPUS STRIATUM :

*CAUDATE NUCLEUS - C-SHAPED NUDEUS - SURROUNDED - LATERAL VENTRICLE
- HEAD, BODY AND TAIL

- RELATION - HEAD - FORM FLOOR - ANTERIOR HORN OF LATERAL VENTRICLE
-MEDIAL WALL OF ANT. LIMB OF INTERNAL CAPSULE

MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
FIRST YEAR MBBS 2021
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE ARTERIAL SUPPLY AND VENOUS DRAINAGE OF HEART — ANS (1) – PAGE NO.75 V1
2. DESCRIBE PALATINE TONSIL UNDER THE FOLLOWING HEADINGS: GROSS FEATURES INCLUDING DEEP RELATIONS, HISTOLOGY AND APPLIED ANATOMY — ANS (2) – PAGE NO.90 V2
3. A 30 YEAR OLD MALE PRESENTED TO THE EMERGENCY DEPARTMENT AFTER A ROAD TRAFFIC ACCIDENT WITH COMPLAINING OF RIGHT ARM PAIN AND INABILITY TO MOVE HIS EXTREMITY. THE FOLLOWING MORNING, HE DEVELOPED CLAWING. NERVE CONDUCTION STUDIES ON THE PERIPHERAL NERVES OF HIS ARM CONFIRMED THE DIAGNOSIS OF INJURY TO THE ULNAR NERVE. KEEPING THIS SCENARIO IN MIND, ANSWER THE FOLLOWING QUESTIONS: ORIGIN AND ROOT VALUE OF THE NERVE, COURSE AND DISTRIBUTION, BRANCHES, EXPLAIN REASON FOR DEVELOPING THE CLAW HAND — ANS (3) – PAGE NO.13 V1

Q2 SHORT

1. CAROTID TRIANGLE — ANS (1) – PAGE NO.61 V2
2. FACIAL NERVE — ANS (2) – PAGE NO. V2
3. DESCRIBE MICROSCOPIC FEATURES OF TRACHEA — ANS (3) – PAGE NO.149 V1
4. WELL LABELLED DIAGRAM SHOWING THE TRANSVERSE SECTION OF UPPER PART OF PONS — ANS (4) – PAGE NO.112 V2
5. CUBITAL FOSSA — ANS (5) – PAGE NO.6 V1
6. CLASSIFICATION OF SYNOVIAL JOINT — ANS (6) – PAGE NO.110 V1
7. RESPIRATORY MOVEMENTS OF THE THORACIC WALL — ANS (7) – (EXTRA – PHYSIOLOGY)
8. DERIVATIVES OF FIRST PHARYNGEAL ARHC — ANS (8) – PAGE NO.93 V1
9. OPENINGS OF LATERAL WALL OF NOSE — ANS (9) – PAGE NO.99 V2
10. WRITE A NOTE ON OTIC GANGLION — ANS (10) – PAGE NO.137 V2

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FIRST YEAR MBBS 2021
ANATOMY PAPER 2

Q1 LONG

1. DESCRIBE INGUINAL CANAL UNDER THE FOLLOWING HEADS: BOUNDARIES, CONTENTS AND APPLIED ASPECTS — ANS (1) – PAGE NO.2 V2
2. DESCRIBE KNEE JOINT UNDER THE FOLLOWING HEADINGS: TYPE OF JOINT AND ARTICULAR SURFACE, CAPSULE AND LIGAMENTS, MOVEMENTS AND MUSCLES RESPONSIBLE AND APPLIED ANATOMY — ANS (2) – PAGE NO.52 V1
3. DESCRIBE DUODENUM UNDER THE FOLLOWING HEADINGS: PERITONEAL AND VISCERAL RELATIONS(FIRST, SECOND AND THIRD PART), HISTOLOGY, EXPLAIN ANATOMICAL BASIS OF DUODENAL CAP AND DUODENAL ULCER. — ANS (3) – PAGE NO.12 V2

Q2 SHORT

1. WRITE A NOTE ON VARIOUS POSITIONS OF APPENDIX — ANS (1) – PAGE NO.16 V2
2. LOCKING UNLOCKING MECHANISM OF KNEE JOINT — ANS (2) – PAGE NO.50 V1
3. ISCHIORECTAL FOSSA — ANS (3) – PAGE NO.36 V2
4. ADDUCTOR CANAL — ANS (4) – PAGE NO.37 V1
5. SUPPORT OF UTERUS — ANS (5) – PAGE NO.37 V2
6. GREAT SAPHENOUS VEIN — ANS (6) – PAGE NO.44 V1
7. HISTOLOGY OF OVARY — ANS (7) – PAGE NO.153 V1
8. PORTACAVAL ANASTOMOSIS (THREE IMPORTANT SITES) (DIAGRAM ONLY) AND MENTION THREE EFFECTS OF PORTAL HYPERTENSION — ANS (8) – PAGE NO.30 V2
9. DEVELOPMENT OF TESTIS — ANS (9) – PAGE NO.103 V1
10. MEDIAL LONGITUDINAL ARCH OF FOOT — ANS (10) – PAGE NO.40 V1

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FIRST YEAR MBBS 2020
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE MEDIAN NERVE UNDER FOLLOWING HEADS: ROOT VALUE, COURSE AND RELATIONS, DISTRIBUTION AND CLINICAL ANATOMY — ANS (1) – PAGE NO.15 V1
2. DESCRIBE BRONCHOPULMONARY SEGMENTS WITH ITS APPLIED — ANS (2) – PAGE NO.66 V1
3. CLASSIFY THE DURAL VENOUS SINUSES AND DESCRIBE CAVERNOUS SINUS IN DETAIL — ANS (3) – PAGE NO.85 V2

Q2 SHORT

1. DEVELOPMENT OF PALATE AND ITS ANOMALIES — ANS (1) – PAGE NO.97V1
2. SCALP — ANS (2) – PAGE NO.55 V2
3. MIDDLE EAR CAVITY — ANS (3) – PAGE NO.102 V2
4. SUPINATION AND PRONATION OF FOREARM — ANS (4) – PAGE NO.24 V1
5. HISTOLOGY OF THYROID GLAND — ANS (5) – PAGE NO.149 V1
6. ROTATOR CUFF OF SHOULDER JOINT — ANS (6) – PAGE NO.17 V1
7. TYPICAL INTERCOSTAL SPACE — ANS (7) – PAGE NO.80 V1
8. CAROTID TRIANGLE — ANS (8) – PAGE NO.61 V2
9. LATERAL NASAL WAAL — ANS (9) – PAGE NO.99 V2
10. SECRETOMOTOR PATHWAY OF PAROTID GLAND — ANS (10) – PAGE NO.70 V2

MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
FIRST YEAR MBBS 2020
ANATOMY PAPER 2

Q1 LONG

1. DESCRIBE THE EXTERNAL FEATURES, RELATIONS, LIGAMENTS, BLOOD SUPPLY AND DEVELOPMENTAL ANATOMY OF URINARY BLADDER. ADD A NOTE ON ITS APPLIED ANATOMY. — ANS (1) – PAGE NO.37 V2
2. DESCRIBE THE ROOT VALUE, COURSE, RELATIONS, BRANCHES AND DISTRIBUTION AND APPLIED ANATOMY OF OBTURATOR NERVE — ANS (2) – PAGE NO.39 V1
3. DESCRIBE THE TYPE, LIGAMENTS, RELATION, MOVEMENTS AND MUSCLES PRODUCING THE MOVEMENTS AND APPLIED ANATOMY OF HIP JOINT — ANS (3) – PAGE NO.49 V1

Q2 SHORT

1. LESSER SAC — ANS (1) – PAGE NO.47 V1
2. HISTOLOGY OF SPLEEN — ANS (2) – PAGE NO.150 V1
3. POPLITEAL FOSSA — ANS (3) – PAGE NO.39 V1
4. SUPPORTS OF UTERUS — ANS (4) – PAGE NO.40 V1
5. DESCENT OF TESTIS — ANS (5) – PAGE NO.103 V1
6. BLOOD SUPPLY OF STOMACH — ANS (6) – PAGE NO.9 V2
7. ADDUCTOR CANAL — ANS (7) – PAGE NO.37 V1
8. MEDIAL LONGITUDINAL ARCH OF FOOT — ANS (8) – PAGE NO.43 V1
9. PORTOCAVAL ANASTOMOSIS — ANS (9) – PAGE NO.30 V2
10. PROSTATIC PART OF URETHRA — ANS (10) – (EXTRA – BDC)

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SURAJ KUMAR

MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
FIRST YEAR MBBS 2019
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE MIDDLE EAR UNDER FOLLOWING HEADINGS A) LOCATION B)CONTENTS C) BOUNDARIES D) NERVE SUPPLY E) APPLIED ANS (1) – PAGE NO.102 V2
2. DESCRIBE BRACHIAL PLEXUS WITH ITS APPLIED ANATOMY ANS (2) – PAGE NO.3 V1

Q2 SHORT

1. SENSORY INNERVATION OF FACE ANS (1) – PAGE NO.54 V2
2. HISTOLOGICAL FEATURES OF PAROTID GLAND ANS (2) – PAGE NO.70 V2
3. LATERAL PTERYGOID MUSCLE ANS (3) – PAGE NO.88 V2
4. DEVELOPMENT OF TONGUE ANS (4) – PAGE NO.73 V1
5. SYNAPSE ANS (5) – (EXTRA – PHYSIOLOGY)
6. TRANSVERSE SECTION OF OPEN PART OF MEDULLA ANS (6) – PAGE NO.111 V2
7. MAJOR OPENINGS OF THORACIC DIAPHRAGM ANS (7) – PAGE NO.84 V1
8. THORACIC DUCT ANS (8) – PAGE NO.79 V1
9. AXILLARY ARTERY ANS (9) – PAGE NO.5 V1
10. CORONARY SINUS ANS (10) – PAGE NO.75 V1

MADHYA PRADESH MEDICAL SCIENE UNIVERSITY
FIRST YEAR MBBS 2018
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE CUBITAL FOSSA ANS (1) – PAGE NO.6 V1
2. DESCRIBE SCALP WITH ITS APPLIED ASPECTS ANS (2) – PAGE NO.55 V2

Q2 SHORT

1. TASTE PATHWAY ANS (1) – PAGE NO.92 V2 (TONGUE NERVE)
2. ULNAR NERVE IN HAND ANS (2) – PAGE NO.13 V1
3. MEDIASTINAL SURFACE OF RIGHT LUNG ANS (3) – PAGE NO.62 V1
4. CLASSIFICATION SYNOVIAL JOINTS ANS (4) – PAGE NO.110 V1
5. RESPIRATORY EPITHELIUM ANS (5) – PAGE NO.151 V1
6. MORULA ANS (6) – (EXTRA)
7. BLOOD SUPPLY OF THYROID GLAND ANS (8) – PAGE NO.73 V2
8. ANATOMICAL SNUFF BOX ANS (8) – PAGE NO.19 V1
9. SECTION OF MEDULLA AT THE LEVEL OF MOTOR DECUSSATION ANS (9) – PAGE NO.111 V2
10. THORACIC DUCT ANS (10) – PAGE NO.79 V1

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FIRST YEAR MBBS 2017
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE BROCHOPULMONARY SEGMENTS WITH ITS APPLIED ASPECTS ANS (1) – PAGE NO.66 V1
2. DESCRIBE CAROTID TRIANGLE ANS (2) – PAGE NO.61 V2

Q2 SHORT

1. INTERIOR OF RIGHT ATRIUM ANS (1) – PAGE NO.73 V1
2. SECRETOMOTOR PATHWAY OF PAROTID GLAND ANS (2) – PAGE NO.70 V2
3. DIAGRAM OF FLOOR OF 4TH VENTRICLE OF BRAIN ANS (3) – PAGE NO.147 V2
4. MUSCULOCUTANEOUS NERVE IN ARM ANS (4) – (EXTRA)
5. SUPINATION AND PRONATION OF FOREARM ANS (5) – PAGE NO.24 V1
6. RELATIONS IN CORONAL SECTION OF CAVERNOUS SINUS ANS (6) – PAGE NO.85 V2
7. CIRCLE OF WILLIS ANS (7) – PAGE NO.153 V2
8. SPERMATOGENESIS ANS (8) – (EXTRA)
9. CLASSIFICATION OF SYNOVIAL JOINTS ANS (9) – PAGE NO.110 V1
10. UROTHELIUM ANS (10) – PAGE NO.153 V1

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FIRST YEAR MBBS 2019
ANATOMY PAPER 2

Q1 LONG

1. DESCRIBE THE 2ND PART OF DUODENUM UNDER FOLLOWING HEADS: RELATIONS, DEVELOPMENT, ARTERIAL SUPPLY INTERIOR AND APPLIED ASPECT ANS (1) – PAGE NO.12 V2
2. DESCRIBE URINARY BLADDER UNDER FOLLOWING HEADS: RELATIONS, EXTERNAL FEATURES, SUPPORTS ARTERIAL AND NERVE SUPPLY AND APPLIED ASPECT ANS (2) – PAGE NO.37 V2

Q2 SHORT

1. DEFENSE MECHANISM OF INGUINAL CANAL ANS (1) – PAGE NO.2 V2
2. RECTOUTERINE POUCH ANS (2) – (EXTRA)
3. HISTOLOGY OF PANCREAS ANS (3) – PAGE NO.152 V1
4. KLINEFELTERS SYNDROME ANS (4) – PAGE NO.112 V1
5. PUDENDAL CANALANS (5) – PAGE NO.36 V2 (ISCHIOANAL FOSSA)
6. DEVELOPMENT OF KIDNEY ANS (6) – PAGE NO.81 V1
7. NOTOCHORD ANS (7) – PAGE NO.90 V1
8. MEDIAL LONGITUDINAL ARCH OF FOOT ANS (8) – PAGE NO.42 V1
9. TIBIALIS ANTERIOR MUSCLE
10. ADDUCTOR CANAL ANS (10) – PAGE NO.37 V1

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FIRST YEAR MBBS 2018
ANATOMY PAPER 2

Q1 LONG

1. DESCRIBE SPLEEN UNDER FOLLOWING HEADS: SITUATIONANS (1) – PAGE NO.48 V2 RELATIONS, EXTERNAL FEATURES AND APPLIED ASPECT
2. DESCRIBE UTERUS UNDER FOLLOWING HEADS: SITUATION, RELATIONS, SUPPORT, BLOOD SUPPLY AND APPLIED ASPECT ANS (2) – PAGE NO.40 V2

Q2 SHORT

1. BOUNDARIES AND CONTENTS OF POPLITEAL FOSSA ANS (1) – PAGE NO.39 V1
2. TURNER'S SYNDROME ANS (2) – PAGE NO.112 V1
3. PORTOCAVAL ANASTOMOSIS ANS (3) – PAGE NO.30 V2
4. RELATIONS AND BLOOD SUPPLY OF SUPRARENAL GLAND
5. DEVELOPMENT OF PANCREAS ANS (5) – PAGE NO.99 V1
6. MICROANATOMY OF DUODENUM ANS (6) – PAGE NO.151 V1
7. DEFINE INVERSION AND EVERSION MOVEMENT AND MENTION THE JOINT INVOLVED ANS (7) – PAGE NO.52 V1 (KNEE JOINT)
8. TIBIALIS POSTERIOR
9. STOMACH BED ANS (3) – PAGE NO.9 V2
10. DESCENT OF TESTIS AND ANOMALIES ANS (10) – PAGE NO.103 V1

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FIRST YEAR MBBS 2017
ANATOMY PAPER 2

Q1 LONG

1. DESCRIBE VERMIFORM APPENDIX UNDER THE FOLLOWING HEADINGS: POSITION, RELATIONS, ARTERIAL SUPPLY, LYMPHATIC DRAINAGE AND APPLIED ASPECT. ANS (1) – PAGE NO.16 V2
2. DESCRIBE VENOUS DRAINAGE OF LOWER LIMB AND DISCUSS ITS APPLIED ASPECT ANS (2) – PAGE NO.44 V2

Q2 SHORT

1. OBTURATOR NERVE- ORIGIN, COURSE AND DISTRIBUTION ANS (1) – PAGE NO.39 V1
2. PELVIC DIAPHRAGM ANS (2) – PAGE NO.46 V2
3. INGUINAL CANAL ANS (3) – PAGE NO.5 V2
4. EXTRAHEPATIC BILIARY APPARATUS ANS (4) – PAGE NO.28 V2
5. MICRO ANATOMY OF THE PROSTATE ANS (5) – PAGE NO.153 V1
6. FEMORAL ARTERY-ORIGIN, COURSE AND DISTRIBUTION ANS (6) – PAGE NO.35 V1
7. DEVELOPMENT OF THE ANAL CANAL ANS (7) – PAGE NO.81 V1
8. ISCHIORECTAL FOSSA ANS (8) – PAGE NO.36 V2
9. RECTO-UTERINE POUCH ANS (9) – (EXTRA)
10. FALLOPIAN TUBEANS (10) – PAGE NO.40 V2

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